Probing valence electron dynamics in gas and condensed-phase matter with attosecond X-ray Free Electron Lasers

Felix Egun

Supervisor: Professor Jonathan P. Marangos

Extreme Light Consortium

Department of Physics

Faculty of Natural Sciences

Imperial College London

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IMPERIAL

"Das ist nicht nur nicht richtig; es ist nicht einmal falsch!"

→ Translation: *"That is not only not right; it is not even wrong!"*

~ W. Pauli, the Swiss-Austrian Physicist dismissing a result he could not make sense of, and my own thoughts many times during my doctorate...

W. Pauli in 1953, during his appointment to the Royal Society of London. Famously direct and indelicate with his criticism what he considered poor scientific arguments, he proposed the neutrino to resolve apparent inconsistencies between predictions and observations regarding the weak interaction and the "Pauli-Ausschlussprinzip" or Exclusion Principle for fermions. For the latter, he was awarded the Nobel Prize in Physics in 1945.

Abstract

In both isolated and condensed-phase systems, molecular interactions are largely governed by the behavior of valence electrons. Understanding the physics of how these electrons behave is therefore central to explaining the most fundamental chemical processes - bond formation and bond cleavage. When a molecule interacts with a photon, it can result in a disturbance from its stable equilibrium configuration. This thesis presents two approaches to observe and characterize the subsequent evolution of valence electrons following such a sudden perturbation.

One approach, impulsive stimulated x-ray Raman scattering (ISXRS), enables the creation of valence-excited states in neutral systems, with spectral modifications to the photon field serving as an observable.

The second approach employs an x-ray pump–x-ray probe scheme, which can generate coherent superpositions of cationic valence-hole states. Using x-ray absorption spectroscopy (XAS), the resulting electron-hole density can be measured in a site-specific manner. Both transmission XAS and electron-yield XAS, the latter accessed via the resonant Auger–Meitner process, are used as experimental observables.

To resolve the underlying electron dynamics, x-ray pulses with durations comparable to or less than the natural timescale of valence electron motion are required. This work makes use of attosecond to few-femtosecond special operating modes of x-ray free-electron lasers (XFELs). These facilities are essential for this work, as they offer a unique combination of spectral brightness, intensity, and pulse duration necessary for probing these dynamics.

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Declaration of Originality

The work contained in this thesis was all performed by me except where credit is given to explicitly referenced papers or disclaimers are given. This thesis has not previously been submitted for a degree or diploma in any university or institution.

Contributions

This experimental work presented in this thesis was collected at three XFEL beamtimes. The participants and contributions made to each experiment are listed here.

The data measuring Impulsive Stimulated X-ray Raman scattering in condensed phase in sections V.2, V.3 and V.4 was conducted at the ChemRIXS beamline at the Linac Coherent Light source (LCLS) at the Stanford Linear Accelerator center (SLAC) National Accelerator Laboratory in experimental run 22.. This experiment was conducted in May 2022. List of collaborators: Agostino Marinelli, Ryan N. Coffee, Georgi L. Dakovski, Daniel DePonte, Douglas Garratt, Daniel J Haxton, Emily Thierstein, Jacob Peter Lee, James P. Cryan, Jordan Tyler Oneal, Jonathan Marangos, Jacob Lee Scott, James L. Turner, James Turner, Kaiyu Hu, Kasia Marianna Kowalczyk, Toby Barnard, Kaixiang Zhao, Kirk A Larsen, Laura Rego Cabezas, Ryan N. Coffee, Ryan N. Coffee, Leszek J. Frasinski, Ming-Fu Lin, Nora Berrah, Oliver George Alexander, Philip H. Bucksbaum, Ru-Pan Wang, Sandra Beauvarlet, Stefan P. Moeller, Taran Driver and Xu Yao. In this experiment the analysis was performed by Oliver Alexander, Douglas Garett and I.

The data presented in section VI.2 on the para-aminophenol system was measured at the Time-resolved Atomic and Molecular physics instrument also at the Linac Coherent Light source (LCLS) at the Stanford Linear Accelerator center (SLAC) National Accelerator Laboratory during an LCLS-II early science. This experiment was conducted in June 2024. List of collaborators: Agostino Marinelli, Amanda Jean Shackelford, Adam M Summers, Cameron Brady, David B. Cesar, Emily Thierstein, Gilles Doumy, Erik Isele, James P. Cryan, Debadarshini Mishra, Jonathan Marangos, Jun Wang, Kurtis Borne, Christopher Anthony Lantigua, Kirk A Larsen, Ming-Fu Lin, Madhu Trivikram Talluri, Oliver George Alexander, Oliver Gessner, Paris Lee Franz, Philip H. Bucksbaum, River Robles, Razib Obaid, Artem Rudenko, Sandra Beauvarlet, Sarvesh Kumar, Smita Oman Karnath Ganguly, Thomas Jacob Arcangelo Wolf, Thorsten

Weber, Xiang Li and James Zhong Manis. The analysis shown is purely done my myself.

The data presented in section VI.3 on liquid phase glycine as conducted at the FL.24 beamline at the FLASH-II XFEL at Deutsches Elektronen-Synchrotron (DESY) in September 2023. List of collaborators: Jacob Lee, Oliver Alexander, Andreas Przystawik, Dian Diaman, David Schwickert, Slawomir Skruszewicz, Jonathan Marangos, Rose Picciuto, Hortense Allegre, Katarzyna Kowalczyk, Stefan Düsterer, Marcel Brockmann and Sven Toleikis and Tim Laarmann. The analysis shown is purely done by myself.

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Chapter I

Fundamentals of Quantum Mechanics and motivations for attosecond physics

I.1 The semiclassical Bohr atomic model

In 1913, Niels Bohr proposed a model of the atom that was the first to explain the discrete nature of the hydrogen atom's energy spectrum.[1] He combined aspects of classical mechanics and some early notions of quantum mechanics, namely that light may be described as discrete packets, or quanta, known as photons. His semi-classical model was a large step forward in the understanding of atomic spectroscopy and therefore, the development of the quantum theory of the atom.[2]

At the time, it was thought that atoms were composed of a dense, positively-charged nuclear core with a negatively charged electron situated in some orbits, bound together via the Coulomb interaction.[3]

Bohr assumed the electron orbits to be circular centered on the nuclear mass. He found that the electron orbital radii must be constrained as

$$r_e = \frac{1}{4\pi\epsilon_0}\frac{Ze^2}{\mu v^2}, \tag{I.1}$$

where v is the orbital speed, $\mu = m_n m_e/(m_n + m_e)$ the reduced mass with m_e and m_n the electron and nuclear mass respectively and the approximation $\mu \approx m_e$ has been used.[4] The electron's kinetic energy is given by $\mu v^2/2$ and is proportional to $1/r_e$. This implies that if the electron kinetic energy can take any value, then so can the radius and vice versa.

It was known however, that charges undergoing acceleration emit radiation, and thus lose energy. This effect would suggest that no atom could be stable, in conflict with observation. Bohr then added a constraint to his model. He claims the electron may be described by a standing wave with wavelength $\lambda_{\text{electron}} = \hbar/(\mu v)$, and that the electron orbital circumference must be an integer n times $\lambda_{\text{electron}}$. Any constraint on the permitted radii of the electron orbit, directly translates into a constraint on its kinetic energy and hence its orbital speed v_n.[5] Therefore, the allowed orbital radii $r_e^{(n)}$ and corresponding speeds v_n read

$$r_e^{(n)} = \frac{1}{Z}\frac{\hbar^2\epsilon_0}{\pi\mu e^2} \times n^2 = \frac{1}{Z}a_0 \times n^2. \qquad v_n = \frac{Ze^2\mu}{2\epsilon_0\hbar m_e} \times \frac{1}{n} \approx \frac{Z\hbar}{2\pi a_0 m_e}\frac{1}{n}. \tag{I.2}$$

The electron orbital time period $\tau_n = 2\pi r_e^{(n)}/v_n$ is therefore given by

$$\tau_n = \frac{4\pi^2 a_0^2 m_e}{Z^2\hbar} \times n^3. \tag{I.3}$$

For the case of, for example, a hydrogen atom $Z = 1$ and $r_e^{(n)} \approx a_0$, the orbit time is 152 as for the lowest lying orbit with $n = 1$.

Bohr's work uses classical mechanics and the insight that light can be composed of photons with discrete amounts of energy proportional to their frequency.[6] From this simplistic treatment of the electron-proton system, we can estimate the typical timescale of electron motion. While this analysis incorrectly treats the electron with a well-defined position and momentum, it enables us to understand the characteristic length and timescales of electrons when confined in atomic systems due to the Coulomb interaction. However, for a more in depth treatment of the physics required for this thesis, we must transition to a quantum mechanical treatment of the problem. We will now see how ultrafast electron motion can be understood by treating the electron-nuclear ensemble as a quantum mechanical system.

I.2 Fundamental properties of the wavefunction

The central object in quantum mechanics is the wavefunction, most commonly denoted $|\Psi\rangle$. This object contains all of the information pertaining to the system it describes. It describes the quantum state of a system, allowing observables (such as position, momentum, energy etc...) to be calculated. The central object of quantum mechanics is to predict the time evolution

of the wavefunction and extract the predicted value of measurable quantities of the system, known as observables. These observables are represented mathematically by objects called operators.

The Born rule states how given a system prepared in a state $|\Psi(t)\rangle$, an observable Q may be calculated. The operator $\mathbf{Q}$ corresponding to the observable Q will have corresponding eigenstates $|Q_i\rangle$ with the property

$$\mathbf{Q}|Q_i\rangle = Q_i|Q_i\rangle, \tag{I.4}$$

where Q_i is an eigenvalue of the state $|Q_i\rangle$. Assuming $|\Psi(t)\rangle$ is normalized such that $\langle\Psi(t)|\Psi(t)\rangle = 1$, the probability of a measurement of Q on the system is

$$\mathcal{P}(Q = Q_i) = a^*_{Q_i} a_{Q_i} \tag{I.5}$$

and we expect $\sum_i a^*_{Q_i} a_{Q_i} = 1$. Here, a_{Q_i} is the inner product $a_{Q_i} = \langle Q_i|\Psi(t)\rangle$. Expressing $|\Psi(t)\rangle$ in a coordinate basis $|\mathbf{x}\rangle$ is achieved by $\langle\mathbf{x}|\Psi(t)\rangle$ denoted here as $\psi(\mathbf{x}, t)$. This generally complex-valued object from here onwards will be referred to as the *wavefunction*. In quantum mechanics $\psi(\mathbf{x}, t)$ may be used to describe an electron wavefunction.[7] Given that $|\Psi(t)\rangle$ is normalizable, the coordinate representation of the wavefunction defines the spatial probability density $\mathcal{P}(\mathbf{x}, t)$ as

$$\mathcal{P}(\mathbf{x}, t) = \psi(\mathbf{x}, t)^*\psi(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2. \tag{I.6}$$

If a system may be prepared in normalized states $\psi_j(\mathbf{x})$ which form the eigenbasis of some operator $\mathbf{Q}$ such that $\mathbf{Q}|\psi_j\rangle = Q_j|\psi_j\rangle$, then a system may also be prepared in a linear combination of these states. Such a linear combination is formally known as a superposition. Such a superposition state $\Psi(\mathbf{x})$ can be written as

$$\Psi(\mathbf{x}) = \sum_j c_j\psi_j(\mathbf{x}), \tag{I.7}$$

where c_j are complex valued prefactors which contain information as to the relative contribution of each $\psi_j(\mathbf{x})$ to the sum and their phase-relationship. The superposition state will have its own associated observables whose probability of measurement may be computed by

a procedure similar to that described in equation (I.5) as

$$\begin{aligned} a_{Q_j} = \langle \psi_j | \Psi \rangle &= \int d^3\mathbf{x} \, \langle \psi_j | \mathbf{x} \rangle \langle \mathbf{x} | \Psi \rangle \\ &= \int d^3\mathbf{x} \, \psi_j^*(\mathbf{x}) \Psi(\mathbf{x}) \\ \implies \mathcal{P}(Q = Q_j) &= \left| \int d^3\mathbf{x} \, \psi_j^*(\mathbf{x}) \Psi(\mathbf{x}) \right|^2, \end{aligned} \tag{I.8}$$

where completeness over the basis $|\mathbf{x}\rangle$ has been leveraged.[8] Probability amplitudes a_{Q_i} found for superpositions of states using equation (I.8) read as

$$\begin{aligned} a_{Q_j} &= \int d^3\mathbf{x} \, \psi_j^*(\mathbf{x}) \sum_\lambda c_\lambda \psi_\lambda(\mathbf{x}) \\ &= \sum_\lambda c_\lambda \underbrace{\int d^3\mathbf{x} \, \psi_j^*(\mathbf{x}) \psi_\lambda(\mathbf{x})}_{\delta_{\lambda j}} \\ &= c_j \\ \implies \mathcal{P}(Q = Q_j) &= |c_j|^2, \end{aligned} \tag{I.9}$$

where $\Psi(x)$ has been re-expressed using its definition given in equation (I.7) and the orthonormalized nature of the functions $\psi_j(x)$ have been exploited.[9] The consequence of this result shows that the weighting of ψ_j in the superposition state, c_j, directly informs the probability of observing the value Q_j. In this way, information about the composition of superposition state may be obtained by measuring $\mathcal{P}(Q = Q_j)$.

I.3 Time evolution of quantum states

In non-relativistic systems, the time-dependent Schrödinger Equation describes the time evolution of a wavefunction. The Schrödinger equation formally reads

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \mathcal{H} |\Psi(t)\rangle, \tag{I.10}$$

where $\mathcal{H}$ is the total energy operator of the system, known as the Hamiltonian. This Hermitian operator includes a kinetic description of the state and any potentials the system may be subject

to and typically reads

$$\mathcal{H} = \frac{1}{2m}\mathbf{P}^2 + V(\mathbf{r}), \tag{I.11}$$

where $\mathbf{P}$ is the momentum operator and V is a potential.[10] We consider a separable wavefunction written in the coordinate representation $\Psi(\mathbf{x}, t)$ that may be expressed by the product

$$\Psi(\mathbf{x}, t) = \phi(\mathbf{x}) \times \mathcal{T}(t). \tag{I.12}$$

Using standard techniques of separable partial differential equations we arrive at a solution to the time dependent component of the wavefunction $\mathcal{T}(t)$ is

$$\mathcal{T}(t) \propto \exp\left(-\frac{iEt}{\hbar}\right). \tag{I.13}$$

Using this we obtain

$$i\hbar\frac{\partial}{\partial t}(\phi\mathcal{T}) = \mathcal{H}(\phi\mathcal{T}) \tag{I.14}$$

$$\mathcal{H}\phi = E\phi, \tag{I.15}$$

where we have assumed that $\mathcal{H}$ is time-independent.[11] Here, we can recognize equation (I.15) as the time-independent variant of the Schrödinger Equation, with E as an eigenvalue of the Hamiltonian $\mathcal{H}$ corresponding to the energy associated with eigenstate $\phi(\mathbf{x})$. We can therefore ascertain that the full time-dependent wavefunction in equation (I.12) for a time-independent Hamiltonian may be expressed as

$$\Psi(\mathbf{x}, t) = \phi(\mathbf{x}) \times \exp\left(-\frac{iEt}{\hbar}\right). \tag{I.16}$$

Now we consider an n-level atomic system, which may be described by a superposition of up to n energy eigenstates of a Hamiltonian. These eigenstates will be described by wavefunctions $\{\phi_j(\mathbf{x})\}$ and have associated energy eigenvalues $\{E_j\}$. The time-dependent Schrödinger equation has solutions for energy eigenstates $\phi(\mathbf{x})e^{-iEt/\hbar}$. Using its linearity property, we can then write the solution for a superposition of n states $\Psi(\mathbf{x}, t)$ as

$$\Psi(\mathbf{x}, t) = \sum_{j=1}^{n} c_j\phi_j(\mathbf{x}) \exp\left(-\frac{iE_jt}{\hbar}\right). \tag{I.17}$$

We can see from this expression that every energy eigenstate has a phase at time $t = 0$, given by a complex number c_j. This phase then evolves in time with a period of $2\pi\hbar/E_j$ for an eigenstate $\phi_j(\mathbf{x})$. Suppose a measurement of some observable $\mathcal{O}$ with corresponding operator $\mathbf{O}$ is made on the system. The expectation value for this observable in the state $\Psi(t)$ in the Schrödinger picture may be calculated formally as

$$\langle \mathbf{O} \rangle_\Psi(t) = \langle \Psi(t) | \mathbf{O} | \Psi(t) \rangle. \tag{I.18}$$

As we are working in the positions-space representation, this reduces to the integral

$$\begin{aligned} \langle \mathbf{O} \rangle_\Psi(t) &= \int d^3\mathbf{x}\, \Psi^*(\mathbf{x}, t) \mathbf{O}_{\mathbf{x}\mathbf{x}} \Psi(\mathbf{x}, t) \\ &= \int d^3\mathbf{x} \sum_{j,j'}^{n} c_j^* c_{j'} \exp\left(-\frac{i(E_{j'} - E_j)t}{\hbar} \right) \phi_j^*(\mathbf{x}) \mathbf{O}_{\mathbf{x}\mathbf{x}} \phi_{j'}(\mathbf{x}), \end{aligned} \tag{I.19}$$

where we define $\mathbf{O}_{\mathbf{x}\mathbf{x}'} = \langle \mathbf{x} | \mathbf{O} | \mathbf{x}' \rangle$ and assume its non-diagonal components (where $\mathbf{x} \neq \mathbf{x}'$) are all zero.[12] Now suppose the system contains two states; namely that $n = 2$. This is the simplest case from which time dependence emerges. Equation (I.19) becomes

$$\begin{aligned} \langle \mathbf{O} \rangle_\Psi(t) = \int d^3\mathbf{x} \Big\{ & \underbrace{|c_1|^2 \phi_1^*(\mathbf{x}) \mathbf{O}_{\mathbf{x}\mathbf{x}} \phi_1(\mathbf{x}) + |c_2|^2 \phi_2^*(\mathbf{x}) \mathbf{O}_{\mathbf{x}\mathbf{x}} \phi_2(\mathbf{x})}_{\text{diagonal terms}} \\ & + \underbrace{c_1^* c_2 \exp\left[-i\Delta E t/\hbar\right] \phi_1^*(\mathbf{x}) \mathbf{O}_{\mathbf{x}\mathbf{x}} \phi_2(\mathbf{x}) + c_1 c_2^* \exp\left[+i\Delta E t/\hbar\right] \phi_1(\mathbf{x}) \mathbf{O}_{\mathbf{x}\mathbf{x}} \phi_2^*(\mathbf{x})}_{\text{off-diagonal terms}} \Big\}. \end{aligned} \tag{I.20}$$

Here, the energy spacing between the eigenstates is abbreviated to $\Delta E \equiv |E_2 - E_1|$. Within the integral we can see two types of contributions arising from the eigenstates. The diagonal terms correspond to the weighting each eigenstate $\phi_j(\mathbf{x})$ has in the superposition. Note that these terms are time-independent in this example. We also notice the presence of a non-diagonal term which after simplification reads

$$\text{off diagonal terms} = 2\Re \left\{ c_1^* c_2 \int d^3\mathbf{x}\, \phi_1^*(\mathbf{x}) \mathbf{O}_{\mathbf{x}\mathbf{x}} \phi_2(\mathbf{x}) \right\} \times \cos\left[\Delta E t/\hbar\right]. \tag{I.21}$$

This term has a time-dependent oscillating contribution to the measured value. We can asso-

ciate the period of this oscillation τ as

$$\tau = \frac{2\pi}{\omega} = \frac{2\pi\hbar}{\Delta E}, \tag{I.22}$$

where ω is the angular frequency of the oscillation. We note that this period scales inversely with the energy separation ΔE. This result is central to the field of attosecond science. It states that for any system with a set of states with some separation ΔE, it has an associated characteristic time τ. An interesting example to consider is if $\mathbf{O}$ is a position operator, and $\phi(\mathbf{x})$ are electronic molecular eigenstates. In this case, the off-diagonal terms in equation (I.21) will contribute towards a time-dependent beating, with an oscillating expectation value i.e. periodic motion of electron density.

For a numerical example, we consider examples of the furanone molecule shown in figure I.1 and helium atom. The plot in Figure I.1 shows two peaks corresponding to molecular

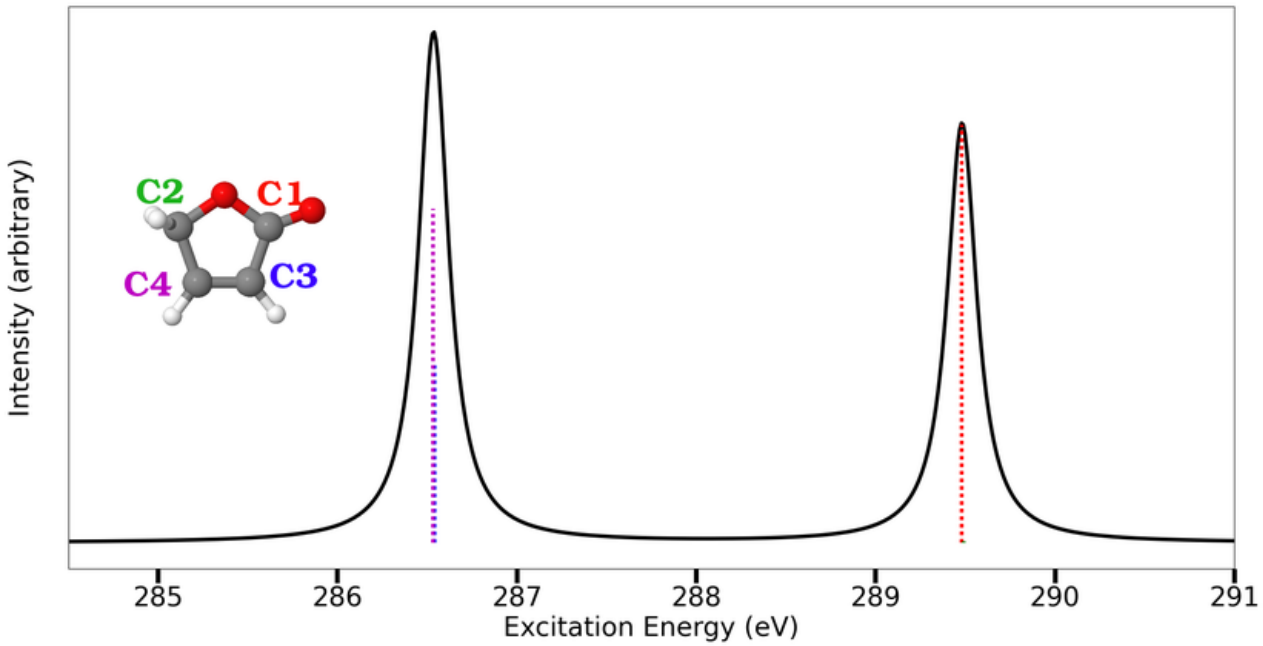

Figure I.1: An numerical calculation of the energy structure of furanone, an organic molecule, adapted from Garner, S. M., Haugen, E. A., Leone, S. R., & Neuscamman, E. (2023) arXiv.[13] The peaks are known as resonances, and indicate the position in energy space of unoccupied or partially occupied states in a molecular system. These resonance correspond to carbon K-shells, the meaning of which will be discussed later.

valence orbitals (energy states), separated in energy by 3 eV. Using the result of equations (I.20) and (I.22) we can calculate that a superposition of both of these states will have a period

of approximately 1.4 fs.

In the case of the helium atom, $\phi_1(\mathbf{x})$ and $\phi_2(\mathbf{x})$ may correspond to the 2s2p and 2s3p multi-electron excited states.[14] These states have excitation energies of 60.15 eV and 63.66 eV respectively and therefore an energy separation of 3.51 eV[15,16], corresponding to a characteristic period of ~1.2 fs per equation (I.22). There is a difference in average radii between these doubly-excited excited states of helium.[17,18] Therefore, any time-dependent population transfer between these states will lead to a expanding and contracting motion, and thus a time-dependent expectation value for the radial distribution of the electron cloud.

The spatial distributions of the two molecular orbitals shown in figure I.1 and the excited states of helium are different. In the helium example, treating the system as a hydrogen-like in character, the expectation values for the radius in a given state with quantum numbers n, l and m is given by

$$\langle r \rangle_{nlm} \equiv \int d^3\mathbf{r}\; r\,|\Psi_{nlm}(\mathbf{r})|^2 = \frac{1}{2}a_0\left[3n^2 - l(l+1)\right], \tag{I.23}$$

where $\Psi_{nlm}(\mathbf{r})$ is the wavefunction describing an energy eigenstate of the hydrogen-like system.[10,19,20] Therefore, as populations oscillate between these eigenstates, their weighting (found in equation (I.9) in their total system superposition varies leading to changing in the electron density over time.

I.4 Introduction to charge migration

The coherent motion of electronic charge in a species (which may occur in isolated gas-phase molecules, or bulk systems in the condensed phase) is commonly referred to as charge migration. Due to the typical energy separation between states in such systems, this corresponds to motion on the sub-femtosecond timescale, in accordance with equation (I.22). In the case of atoms, the eigenstates of electrons differ in their average distance from the nucleus. Therefore, oscillations of the type shown in equation (I.21) take the form periodic changes in the expectation value of the size of the electron cloud. This is sometimes known as a breathing mode.[19] An example of this is described by equation (I.23). This is in contrast to molecules, whose valence states are spatially distributed across the molecule. This can lead to electron density movement across the molecule.

Normally occurring on timescales faster than vibrational motion (a type of nuclear motion), charge migration plays a crucial role in influencing chemical reactivity and energy transfer processes in molecular systems. This motion can follow after a triggering process that initiates the dynamics such as the ionization or excitation of a system into a superposition of excited states. Figure I.2 shows an example of the behaviour of a system responding to ion-

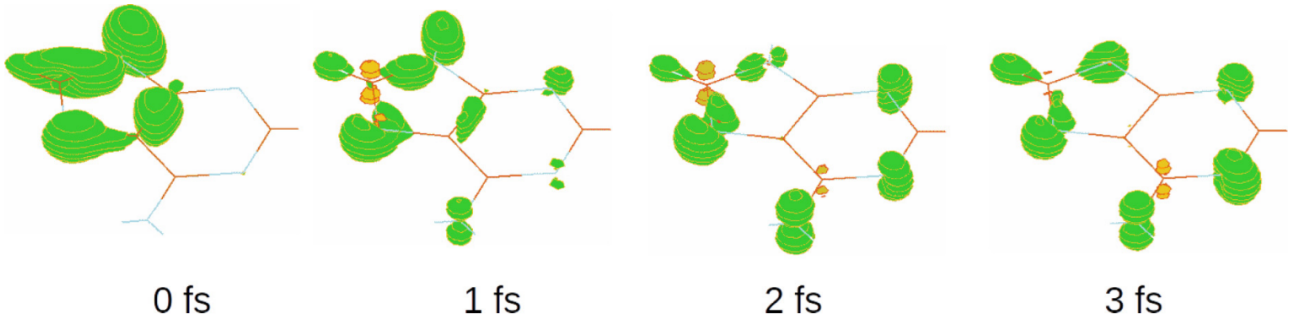

Figure I.2: Calculated charge dynamics for the organic molecule adenine, adapted from Despré, V., & Kuleff, A. I. *Physical Review A*, **106**(2), L021501 (2022).[21] The results are computed at the ADC(3) level of theory.[22] These simulated dynamics are intended to replicate the behaviour of the system after ionisation from the HOMO-16 occurs. Regions with excess positive electron-hole density are shown in green and excess electron density shown in orange. Both colours are with respect to the ground state electron density distribution.

isation from a valence state. It shows the time-dependent valence electron density following the removal of an electron from the 16th highest level below the highest occupied molecular orbital (HOMO), often abbreviated to HOMO-16. The diagram indicates the electron density relative to the ground state configuration. The spatial distribution of the HOMO-16 state from which an electron is removed is spatially distributed over the imidazole group [$(CH)_3(NH)N$, the pentagonal ring in the upper-left side of the molecule illustration in figure I.2], as reflected in the hole density in the plot at t = 0 fs which shows σ character. The hole density appears to rapidly (within ~ 1 fs) spread throughout the entire molecule and changes from σ to π character. It has been argued that the changes in the population of σ orbitals in particular can lead to bond elongation[23] and in extreme cases, bond cleavage over the tens of femtosecond timescale.[24] In both cases following initial charge dynamics showing oscillations of 2 fs period.[25] This provides evidence of initial attosecond electronic dynamics coupling to subsequent bond-breaking, leading ultimately to the fragmentation of the parent molecule. As bond breaking is a fundamental step in chemical processes, this motivates the study of electronic dynamics due to its potentially macroscopic consequences in chemical reactions.

In an experiment using extreme ultraviolet pump pulses (15 - 35 eV), and 700 nm NIR / visible wavelength probe pulses, the fragmentation dynamics of phenylalanine were able to be studied.[26] It was shown that following the initial creation of singly-charged cations by the pump, dications were formed by the probe pulse, in a second ionization event. The probe excites electrons from the excited states of the cation to the lowest excited states that may be populated in the resultant dication.

The observable was the dication yield, distinguishable from the cation via its mass-to-charge ratio. This yield, resolved over pump-probe delay is shown in figure I.3. Note that this measurement does not measure the motion of electronic charge directly. Rather, it relies on the relation between the hole density in the singly-charged cation seen by the probe pulse and the recorded phenylalanine dication yield.

Supporting theoretical calculations on this system were also performed.[27] They report electron motion between the N site and the phenyl group, with a period of 4 fs, preceding the influence of vibrational motion. The vibrational dynamics of phenylalanine were also studied. The fastest such motion occurred with a period of 9 fs, leading the authors to conclude that the observed dication yield oscillations manifest primarily due to electronic motion rather than nuclear motion. However, they stress that nuclear motion still likely plays a minor role in the hole migration, as C-C bond stretching has been predicted to occur on the few-femtosecond timescale in this molecule.[23]

I.5 Motivation to study electron dynamics

Such short timescale oscillations arising due to by electronic dynamics motivate the development and use of techniques that are capable of measuring observables with attosecond time resolution. These observables can shed light on preferred fragmentation pathways and relaxation times for excited state systems.[28] In addition to enhancing our understanding on the guiding process of reaction pathways, direct measurements of electron motion can be used to help the development of theoretical methods. Computing full solutions to the time-dependent Schrödinger Equation is not viable over all geometries a molecule can exhibit, for non-trivial molecules. For this reason, approximation methods are needed.

Time-dependent density functional theory (TDDFT) is an example of an approximation scheme. TDDFT utilizes a so-called simplified exchange-correlation potential.[29] This simplifi-

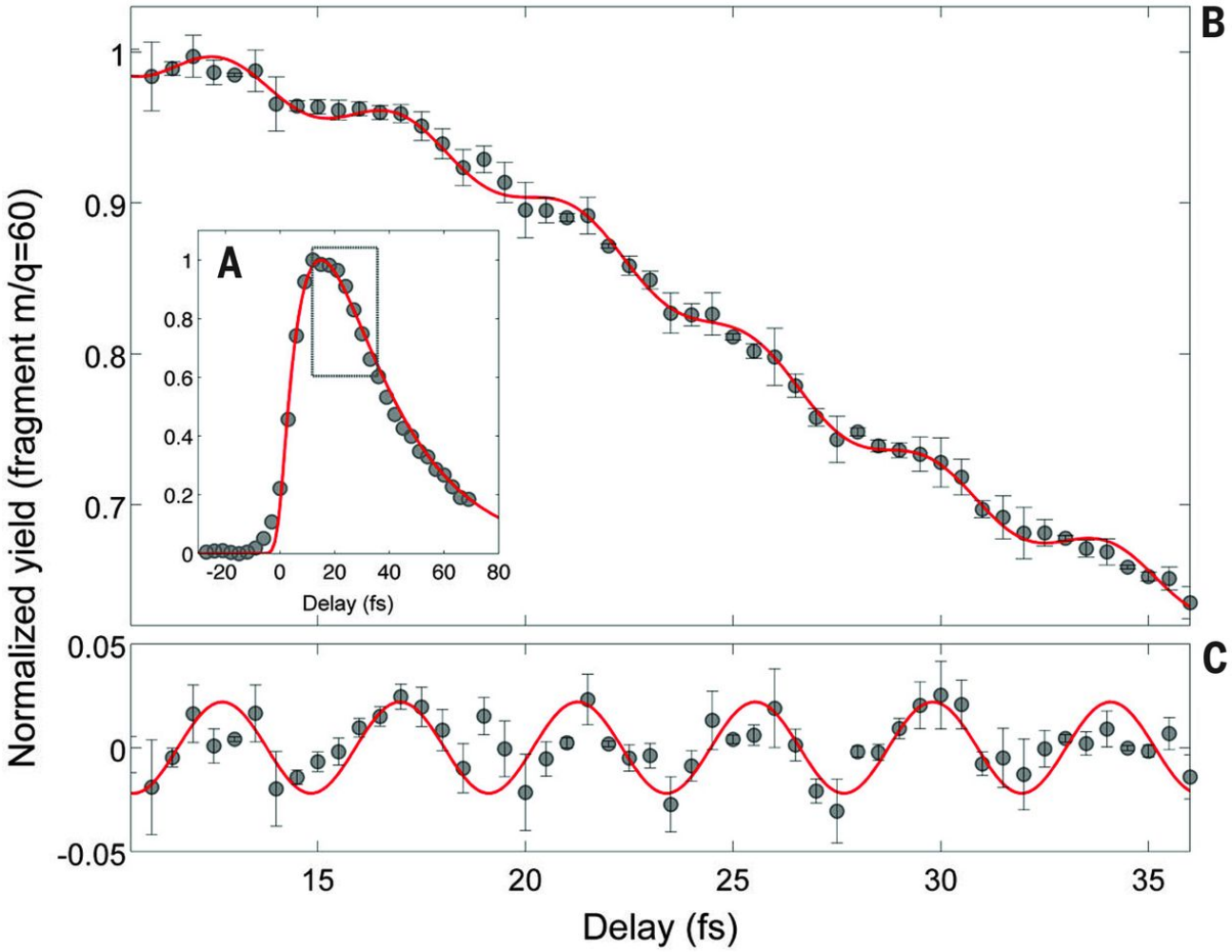

Figure I.3: Experimental data taken in the phenylalanine time-resolved experiment, adapted from F. Calegari et al. *Science* **346**, 336-339 (2014).[26] (**A**) The phenylalanine dication yield as a function of pump-probe delay. The authors find a characteristic exponential decay time of 25 fs. (**B**) Another dataset taken with a higher time resolution, taken in the region indicated by the gray box in (**A**). The authors identify the trend as a combination of an exponential decay and a sinusoidal periodic oscillation. (**C**) The same data as (**B**), but with the exponential decay contribution subtracted away resulting in only the sinusoidal contribution remaining. The oscillation period was identified as 4.3 fs, or 0.234 PHz.

cation (sometimes termed an adiabatic approximation[30]) simplifies the treatment of electron-electron interactions in a many-electron system to the advantage of computational cost, typically of order $\mathcal{O}(N^4)$, where N is the number of interaction components of the system being considered.[31] This simplified exchange-correlation potential limits TDDFT's ability to fully capture electron correlation effects when the system undergoes rapid (diabatic) behaviour. An example of TDDFT being used to understand the properties of an excited state in a molecular system is given in figure I.4.

An example of more computationally expensive approximation schemes are the Algebraic diagrammatic construction (ADC) methods. These schemes use a series expansion of propagators (mathematical objects used to encode the response of a system to some external change i.e. an time-dependent electric field).[22] Different ADC schemes truncate this series expansion after e.g. 2 or 3 terms, leading to ADC(2) and ADC(3) respectively. These methods can account for electron-electron correlation and are commonly used to calculate the energies of systems in their excited states. The computational cost of these schemes can vary up to $\mathcal{O}(N^6)$ complexity[32], which decreases its suitability to be applied to larger molecular systems. Gaining an understanding of charge migration, like any physical process, requires developing a mathematical procedure for predicting the measurement outcomes of experiments. If agreement between theory and experiment is achieved, then information about the system being measured may be gained. One example might be the importance of electron-electron correlation when studying electron-hole migration.[34] For this reason, creating and employing experimental methods with reliable ways of extracting observables are essential for making progress towards understanding the behaviour of electrons in molecules on their natural sub-femtosecond timescale. It is detailing these methods of extracting critical information from complex measurement schemes that will form the primary content of this thesis.

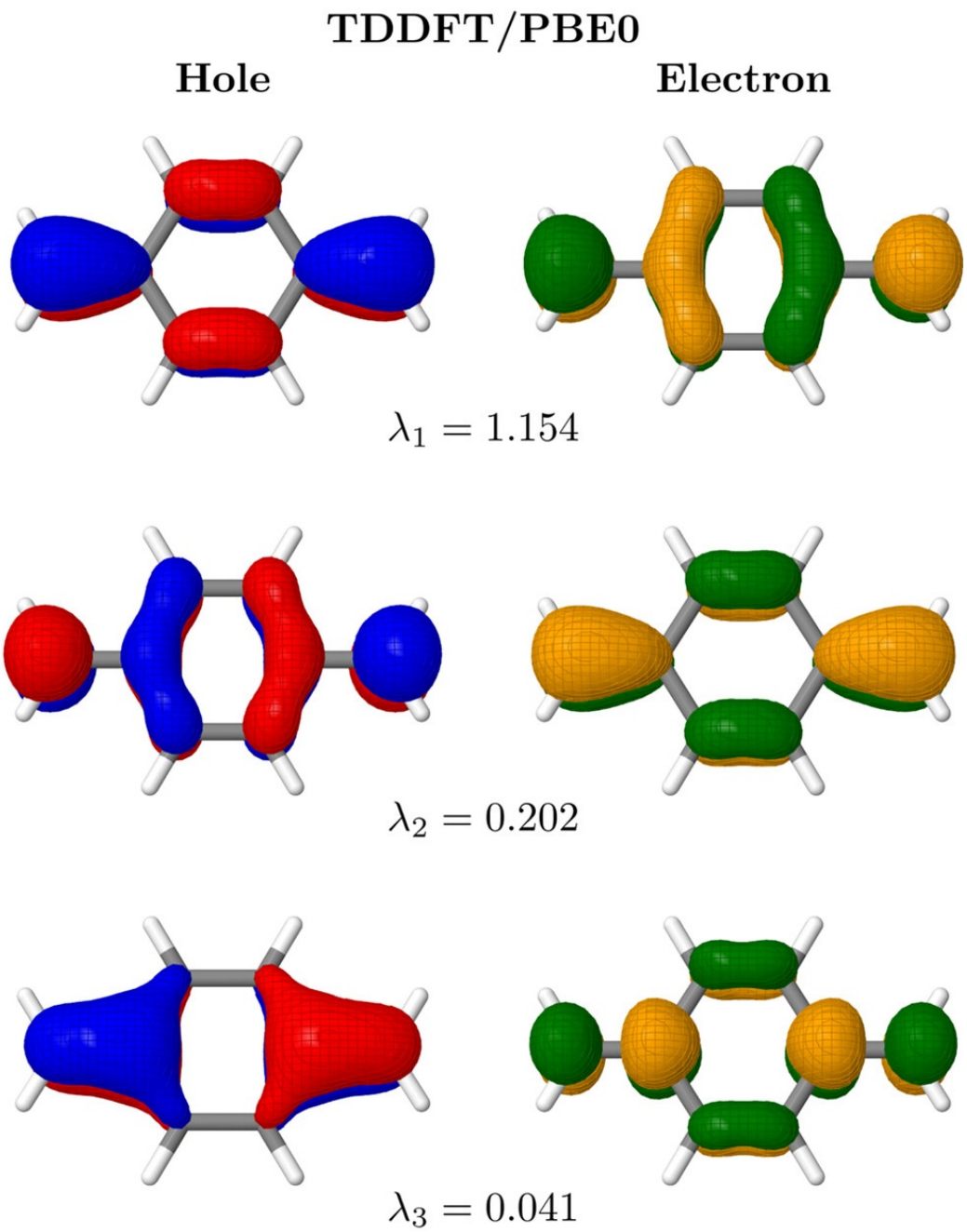

Figure I.4: TDDFT calculations for the para-quinodimethane molecule, a benzene derivative formed by attaching two methylene groups to a benzene molecule, adapted from Plasser, F. (2025). *Journal of Computational Chemistry*, **46**(8), e70072.[33] Shown are the natural transition orbitals of the lowest energy triplet state of the molecule in its planar geometry for a LUMO $\rightarrow$ HOMO de-excitation. The amplitudes of each orbital involved in the transition being studied are given as λ_i, $i = 1, 2, 3$ in the diagram. The hole and electron wavefunctions are illustrated on the left and right-hand columns respectively.

References

[1] N. Bohr, "I. On the constitution of atoms and molecules", The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science **26**, 1–25, ISSN: 1941-5990 (1913) 10.1080/14786441308634955.

[2] F. Schwabl, *Quantum Mechanics*, 4th ed. (Springer Berlin Heidelberg, 2007), ISBN: 978-3-540-71933-5, 10.1007/978-3-540-71933-5.

[3] E. Rutherford, "LXXIX. The scattering of α and β particles by matter and the structure of the atom", The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science **21**, 669–688, ISSN: 1941-5990 (1911) 10.1080/14786440508637080.

[4] B. C. Reed, *The Bohr Atom* (IOP Publishing Ltd, 2020), ISBN: 9780750336116, 10.1088/978-0-7503-3611-6.

[5] W. Demtröder, *Atoms, Molecules and Photons: An Introduction to Atomic-, Molecular- and Quantum Physics* (Springer Berlin Heidelberg, 2018), ISBN: 9783662555231, 10.1007/978-3-662-55523-1.

[6] M. Planck, "Zur Theorie des Gesetzes der Energieverteilung im Normalspectrum", German, Verhandlungen der Deutschen Physikalischen Gesellschaft **2**, Introduced the quantization of energy, $E = hf$, 237–245 (1900).

[7] D. Tong, *Quantum Mechanics: Lectures on Theoretical Physics* (Cambridge University Press, June 2025), ISBN: 9781009594851, 10.1017/9781009594806.

[8] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics* (Cambridge University Press, Aug. 2018), ISBN: 9781107189638, 10.1017/9781316995433.

[9] R. Shankar, *Principles of Quantum Mechanics* (Springer US, 1994), ISBN: 9781475705768, 10.1007/978-1-4757-0576-8.

[10] J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics* (Cambridge University Press, Sept. 2017), ISBN: 9781108499996, 10.1017/9781108499996.

[11] B. C. Reed, *Quantum Mechanics: An Enhanced Primer* (Springer International Publishing, 2022), ISBN: 9783031140204, 10.1007/978-3-031-14020-4.

[12]F. Mandl, *Quantum Mechanics*, en, Manchester Physics Series (John Wiley & Sons, Chichester, England, Mar. 1992).

[13]S. M. Garner, E. A. Haugen, S. R. Leone, and E. Neuscamman, *Spin Coupling Effect on Geometry-Dependent X-ray Absorption of Diradicals*, 2023, 10.48550/ARXIV.2307.15207.

[14]R. P. Madden and K. Codling, "New Autoionizing Atomic Energy Levels in He, Ne, and Ar", Physical Review Letters **10**, 516–518, ISSN: 0031-9007 (1963) 10.1103/physrevlett.10.516.

[15]S. M. Silverman and E. N. Lassettre, "Additional Collision Cross Sections for Helium Especially in the Ionized Continuum", The Journal of Chemical Physics **40**, 1265–1271, ISSN: 1089-7690 (1964) 10.1063/1.1725307.

[16]P. Rupprecht, N. G. Puskar, D. M. Neumark, and S. R. Leone, "Extracting doubly excited state lifetimes in helium directly in the time domain with attosecond noncollinear four-wave-mixing spectroscopy", Physical Review Research **6**, ISSN: 2643-1564 (2024) 10.1103/physrevresearch.6.043100.

[17]J. S. Briggs and V. Schmidt, "Differential cross sections for photo-double-ionization of the helium atom", Journal of Physics B: Atomic, Molecular and Optical Physics **33**, R1–R48, ISSN: 1361-6455 (1999) 10.1088/0953-4075/33/1/201.

[18]G. W. F. Drake, "High Precision Theory of Atomic Helium", Physica Scripta **T83**, 83, ISSN: 0031-8949 (1999) 10.1238/physica.topical.083a00083.

[19]B. H. Bransden and C. J. Joachain, *Physics of atoms and molecules*, en, 2nd ed. (Prentice-Hall, London, England, Apr. 2003).

[20]H. A. Bethe and E. E. Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms* (Springer Berlin Heidelberg, 1957), ISBN: 9783662128695, 10.1007/978-3-662-12869-5.

[21]V. Despré and A. I. Kuleff, "Correlation-driven charge migration as an initial step of the dynamics in correlation bands", Physical Review A **106**, ISSN: 2469-9934 (2022) 10.1103/physreva.106.l021501.

[22]J. Schirmer, A. B. Trofimov, and G. Stelter, "A non-Dyson third-order approximation scheme for the electron propagator", The Journal of Chemical Physics **109**, 4734–4744, ISSN: 1089-7690 (1998) 10.1063/1.477085.

[23]S. Lünnemann, A. I. Kuleff, and L. S. Cederbaum, "Ultrafast charge migration in 2-phenylethyl-N, N-dimethylamine", Chemical Physics Letters **450**, 232–235, ISSN: 0009-2614 (2008) 10.1016/j.cplett.2007.11.031.

[24]L. Lehr, T. Horneff, R. Weinkauf, and E. W. Schlag, "Femtosecond Dynamics after Ionization: 2-Phenylethyl-N, N-dimethylamine as a Model System for Nonresonant Downhill Charge Transfer in Peptides", The Journal of Physical Chemistry A **109**, 8074–8080, ISSN: 1520-5215 (2005) 10.1021/jp0210935.

[25]A. I. Kuleff and L. S. Cederbaum, "Ultrafast correlation-driven electron dynamics", Journal of Physics B: Atomic, Molecular and Optical Physics **47**, 124002, ISSN: 1361-6455 (2014) 10.1088/0953-4075/47/12/124002.

[26]F. Calegari, D. Ayuso, A. Trabattoni, L. Belshaw, S. De Camillis, S. Anumula, F. Frassetto, L. Poletto, A. Palacios, P. Decleva, J. B. Greenwood, F. Martín, and M. Nisoli, "Ultrafast electron dynamics in phenylalanine initiated by attosecond pulses", Science **346**, 336–339, ISSN: 1095-9203 (2014) 10.1126/science.1254061.

[27]D. Mendive-Tapia, M. Vacher, M. J. Bearpark, and M. A. Robb, "Coupled electron-nuclear dynamics: Charge migration and charge transfer initiated near a conical intersection", The Journal of Chemical Physics **139**, ISSN: 1089-7690 (2013) 10.1063/1.4815914.

[28]D. C. Haynes, M. Wurzer, A. Schletter, A. Al-Haddad, C. Blaga, C. Bostedt, J. Bozek, H. Bromberger, M. Bucher, A. Camper, S. Carron, R. Coffee, J. T. Costello, L. F. DiMauro, Y. Ding, K. Ferguson, I. Grguraš, W. Helml, M. C. Hoffmann, M. Ilchen, S. Jalas, N. M. Kabachnik, A. K. Kazansky, R. Kienberger, A. R. Maier, T. Maxwell, T. Mazza, M. Meyer, H. Park, J. Robinson, C. Roedig, H. Schlarb, R. Singla, F. Tellkamp, P. A. Walker, K. Zhang, G. Doumy, C. Behrens, and A. L. Cavalieri, "Clocking Auger electrons", Nature Physics **17**, 512–518, ISSN: 1745-2481 (2021) 10.1038/s41567-020-01111-0.

[29]K. Lopata and N. Govind, "Modeling Fast Electron Dynamics with Real-Time Time-Dependent Density Functional Theory: Application to Small Molecules and Chromophores", Journal of Chemical Theory and Computation **7**, 1344–1355, ISSN: 1549-9626 (2011) 10.1021/ct200137z.

[30]E. K. U. Gross and N. T. Maitra, "Introduction to TDDFT", in *Fundamentals of Time-Dependent Density Functional Theory* (Springer Berlin Heidelberg, 2012), pp. 53–99, ISBN: 9783642235184, 10.1007/978-3-642-23518-4_4.

[31]F. Jensen, *Introduction to Computational Chemistry*, 3rd (Wiley, Chichester, UK, 2017), ISBN: 978-1-118-82599-0.

[32]J. Schirmer, L. S. Cederbaum, and O. Walter, "New approach to the one-particle Green's function for finite Fermi systems", Physical Review A **28**, 1237–1259, ISSN: 0556-2791 (1983) 10.1103/physreva.28.1237.

[33]F. Plasser, "On the Meaning of De-Excitations in Time-Dependent Density Functional Theory Computations", Journal of Computational Chemistry **46**, ISSN: 1096-987X (2025) 10.1002/jcc.70072.

[34]L. Cederbaum and J. Zobeley, "Ultrafast charge migration by electron correlation", Chemical Physics Letters **307**, 205–210, ISSN: 0009-2614 (1999) 10.1016/s0009-2614(99)00508-4.

Chapter II

Molecular States and interactions with the radiation field

Molecules are complex systems composed of interacting electrons and nuclei. Even the simplest molecule, H_2, has four interacting components, two electrons and two protons. Therefore it has ${}^4C_2 = 6$ terms[II.1] in the Coulomb potential describing the interaction between both electrons and both nuclei.[1] For larger systems, the number of ways a molecule can move (known as degrees of freedom) can become very large for even relatively small systems. This motivates approaches that can simplify the involved physics. This chapter will lay out the technical basis for some of the fundamental assumptions essential to the interpretation of experimental observables in molecular physics. In particular the interaction of molecules with radiation fields. The results shown here will be used extensively to understand the original work done in this thesis.

II.1 The Born-Oppenheimer approximation

To describe the Born-Oppenheimer approximation, we start by considering the problem of the simplest molecule, the hydrogen cation, H_2^+.[2] The total energy of the system represented by

[II.1] Here I use the notation $nCr \equiv \frac{n!}{(n-r)!r!}$.

the Hamiltonian of the complete system is given by

$$\mathcal{H} = \underbrace{-\frac{\hbar^2}{2m_e}\nabla^2}_{\mathcal{T}_e} \underbrace{-\sum_{j=1,2}\frac{\hbar^2}{2M_j}\frac{\partial^2}{\partial Z_j^2}}_{\mathcal{T}_N} + \mathcal{V}(\mathbf{r}, Z_1, Z_2), \tag{II.1}$$

where $\mathbf{r}$ and Z_j are the electron and nuclear positions respectively and are treated initially as variables that may change with respect to time. The masses of the nuclei are labeled as M_j. Clearly the structure of this full Hamiltonian can be broken down into contributions

$$\mathcal{H} = \mathcal{T}_e + \mathcal{T}_N + \mathcal{V}. \tag{II.2}$$

Here, $\mathcal{T}$ denotes kinetic energy contributions to the Hamiltonian. The full time independent Schrödinger equation may be written as

$$\mathcal{H}\Psi(\mathbf{r}, Z_1, Z_2) = E\Psi(\mathbf{r}, Z_1, Z_2) \tag{II.3}$$

and its solutions are the desired goal of this problem. We then parametrize the nuclear coordinates by attempting to find a partially separable solution to equation (II.3) of the form

$$\Psi(\mathbf{r}, Z_1, Z_2) = \psi_e(\mathbf{r}|Z_1, Z_2) \times \chi(Z_1, Z_2), \tag{II.4}$$

where $\psi_e(\mathbf{r}|Z_1, Z_2)$ is a fully electronic wavefunction describing the electron for a set of fixed coordinates Z_1 and Z_2.[3] This is sometimes referred to as a parametric dependence on Z_1 and Z_2.[4] $\chi(Z_1, Z_2)$ is a nuclear wavefunction for the set of two nuclei. Assuming a solution exists of the separable form written in equation (II.4), we can now re-express equation (II.3) as

$$\begin{aligned} \mathcal{H}(\psi_e\chi) &= \mathcal{T}_e(\psi_e\chi) + \mathcal{T}_N(\psi_e\chi) + \mathcal{V}\psi_e\chi \\ &= \chi\mathcal{T}_e\psi_e + \mathcal{T}_N(\psi_e\chi) + \mathcal{V}\psi_e\chi, \end{aligned} \tag{II.5}$$

where the fact that χ is independent of $\mathbf{r}$ has been leveraged. Further details lie in the $\mathcal{T}_N(\psi_e\chi)$ term as the electronic wavefunction ψ_e is not fully independent of the nuclear coordinates. We

find that

$$\begin{aligned}
\mathcal{T}_N(\psi_e\chi) &= -\sum_j \frac{\hbar^2}{2M_j}\partial^2_{Z_j}(\psi_e\chi) \\
&= -\sum_j \frac{\hbar^2}{2M_j}\left[(\partial_{Z_j}\chi)(\partial_{Z_j}\psi_e) + \chi\partial^2_{Z_j}\psi_e + (\partial_{Z_j}\psi_e)(\partial_{Z_j}\chi) + \psi_e\partial^2_{Z_j}\chi\right] \\
&= -\sum_j \frac{\hbar^2}{2M_j}\left[\psi_e\partial^2_{Z_j}\chi + \underbrace{\chi\partial^2_{Z_j}\psi_e + 2(\partial_{Z_j}\psi_e)(\partial_{Z_j}\chi)}_{\text{diabatic term}}\right] \\
&= \psi_e\mathcal{T}_N\chi + W.
\end{aligned} \tag{II.6}$$

This follows steps detailed in Atkins.[2] The partial derivative operators have been abbreviated as $\partial/\partial Z_j \to \partial_{Z_j}$ and the shorthand

$$W = -\sum_j \frac{\hbar^2}{2M_j}\left[\chi\partial^2_{Z_j}\psi_e + 2(\partial_{Z_j}\psi_e)(\partial_{Z_j}\chi)\right] \tag{II.7}$$

was used. The central assumption of the Born-Oppenheimer Approximation is to reason that $M_j \gg m_e$. Hence in equation (II.6), compared to the $\chi\mathcal{T}_e\psi_e$ term, W does not make a significant contribution to the solutions Ψ and will be neglected, enabling equation (II.5) to be approximated as

$$\mathcal{H}(\psi_e\chi) \approx \chi\mathcal{T}_e\psi_e + \psi_e\mathcal{T}_N\chi + \mathcal{V}\psi_e\chi. \tag{II.8}$$

Critically, unlike equation (II.5), the kinetic energy operators for each part of the system only act on the wavefunction for its own subsystem (i.e. $\mathcal{T}_e$ acting on ψ_e, $\mathcal{T}_N$ acting on χ). Using this simplified approximate form of the full system Hamiltonian, we can rewrite the eigenequation in (II.3) as

$$\psi_e\mathcal{T}_N\chi + [\mathcal{T}_e\psi_e + \mathcal{V}\psi_e]\,\chi = E\psi_e\chi. \tag{II.9}$$

We now consider fixed values for the nuclear positions Z_j. From its definition in equation (II.1) we can see that this results in the $\psi_e\mathcal{T}_N\chi$ term vanishing. Identifying that this equation will hold for any physical (nonzero) χ, we can pull out all factors of the nuclear wavefunction

χ to write

$$\mathcal{T}_e\psi_e(\mathbf{r}|Z_1, Z_2) + \mathcal{V}(\mathbf{r}, Z_1, Z_2)\psi_e(\mathbf{r}|Z_1, Z_2) = E_e(Z_1, Z_2) \times \psi_e(\mathbf{r}|Z_1, Z_2). \tag{II.10}$$

This equation is the standard time independent Schrödinger equation introduced in equation (I.15). However, it has an embedded parametric dependence on the nuclear coordinates. $\mathcal{V}$ encodes the Coulomb potential seen by the electron as a function of its position for a given choice of Z_1 and Z_2. The eigenvalue $E_e(Z_1, Z_2)$ corresponds to the energy of the electron, for a given $\psi_e(\mathbf{r}|Z_1, Z_2)$ exploring a potential $\mathcal{V}$ for the same fixed choice of nuclear coordinates and nuclear-nuclear repulsion.[5] Back substituting equation (II.10) into (II.9) and pulling out factors of the electronic wavefunction ψ_e this time, we find

$$\mathcal{T}_N\chi + E_e(Z_1, Z_2)\chi = E\chi. \tag{II.11}$$

This eigenequation may also be identified as a time-independent Schrödinger equation, but this time for the nuclear system. Here, the $E_e(Z_1, Z_2)\chi$ term plays the role of an interaction potential which includes both the electron-nuclear contribution as well as the nuclear-nuclear contribution. E represents the total energy of the system. Unlike $E_e(Z_1, Z_2)$, the total energy E may contain kinetic energy contributions from the nuclei, as it includes a contribution from the $\mathcal{T}_N$ operator. Here, we have managed to separate the electronic degrees of freedom from the nuclear kinetic energy terms. This can be seen in equation (II.10) as the nuclear coordinates Z_1 and Z_2 only appear as fixed parameters, not variables. Accordingly, all terms involving ∂_{Z_j} vanish.

Due to differences in the electron and nuclear mass, we may typically expect to see the electrons respond much more quickly to sudden changes in a molecule compared to the nuclei.[II.2] For this reason, under this parametric view of the nuclear coordinates, if the electron starts in a particular eigenstate of the Hamiltonian for a given choice of Z_1 and Z_2, it must remain in this instantaneous eigenstate up to a phase factor. This is known as the *adiabatic theorem* in quantum mechanics, and is distinct from the use of the same term in thermodynamics.[7] The origin of this phase factor is shown in equation (I.16).

[II.2] Note that this statement is potentially false if the contribution of the diabatic term W to $\mathcal{T}_N(\psi_e\chi)$ is not small. See equation (II.6).

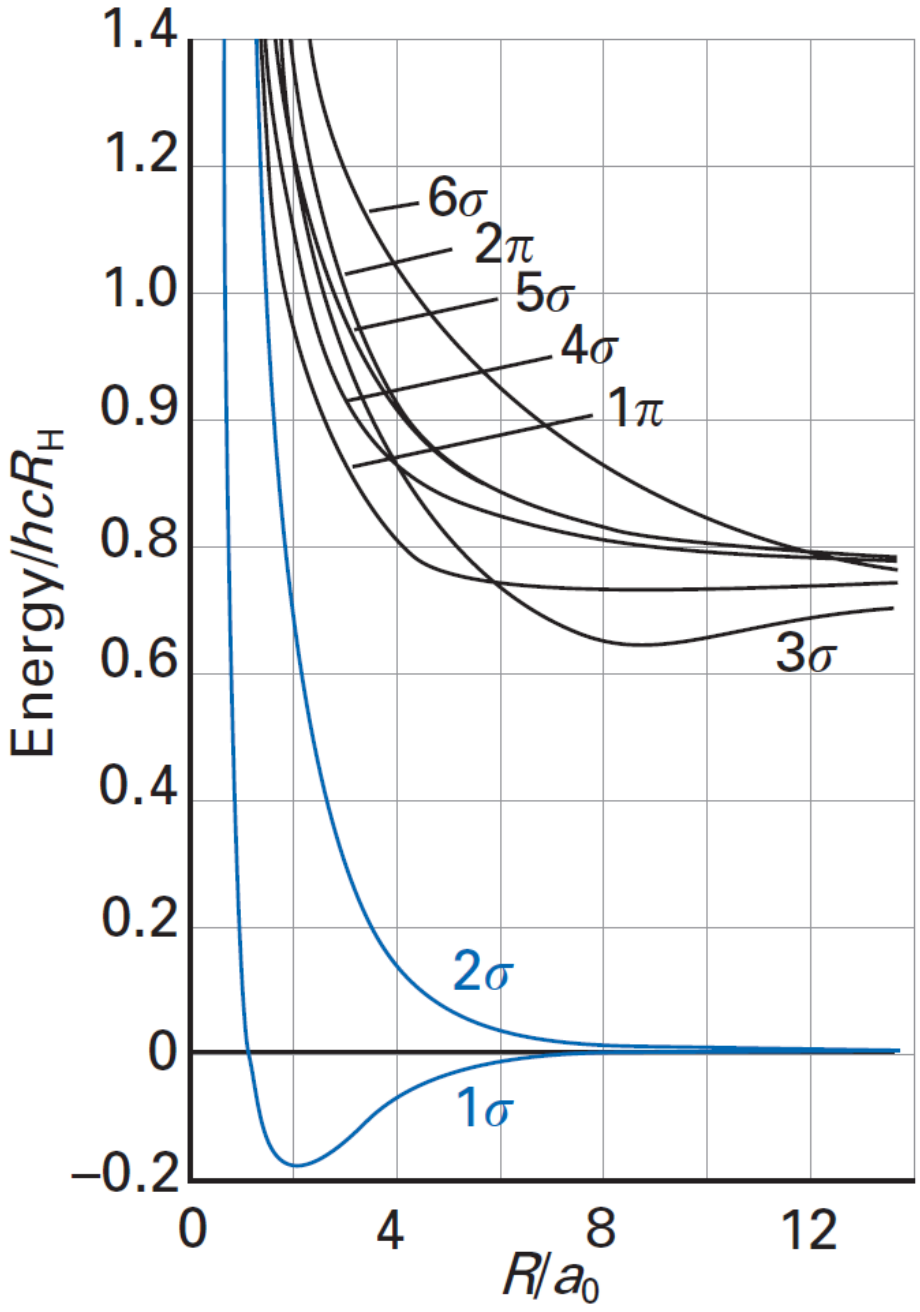

Figure II.1: A calculation of the parameterized eigenvalues $E_e(Z_1, Z_2)$ defined in equation (II.10) versus the internuclear separation $R = |Z_2 - Z_1|$ in units of Bohr radii, from P. W. Atkins and R. S. Friedman, Molecular Quantum Mechanics, 5th (*Oxford University Press*, 2011).[2] R_H is the Rydberg constant. The 8 lowest lying energy states are illustrated. Here, states labeled σ have cylindrical symmetry about the bond-axis. States labeled π exist in lobe-like forms orthogonal to the bond axis. Notice the unique character of the lowest-lying state. It has a minimum at approximately $2a_0$ and an energy of -0.2hcR_H, in good agreement with experiment.[6] In the limit $R \to \infty$ we see the energy of the fully dissociated system is zero, indicating that the 1σ energy level is a bound state with a local minimum energy < 0.

II.2 The multipolar electrostatic series

Now having discussed the separation of electronic and nuclear degrees of freedom within a molecule, we now consider the most important results which describe how a complex charge distribution may be broken down into a sum of simple charge configurations.

We consider an arbitrary charge distribution ξ, and the potential it creates at a generalized point P shown in figure II.2. Formally, according to the definition of an electric potential, at

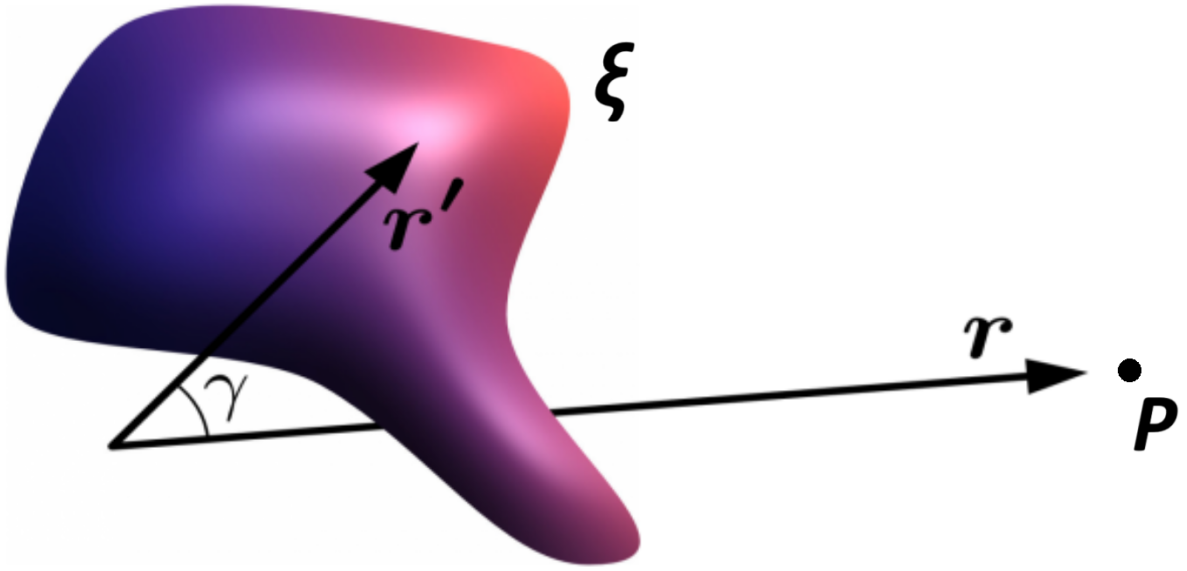

Figure II.2: An arbitrary charge distribution ξ, induces an electric potential at a general point ***P*** whose position is given by a vector **r**. Figure from Department of Physics webpage, University of Guelph.[8] The dummy variable $\mathbf{r}'$ will be used to describe all points within the distribution ξ.

the point P the potential is

$$\Phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int_{\xi} d^3\mathbf{r}' \frac{1}{|\mathbf{r} - \mathbf{r}'|} \rho(\mathbf{r}', t). \tag{II.12}$$

Here, the charge distribution is allowed time-dependence, but this treatment shall be non-relativistic. This can be justified by referring back to the simplistic Bohr atomic model, where the speed of an orbit in the ith orbit is given by

$$\beta = \frac{v}{c} = \frac{Z\alpha}{i}, \tag{II.13}$$

where $\alpha \approx 1/137$ is the fine structure constant.[9,10] We can see that for light atoms, relativistic effects can be safely ignored. Hence, there will be no involvement of retarded time.

By geometry the factor $|\mathbf{r} - \mathbf{r}'|$ may be re-expressed[11]

$$|\mathbf{r} - \mathbf{r}'| = r\sqrt{(1+\epsilon)}, \quad \text{where} \quad \epsilon = \frac{r'}{r} \times \left(\frac{r'}{r} + 2\cos\gamma\right). \tag{II.14}$$

In equation (II.14), we have used $r \equiv |\mathbf{r}|$ and $r' \equiv |\mathbf{r}'|$. In the far field case ($r' \ll r$), clearly $\epsilon \ll 1$. It is therefore treated as a perturbation. We can write the $1/|\mathbf{r} - \mathbf{r}'|$ term via the sum

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \sum_{\lambda=0}^{\infty} \frac{(-1)^\lambda}{\lambda!} (\mathbf{r}' \cdot \nabla)^\lambda \left(\frac{1}{r}\right) = \frac{1}{r} \sum_{\lambda=0}^{\infty} P_\lambda(\cos\gamma), \tag{II.15}$$

which includes the Legendre polynomials P_λ[12] and the radial derivatives of the $1/r$ term[13]. Substituting this series back into equation (II.12) we find that

$$\Phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \times \sum_{\lambda=0}^{\infty} \frac{1}{r^{\lambda+1}} \int_\xi d^3\mathbf{r}' \, (r')^\lambda P_\lambda(\cos\gamma) \rho(\mathbf{r}', t), \tag{II.16}$$

noting that this expression for Φ may not converge in the near field, as in this region our assumption that $\epsilon \ll 1$ will be violated. This series expansion of Φ may be considered a sum of potentials for different charge configurations. Writing these out directly yields[14]

$$\begin{aligned}
\Phi(\mathbf{r}, t) = & \underbrace{\frac{1}{4\pi\epsilon_0} \frac{1}{r} \int_\xi d^3\mathbf{r}' \, \rho(\mathbf{r}', t)}_{\text{Monopole}} + \underbrace{\frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \int_\xi d^3\mathbf{r}' \, r' \cos\gamma \rho(\mathbf{r}', t)}_{\text{Dipole}} \\
& + \underbrace{\frac{1}{4\pi\epsilon_0} \frac{1}{r^3} \int_\xi d^3\mathbf{r}' \, (r')^2 \left(\frac{3\cos^2\gamma - 1}{2}\right) \rho(\mathbf{r}', t)}_{\text{Quadrupole}} \\
& + \underbrace{\frac{1}{4\pi\epsilon_0} \frac{1}{r^4} \int_\xi d^3\mathbf{r}' \, (r')^3 \left(\frac{5\cos^3\gamma - 3\cos\gamma}{2}\right) \rho(\mathbf{r}', t)}_{\text{Octupole}} \\
& + \text{ Higher order charge configurations} \\
= & \sum_{\lambda=0}^{\infty} \Phi^{(\lambda)}(\mathbf{r}, t).
\end{aligned} \tag{II.17}$$

This is the **electric multipole series expansion** due to the arbitrary distribution ξ, for the potential at P and may be viewed as a series of contributions of increasing complexity, but decreasing

influence in the far-field. See figure II.3. We note that, the total charge may be written

$$Q_{\text{total}} = \int_{\xi} d^3\mathbf{r}'\, \rho(\mathbf{r}', t) \tag{II.18}$$

and forms part of the monopole contribution to the scalar potential derived in equation (II.17). Hence, if we are wanting to understand the potential due to a neutral system with $Q_{\text{total}} = 0$ (e.g. a molecule) at large distances $r \gg 1$, the dipolar term is the leading non-zero (and therefore dominant) contribution. We also introduce the electric dipole moment $\mathbf{p}$ here as

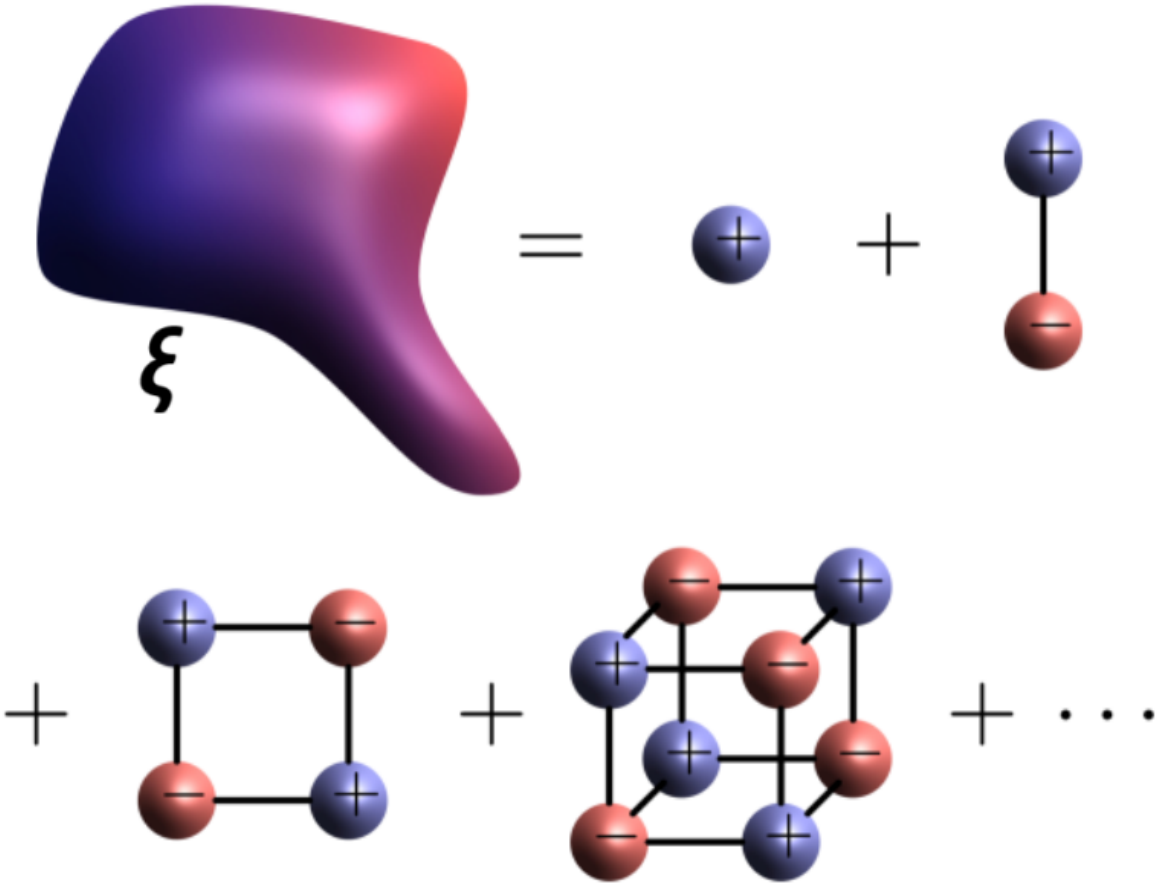

Figure II.3: An arbitrary charge distribution ξ may be viewed as a sum of simplistic charge configurations as spelled out by equation (II.17). Figure from Department of Physics webpage, University of Guelph.[8] Shown in order are the monopole, dipole, quadrupole and octupole charge configurations. The positive and negative charges in each term do not have to be of equal magnitude.

$$\mathbf{p} = \int d^3\mathbf{r}'\ \mathbf{r}'\rho(\mathbf{r}'). \tag{II.19}$$

This allows the dipole contribution Φ_{dipole} to the total field Φ to be written as

$$\Phi_{\text{dipole}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\mathbf{r}}{r^3} = \frac{1}{4\pi\epsilon_0}\frac{1}{r^2}(\mathbf{p}\cdot\hat{\mathbf{r}}). \tag{II.20}$$

In order to describe a single charge q at location $\mathbf{r}$ the charge density $\rho(\mathbf{r}) = q\delta^{(3)}(\mathbf{r}-\mathbf{r}')$ can be used. With this charge density equation (II.19) simplifies to $\mathbf{p} = q\mathbf{r}$. This form of the electric dipole moment will be used in the next section. An analogous expansion for the vector potential defined in equation (II.21) may be carried out as its form is very similar to the scalar potential Φ detailed in equation (II.12). This expansion will not be detailed in this thesis, but a thorough treatment may be found in textbooks.[14]

$$A(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int d^3\mathbf{r}'\,\frac{1}{|\mathbf{r}-\mathbf{r}'|}\mathbf{J}(\mathbf{r}',t). \tag{II.21}$$

II.3 The dipole approximation and the interaction Hamiltonian

In order to derive an appropriate Hamiltonian to describe the interaction of a collection of charges with radiation, the principle of minimum coupling is often used due to its relative simplicity.[15] In classical mechanics, the principle of minimum coupling involves constructing a Lagrangian designed to yield Newton's equation of motions.[16] In the case of a charged particle in a radiation field, the force acting should be the Lorentz force. It is also required that the system be gauge invariant, similar to in classical electrodynamics.[17] Such a Lagrangian may be related to a Hamiltonian via a Legendre Transformation. Using the correspondence principle, we arrive at

$$\begin{aligned}\mathcal{H} &= \frac{1}{2m}[\mathbf{P} - q\mathbf{A}(\mathbf{r},t)]^2 + V(\mathbf{r}) \\ &= \underbrace{\frac{1}{2m}\mathbf{P}^2 + V(\mathbf{r})}_{\text{Field Independent, } \mathcal{H}_0} \underbrace{-\frac{q}{2m}[\mathbf{P}\cdot\mathbf{A}(\mathbf{r},t) + \mathbf{A}(\mathbf{r},t)\cdot\mathbf{P}] + \frac{q^2}{2m}\mathbf{A}^2(\mathbf{r},t)}_{\text{Field Dependent, } \mathcal{H}_{\text{int}}}\end{aligned} \tag{II.22}$$

for the Hamiltonian of a particle of charge q in a radiation field. Here, the standard canonical momentum operator $\mathbf{P}$ seen in equation (I.11) has been replaced by field dependent kinetic

momentum $\mathbf{P} - q\mathbf{A}(\mathbf{r}, t)$ to allow for gauge invariance.[II.3] Minimum coupling is termed 'minimal' as simply demanding gauge invariance of the Hamiltonian allows higher order field terms $\mathbf{E}^2$ and $\mathbf{B}^2$ to be accounted for, without explicitly forcing them into the Hamiltonian.[19]

If we expand the Hamiltonian $\mathcal{H}$ defined in equation (II.22), we find two terms of different orders involving the vector potential. One linear term $\propto \mathbf{A}$ and one nonlinear[II.4] term $\propto \mathbf{A} \cdot \mathbf{A}$. These terms form the field dependent part of the Hamiltonian and are collectively known as the interaction Hamiltonian $\mathcal{H}_{\text{int}}$. The interaction Hamiltonian describes how the field couples to the atom. The nonlinear term contributes most for non-resonant (above edge) interactions. The linear term dominates when the energy of radiation field corresponds to photons close to resonance with states of the system.[21] For this reason we focus on the linear term in $\mathbf{A}$. The interaction Hamiltonian now reads

$$\mathcal{H}_{\text{int}} = -\frac{1}{2m} q \left[\mathbf{P} \cdot \mathbf{A}(\mathbf{r}, t) + \mathbf{A}(\mathbf{r}, t) \cdot \mathbf{P}\right], \tag{II.23}$$

where care has to be taken over the detail and $\mathbf{A}$ and $\mathbf{P}$ in general do not commute. Now we consider light, represented explicitly by a plane wave. The vector potential for such a plane wave reads as

$$\begin{aligned} \mathbf{A}(\mathbf{r}, t) &= A_0 \hat{\mathbf{e}} \exp(i[\mathbf{k} \cdot \mathbf{r} - \omega t]) \\ &= A_0 \hat{\mathbf{e}} \exp(i\mathbf{k} \cdot \mathbf{r}) \times \exp(-i\omega t), \end{aligned} \tag{II.24}$$

where $\mathbf{k}$ is the wavevector with magnitude $|\mathbf{k}| = 2\pi/\lambda$ and whose orientation defines the propagation direction of the wave. A_0 describes the magnitude of the potential and $\hat{\mathbf{e}}$ is its polarization vector and defines the orientation of the resulting electric field vector.

We now focus on the $\exp(i\mathbf{k} \cdot \mathbf{r})$ term in equation (II.24). We may expand it in the standard

[II.3] Similarly to Classical Electrodynamics, this formulation of the Hamiltonian results in physics invariant under the gauge transformation $\mathbf{A} \to \mathbf{A}' = \mathbf{A} + \nabla\Theta$ and $\Phi \to \Phi' = \Phi - \partial_t\Theta$. Equation (II.22) uses a gauge convention known as the $\mathbf{A} \cdot \mathbf{P}$ or "velocity" gauge.[18]

[II.4] The $\mathbf{A} \cdot \mathbf{A}$ term is sometimes known as the diamagnetic term. It is important in strong field interactions and the simultaneous absorption/emission of two photons **from the same field mode** in first order perturbation theory.[20] Nether of these cases are relevant to the interactions measured in this thesis, so I will not being considering them here.

exponential series as

$$\begin{aligned}\exp(i\mathbf{k}\cdot\mathbf{r}) &= \sum_{j=0}^{\infty}\frac{1}{j!}(i\mathbf{k}\cdot\mathbf{r})^j \\ &= 1 + (i\mathbf{k}\cdot\mathbf{r}) + \frac{1}{2!}(i\mathbf{k}\cdot\mathbf{r})^2 + \frac{1}{3!}(i\mathbf{k}\cdot\mathbf{r})^3\ldots\end{aligned} \tag{II.25}$$

This power series in $\mathbf{k}\cdot\mathbf{r}$ motivates us to rewrite the interaction Hamiltonian as the series

$$\mathcal{H}_{\text{int}} = -\frac{1}{2m}qA_0\exp(-i\omega t)\times\{\mathbf{P}\cdot\hat{\mathbf{e}}[1+(i\mathbf{k}\cdot\mathbf{r})+\ldots] + [1+(i\mathbf{k}\cdot\mathbf{r})+\ldots]\hat{\mathbf{e}}\cdot\mathbf{P}\}. \tag{II.26}$$

This general form simplifies significantly if we consider the size of a charge distribution of an atom. Suppose the charge radius of the atom is on a characteristic length scale a which is smaller than the wavelength of the light λ then $|\mathbf{k}|a \ll 1$, where $|\mathbf{k}| \equiv 2\pi/\lambda$. For light with wavelengths larger than hard x-rays, this is typically the case.[22] In this case, we are justified to truncate the series in equation (II.25) after just the first term approximating $\exp(i\mathbf{k}\cdot\mathbf{r}) \approx 1$. We can substitute this expression back into the vector potential to find that

$$\mathbf{A}(\mathbf{r},t) \approx A_0\hat{\mathbf{e}}\times\exp(-i\omega t) \equiv \mathbf{A}(t). \tag{II.27}$$

Namely, we assume the potential is approximately constant over the spatial extent of the charge distribution. This assumption is known as the **Dipole Approximation**. This enables us to simplify the mathematics extensively. In the Dipole approximation, the operators $\mathbf{A}$ and $\mathbf{P}$ can be considered to commute.[II.5] We can now rewrite the linear part of the interaction Hamiltonian in equation (II.23) to read

$$\mathcal{H}_{\text{int}} = -\frac{q}{m}\mathbf{A}(t)\cdot\mathbf{P}. \tag{II.28}$$

Using this form of the interaction Hamiltonian we can write an expression for the transition matrix element between eigenstates of the field independent Hamiltonian defined in equation (II.22). For an initial state $|i\rangle$ and a final state $|f\rangle$, the matrix element reads

$$\mathcal{M}_{fi} \propto \langle f|\mathcal{H}_{\text{int}}|i\rangle = -\frac{q}{m}\mathbf{A}(t)\cdot\langle f|\mathbf{P}|i\rangle. \tag{II.29}$$

[II.5] $\mathbf{P}$ is the generator of spatial translations. We have assumed the potential $\mathbf{A}$ is constant over the domain of the charge density. Hence within the dipole approximation, we assume $\mathbf{A}$ has spatial translational symmetry over the system of interest, and hence $[\mathbf{P},\mathbf{A}] = [\mathbf{A},\mathbf{P}] = 0$.

This object forms the off-diagonal elements of $\mathcal{H}_{\text{int}}$. We can utilize the commutator properties of the field-free Hamiltonian to derive the relations

$$[\mathbf{r}, \mathbf{P}] = i\hbar, \quad [\mathbf{r}, V(\mathbf{r})] = 0 \implies \mathbf{P} = -\frac{im}{\hbar}[\mathbf{r}, \mathcal{H}_0], \tag{II.30}$$

which allow us to switch from the $\mathbf{P}$ to the $\mathbf{r}$ operator.[23] From this we can arrive at the familiar form

$$\begin{aligned} \mathcal{M}_{fi} &\propto -\frac{iq}{\hbar}\int dt' \, \mathbf{A}(t') \exp(i\omega_{fi}t') \cdot \langle f|[\mathbf{r}, \mathcal{H}_0]]|i\rangle \\ &\propto -2\pi i\omega_{fi}A_0\langle f|\mathbf{p}\cdot\hat{\mathbf{e}}|i\rangle\delta(\omega - \omega_{fi}), \end{aligned} \tag{II.31}$$

where we have used $\omega_{fi} = (E_f - E_i)/\hbar$, $\mathbf{p} = q\mathbf{r}$ as the electric dipole moment operator and the definition of the Dirac delta distribution δ in terms of exponentials[II.6]

$$2\pi\delta(\omega - \omega_{fi}) = 2\pi\delta(\omega_{fi} - \omega) \equiv \int dt' \, \exp(-i\omega t') \times \exp(i\omega_{fi}t'). \tag{II.32}$$

Equation (II.31) contains the familiar product $\mathbf{p}\cdot\hat{\mathbf{e}} \propto \mathbf{p}\cdot\mathbf{E}$ term commonly associated with dipole interactions and the $\delta(\omega - \omega_{fi})$ term enforces energy conservation.

II.4 Selection rules for dipole transitions

If radiation is incident on an atomic or molecule system, there is a probability for absorption. This probability is found from the transition matrix element shown in equation (II.31) in the dipole approximation.[II.7] Looking at the spatially dependent terms only, we have

$$\begin{aligned} \mathcal{M}_{fi} &\propto \langle f|\mathbf{p}|i\rangle \\ &\propto \langle f|\mathbf{r}|i\rangle. \end{aligned} \tag{II.33}$$

Expressing this object in coordinate-space representation, it can be re-written as the integral[24]

$$\mathcal{M}_{fi} \propto \int d^3\mathbf{r} \, \phi_f^*(\mathbf{r}) \, \mathbf{r} \, \phi_i(\mathbf{r}), \tag{II.34}$$

[II.6] It follows from the definition of $\delta(x)$ that it is symmetric about $x = 0$, i.e. $\delta(x) = \delta(-x)$.
[II.7] This probability is calculated as $\mathcal{P}(i \to f) \propto |\mathcal{M}_{fi}|^2$.

where $\phi_i(\mathbf{r})$ and $\phi_f(\mathbf{r})$ are energy eigenstates of the field independent Hamiltonian $\mathcal{H}_0$ as defined in equation (II.22). Note that the dipole operator is anti-symmetric under parity inversion.

In the case of atoms, eigenstates of $\mathcal{H}_0$ have very well defined symmetries. As an example, we may consider the dipole transition matrix element for the excitation 1s $\rightarrow$ 2s sublevels of atomic hydrogen. This transition corresponds to an excitation of 10.2 eV. If we compute the matrix element we obtain

$$\mathcal{M}_{2s,1s} \propto \int d^3\mathbf{r}\ \phi^*_{2s}(\mathbf{r})\, \mathbf{r}\, \phi_{1s}(\mathbf{r}). \tag{II.35}$$

However, as both of these states are symmetric under a parity inversion[II.8], the integral is zero. Therefore the transition from the 1s $\rightarrow$ 2s sublevel is said to be **electric dipole forbidden**. Alternatively, an analogous reasoning with the 1s $\rightarrow$ 2p sublevel results in $\mathcal{M}_{2p,1s} \neq 0$ due to the 2p wavefunction being anti-symmetric under parity inversion.

This is a general property of hydrogen-like atomic sublevels and forms the Laporte rule. Only transitions where the initial and final state have opposite parities have nonzero transition matrix elements. This then means that the orbital angular momentum quantum number l must change by an odd number during the transition.

Additionally, due to conservation of angular momentum, constraints such as $\Delta l = \pm 1$ and the magnetic orbital angular momentum quantum number $\Delta m_l = 0, \pm 1$ exist.[25] The transitions that respect all of these conditions simultaneously are then known as **electric dipole allowed**. In most parts of the electromagnetic spectrum, these are the strongest observed transitions.

Transitions which are electric dipole forbidden are physically allowed. Higher order terms in the expansion shown in equation (II.25) give rise to other contributions to the transition matrix element. However, they are typically much weaker than the electric dipole allowed transitions. This is because they are only nonzero outside of the dipole approximation, and arise due to spatial variation of the radiation field over the system. Only for energies in or above the hard x-ray range, is this notable for atoms and isolated molecules as then we leave the regime in which $\lambda \gg a$ and the dipole approximation may no longer be applied.

[II.8] Parity inversion refers to a coordinate negation. I.e. $\phi(\mathbf{r}) \rightarrow \phi'(\mathbf{r}) = \phi(-\mathbf{r})$.

References

[1] F. Mandl, *Statistical Physics*, 2nd ed. (John Wiley & Sons, 1988), ISBN: 9781118723432.

[2] P. W. Atkins and R. S. Friedman, *Molecular Quantum Mechanics*, 5th (Oxford University Press, 2011), ISBN: 9780199541423.

[3] P. K. Mallick, *Fundamentals of Molecular Spectroscopy* (Springer Nature Singapore, 2023), ISBN: 9789819907915, 10.1007/978-981-99-0791-5.

[4] C. Cohen-Tannoudji, B. Diu, and F. Laloë, *Quantum Mechanics*, 1st, English translation (Wiley, 1973), ISBN: 9780471164333.

[5] A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory*, First (Dover Publications, Inc., Mineola, 1996), ISBN: 9780486691862.

[6] D. Bressanini, M. Mella, and G. Morosi, "Nonadiabatic wavefunctions as linear expansions of correlated exponentials. A quantum Monte Carlo application to H2+ and Ps2", Chemical Physics Letters **272**, 370–375, ISSN: 0009-2614 (1997) 10.1016/s0009-2614(97)00571-x.

[7] S. J. Gustafson and I. M. Sigal, "Born–Oppenheimer Approximation and Adiabatic Dynamics", in *Mathematical Concepts of Quantum Mechanics* (Springer International Publishing, Cham, 2020), pp. 151–170, ISBN: 9783030595623, 10.1007/978-3-030-59562-3_12.

[8] University of Guelph, Department of Physics, *Chapter 3: Legendre Polynomials*, Accessed: 2025-08-09, (2025) https://www.physics.uoguelph.ca/chapter-3-legendre-polynomials.

[9] I. P. Grant, ed., *Relativistic Quantum Theory of Atoms and Molecules: Theory and Computation*, Vol. 40, Springer Series on Atomic, Optical, and Plasma Physics (Springer, New York, 2007), ISBN: 978-0-387-34671-7, 10.1007/978-0-387-35069-1.

[10] W. Greiner, *Relativistic Quantum Mechanics. Wave Equations* (Springer Berlin Heidelberg, 2000), ISBN: 9783662042755, 10.1007/978-3-662-04275-5.

[11] D. J. Griffiths, *Introduction to Electrodynamics* (Cambridge University Press, Oct. 2023), ISBN: 9781009397759, 10.1017/9781009397735.

[12] M. L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed. (John Wiley & Sons, Hoboken, NJ, 2006), ISBN: 978-0-471-19826-0.

[13]M. Chaichian, I. Merches, D. Radu, and A. Tureanu, *Electrodynamics* (Springer Berlin Heidelberg, 2016), ISBN: 9783642173813, 10.1007/978-3-642-17381-3.

[14]J. D. Jackson, *Classical Electrodynamics* (Wiley, Apr. 2003), ISBN: 9783527600434, 10.1002/3527600434.eap109.

[15]F. Grossmann, *Theoretical Femtosecond Physics* (Springer Berlin Heidelberg, 2008), ISBN: 9783540778974, 10.1007/978-3-540-77897-4.

[16]R. Santra, "Concepts in x-ray physics", Journal of Physics B: Atomic, Molecular and Optical Physics **42**, 023001, ISSN: 1361-6455 (2008) 10.1088/0953-4075/42/2/023001.

[17]J. B. Marion, *Classical dynamics of particles and systems*, 4th ed. (Brooks/Cole, Florence, KY, Feb. 1995), ISBN: 9780030973024.

[18]B. H. Bransden and C. J. Joachain, *Physics of atoms and molecules*, en, 2nd ed. (Prentice-Hall, London, England, Apr. 2003).

[19]C. Cohen-Tannoudji, J. Dupont-Roc, and G. Grynberg, *Photons and Atoms: Introduction to Quantum Electrodynamics* (Wiley, Mar. 1997), ISBN: 9783527618422, 10.1002/9783527618422.

[20]L. D. Barron, "Molecules in electric and magnetic fields", in *Molecular Light Scattering and Optical Activity* (Cambridge University Press, 2004), pp. 53–122, 10.1017/cbo9780511535468.004.

[21]N. Rohringer, "X-ray Raman scattering: a building block for nonlinear spectroscopy", Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences **377**, 20170471, ISSN: 1471-2962 (2019) 10.1098/rsta.2017.0471.

[22]I. Angeli and K. Marinova, "Table of experimental nuclear ground state charge radii: An update", Atomic Data and Nuclear Data Tables **99**, 69–95, ISSN: 0092-640X (2013) 10.1016/j.adt.2011.12.006.

[23]J. J. Sakurai and J. J. Napolitano, *Modern Quantum Mechanics*, 2nd ed. (Pearson, Upper Saddle River, NJ, Oct. 2007), ISBN: 9780321503367.

[24]R. Shankar, *Principles of Quantum Mechanics* (Springer US, 1994), ISBN: 9781475705768, 10.1007/978-1-4757-0576-8.

[25]S. Svanberg, *Atomic and Molecular Spectroscopy: Basic Aspects and Practical Applications* (Springer International Publishing, 2022), ISBN: 9783031047763, 10.1007/978-3-031-04776-3.

Chapter III

X-ray Spectroscopies and Measurement Techniques

X-rays form part of the electromagnetic spectrum. Their discovery is largely credited to Wilhelm Conrad Röntgen, of Universität Würzburg in 1895.[1] They are typically classed as having wavelengths from 10^{-9} m to 10^{-12} m. This corresponds to photons of energy 100 eV to 100 keV. For comparison, optical light in the range 400 nm $-$ 700 nm has photon energies of around 2-3 eV.

Many elements that are commonly found in organic molecules (e.g. carbon, nitrogen and oxygen) have binding energies for their inner-most electrons that correspond to the few hundred electronvolt range. This makes x-rays an ideal wavelength to study systems containing such atoms. When studying the electronic structure of molecules, two different kinds of electronic states are referred to.

Core states are occupied by strongly-bound electrons with high binding energies. The wavefunctions of core states are highly localised to a specific nucleus within the molecule and have small characteristic sizes. The binding energy of core states depends on the atomic number of the nucleus and therefore are element specific. Due to their small size, there is little overlap between core orbitals and other molecules, and therefore they do not normally contribute towards chemical bonding.

Valence states are the high-lying, weakly bound states with low binding energies (typically tens of electronvolts). Valence states can be (but not in all cases) delocalised across a molecule.[2] This tendency for delocalization makes them primarily responsible for interactions

with neighboring atoms, the chemical environment (e.g. solute-solvent interactions) and general chemical reactions. This chapter will focus on the relevant techniques used to probe excitations of valence states in this thesis, taking advantage of the special capabilities that x-rays allow.

III.1 X-ray Absorption Spectroscopy

One technique often used for extracting information about the internal structure of system is X-ray Absorption Spectroscopy. It is used extensively on a variety of samples. XAS can be used in all main phases of matter with many examples in solid[3–6], liquid[7–11] and gas phase.[12–16]

Measuring XAS experimentally involves impinging x-rays onto a target sample. When these incident x-rays interact with the sample, they may excite electrons into a higher energy unoccupied state, similar to in optical and UV spectroscopy. However, due to the higher energy of x-ray photons, x-rays allow for the excitation of core-level electrons into high-lying valence states and into the continuum (unbound states). These are explained in section III.4 and their relative energy positions are illustrated in figure III.6.

During an XAS measurement, the wavelength of the photons is varied across an energy region of interest. This region of interest normally corresponds to a particular transition of the molecule. Changes in the strength of a transition can give information about population transfer into and out of a given state.[17,18] Energy shifts of a transition can give information about a changing chemical environment experienced by the electrons being excited.[19–22]

A key advantage of using x-rays to probe structure over UV / visible light, is site specificity. Consider exciting an electron from a core state e.g. a K-shell (1s) sublevel of a specific atom within a molecule to an unoccupied delocalised valence state. The energy difference between these states is well-defined and unique to the particular atom involved. For example, the 1s sublevels of carbon, nitrogen and oxygen sit at approximately 280 eV, 410 eV and 540 eV below the continuum threshold.[23]

Core states are highly localised about the nucleus in nature.[24] Due to the dependence of the cross-section for an absorption event on the overlap between the initial and final states mediated by the dipole operator (see equation II.31), x-ray absorption of this energy contains information about chemical environment of the corresponding atomic-site and the population

of the K shell.[III.1]

This is in contrast to UV / visible absorption spectroscopy. As these techniques involve photons of lower energy, they are not capable of core-valence transitions. Instead, they promote electrons between valence states, for example from the highest occupied molecular orbital (HOMO) $\rightarrow$ the lowest unoccupied molecular orbital (LUMO). These states are typically delocalised. Therefore they do not ordinarily contain information about any specific well-localised atomic site of the molecular system.

In literature, XAS is divided into two primary regions. X-ray Absorption Near Edge Structure (XANES) also sometimes referred to as Near-edge X-ray Absorption Fine Structure (NEXAFS) and Extended X-ray Absorption Fine Structure (EXAFS). See figure III.1. The term "edge" here refers to a sudden increase in absorption with respect to increasing photon energy, due to x-ray photons promoting core electrons to unoccupied states (both bound and continuum types).

The interference effect in the EXAFS region illustrated in figure III.1 can arise from the interaction of the outgoing electron wavepacket with the structure of the sample. They can be used in the condensed phase to understand the spatial structure of solids, such as atomic spacing[26], coordination numbers in solution-phase[27] and in transition metal complexes.[28,29]

Of particular interest to the work in this thesis will be the region of the spectrum with energies in the vicinity to the pre-edge. In particular, approximately ~ 10 eV below the edge. This special area is known as the pre-edge. How pre-edge features arise in XAS will be covered in later sections about more advanced x-ray techniques.

It is important to note that there are many different techniques which are used to measure XAS. The technique chosen often depends on what is most suitable for the phase of the sample among other factors. Some are direct methods, whereas others use indirect methods that use an observable which correlates with absorption.

XAS can be numerically quantified via the optical density (OD) given by the equation

$$\mathrm{OD}(\omega) = \log_{10}\left[\frac{I_0(\omega)}{I_T(\omega)}\right], \tag{III.1}$$

where ω stands for the x-ray frequency, and I_0 and I_T the incident and transmitted flux respec-

[III.1] This is because the valence states and their energies can change depending on chemical environment. Consequently, the dipole transition strength as a function of x-ray wavelength serves as a measure of chemical shifts in the vicinity of the resonant atom.

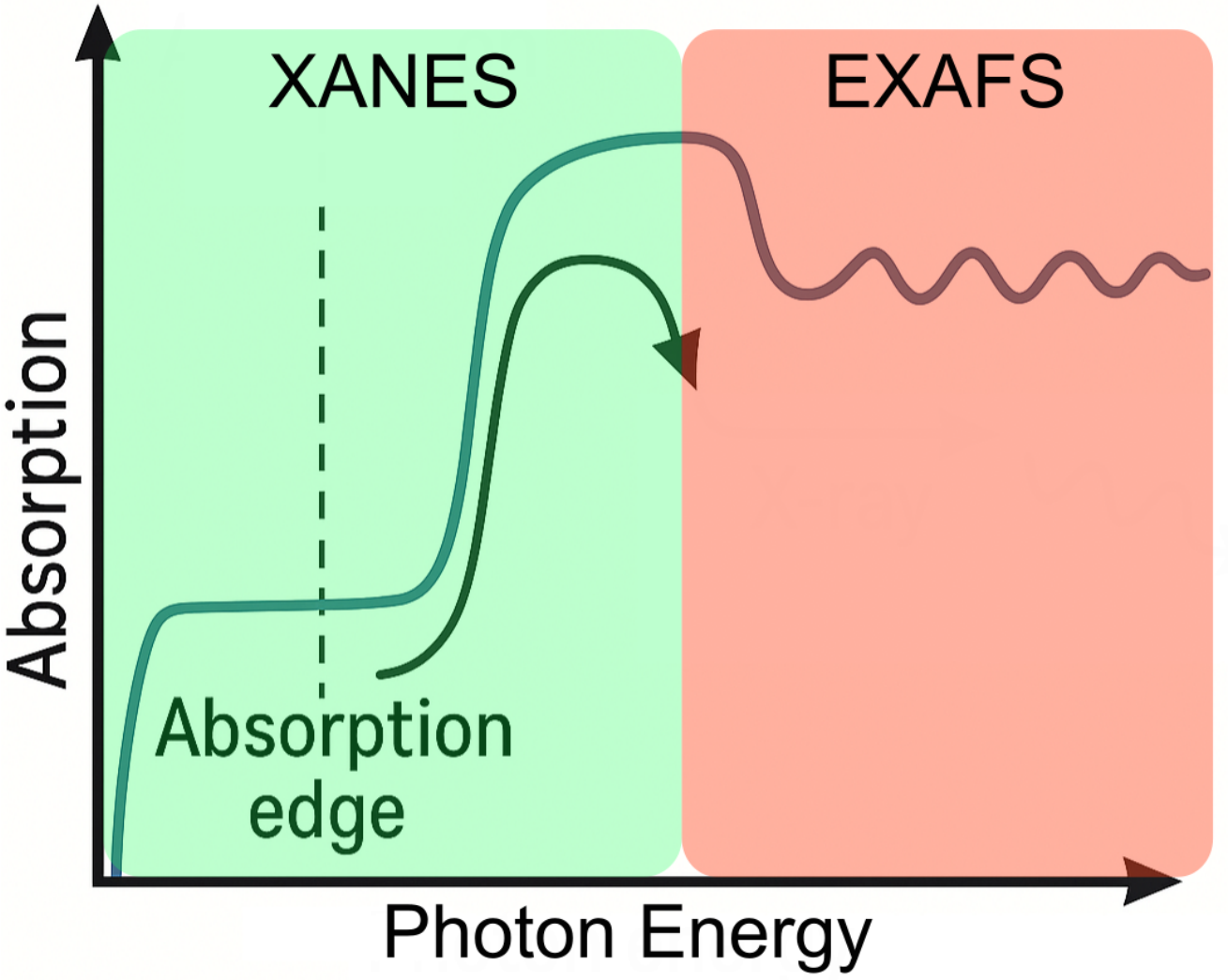

Figure III.1: An illustration of XAS in the vicinity of an edge. The X-ray Absorption Near Edge Structure (XANES) and Extended X-ray Absorption Fine Structure (EXAFS) are labeled. The edge forms the sharp rise indicated by the vertical dashed line in the XANES region. XANES (green region) is normally considered to be within ± 30 eV of the edge. It contains information about structure (both electronic and vibrational) and chemical environment (e.g. oxidation state.) EXAFS (red region) can extend to far above the edge by hundreds (or in some cases thousands) of eV[25] and can contain constructive and destructive interference features as drawn in the top right.

tively.[III.2] This is known as the Beer-Lambert Law.[30] The term absorbance is sometimes used and is synonymous with optical density. OD is a unit-less quantity but its value is not arbitrary. A sample with OD = 0 implies no x-ray photons absorbed. See figure III.2 for from the resonance at $\hbar\omega_{\text{resonance}}$. If all photons are absorbed, namely $I_T = 0$, the sample has an infinite optical density.

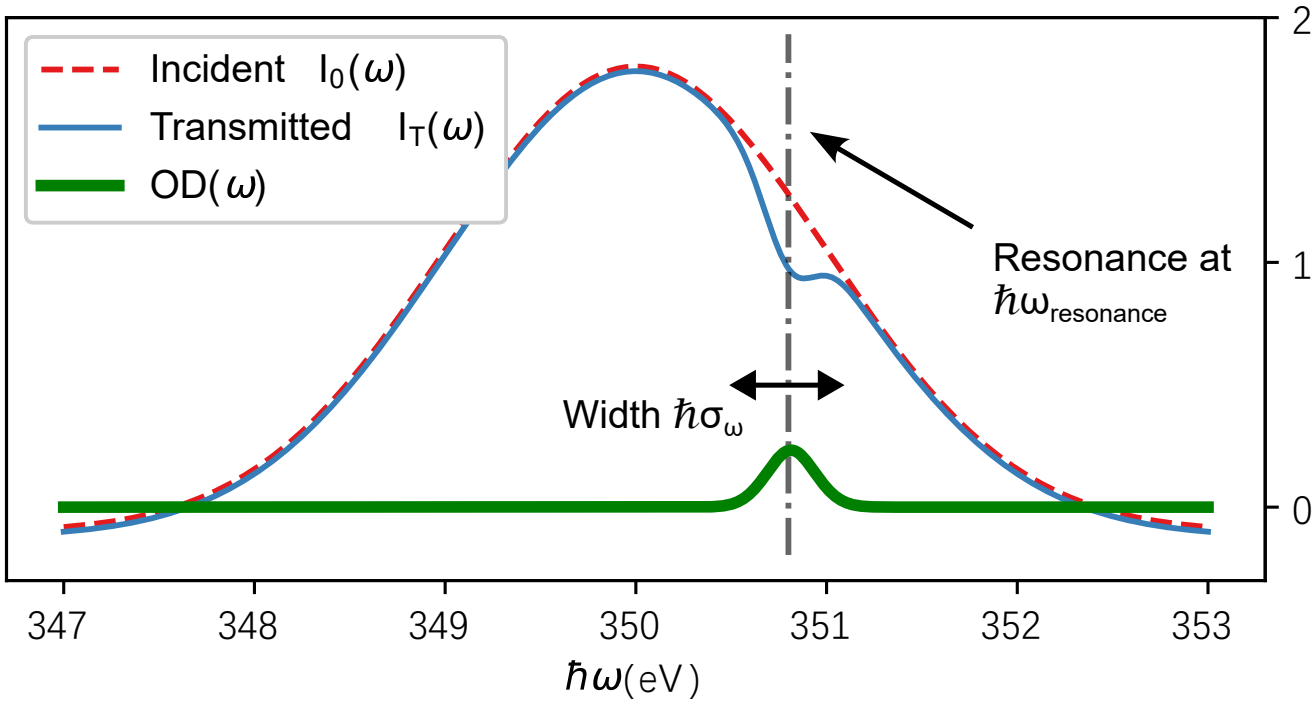

Figure III.2: A plot of the components of a transmission XAS measurement as a function of photon energy. The figure shows the signature of an absorption resonance located at $\hbar\omega_{\text{resonance}} = 350.8$ eV with a shape width of ≈ 0.13 eV. The incident $I_0(\omega)$ (red dashed) and transmitted $I_T(\omega)$ (blue solid) spectra are shown. The resulting optical density from equation (III.1) is computed and displayed.

Consider the case of x-rays interacting with an isolated molecule. We choose this case as isolated molecules typically have more well-defined absorption features compared with the condensed phase due to lack of intermolecular interactions.[31] Suppose the molecule has a core-valence transition resonance with an associated energy $\hbar\omega_{\text{resonance}}$. If x-rays are absorbed by the sample due to this resonance, then photons will be absorbed at this energy. Therefore, the transmitted light will show less photons at energy $\hbar\omega_{\text{resonance}}$. Electronic transition resonances also have widths in the energy domain $\hbar\sigma_\omega$, typically on the order of ~1 eV. Therefore, absorption due to a transition resonance will result in a transmitted spectrum with less intensity than

[III.2]The optical density is occasionally seen using the natural logarithm in literature, but this thesis will always use the definition shown in (III.1).

the incident spectrum over a range $\hbar\sigma_\omega$. This is illustrated in figure III.2.

The most simplistic method to measure XAS is to measure the incident I_0 and transmitted I_T spectra as a function of photon energy. This is sometimes known as transmission XAS. This can be used with very thin liquid samples.[32–35] Due to the number density of absorbing atoms in condensed phase, a too thick sample can result in no transmitted photons.[III.3] Transmission XAS has also been demonstrated in gas phase.[36]

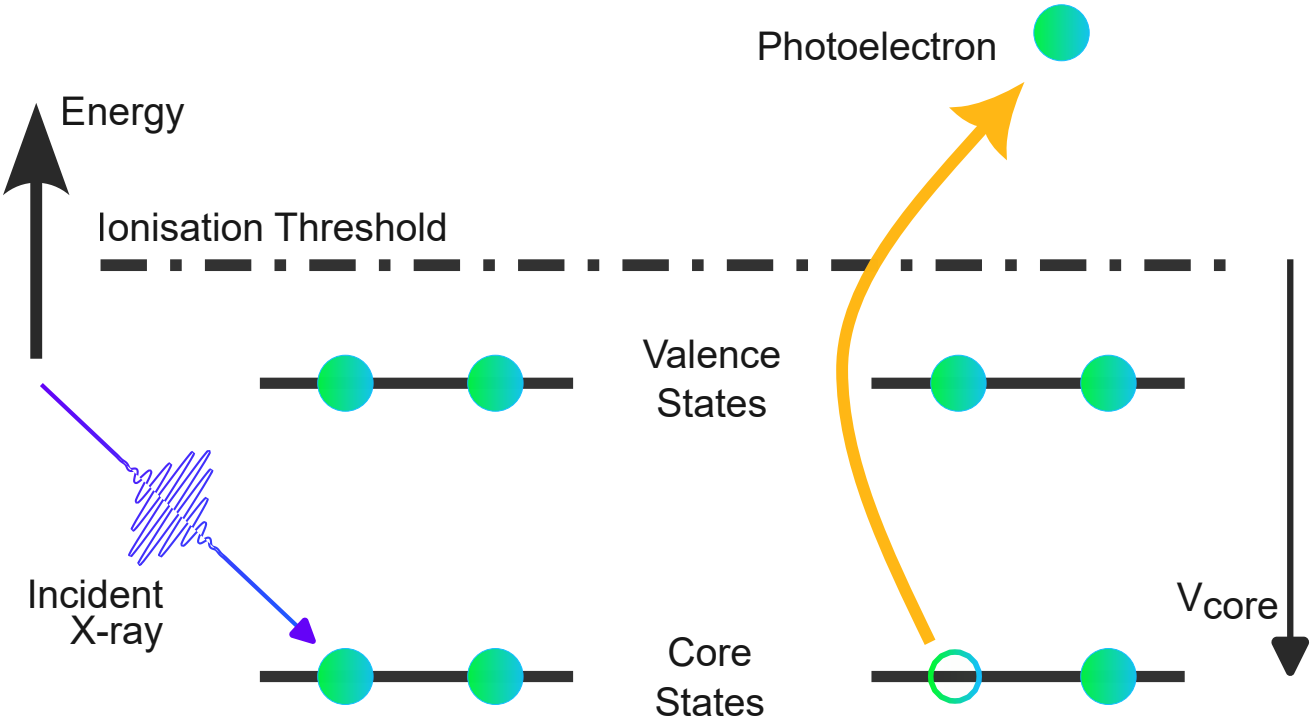

Figure III.3: A schematic illustration of electron yield x-ray aborption spectroscopy (EY-XAS). *Left:* Incident x-rays with energy $\hbar\omega$ excite electrons from the core states (e.g. K-shell or L-shell) with binding energy V_{core} to the continuum, leaving a core-hole. *Right:* In this case, the result is direct photoionization of the molecule and the production of a photoelectron. In the case of direct photoionsiation of the core electrons, the valence electrons do not directly participate. Note how this process is sensitive to the core-state population.

In the gas-phase, the number density of absorbing atoms is orders of magnitude lower than in condensed-phase. Therefore, only a very small fraction of the incident photons are absorbed, even on resonance. This leads to the condition $I_0 \approx I_T$ in equation (III.1). This can lead to a low signal-to-noise ratio in transmission XAS measurements in gas-phase.

Alternatively, a correlated observable can be measured to quantify XAS. One example often used in gas-phase measurements is electron-yield x-ray absorption spectroscopy (EY-XAS).

[III.3] When no photons are transmitted through the sample, the absorption is said to be saturated.

The principle is shown in figure III.3. Following the absorption of a photon by a core-electron, a photoelectron is emitted. The number of these photoelectrons (electron yield) as a function of incident photon energy forms EY-XAS. Note that although the valence electrons do not interact with the photon in this model, their contribution to electron shielding effect may have a minor role in determining V_{core}.[37]

Additionally, following a sudden ionization of a core-electron, there is a rapid change in the potential seen by the remaining bound electrons. This can result in additional electrons being ionized from the valence. This electron-correlation effect forms a type of shake-off process[38–40] Due to energy conservation, the photoelectron and the secondary electron must have total energy $\hbar\omega - V_{core}$.[41]

If the electrons are resolved in energy, the data collected is referred to as X-ray Photoelectron Spectroscopy (XPS). EY-XAS involves summing the total yield of electrons in some energy region of interest.[III.4]

Transmission XAS and EY-XAS are both sensitive to electronic structure and state populations. However, due to the difference in the observable, different absorption resonances may show different strengths between the two techniques. This is because a single-photon absorption event in transmission XAS may show up as a single ionization event (direct photoionization as shown in figure III.3) or possibly a direct ionization with a shake-off effect resulting in double ionization. The electron yield may therefore show different absorption strengths depending on the propensity of the molecule to undergo shake-off processes following x-ray absorption. This is because following photo-absorption, single electron and multielectron emission form competing pathways.

Following an excitation or absorption event of a core electron, there will be an electron-hole in the tightly bound core state. This electron configuration is referred to as a core-hole. A core-hole configuration is an electronically excited state of a system, that may decay via a number of pathways. One pathway is via a radiative mechanism known as fluorescence, where a weakly bound electron from the valence drops down to fill the core hole, resulting in a valence-hole configuration.

This relaxation process can result in the emission of a photon.[III.5] The number of emitted

[III.4]The region of interest chosen depends on the system being studied and what information one wishes to extract from the XAS. The region of interest may be all possible energies from 0 eV to $h\nu - V_{core}$.

[III.5]This process dominates in heavy atoms. In light atoms, the Auger-Meitner process has a shorter lifetime and therefore dominates.[42] This is discussed in the next section.

photons (fluorescence yield) also forms a method of XAS measurement. Similarly to EY-XAS, fluorescence yield measurements can count photons of all energies or count photons in a specific energy range.[43,44] These are known as total (TFY) and partial (PFY) fluorescence yield respectively and are often used to understand condensed phase systems due to the large number density of target molecules.[45,46]

III.2 Auger-Meitner Spectroscopy

Auger-Meitner Decay is a non radiative process by which an atom or molecule can relax from a core-hole configuration.[47] It is an example of an autoionization process. It was first experimentally observed independently by Lisa Meitner in 1922[48,49], and Pierre Auger in 1925.[50,51]

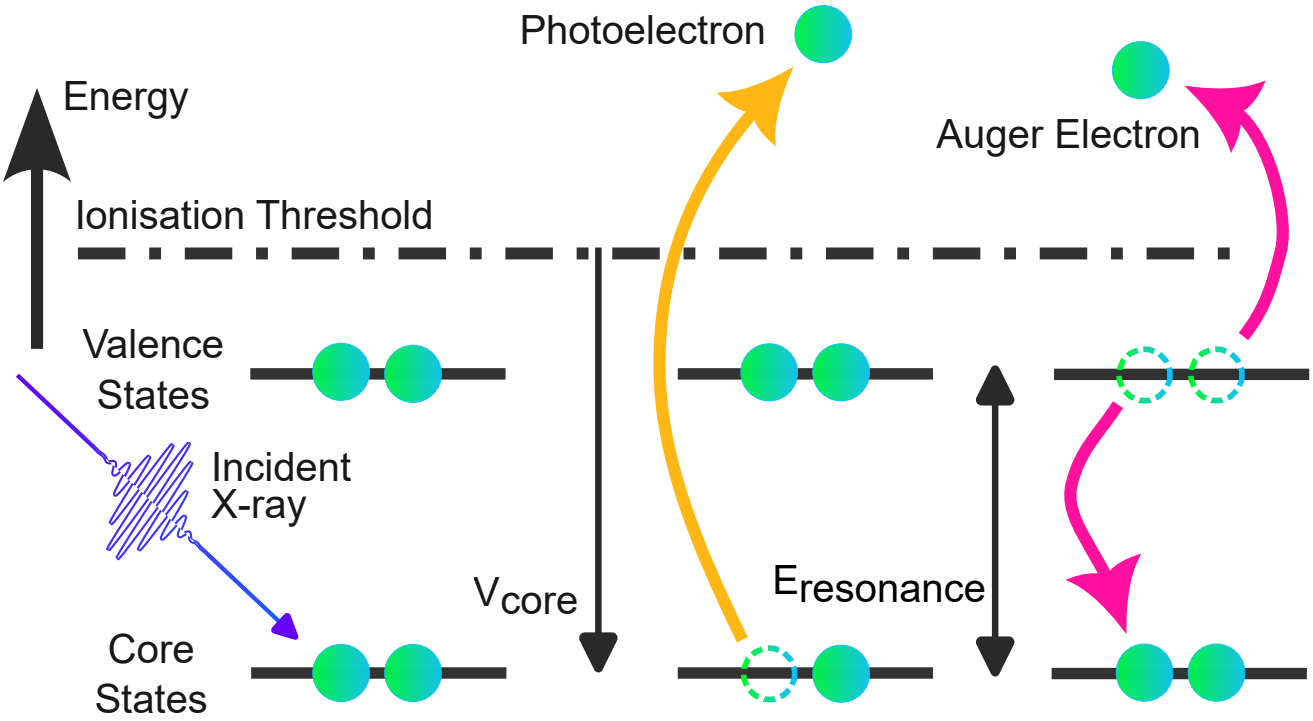

Figure III.4: A sketch illustration of the non-resonant Auger-Meitner relaxation process. *Left*: An incident x-ray photon ionizes a core electron, from a neutral atom or molecule. *Center*: The photoionization process results in a singly-charged cation in a core-hole configuration. *Right*: A valence electron relaxes into the core-hole. Via the electrostatic interaction, energy $E_{resonance}$ is transferred to another valence electron. This valence electron is then promoted to an unbound state and undergoes ionization, resulting in a doubly-charged cation.

The normal (non-resonant) Auger-Meitner process is illustrated schematically in figure III.4. A photoionization event followed by an Auger-Meitner process results in a doubly-

charged cation. Two electrons are emitted. The initial photoelectron will have kinetic energy given by

$$T_{\text{photoelectron}} = \hbar\omega + E_{\text{resonance}} - V_{\text{core}}, \tag{III.2}$$

where $V_{\text{core}} - E_{\text{resonance}}$ can be seen as the valence-state binding energy and are defined in figure III.4. The Auger-Meitner electron will be subsequently ionized and will have kinetic energy

$$T_{\text{auger}} = 2E_{\text{resonance}} - V_{\text{core}}. \tag{III.3}$$

However, this simple picture makes the assumption that the energy distribution of the bound states of the system is the same in both the neutral system and the cation formed after photoelectron emission. However, it is known that the binding energy of the valence can exhibit shifts of few electron volts following the formation of a core-hole.[52] This occurs as the introduction of a core-hole creates a localised effective positive charge, inducing electron rearrangement. This rearrangement can then alter spectroscopic observables, such as the energy of Auger-Meitner electrons and UV absorption.[53–55] Without generalization, the energy of the Auger-Meitner electron can be expressed as

$$T_{\text{auger}} = E^{*}_{\text{cation}} - E_{\text{dication}}, \tag{III.4}$$

where E^{*}_{cation} is the highly-excited core-hole configuration and E_{dication} is the energy of the resultant doubly-charged cation.

Photoionization and the non-resonant Auger-Meitner emission are both threshold processes, as in the model presented in figure III.4, none can occur if $\hbar\omega < V_{\text{core}}$.[III.6] Note that unlike the initial photoelectron emission which is dispersive, the Auger-Meitner emission is non-dispersive.[III.7] This property of the Auger-Meitner process makes Auger electrons easy to distinguish from photoelectrons, as during a measurement, electrons for many different photon energies are recorded.

Following the creation of the core-hole state and the photoelectron emission, there is a characteristic delay before the Auger-Meitner electron is emitted. This time delay is typically

[III.6] Resonant Auger-Meitner relaxation is also subject to a threshold, but in this case, the incident x-ray photon must have sufficient energy to promote a core electron to an unoccupied bound state, instead of the continuum.

[III.7] Dispersive refers to a process where kinetic energy of the emitted electron(s) change as a function of incident photon energy.

on the order of 2-6 fs. This time known as the Auger-Meitner lifetime has been measured via streaking techniques[56,57] and pump-probe measurements.[58]

Another type of Auger-Meitner decay is known as the resonant Auger-Meitner process. This process is subtly different from the non-resonant case depicted in figure III.4. In the resonant case, a core-hole is induced by an x-ray photon via an excitation, instead of ionization. An x-ray photon promotes a core-electron into an unoccupied valence state. This core-hole then decays via the center and right diagrams in figure III.4. Therefore, due to the lack of photoelectron emission, a resonant Auger-Meitner decay results in a singly-charged cation.[59]

Within the resonant Auger-Meitner process, there are two possible cases, referred to as participator and spectator. In the participator case, the same electron which is promoted to an unoccupied state relaxes back into the core-hole. In the spectator case, a different electron relaxes into the core-hole. From both cases, information about the electronic energy structure can be obtained. In the participator (spectator) case the Auger-Meitner emission is dependent on a relaxation from a ground state unoccupied (occupied) valence state to a core state.

Complications to the Auger-Meitner relaxation process may exist if the initial photoelectron leaving the ion has a low kinetic energy. An example of this is known as post collision interaction and applies to auto-ionizing states of a system.[60,61] This occurs when the photoelectron is promoted to a low energy continuum state and does not leave the vicinity of the cation before the cation's Auger-Meitner lifetime. Then, an Auger-Meitner electron may be emitted causing the initial photoelectron to be subjected to a new potential due to the change in the cation's charge state.[62]

This can cause distortions to the Auger-Meitner emission lineshape as energy can be redistributed between the photoelectron and the Auger-Meitner electrons.[63,64] In our experiments using Auger-Meitner decay as an observable, the pump pulse used to initiate the electron dynamics will create a photoelectron of hundreds of electronvolts, far above the threshold for post collision interaction. Therefore, no modification to the Auger-Meitner lineshape due to this effect is anticipated.

III.3 X-ray Pump X-ray Probe Spectroscopy

We now come to time resolved all x-ray techniques. In particular focusing on a measurement scheme involving both XAS and the Resonant Auger-Meitner process.

In a pump-probe experiment two time separated pulses are used to understand a time-dependent quantity.[65] The first pulse is known as the pump pulse. This initial pulse introduces a disturbance into the system sending it into a non-equilibrium state. This is done in order to trigger a dynamical change within the system.

The second pulse is known as the probe pulse. The probe pulse is time delayed with respect to the pump pulse. The probe pulse is used to interrogate the state of the system by mapping its properties onto an observable. This observable may be a transmitted spectrum[66,67], inelastic x-ray scattering[68,69], a diffraction pattern[70,71], or electron emission.[72,73] This observable can be used to understand relaxation pathways, structural rearrangements and the motion of electronic charge. In this work, the observable created by the probe pulse is resonant Auger-Meitner electron emission. In scanning the photon energy across the pre-edge, the hole densities of different valance states is encoded into the Auger-Meitner yield. This is therefore an EY-XAS scheme of pump-probe type. It is also possible to perform this measurement via transmission XAS by measuring the optical density of the sample at different probe photon energies.

The x-ray pump x-ray probe detection scheme is illustrated in figure III.5. The pump pulse is tuned to a photon energy that is insufficient to ionize core electrons, but sufficient to ionize valence electrons via non-resonant absorption. Due to the bandwidth of an XFEL pulse[75] and the typical separation of valence states[76], a coherent superposition of many different valence hole energy eigenstates is created.

A system in a superposition of energy eigenstates will naturally undergo oscillatory behaviour as detailed for the case of a two state system in equation (I.20). In the case of an isolated gas molecule, this manifests as coherent electronic motion.

The photon energy of probe pulse is chosen to be on resonance with the core-valence transition of a specific atom within the molecule, depicted in figure III.5 as $E_{resonance}$. Consequently, the cross-section for the absorption of the probe pulse is dependent on the hole-density in the valence manifold in the vicinity of the atom the probe is resonant with.[III.8] E.g. if the molecule contains oxygen and the probe photon energy is tuned to the oxygen core-valence resonance (~540 eV[78]), the probe absorption cross section reveals if and how localised a hole wavepacket is to the oxygen site.[79]

[III.8] Note that figure III.5 draws the valence states in the neutral species and the singly and doubly ionized cation with the same energy separation from the continuum. However, due to changes in the electron-electron interaction, the energies of the valence for these different ion charge states can show small differences.[77]

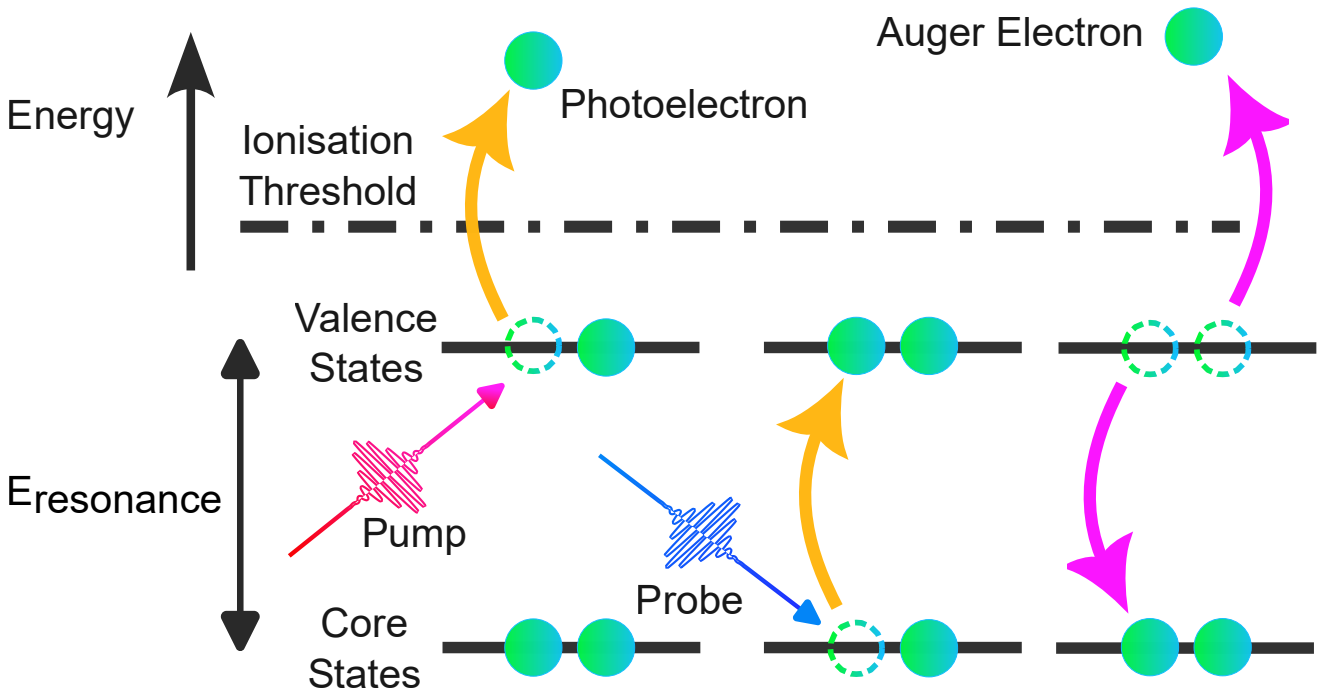

Figure III.5: A pump-probe scheme proposed for investigating charge migration in molecular systems in X-ray Free electron lasers.[74] *Left*: A pump pulse ionizes an electron from the valence, thus creating a valence-hole and a resulting cation. *Center*: After a time delay, a subsequent probe pulse resonantly excites a core electron into the newly unoccupied valence hole, resulting in highly excited core-hole configuration. *Right*: The core-hole decays via resonant Auger-Meitner decay. The parent molecule is now a doubly-charged cation.

The directly detected observable in this experimental scheme is the resulting Auger-Meitner emission. This electron will have a non-dispersive and characteristic energy dictated by equation (III.4). Following the probe absorption promoting an electron into the valence state and creation of a core-hole, either participator or spectator resonant Auger-Meitner decay can occur.

To reveal the hole dynamics, the Auger-Meitner electron yield as a function of pump-probe delay is analyzed. Note that this experimental scheme relies on the pump pulse ionizing a valence electron in the first step, as shown in figure III.5. This means that the dynamics encoded in the Auger-Meitner electron yield contains information about hole migration in the cation, not the original parent molecule.

III.4 X-ray Raman Scattering

In section III.3, a molecule is disturbed from equilibrium by a photoionization event. This disturbance is caused by the pump pulse and results in the creation of a valence-hole state and the removal of an electron. Therefore, the probe allows for the observation of hole migration in cations.

In order to study the response of a neutral molecular system to a disturbance from equilibrium, we are motivated to consider a technique whose pump step relies on excitation, rather than ionization. Since the chemistry of molecules is normally determined by valence electrons, we consider a scheme that can produce excitations of valence states. Such configurations are known as valence-excited states and are depicted on the right-hand side of figure III.6.

In order to send a system into a valence-excited state, x-ray Raman scattering has been employed. This technique has already been employed successfully to cause vibrational excitations using other parts of the electromagnetic spectrum.[80–82] Due to its use of x-ray light, x-ray Raman scattering is of sufficient photon energy to excite electronic states.

An X-ray Raman scattering event is an interaction in which an incident photon couples to a system, sending it into an excited state. The system then in turn emits a photon of different energy.[III.9] To demonstrate this, we consider the case where a system starts in its ground state. It initially absorbs a photon with a wavelength corresponding to a resonant transition, promoting an electron into an unoccupied valence state, resulting in a core-excited state. Such a state is illustrated in the center of figure III.6. Now, the system has a core vacancy.

At this point, the system could decay via a resonant auger process as discussed back in section III.2. An alternative decay pathway is also possible. An electron from the LUMO or HOMO electron may undergo a radiative decay in order to fill the core vacancy, emitting a photon in the process. If the LUMO electron decays, the system returns to its ground state. However, in the case where an electron from the HOMO fills the hole, the emitted photon will have less energy than the incident photon. Hence in this case, from the perspective of the photon field, the process is inelastic.

The scattering process described here is often known as resonant inelastic x-ray scattering (RIXS). RIXS is a well-established technique[83,84] relying on the spontaneous decay of states in and energetically below the HOMO. Photons from this process are emitted isotropically in

[III.9] The term "Rayleigh scattering" is used for the case of an elastic interaction.

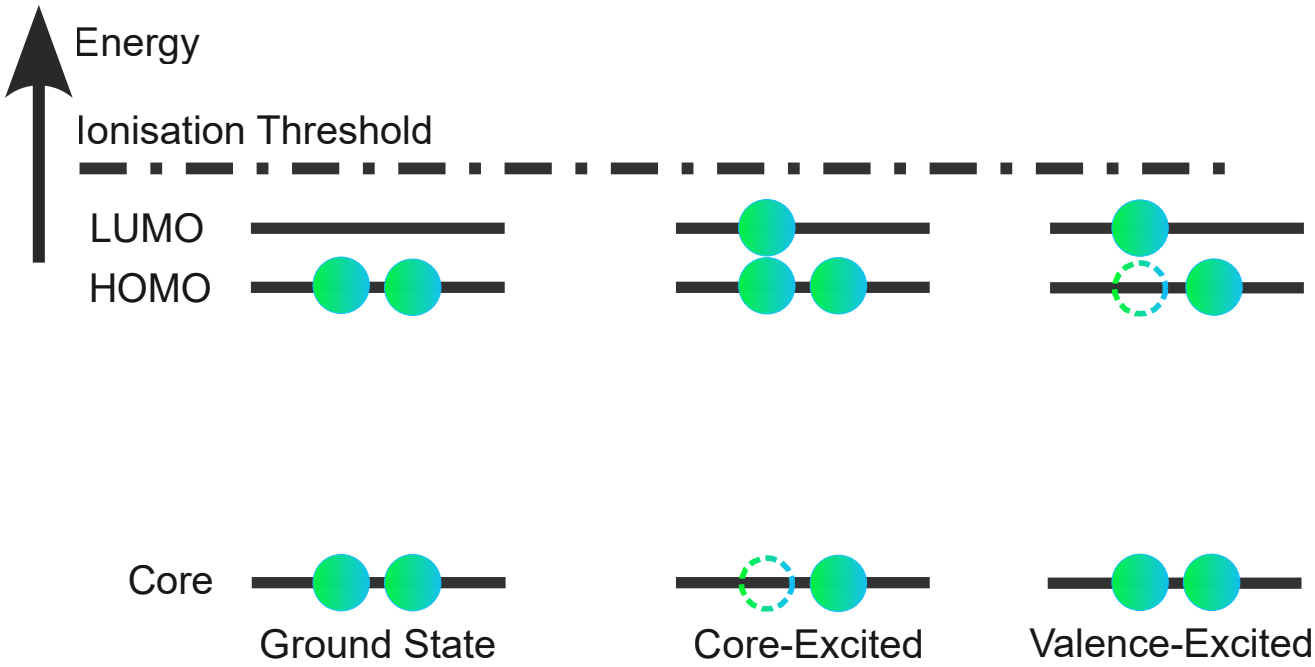

Figure III.6: An energy level illustration describing the important multi-electron configurations, relevant to molecular excitations in X-ray Raman Scattering. *Left*: The ground state has all electrons in their lowest possible energy state. *Center*: A core excited state can be obtained by starting from the ground state and promoting an electron into an unoccupied state in the valence. An excitation into the lower unoccupied molecular orbital (LUMO) is depicted. In principle, an excitation into a higher lying state (e.g. LUMO+1, LUMO+2) is also a core-excited state. *Right*: A valence-excited state; obtained by an intra-valence excitation. Note the vacancy (shown as a dashed circle) in this case sits in the valence.

simple systems, such as gas-phase atoms[85], however exceptions arise for more complex systems.[III.10]

Another option rather than relying on spontaneous decay to reach a valence-excited state is to drive the system into the desired final state. This can be done by using two incident photons of different energy. These energies are chosen to correspond to transition resonances for the ground $\rightarrow$ core-excited and core-excited $\rightarrow$ valence-excited states. The former is known as the "pump" transition and the latter is known as the "dump" transition. By providing incident photons at dump transition, the process is seeded. This enhancement by the field has the advantage of increasing the cross-section of the process, resulting in orders of magnitude more photons emitted in the forward scattering direction. Enhancements of 10^6 have been reported over the spontaneous RIXS process.[87]

Stimulated RIXS, also known as Stimulated X-ray Raman Scattering (SXRS), has been suc-

[III.10]This isotropic emission can be compromised in solid-phase systems, where highly-ordered layers and cause preferential emission angles for the emitted photons with respect to incident photons.[86]

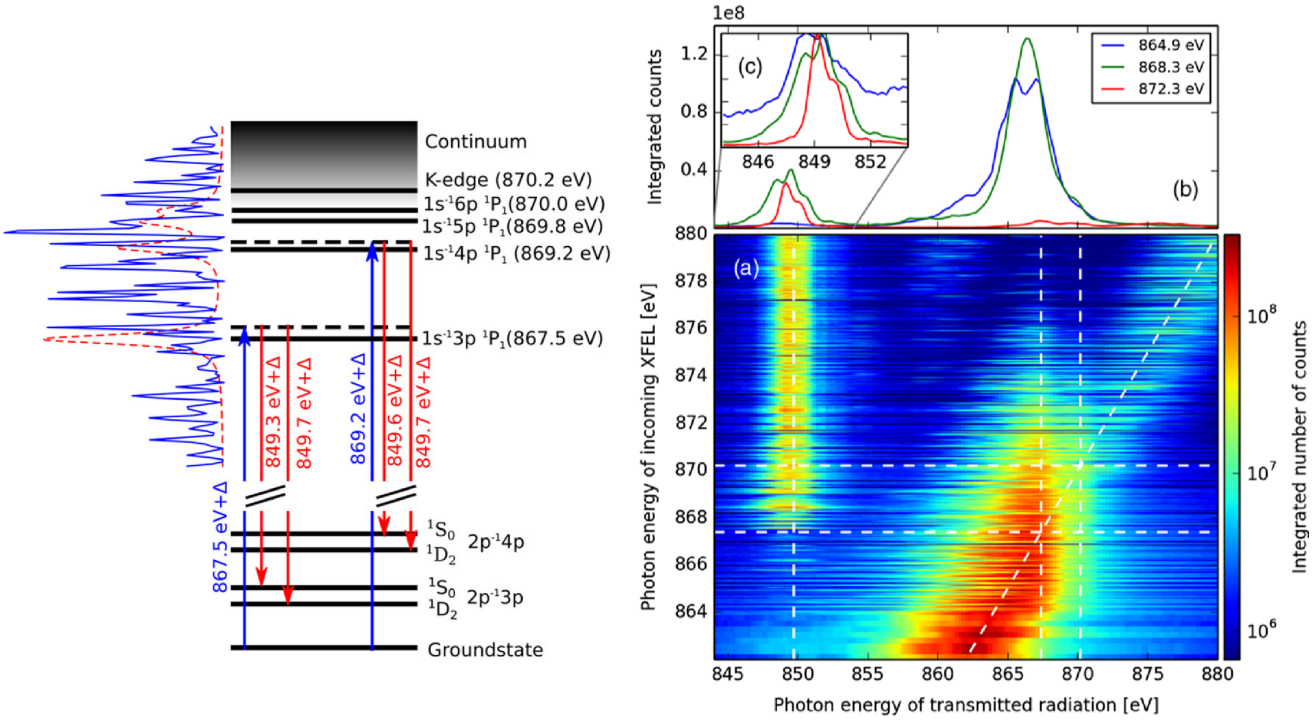

Figure III.7: The energy level scheme and resulting measurements taken to measure Stimulated RIXS in gas-phase by using an X-ray Free Electron Laser, adapted from Weninger, C. et al. (2013). *Physical Review Letters*, **111**(23), 233902.[88] *Left*: A energy level diagram for the system showing the ground state, valence excited states ($2p^{-1}Xp$) and core-excited state (involving $1s^{-1}Xp$). The blue line shows a simulated SASE XFEL pulse, similar to what was used in the experiment. The dashed red line shows the resonance locations and widths for the core-excited states. *Right*: Spectrally resolved x-rays after passing interacting with the sample. The photon energy of the incident x-rays are shown on the vertical axis of the colourmap. The dispersive (diagonal) white dashed line shows the elastically scattered photons (or those that do not interact with the atoms). The vertical line at ~850 eV is evidence of inelastically scattered photons.

cessfully demonstrated in gas-phase systems such as Neon atoms.[88,89] In this experiment, a self amplified spontaneous emission (SASE) beam was focused into a gas cell. The outgoing photon spectrum was used as the observable. The SASE pulse (≈40 fs duration) contained photons on resonance with both the pump (dump) transition blue (red) on the left side of figure III.7. The core-hole lifetime in neon is of order 2-3 fs.[56] This is similar to the femtosecond timescale of population transfer in Neon. Despite the fact that Auger-Meitner decay is the dominant relaxation pathway for a Neon core-hole state[88,90], the radiative decay pathway is still measurable in this experiment, as the process is amplified by stimulated emission.[III.11]

In the false colourmap on the right side of figure III.7, there is a threshold feature at a transmitted energy of 850 eV, which shows up when the incident photons exceed ~ 868 eV. We note

[III.11] In Weninger et al.[88], the ratio of radiative versus non-radiative decay is quoted as approximately 1:50.

that this feature is less intense than the dispersive feature and is itself non-dispersive. The upper panel of the colourmap shows three single shot spectra at different central photon energies. In the zoomed-in inset, we can see that the energy of the inelastically scattered photons does not change. This indicates this emission energy corresponds to a property of the atom, not the changing energy of the incident x-rays. This is the expected properties of a inelastic scattering event from a core-excited to a valence-excited state.

When attempting to demonstrate SXRS in isolated molecules rather than atoms, complications can occur. The SXRS process is more difficult to detect in molecules. In a molecule, there are vibrational degrees of freedom due to nuclear motion. This reduces the cross-section for the required transitions necessary for SXRS.[91,92]

Additionally, the cross-section for x-ray interactions with isolated molecular systems can be reduced compared to their atomic counterparts. This is as it depends on the orientation of the polarization vector of the x-ray compared to the transition dipole of the molecule.[93] See equation (II.31). Only a fraction of the sample will have an alignment for a sizable cross-section in a randomly orientated sample in the gas phase.

III.5 Stimulated X-ray Raman scattering in the impulsive limit

Due to recent advances in XFEL electron beam manipulation, it is now possible to prepare a system in coherent superposition of valence-excited states. This electronic excitation is done by exciting the system with one very wide bandwidth, ultrashort pulse. This type of XFEL mode has been developed at LCLS and is known as X-ray Laser-Enhanced Attosecond Pulse generation (XLEAP), and has been demonstrated to produce pulses of 400 as duration at 570 eV.[75] As a result, the ultrashort pulses produced by the XFEL are less than two times the Fourier transform limit.

Unlike SASE XFEL pulses which can have a durations of tens of femtoseconds,[94,95] XLEAP pulses offer the opportunity to drive a system from its ground state to a coherent superposition of valence-excited state over a timescale on the order of or faster than the characteristic timescale of the system. Driving the system in this manner is referred to as impulsive.

The impulsive limit has been demonstrated and used for some time in impulsive stimulated Raman scattering to generate coherent vibrational wave packets using optical wavelength light.[96] We discuss here its x-ray analogue, impulsive stimulated x-ray Raman scattering

(ISXRS).

In ISXRS, an x-ray pulse with a broad bandwidth is required. This bandwidth must be simultaneously on resonance with the pump (ground state → core-excited state) and Stokes-Raman (core-excited → valence-excited) transitions. See figure III.8.

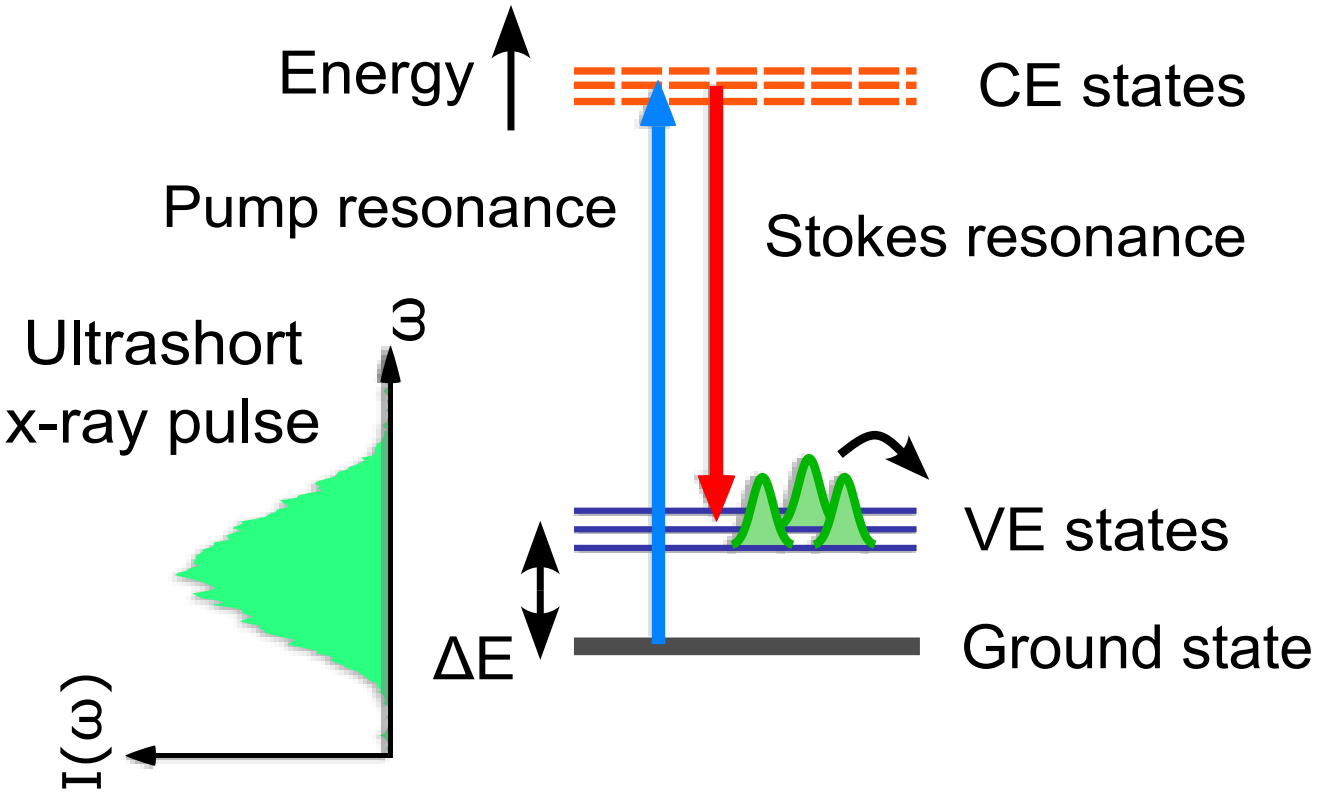

Figure III.8: The energy level scheme and transitions involved in the impulsive stimulated x-ray Raman process, adapted from the supplementary material of Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[97] Unlike previous diagrams, here the states are not orbitals that individual electrons can inhabit, but rather states the molecule can occupy. An ultrashort pulse has its central photon energy tuned to the pre-edge region of the molecule, slightly below the ionization edge. A real XLEAP pulse is shown on left side, and its coherent bandwidth is sufficient to allow it to span both the pump and Stokes transition resonance simultaneously. Population is then driven from the ground state to the valence-excited (VE in figure) states. This two-photon process is resonantly enchanced by the intermediate core-excited (CE in figure) states.

As core-excited states are highly localised to their corresponding atomic site, the Stokes-Raman transition produces a valence-excited electronic electronic wavepacket that is initially well localised. Therefore, the ISXRS process can create an excitation in a site specific manner. However, unlike in the x-ray pump x-ray probe scheme described back in section III.3, the disturbance is created in the neutral parent molecule.

The observable that corresponds to successful population of valence-excited states is the

emission of photons at the Stokes resonance. This is dissimilar to the x-ray pump x-ray probe scheme described in section III.3, where the observable is the emission of Auger-Meitner electrons. Electrons can undergo internal and surface scattering in condensed-phase bulk systems.[98] They can also be subject to space charge effects for high sample number densities[99,100] when propagating to a detector. This makes Auger-Meitner spectroscopy more difficult than detecting scattered photons for condensed-phase targets.[101] This makes ISXRS more ideal in comparison for investigating electron motion in all phases of matter.

In order to describe the ISXRS process, we return to the minimum coupling Hamiltonian defined in equation (II.22). Its interaction Hamiltonian describing the matter-radiation field coupling reads

$$\mathcal{H}_{\text{int}} = -\frac{1}{2m} q \left[\mathbf{P}\cdot\mathbf{A} + \mathbf{A}\cdot\mathbf{P} + \mathbf{A}\cdot\mathbf{A}\right]. \tag{III.5}$$

To first order, the $\mathbf{P}\cdot\mathbf{A}$ term describes single photon absorption and emission, hence its use for x-ray absorption processes.[102] It is therefore inappropriate to describe two-photon scattering. In first order, the $\mathbf{A}\cdot\mathbf{A}$ term describes elastic scattering interactions, where the incident and emitted photon occupy the same field mode.[III.12] Consequently, it is also inappropriate to describe the ISXRS process.

Therefore, we are forced resort to second-order in the interaction Hamiltonian and consider only the contribution of the $\mathbf{P}\cdot\mathbf{A}$ term. Due to the nuclear mass, the response of the nuclei to the radiation field is neglected. Writing the interaction Hamiltonian under second quantization reads as

$$\mathcal{H}_{\text{int}} = \int d^3\mathbf{x}\ \phi^\dagger(\mathbf{x})[\mathbf{A}\cdot\mathbf{P}]\phi(\mathbf{x}), \tag{III.6}$$

where the field operators $\phi(\mathbf{x})$ and $\phi^\dagger(\mathbf{x})$ create and annihilate electrons respectively.[III.13] In its quantized form the general expression for a radiation field is

$$\mathbf{A}(\mathbf{x},t) = \sum_{\mathbf{k},\lambda}\left(\frac{\hbar}{2\epsilon_0 V\omega_{\mathbf{k}}}\right)^{\frac{1}{2}}\left(\epsilon_{\mathbf{k},\lambda}\, a_{\mathbf{k},\lambda}\, \exp\left[i(\mathbf{k}\cdot\mathbf{x}-\omega t)\right] + \epsilon^*_{\mathbf{k},\lambda}\, a^\dagger_{\mathbf{k},\lambda}\, \exp\left[-i(\mathbf{k}\cdot\mathbf{x}-\omega t)\right]\right), \tag{III.7}$$

[III.12] Field mode here refers to a specific energy and momentum of a photon. I.e. a given wavevector $\mathbf{k}$. The $\mathbf{A}\cdot\mathbf{A}$ describes the elastic forward scattering interaction i.e. where $\mathbf{k}_{\text{in}} = \mathbf{k}_{\text{out}}$. For ISXRS, $\mathbf{k}_{\text{in}} \propto \mathbf{k}_{\text{out}}$ but $|\mathbf{k}_{\text{in}}| \neq |\mathbf{k}_{\text{out}}|$. Therefore, the $\mathbf{A}\cdot\mathbf{A}$ cannot describe the ISXRS interaction.

[III.13] We are forced to invoke second quantization here as we need to describe the absorption and emission of photons in the scattering process. Second quantization allows for the description of interactions which do not have a fixed number of particles.

where similar to before, $a^{\dagger}_{\mathbf{k},\lambda}$ and $a_{\mathbf{k},\lambda}$ are photon field operators that create and annihilate photons of wavevector $\mathbf{k}$ and polarization λ and $\epsilon_{\mathbf{k},\lambda}$, $\epsilon^{*}_{\mathbf{k},\lambda}$, are polarization vectors.[103]

In second-order perturbation theory, the transition matrix element reads

$$\mathcal{M}^{(2)}_{fi} = \sum_{M} \frac{\langle f|\mathcal{H}_{int}|M\rangle \, \langle M|\mathcal{H}_{int}|i\rangle}{E_i - E_M + i\epsilon}. \tag{III.8}$$

from an initial state $|i\rangle$ to a final state $|f\rangle$.[III.14] From the structure of this expression we notice that $\mathcal{H}_{int}$ appears twice. The right (left) one represents absorption (emission) at the pump (Stokes) resonance.

Without detailed calculation we can see that from the definition given of the vector potential given in (III.7) there will be four terms involving photon-field operators. Considering the photon field component of the initial (1 pump photon, no Stokes photons) and final (no pump photons, 1 Stokes photon) Fock states we can write

$$\begin{aligned}\mathcal{M}^{(2)}_{fi} \sim \sum_{\mathbf{k},\lambda} \langle 1_{Stokes}, 0_{pump}| \Big\{ & a^{(Stokes)}_{\mathbf{k},\lambda} a^{(pump)}_{\mathbf{k},\lambda} + a^{(Stokes)}_{\mathbf{k},\lambda} a^{(pump)\dagger}_{\mathbf{k},\lambda} \\ & + a^{(Stokes)\dagger}_{\mathbf{k},\lambda} a^{(pump)}_{\mathbf{k},\lambda} + a^{(Stokes)\dagger}_{\mathbf{k},\lambda} a^{(pump)\dagger}_{\mathbf{k},\lambda} \Big\} |0_{Stokes}, 1_{pump}\rangle.\end{aligned} \tag{III.9}$$

The $a^{(Stokes)} a^{(pump)}$ and $a^{(Stokes)} a^{(pump)\dagger}$ terms vanish as they remove a photon from the already empty Stokes channel, resulting in a null state. The final $a^{(Stokes)\dagger} a^{(pump)\dagger}$ term results in the wrong number of photons in the final state. We reason that, **the only relevant and surviving term is** $a^{(Stokes)\dagger} a^{(pump)}$. This enables us to remove the photon field operators and all modes of the field $\mathbf{k}, \lambda$ that do not correspond to the pump and Stokes resonances.

Consider an ISXRS process starting with a system in its multi-particle ground state $|\Psi^{N_{el}}_0\rangle$ and ending in a valence-excited state $|\Psi^{N_{el}}_{VE}\rangle$. Suppose it does this by a two-photon scattering process, via an intermediate core-excited state $|\Psi^{N_{el}}_{M}\rangle$. A simplified expression[III.15] for transition

[III.14] This is a direct analogue of what was shown in equation (II.31), but allowing more than one coupling to the photon field.

[III.15] We do not consider the effect of the photon flux on the transition rate, however it has quadratic dependence.

rate for this interaction can now be written following Rohringer[104] as

$$\Gamma_{0\to VE}^{(ISXRS)} \propto 2\pi\delta(E_{VE}^{N_{el}} - E_0^{N_{el}} - \hbar\omega_{pump,\,Stokes})\frac{1}{\omega_{Stokes}\omega_{pump}} \times \left| \int d^3\mathbf{x}\, d^3\mathbf{x}' \sum_M \frac{\mathcal{K}_{VE,M}(\mathbf{x})}{E_M^{N_{el}} - E_{VE}^{N_{el}} - \hbar\omega_{Stokes} + i\epsilon_2} \frac{\mathcal{K}_{M,0}(\mathbf{x}')}{E_M^{N_{el}} - E_0^{N_{el}} - \hbar\omega_{pump} + i\epsilon_1} \right|^2 . \tag{III.10}$$

Here, $E_{VE}^{N_{el}}$ and $E_0^{N_{el}}$ are the eigenenergies of the ground & final valence-excited states and $\hbar\omega_{pump,\,Stokes} = \hbar\omega_{pump} - \hbar\omega_{Stokes} = \Delta E$ as shown in figure III.8. The summation index M counts intermediate states. In our case, the sum is over many different core-excited state. The inclusion of the $i\epsilon_1$ and $i\epsilon_2$ terms in the denominator is a standard procedure used in scattering theory and perturbation theory to ensure the spatial integral always convergences.[105] Additionally, equation (III.10) contains the shorthand for the inner products

$$\begin{aligned} \mathcal{K}_{VE,M}(\mathbf{x}) &= \langle \Psi_{VE}^{N_{el}} | \hat{\phi}^\dagger(\mathbf{x}) \exp(-i\mathbf{k}_{Stokes}\mathbf{x}) \epsilon_{\mathbf{k}_{Stokes}}^* \cdot \mathbf{P}\hat{\phi}(\mathbf{x}) | \Psi_M^{N_{el}} \rangle \\ \mathcal{K}_{M,0}(\mathbf{x}') &= \langle \Psi_M^{N_{el}} | \hat{\phi}^\dagger(\mathbf{x}') \exp(i\mathbf{k}_{pump}\mathbf{x}') \epsilon_{\mathbf{k}_{pump}} \cdot \mathbf{P}'\hat{\phi}(\mathbf{x}') | \Psi_0^{N_{el}} \rangle . \end{aligned} \tag{III.11}$$

In these expressions $|\Psi_M^{N_{el}}\rangle$ represents a core-excited state. In both (III.10) and (III.11) the label N_{el} is attached to the state to reiterate the fact the that an electron is never removed from the molecule, only excited or de-excited. $\mathbf{P} = -i\hbar\nabla$ refers to the canonical momentum. $\mathbf{P}' = -i\hbar\nabla'$ refers to primed coordinates.

This complex expression can give useful insights into the ISXRS process. The inner product terms $\mathcal{K}_{M,0}(\mathbf{x}')$ and $\mathcal{K}_{VE,\,M}(\mathbf{x})$ can be interpreted as representing the individual steps indicated by the blue and red arrows drawn in figure III.8 respectively, as they describe the coupling of the many-bony molecular states to the radiation field.

Consider $\mathcal{K}_{M,0}(\mathbf{x}')$. We start in the ground state $|\Psi_0^{N_{el}}\rangle$. The annihilation operator $\hat{\phi}(\mathbf{x}')$ acts on this state, removing an electron from a core shell. A photon is annihilated in the channel $\mathbf{k}_{pump}$ by an annihilation operator attached to $\exp(i\mathbf{k}_{pump}\mathbf{x}')\epsilon_{\mathbf{k}_{pump}}$.[III.16] The electron creation operator $\hat{\phi}^\dagger(\mathbf{x}')$ acts to the right to now add back an electron, but in for example the LUMO. The result is that the whole molecular system now enters a core-excited state $|\Psi_M^{N_{el}}\rangle$, an intermediate state to the ISXRS process.

A similar story applies to $\mathcal{K}_{VE,\,M}(\mathbf{x})$. The system starts this time in a core-excited state $|\Psi_M^{N_{el}}\rangle$.

[III.16] The annihilation operator for the photon field is not explicitly written as the reasoning presented in equation (III.9) enables us replace it with a scalar in the transition rate expression (III.10).

An electron annihilation operator $\hat{\phi}(\mathbf{x})$ acts to the right to remove an electron from, for example, the HOMO. A photon creation operator attached to $\exp(-i\mathbf{k}_{\text{Stokes}}\mathbf{x})\epsilon^*_{\mathbf{k}_{\text{Stokes}}}$ acts to add a photon into the Stokes channel.[III.17] $\hat{\phi}^\dagger(\mathbf{x})$ then creates an electron to fill the core-hole.

Interestingly, in order to drive population into a coherent superposition of valence-excited states $|\Psi^{N_{el}}_{VE}\rangle$, it is not necessary to drive population into intermediate core-excited states. It is instead possible to drive population into the desired valence-excite states via virtual states.[III.18]

This can be accomplished by ensuring the denominator in equation (III.10) meets the condition

$$E^{N_{el}}_M - E^{N_{el}}_0 - \hbar\omega_{\text{pump}} \neq 0. \tag{III.12}$$

Namely, tuning the pump to be slightly off-resonance with the core-excited states. This way, the population transfer from $|\Psi^{N_{el}}_0\rangle$ to $|\Psi^{N_{el}}_{VE}\rangle$ benefits from resonant enhancement. Avoiding driving population into real intermediate states prevents the system evolving in the $|\Psi^{N_{el}}_M\rangle$ states, simplifying the time evolution of the created excitation.[97]

[III.17] For the same reason, the photon field creation operator is not explicitly written.
[III.18] Virtual state here refers to a state which is not an energy eigenstate of the molecular Hamiltonian, and therefore, does not possess a well-defined energy.

References

[1]W. C. Röntgen, "Ueber eine neue Art von Strahlen", Annalen der Physik **300**, 12–17, ISSN: 1521-3889 (1898) 10.1002/andp.18983000103.

[2]T. J. A. Wolf, R. H. Myhre, J. P. Cryan, S. Coriani, R. J. Squibb, A. Battistoni, N. Berrah, C. Bostedt, P. Bucksbaum, G. Coslovich, R. Feifel, K. J. Gaffney, J. Grilj, T. J. Martinez, S. Miyabe, S. P. Moeller, M. Mucke, A. Natan, R. Obaid, T. Osipov, O. Plekan, S. Wang, H. Koch, and M. Gühr, "Probing ultrafast $\pi\pi^*/n\pi^*$ internal conversion in organic chromophores via K-edge resonant absorption", Nature Communications **8**, ISSN: 2041-1723 (2017) 10.1038/s41467-017-00069-7.

[3]M. L. Knotek and P. J. Feibelman, "Ion Desorption by Core-Hole Auger Decay", Physical Review Letters **40**, 964–967, ISSN: 0031-9007 (1978) 10.1103/physrevlett.40.964.

[4]R. OIKE, Y. OKAMOTO, T. TOKUSHIMA, T. NAKAMURA, and K. AMEZAWA, "In-situ Simultaneous Soft X-ray Absorption and Emission Spectroscopy under Controlled Atmosphere and Temperature", Electrochemistry **84**, 793–796, ISSN: 2186-2451 (2016) 10.5796/electrochemistry.84.793.

[5]K. Amezawa, Y. Fujimaki, K. Mizuno, Y. Kimura, T. Nakamura, K. Nitta, Y. Terada, F. Iguchi, H. Yugami, K. Yashiro, and T. Kawada, "(Invited) Triple Phase Boundary Reaction in a Mixed-Conducting SOFC Cathode", ECS Transactions **77**, 41–47, ISSN: 1938-5862 (2017) 10.1149/07710.0041ecst.

[6]F. Giannici, G. Canu, A. Chiara, M. Gambino, C. Aliotta, A. Longo, V. Buscaglia, and A. Martorana, "Cation Diffusion and Segregation at the Interface between Samarium-Doped Ceria and LSCF or LSFCu Cathodes Investigated with X-ray Microspectroscopy", ACS Applied Materials &; Interfaces **9**, 44466–44477, ISSN: 1944-8252 (2017) 10.1021/acsami.7b13377.

[7]K. R. Wilson, B. S. Rude, T. Catalano, R. D. Schaller, J. G. Tobin, D. T. Co, and R. J. Saykally, "X-ray Spectroscopy of Liquid Water Microjets", The Journal of Physical Chemistry B **105**, 3346–3349, ISSN: 1520-5207 (2001) 10.1021/jp010132u.

[8]D. Fischer, U. Döbler, D. Arvanitis, L. Wenzel, K. Baberschke, and J. Stöhr, "Carbon K-edge structure of chemisorbed molecules by means of fluorescence detection", Surface Science **177**, 114–120, ISSN: 0039-6028 (1986) 10.1016/0039-6028(86)90261-x.

[9]J. Holburg, S. Figul, A. Charvat, H. Bluhm, B. Abel, G. Marowsky, D.-D. Mai, and K. Mann, "Soft X-Ray Absorption Spectroscopy With a Flat Liquid Jet in Vacuum Using a Table-Top Laser-Induced Plasma Source", X-Ray Spectrometry **54**, 394–400, ISSN: 1097-4539 (2025) 10.1002/xrs.3474.

[10]K. R. Wilson, M. Cavalleri, B. S. Rude, R. D. Schaller, T. Catalano, A. Nilsson, R. J. Saykally, and L. G. M. Pettersson, "X-ray Absorption Spectroscopy of Liquid Methanol Microjets: Bulk Electronic Structure and Hydrogen Bonding Network", The Journal of Physical Chemistry B **109**, 10194–10203, ISSN: 1520-5207 (2005) 10.1021/jp049278u.

[11]R. K. Lam, J. W. Smith, and R. J. Saykally, "Communication: Hydrogen bonding interactions in water-alcohol mixtures from X-ray absorption spectroscopy", The Journal of Chemical Physics **144**, ISSN: 1089-7690 (2016) 10.1063/1.4951010.

[12]T. A. Carlson and R. M. White, "Measurement of the Relative Abundances and Recoil-Energy Spectra of Fragment Ions Produced as the Initial Consequences of X-Ray Interaction with CH_3I, HI, and DI", The Journal of Chemical Physics **44**, 4510–4520, ISSN: 1089-7690 (1966) 10.1063/1.1726667.

[13]J. Leroux, J.-Y. Chesnel, C. Ortiz-Mahecha, A. Nair, B. Oostenrijk, L. Pille, F. Trinter, L. Schwob, and S. Bari, "Structures of Gas-Phase Hydrated Phosphotyrosine Revealed by Soft X-ray Action Spectroscopy", Chemistry – A European Journal **31**, ISSN: 1521-3765 (2025) 10.1002/chem.202403665.

[14]R. A. Ingle, A. Banerjee, C. Bacellar, T. R. Barillot, L. Longetti, M. Coreno, M. de Simone, F. Zuccaro, L. Poletto, P. Miotti, A. Röder, A. Stolow, M. S. Schuurman, M. Odelius, and M. Chergui, "Carbon K-edge x-ray emission spectroscopy of gas phase ethylenic molecules", Journal of Physics B: Atomic, Molecular and Optical Physics **55**, 044001, ISSN: 1361-6455 (2022) 10.1088/1361-6455/ac4e66.

[15]Y. Hikosaka, Y. Velkov, E. Shigemasa, T. Kaneyasu, Y. Tamenori, J. Liu, and F. Gel'mukhanov, "X-Ray Absorption Measured in the Resonant Auger Scattering Mode", Physical Review Letters **101**, ISSN: 1079-7114 (2008) 10.1103/physrevlett.101.073001.

[16]O. S. Ablyasova, M. Guo, V. Zamudio-Bayer, M. Kubin, T. Gitzinger, M. da Silva Santos, M. Flach, M. Timm, M. Lundberg, J. T. Lau, and K. Hirsch, "Electronic Structure of the Complete Series of Gas-Phase Manganese Acetylacetonates by X-ray Absorption Spectroscopy", The Journal of Physical Chemistry A **127**, 7121–7131, ISSN: 1520-5215 (2023) 10.1021/acs.jpca.3c02794.

[17]P. W. Hillyard, S. V. N. T. Kuchibhatla, T. E. Glover, M. P. Hertlein, N. Huse, P. Nachimuthu, L. V. Saraf, S. Thevuthasan, and K. J. Gaffney, "Atomic resolution mapping of the excited-state electronic structure of Cu_2O with time-resolved x-ray absorption spectroscopy", Physical Review B **80**, ISSN: 1550-235X (2009) 10.1103/physrevb.80.125210.

[18]C. H. M. van Oversteeg, H. Q. Doan, F. M. F. de Groot, and T. Cuk, "In situ X-ray absorption spectroscopy of transition metal based water oxidation catalysts", Chemical Society Reviews **46**, 102–125, ISSN: 1460-4744 (2017) 10.1039/c6cs00230g.

[19]M. Kubin, M. Guo, T. Kroll, H. Löchel, E. Källman, M. L. Baker, R. Mitzner, S. Gul, J. Kern, A. Föhlisch, A. Erko, U. Bergmann, V. Yachandra, J. Yano, M. Lundberg, and P. Wernet, "Probing the oxidation state of transition metal complexes: a case study on how charge and spin densities determine Mn L-edge X-ray absorption energies", Chemical Science **9**, 6813–6829, ISSN: 2041-6539 (2018) 10.1039/c8sc00550h.

[20]T.-J. Kühn, J. Hormes, N. Matoussevitch, H. Bönnemann, and P. Glatzel, "Site-Selective High-Resolution X-ray Absorption Spectroscopy and High-Resolution X-ray Emission Spectroscopy of Cobalt Nanoparticles", Inorganic Chemistry **53**, 8367–8375, ISSN: 1520-510X (2014) 10.1021/ic5000183.

[21]J. Yano and V. K. Yachandra, "X-ray absorption spectroscopy", Photosynthesis Research **102**, 241–254, ISSN: 1573-5079 (2009) 10.1007/s11120-009-9473-8.

[22]D. Joseph, S. Basu, S. Jha, and D. Bhattacharyya, "Chemical shifts of K-X-ray absorption edges on copper in different compounds by X-ray absorption spectroscopy (XAS) with Synchrotron radiation", Nuclear Instruments and Methods in Physics Research Section B: Beam

Interactions with Materials and Atoms **274**, 126–128, ISSN: 0168-583X (2012) 10.1016/j.nimb.2011.12.011.

[23]B. Henke, E. Gullikson, and J. Davis, "X-Ray Interactions: Photoabsorption, Scattering, Transmission, and Reflection at E = 50-30, 000 eV, Z = 1-92", Atomic Data and Nuclear Data Tables **54**, 181–342, ISSN: 0092-640X (1993) 10.1006/adnd.1993.1013.

[24]R. M. Martin, *Electronic Structure: Basic Theory and Practical Methods* (Cambridge University Press, Apr. 2004), ISBN: 9780511805769, 10.1017/cbo9780511805769.

[25]A. Iglesias-Juez, G. L. Chiarello, G. S. Patience, and M. O. Guerrero-Pérez, "Experimental methods in chemical engineering: X-ray absorption spectroscopy - XAS, XANES, EXAFS", The Canadian Journal of Chemical Engineering **100**, 3–22, ISSN: 1939-019X (2021) 10.1002/cjce.24291.

[26]C. M. B. Henderson, J. M. Charnock, G. Cressey, and D. T. Griffen, "An EXAFS study of the local structural environments of Fe, Co, Zn and Mg in natural and synthetic staurolites", Mineralogical Magazine **61**, 613–625, ISSN: 1471-8022 (1997) 10.1180/minmag.1997.061.408.01.

[27]A. Romanchuk, A. Trigub, T. Plakhova, A. Kuzenkova, R. Svetogorov, K. Kvashnina, and S. Kalmykov, "Effective coordination numbers from EXAFS: general approaches for lanthanide and actinide dioxides", Journal of Synchrotron Radiation **29**, 288–294, ISSN: 1600-5775 (2022) 10.1107/s160057752101300x.

[28]H. Guesmi and P. Massiani, "A combined EXAFS and DFT study of the Ni2+ environment in dehydrated Ni/NaX", Catalysis Today **177**, 25–30, ISSN: 0920-5861 (2011) 10.1016/j.cattod.2011.04.047.

[29]D. Lundberg and I. Persson, "On solvated tin(<scp>iv</scp>) ions and the coordination chemistry of high-valent d10 metal ions", Dalton Transactions **48**, 9089–9093, ISSN: 1477-9234 (2019) 10.1039/c9dt01097a.

[30]P. Atkins, J. De Paula, and J. Keeler, *Atkins' physical chemistry*, en, 11th ed. (Oxford University Press, London, England, Dec. 2017), ISBN: 9780198847816.

[31]J. A. Sellberg, S. Kaya, V. H. Segtnan, C. Chen, T. Tyliszczak, H. Ogasawara, D. Nordlund, L. G. M. Pettersson, and A. Nilsson, "Comparison of x-ray absorption spectra between water and ice: New ice data with low pre-edge absorption cross-section", The Journal of Chemical Physics **141**, ISSN: 1089-7690 (2014) 10.1063/1.4890035.

[32]M. Ekimova, W. Quevedo, M. Faubel, P. Wernet, and E. T. J. Nibbering, "A liquid flatjet system for solution phase soft-x-ray spectroscopy", Structural Dynamics **2**, ISSN: 2329-7778 (2015) 10.1063/1.4928715.

[33]O. Alexander, J. C. T. Barnard, E. W. Larsen, T. Avni, S. Jarosch, C. Ferchaud, A. Gregory, S. Parker, G. Galinis, A. Tofful, D. Garratt, M. R. Matthews, and J. P. Marangos, "Observation of recollision-based high-harmonic generation in liquid isopropanol and the role of electron scattering", Physical Review Research **5**, ISSN: 2643-1564 (2023) 10.1103/physrevresearch.5.043030.

[34]A. D. Smith, T. Balčiūnas, Y.-P. Chang, C. Schmidt, K. Zinchenko, F. B. Nunes, E. Rossi, V. Svoboda, Z. Yin, J.-P. Wolf, and H. J. Wörner, "Femtosecond Soft X-ray Absorption Spectroscopy of Liquids with a Water-Window High-Harmonic Source", The Journal of Physical Chemistry Letters **11**, 1981–1988, ISSN: 1948-7185 (2020) 10.1021/acs.jpclett.9b03559.

[35]D. J. Hoffman, T. B. Van Driel, T. Kroll, C. J. Crissman, E. S. Ryland, K. J. Nelson, A. A. Cordones, J. D. Koralek, and D. P. DePonte, "Microfluidic liquid sheets as large-area targets for high repetition XFELs", Frontiers in Molecular Biosciences **9**, ISSN: 2296-889X (2022) 10.3389/fmolb.2022.1048932.

[36]M. Nagasaka, "Soft X-ray absorption spectroscopy in the low-energy region explored using an argon gas window", Journal of Synchrotron Radiation **27**, 959–962, ISSN: 1600-5775 (2020) 10.1107/s1600577520005883.

[37]D. H. Zadeh, "Electronic structures of elements according to ionization energies", Journal of Molecular Modeling **23**, ISSN: 0948-5023 (2017) 10.1007/s00894-017-3534-2.

[38]M. Kavčič and M. Žitnik, "Shake-up and shake-off processes", in *International Tables for Crystallography* (International Union of Crystallography, Nov. 2023), pp. 653–658, ISBN: 9781119433941, 10.1107/s1574870722005535.

[39]C. J. Clarke, S. Maxwell-Hogg, E. F. Smith, R. R. Hawker, J. B. Harper, and P. Licence, "Resolving X-ray photoelectron spectra of ionic liquids with difference spectroscopy", Physical Chemistry Chemical Physics **21**, 114–123, ISSN: 1463-9084 (2019) 10.1039/c8cp06701e.

[40]O. Travnikova, M. Patanen, J. Söderström, A. Lindblad, J. J. Kas, F. D. Vila, D. Céolin, C. Nicolas, T. Marchenko, G. Goldsztejn, L. Journel, R. Guillemin, T. X. Carroll, K. J. Børve, P. Decleva, J. J. Rehr, N. Mårtensson, M. Simon, S. Svensson, and L. J. Sæthret, "Energy dependent relative cross sections in carbon 1s photoionization", Journal of Physics: Conference Series **1412**, 152050, ISSN: 1742-6596 (2020) 10.1088/1742-6596/1412/15/152050.

[41]G. Greczynski and L. Hultman, "A step-by-step guide to perform x-ray photoelectron spectroscopy", Journal of Applied Physics **132**, ISSN: 1089-7550 (2022) 10.1063/5.0086359.

[42]G. Miloshevsky, V. Tolkach, G. Shani, and S. Rozin, "Calculated gadolinium atomic electron energy levels and Auger electron emission probability as a function of atomic number Z", Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms **192**, 360–364, ISSN: 0168-583X (2002) 10.1016/s0168-583x(02)00489-5.

[43]A. J. Achkar, T. Z. Regier, H. Wadati, Y.-J. Kim, H. Zhang, and D. G. Hawthorn, "Bulk sensitive x-ray absorption spectroscopy free of self-absorption effects", Physical Review B **83**, ISSN: 1550-235X (2011) 10.1103/physrevb.83.081106.

[44]L. Tröger, D. Arvanitis, K. Baberschke, H. Michaelis, U. Grimm, and E. Zschech, "Full correction of the self-absorption in soft-fluorescence extended x-ray-absorption fine structure", Physical Review B **46**, 3283–3289, ISSN: 1095-3795 (1992) 10.1103/physrevb.46.3283.

[45]Y. Tamenori, M. Morita, and T. Nakamura, "Two-dimensional approach to fluorescence yield XANES measurement using a silicon drift detector", Journal of Synchrotron Radiation **18**, 747–752, ISSN: 1600-5775 (2011) 10.1107/s0909049511027531.

[46]J. Wang, C.-S. Hsu, T.-S. Wu, T.-S. Chan, N.-T. Suen, J.-F. Lee, and H. M. Chen, "In situ X-ray spectroscopies beyond conventional X-ray absorption spectroscopy on deciphering dynamic configuration of electrocatalysts", Nature Communications **14**, ISSN: 2041-1723 (2023) 10.1038/s41467-023-42370-8.

[47]H.-E. Mahnke, "Lise Meitner, β-decay and non-radiative electromagnetic transitions", Notes and Records: the Royal Society Journal of the History of Science **76**, 107–116, ISSN: 1743-0178 (2020) 10.1098/rsnr.2020.0036.

[48]L. Meitner, "Über den Zusammenhang zwischen β- und γ-Strahlen", Zeitschrift für Physik **9**, 145–152, ISSN: 1434-601X (1922) 10.1007/bf01326963.

[49]L. Meitner, "Das β-Strahlenspektrum von UX_1 und seine Deutung", Zeitschrift für Physik **17**, 54–66, ISSN: 1434-601X (1923) 10.1007/bf01328663.

[50]P. Auger, "Sur l'effet photoélectrique composé", Journal de Physique et le Radium **6**, 205–208, ISSN: 0368-3842 (1925) 10.1051/jphysrad:0192500606020500.

[51]E. H. S. Burhop, "The Auger effect", Proceedings of the Royal Society of London. Series A - Mathematical and Physical Sciences **148**, 272–284, ISSN: 2053-9169 (1935) 10.1098/rspa.1935.0018.

[52]J.-R. Zhang, S.-Y. Wang, and W. Hua, "Core Hole Effect to Valence Excitations: Tracking and Visualizing the Same Excitation in XPS Shake-Up Satellites and UV Absorption Spectra", Journal of Chemical Theory and Computation **20**, 6125–6133, ISSN: 1549-9626 (2024) 10.1021/acs.jctc.4c00521.

[53]S. A. Chambers and T. D. Thomas, "Satellite structure in the x-ray photoelectron spectra of gaseous furan, pyrrole, and thiophen", The Journal of Chemical Physics **67**, 2596–2603, ISSN: 1089-7690 (1977) 10.1063/1.435200.

[54]D. P. Chong, "Density functional theory study of the vertical ionization energies of the valence and core electrons of cyclopentadiene, pyrrole, furan, and thiophene", Canadian Journal of Chemistry **89**, 1477–1488, ISSN: 1480-3291 (2011) 10.1139/v11-121.

[55]E. E. Rennie, U. Hergenhahn, O. Kugeler, A. Rüdel, S. Marburger, and A. M. Bradshaw, "A core-level photoionization study of furan", The Journal of Chemical Physics **117**, 6524–6532, ISSN: 1089-7690 (2002) 10.1063/1.1504435.

[56]D. C. Haynes, M. Wurzer, A. Schletter, A. Al-Haddad, C. Blaga, C. Bostedt, J. Bozek, H. Bromberger, M. Bucher, A. Camper, S. Carron, R. Coffee, J. T. Costello, L. F. DiMauro, Y. Ding, K. Ferguson, I. Grguraš, W. Helml, M. C. Hoffmann, M. Ilchen, S. Jalas, N. M. Kabach-

nik, A. K. Kazansky, R. Kienberger, A. R. Maier, T. Maxwell, T. Mazza, M. Meyer, H. Park, J. Robinson, C. Roedig, H. Schlarb, R. Singla, F. Tellkamp, P. A. Walker, K. Zhang, G. Doumy, C. Behrens, and A. L. Cavalieri, "Clocking Auger electrons", Nature Physics **17**, 512–518, ISSN: 1745-2481 (2021) 10.1038/s41567-020-01111-0.

[57]K. Wenig, M. Wieland, A. Baumann, S. Walther, A. Dimitriou, M. J. Prandolini, O. Schepp, I. Bermúdez Macias, M. Sumfleth, N. Stojanovic, S. Düsterer, J. Rönsch-Schulenburg, E. Zapolnova, R. Pan, M. Drescher, and U. Frühling, "Electronic decay of core-excited HCl molecules probed by THz streaking", Structural Dynamics **6**, ISSN: 2329-7778 (2019) 10.1063/1.5091756.

[58]T. Shimizu, T. Sekikawa, T. Kanai, S. Watanabe, and M. Itoh, "Time-Resolved Auger Decay in CsBr Using High Harmonics", Physical Review Letters **91**, ISSN: 1079-7114 (2003) 10.1103/physrevlett.91.017401.

[59]S. Sorensen and S. Svensson, "Resonant photoemission and Auger emission from molecules", Journal of Electron Spectroscopy and Related Phenomena **114–116**, 1–14, ISSN: 0368-2048 (2001) 10.1016/s0368-2048(00)00344-3.

[60]A. Russek and W. Mehlhorn, "Post-collision interaction and the Auger lineshape", Journal of Physics B: Atomic and Molecular Physics **19**, 911–927, ISSN: 0022-3700 (1986) 10.1088/0022-3700/19/6/013.

[61]A. Niehaus, "Analysis of post-collision interactions in Auger processes following near-threshold inner-shell photoionization", Journal of Physics B: Atomic and Molecular Physics **10**, 1845–1857, ISSN: 0022-3700 (1977) 10.1088/0022-3700/10/10/014.

[62]G. C. King, F. H. Read, and R. C. Bradford, "Structure near autoionizing energies in the excitation of bound states of helium, neon and argon by electron impact", Journal of Physics B: Atomic and Molecular Physics **8**, 2210–2224, ISSN: 0022-3700 (1975) 10.1088/0022-3700/8/13/013.

[63]H. W. Berry, "Energy Distribution of Electrons from Ionizing Collisions of Heavy Particles", Physical Review **121**, 1714–1719, ISSN: 0031-899X (1961) 10.1103/physrev.121.1714.

[64]N. Velasquez, O. Travnikova, R. Guillemin, I. Ismail, L. Journel, J. B. Martins, D. Koulentianos, D. Céolin, L. Fillaud, M. L. M. Rocco, R. Püttner, M. N. Piancastelli, M. Simon, S. Sheinerman, L. Gerchikov, and T. Marchenko, "Generalization of the post-collision interaction effect from gas-phase to solid-state systems demonstrated in thiophene and its polymers", Physical Review Research **5**, ISSN: 2643-1564 (2023) 10.1103/physrevresearch.5.013048.

[65]M. B. Johnston and J. Lloyd-Hughes, "Pump-Probe Spectroscopy at Terahertz Frequencies", in *Terahertz Spectroscopy and Imaging* (Springer Berlin Heidelberg, 2012), pp. 251–271, ISBN: 9783642295645, 10.1007/978-3-642-29564-5_10.

[66]M. Saes, C. Bressler, R. Abela, D. Grolimund, S. L. Johnson, P. A. Heimann, and M. Chergui, "Observing Photochemical Transients by Ultrafast X-Ray Absorption Spectroscopy", Physical Review Letters **90**, ISSN: 1079-7114 (2003) 10.1103/physrevlett.90.047403.

[67]R. B. Weakly, C. E. Liekhus-Schmaltz, B. I. Poulter, E. Biasin, R. Alonso-Mori, A. Aquila, S. Boutet, F. D. Fuller, P. J. Ho, T. Kroll, C. M. Loe, A. Lutman, D. Zhu, U. Bergmann, R. W. Schoenlein, N. Govind, and M. Khalil, "Revealing core-valence interactions in solution with femtosecond X-ray pump X-ray probe spectroscopy", Nature Communications **14**, ISSN: 2041-1723 (2023) 10.1038/s41467-023-39165-2.

[68]H. Lu, A. Gauthier, M. Hepting, A. S. Tremsin, A. H. Reid, P. S. Kirchmann, Z. X. Shen, T. P. Devereaux, Y. C. Shao, X. Feng, G. Coslovich, Z. Hussain, G. L. Dakovski, Y. D. Chuang, and W. S. Lee, "Time-resolved RIXS experiment with pulse-by-pulse parallel readout data collection using X-ray free electron laser", Scientific Reports **10**, ISSN: 2045-2322 (2020) 10.1038/s41598-020-79210-4.

[69]N. Ignatova, V. V. da Cruz, R. C. Couto, E. Ertan, M. Odelius, H. Ågren, F. F. Guimarães, A. Zimin, S. P. Polyutov, F. Gel'mukhanov, and V. Kimberg, "Infrared-pump–x-ray-probe spectroscopy of vibrationally excited molecules", Physical Review A **95**, ISSN: 2469-9934 (2017) 10.1103/physreva.95.042502.

[70]K. Pande, C. D. M. Hutchison, G. Groenhof, A. Aquila, J. S. Robinson, J. Tenboer, S. Basu, S. Boutet, D. P. DePonte, M. Liang, T. A. White, N. A. Zatsepin, O. Yefanov, D. Morozov, D. Oberthuer, C. Gati, G. Subramanian, D. James, Y. Zhao, J. Koralek, J. Brayshaw, C. Kupitz,

C. Conrad, S. Roy-Chowdhury, J. D. Coe, M. Metz, P. L. Xavier, T. D. Grant, J. E. Koglin, G. Ketawala, R. Fromme, V. Šrajer, R. Henning, J. C. H. Spence, A. Ourmazd, P. Schwander, U. Weierstall, M. Frank, P. Fromme, A. Barty, H. N. Chapman, K. Moffat, J. J. van Thor, and M. Schmidt, "Femtosecond structural dynamics drives the trans/cis isomerization in photoactive yellow protein", Science **352**, 725–729, ISSN: 1095-9203 (2016) 10.1126/science.aad5081.

[71] A. Aquila, M. S. Hunter, R. B. Doak, R. A. Kirian, P. Fromme, T. A. White, J. Andreasson, D. Arnlund, S. Bajt, T. R. M. Barends, M. Barthelmess, M. J. Bogan, C. Bostedt, H. Bottin, J. D. Bozek, C. Caleman, N. Coppola, J. Davidsson, D. P. DePonte, V. Elser, S. W. Epp, B. Erk, H. Fleckenstein, L. Foucar, M. Frank, R. Fromme, H. Graafsma, I. Grotjohann, L. Gumprecht, J. Hajdu, C. Y. Hampton, A. Hartmann, R. Hartmann, S. Hau-Riege, G. Hauser, H. Hirsemann, P. Holl, J. M. Holton, A. Hömke, L. Johansson, N. Kimmel, S. Kassemeyer, F. Krasniqi, K.-U. Kühnel, M. Liang, L. Lomb, E. Malmerberg, S. Marchesini, A. V. Martin, F. R. Maia, M. Messerschmidt, K. Nass, C. Reich, R. Neutze, D. Rolles, B. Rudek, A. Rudenko, I. Schlichting, C. Schmidt, K. E. Schmidt, J. Schulz, M. M. Seibert, R. L. Shoeman, R. Sierra, H. Soltau, D. Starodub, F. Stellato, S. Stern, L. Strüder, N. Timneanu, J. Ullrich, X. Wang, G. J. Williams, G. Weidenspointner, U. Weierstall, C. Wunderer, A. Barty, J. C. H. Spence, and H. N. Chapman, "Time-resolved protein nanocrystallography using an X-ray free-electron laser", Optics Express **20**, 2706, ISSN: 1094-4087 (2012) 10.1364/oe.20.002706.

[72] J. Mahl, P. Aydogan Gokturk, R. Hamlyn, D. English, S. Suzer, J. Qian, and E. J. Crumlin, "Time-resolved electrical potential pump – X-ray photoelectron spectroscopy probe developments for investigating dynamic processes occurring at electrochemical interfaces", Applied Surface Science **713**, 164185, ISSN: 0169-4332 (2025) 10.1016/j.apsusc.2025.164185.

[73] T. Frohnmeyer and T. Baumert, "Femtosecond pump–probe photoelectron spectroscopy on Na2: a tool to study basic coherent control schemes", Applied Physics B **71**, 259–266, ISSN: 1432-0649 (2000) 10.1007/s003400000350.

[74] B. Cooper, P. Kolorenč, L. J. Frasinski, V. Averbukh, and J. P. Marangos, "Analysis of a measurement scheme for ultrafast hole dynamics by few femtosecond resolution X-ray pump–probe Auger spectroscopy", Faraday Discuss. **171**, 93–111, ISSN: 1364-5498 (2014) 10.1039/c4fd00051j.

[75]J. Duris, S. Li, T. Driver, E. G. Champenois, J. P. MacArthur, A. A. Lutman, Z. Zhang, P. Rosenberger, J. W. Aldrich, R. Coffee, G. Coslovich, F.-J. Decker, J. M. Glownia, G. Hartmann, W. Helml, A. Kamalov, J. Knurr, J. Krzywinski, M.-F. Lin, J. P. Marangos, M. Nantel, A. Natan, J. T. O'Neal, N. Shivaram, P. Walter, A. L. Wang, J. J. Welch, T. J. A. Wolf, J. Z. Xu, M. F. Kling, P. H. Bucksbaum, A. Zholents, Z. Huang, J. P. Cryan, and A. Marinelli, "Tunable isolated attosecond X-ray pulses with gigawatt peak power from a free-electron laser", Nature Photonics **14**, 30–36, ISSN: 1749-4893 (2019) 10.1038/s41566-019-0549-5.

[76]F. de Groot, "High-Resolution X-ray Emission and X-ray Absorption Spectroscopy", Chemical Reviews **101**, 1779–1808, ISSN: 1520-6890 (2001) 10.1021/cr9900681.

[77]L. M. Cornetta, L. Kjellsson, R. C. Couto, H. Ågren, V. Carravetta, S. L. Sörensen, M. Kubin, C. Bülow, V. Zamudio-Bayer, B. von Issendorff, J. T. Lau, J. Söderström, M. Agåker, J.-E. Rubensson, and R. Lindblad, "X-ray-absorption spectrum of O_2^{2+}", Physical Review A **111**, ISSN: 2469-9934 (2025) 10.1103/physreva.111.022808.

[78]U. Bergmann, D. Nordlund, P. Wernet, M. Odelius, L. G. M. Pettersson, and A. Nilsson, "Isotope effects in liquid water probed by x-ray Raman spectroscopy", Physical Review B **76**, ISSN: 1550-235X (2007) 10.1103/physrevb.76.024202.

[79]T. Barillot, O. Alexander, B. Cooper, T. Driver, D. Garratt, S. Li, A. Al Haddad, A. Sanchez-Gonzalez, M. Agåker, C. Arrell, M. J. Bearpark, N. Berrah, C. Bostedt, J. Bozek, C. Brahms, P. H. Bucksbaum, A. Clark, G. Doumy, R. Feifel, L. J. Frasinski, S. Jarosch, A. S. Johnson, L. Kjellsson, P. Kolorenč, Y. Kumagai, E. W. Larsen, P. Matia-Hernando, M. Robb, J.-E. Rubensson, M. Ruberti, C. Sathe, R. J. Squibb, A. Tan, J. W. G. Tisch, M. Vacher, D. J. Walke, T. J. A. Wolf, D. Wood, V. Zhaunerchyk, P. Walter, T. Osipov, A. Marinelli, T. J. Maxwell, R. Coffee, A. A. Lutman, V. Averbukh, K. Ueda, J. P. Cryan, and J. P. Marangos, "Correlation-Driven Transient Hole Dynamics Resolved in Space and Time in the Isopropanol Molecule", Physical Review X **11**, ISSN: 2160-3308 (2021) 10.1103/physrevx.11.031048.

[80]A. M. Weiner, G. P. Wiederrecht, K. A. Nelson, and D. E. Leaird, "Femtosecond multiple-pulse impulsive stimulated Raman scattering spectroscopy", Journal of the Optical Society of America B **8**, 1264, ISSN: 1520-8540 (1991) 10.1364/josab.8.001264.

[81]T. P. Dougherty, G. P. Wiederrecht, and K. A. Nelson, "Impulsive stimulated Raman scattering experiments in the polariton regime", Journal of the Optical Society of America B **9**, 2179, ISSN: 1520-8540 (1992) 10.1364/josab.9.002179.

[82]S.-i. Zaitsu, Y. Kida, and T. Imasaka, "Stimulated Raman scattering in the boundary region between impulsive and nonimpulsive excitation", Journal of the Optical Society of America B **22**, 2642, ISSN: 1520-8540 (2005) 10.1364/josab.22.002642.

[83]M. P. M. Dean, Y. Cao, X. Liu, S. Wall, D. Zhu, R. Mankowsky, V. Thampy, X. M. Chen, J. G. Vale, D. Casa, J. Kim, A. H. Said, P. Juhas, R. Alonso-Mori, J. M. Glownia, A. Robert, J. Robinson, M. Sikorski, S. Song, M. Kozina, H. Lemke, L. Patthey, S. Owada, T. Katayama, M. Yabashi, Y. Tanaka, T. Togashi, J. Liu, C. Rayan Serrao, B. J. Kim, L. Huber, C.-L. Chang, D. F. McMorrow, M. Först, and J. P. Hill, "Ultrafast energy- and momentum-resolved dynamics of magnetic correlations in the photo-doped Mott insulator Sr2IrO4", Nature Materials **15**, 601–605, ISSN: 1476-4660 (2016) 10.1038/nmat4641.

[84]F. D. Fuller, A. Loukianov, T. Takanashi, D. You, Y. Li, K. Ueda, T. Fransson, M. Yabashi, T. Katayama, T.-C. Weng, R. Alonso-Mori, U. Bergmann, J. Kern, V. K. Yachandra, P. Wernet, and J. Yano, "Resonant X-ray emission spectroscopy from broadband stochastic pulses at an X-ray free electron laser", Communications Chemistry **4**, ISSN: 2399-3669 (2021) 10.1038/s42004-021-00512-3.

[85]N. Rohringer, "X-ray Raman scattering: a building block for nonlinear spectroscopy", Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences **377**, 20170471, ISSN: 1471-2962 (2019) 10.1098/rsta.2017.0471.

[86]Y.-D. Chuang, X. Feng, A. Cruz, K. Hanzel, A. Brown, A. Spucces, A. Frano, W.-S. Lee, J. Kim, Y.-J. Chen, B. Smith, J. S. Pepper, Y.-C. Shao, S.-W. Huang, L. A. Wray, E. Gullikson, Z.-X. Shen, T. P. Devereaux, A. Tremsin, W. Yang, J. Guo, R. Duarte, and Z. Hussain, "Momentum-resolved resonant inelastic soft X-ray scattering (qRIXS) endstation at the ALS", Journal of Electron Spectroscopy and Related Phenomena **257**, 146897, ISSN: 0368-2048 (2022) 10.1016/j.elspec.2019.146897.

[87]D. J. Higley, Z. Chen, M. Beye, M. Hantschmann, A. H. Reid, V. Mehta, O. Hellwig, G. L. Dakovski, A. Mitra, R. Y. Engel, T. Maxwell, Y. Ding, S. Bonetti, M. Bucher, S. Carron, T.

Chase, E. Jal, R. Kukreja, T. Liu, A. Föhlisch, H. A. Dürr, W. F. Schlotter, and J. Stöhr, "Stimulated resonant inelastic X-ray scattering in a solid", Communications Physics **5**, ISSN: 2399-3650 (2022) 10.1038/s42005-022-00857-8.

[88]C. Weninger, M. Purvis, D. Ryan, R. A. London, J. D. Bozek, C. Bostedt, A. Graf, G. Brown, J. J. Rocca, and N. Rohringer, "Stimulated Electronic X-Ray Raman Scattering", Physical Review Letters **111**, ISSN: 1079-7114 (2013) 10.1103/physrevlett.111.233902.

[89]U. Eichmann, H. Rottke, S. Meise, J.-E. Rubensson, J. Söderström, M. Agåker, C. Såthe, M. Meyer, T. M. Baumann, R. Boll, A. De Fanis, P. Grychtol, M. Ilchen, T. Mazza, J. Montano, V. Music, Y. Ovcharenko, D. E. Rivas, S. Serkez, R. Wagner, and S. Eisebitt, "Photon-recoil imaging: Expanding the view of nonlinear x-ray physics", Science **369**, 1630–1633, ISSN: 1095-9203 (2020) 10.1126/science.abc2622.

[90]G. Goldsztejn, R. Püttner, L. Journel, R. Guillemin, O. Travnikova, B. Cunha de Miranda, I. Ismail, S. Carniato, P. Selles, D. Céolin, A. F. Lago, R. Feifel, P. Lablanquie, F. Penent, M. N. Piancastelli, M. Simon, and T. Marchenko, "Experimental and theoretical study of the double-core-hole hypersatellite Auger spectrum of Ne", Physical Review A **96**, ISSN: 2469-9934 (2017) 10.1103/physreva.96.012513.

[91]V. Kimberg and N. Rohringer, "Amplified X-Ray Emission from Core-Ionized Diatomic Molecules", Physical Review Letters **110**, ISSN: 1079-7114 (2013) 10.1103/physrevlett.110.043901.

[92]V. Kimberg, S. B. Zhang, and N. Rohringer, "X-ray lasing in the CO molecule", Journal of Physics B: Atomic, Molecular and Optical Physics **46**, 164017, ISSN: 1361-6455 (2013) 10.1088/0953-4075/46/16/164017.

[93]V. Kimberg, S. B. Zhang, and N. Rohringer, "X-ray lasing in diatomic molecules", Journal of Physics: Conference Series **488**, 012025, ISSN: 1742-6596 (2014) 10.1088/1742-6596/488/1/012025.

[94]R. Riedel, A. Al-Shemmary, M. Gensch, T. Golz, M. Harmand, N. Medvedev, M. J. Prandolini, K. Sokolowski-Tinten, S. Toleikis, U. Wegner, B. Ziaja, N. Stojanovic, and F. Tavella, "Single-shot pulse duration monitor for extreme ultraviolet and X-ray free-electron lasers", Nature Communications **4**, ISSN: 2041-1723 (2013) 10.1038/ncomms2754.

[95]O. Y. Gorobtsov, G. Mercurio, G. Brenner, U. Lorenz, N. Gerasimova, R. P. Kurta, F. Hieke, P. Skopintsev, I. Zaluzhnyy, S. Lazarev, D. Dzhigaev, M. Rose, A. Singer, W. Wurth, and I. A. Vartanyants, "Statistical properties of a free-electron laser revealed by Hanbury Brown–Twiss interferometry", Physical Review A **95**, ISSN: 2469-9934 (2017) 10.1103/physreva.95.023843.

[96]L. Dhar, J. A. Rogers, and K. A. Nelson, "Time-resolved vibrational spectroscopy in the impulsive limit", Chemical Reviews **94**, 157–193, ISSN: 1520-6890 (1994) 10.1021/cr00025a006.

[97]O. Alexander, F. Egun, L. Rego, A. M. Gutierrez, D. Garratt, G. A. Cárdenas, J. J. Nogueira, J. P. Lee, K. Zhao, R.-P. Wang, D. Ayuso, J. C. T. Barnard, S. Beauvarlet, P. H. Bucksbaum, D. Cesar, R. Coffee, J. Duris, L. J. Frasinski, N. Huse, K. M. Kowalczyk, K. A. Larsen, M. Matthews, S. Mukamel, J. T. O'Neal, T. Penfold, E. Thierstein, J. W. G. Tisch, J. R. Turner, J. Vogwell, T. Driver, N. Berrah, M.-F. Lin, G. L. Dakovski, S. P. Moeller, J. P. Cryan, A. Marinelli, A. Picón, and J. P. Marangos, "Attosecond impulsive stimulated X-ray Raman scattering in liquid water", Science Advances **10**, ISSN: 2375-2548 (2024) 10.1126/sciadv.adp0841.

[98]L. Ban, T. E. Gartmann, B. L. Yoder, and R. Signorell, "Low-Energy Electron Escape from Liquid Interfaces: Charge and Quantum Effects", Physical Review Letters **124**, ISSN: 1079-7114 (2020) 10.1103/physrevlett.124.013402.

[99]G. Greco, A. Verna, F. Offi, and G. Stefani, "Space-charge effect in electron time-of-flight analyzer for high-energy photoemission spectroscopy", Journal of Electron Spectroscopy and Related Phenomena **212**, 86–93, ISSN: 0368-2048 (2016) 10.1016/j.elspec.2016.09.004.

[100]A. Verna, G. Greco, V. Lollobrigida, F. Offi, and G. Stefani, "Space-charge effects in high-energy photoemission", Journal of Electron Spectroscopy and Related Phenomena **209**, 14–25, ISSN: 0368-2048 (2016) 10.1016/j.elspec.2016.03.001.

[101]D. Baer, A. Lea, J. Geller, J. Hammond, L. Kover, C. Powell, M. Seah, M. Suzuki, J. Watts, and J. Wolstenholme, "Approaches to analyzing insulators with Auger electron spectroscopy: Update and overview", Journal of Electron Spectroscopy and Related Phenomena **176**, 80–94, ISSN: 0368-2048 (2010) 10.1016/j.elspec.2009.03.021.

[102]R. Santra, "Concepts in x-ray physics", Journal of Physics B: Atomic, Molecular and Optical Physics **42**, 023001, ISSN: 1361-6455 (2008) 10.1088/0953-4075/42/2/023001.

[103]L. J. P. Ament, M. van Veenendaal, T. P. Devereaux, J. P. Hill, and J. van den Brink, "Resonant inelastic x-ray scattering studies of elementary excitations", Reviews of Modern Physics **83**, 705–767, ISSN: 1539-0756 (2011) 10.1103/revmodphys.83.705.

[104]N. Rohringer, *Introduction to the theory of x-ray matter interaction*, 2018, 10.48550/ARXIV.1811.12052.

[105]J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics*, 2nd ed. (Addison-Wesley, San Francisco, 2011), ISBN: 978-0-8053-8291-4.

Chapter IV

X-ray Free-Electron Lasers

When conducting experiments in x-ray science, it is necessary to use a lightsource that can produce x-ray light. We are motivated to use the measurement schemes described in section III and apply them to to organic systems containing common atoms such as carbon, nitrogen and oxygen.

Tabletop sources such as high harmonic generation (HHG) in practice generate x-rays too low in flux to support the multiple delay points needed for time resolved measurements above the carbon K edge. This is due to the poor up-conversion efficiency of the HHG process in the soft x-ray range.[1–3] The result is multiple orders of magnitude less x-ray flux in the x-ray range compared to an XFEL, as shown in figure IV.1. This is especially the case with the experimental schemes outlined in sections III.4 and III.5 as they rely on driving two-photon interactions. For this reason, the experiments containing the original work in this thesis were performed at special purpose facilities, known as X-ray Free-Electron Lasers (XFELs).

X-ray free-electron lasers (XFELs) are accelerator-based light sources that provide intense, ultrashort pulses of x-rays and high peak spectral brightness.[IV.1] Free electron lasers designed to generate x-rays have been demonstrated to produce x-rays of photon energies down to 200 eV and up to 25 keV.[5]

Unlike conventional lasers, which rely on creating population inversions and stimulating emission between bound states in a medium, XFELs lase using relativistic bunches of electrons radiating in magnetic fields via a process known as self-amplified spontaneous emission (SASE) or its derivatives.[6] The resulting pulses of x-ray light have durations on the femtosec-

[IV.1] Spectral brightness (also known as brilliance) is defined as photon flux / ($mm^2 \cdot mrad^2 \cdot$ relative bandwidth).

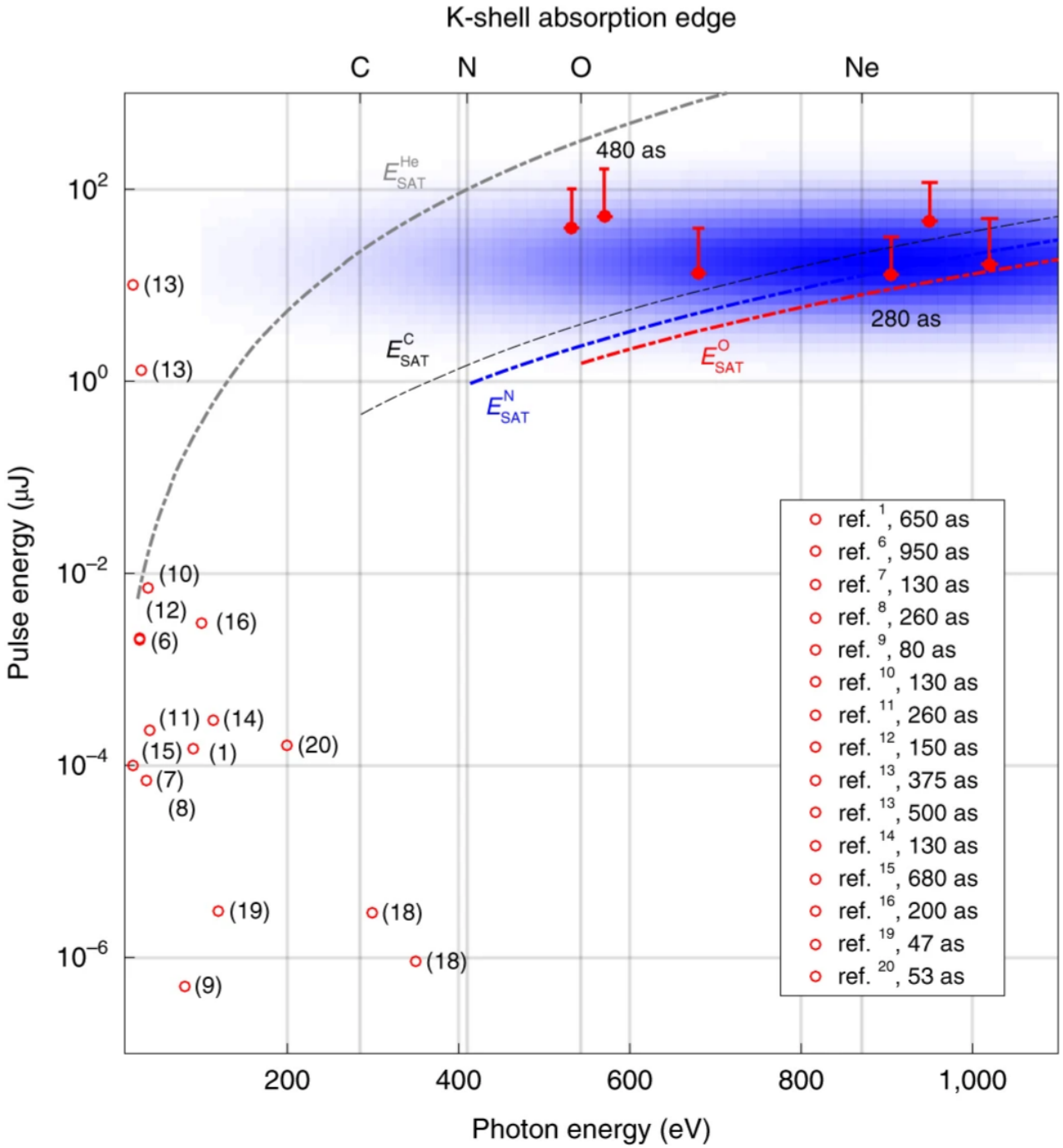

Figure IV.1: A logarithmic-linear plot of the pulse energy / number of photons versus photon energy over the x-ray range for both tabletop HHG sources and an attosecond XFEL, from Duris, J., Li, S., Driver, T. et al. *Nat. Photonics* **14**, 30–36 (2020).[4] The citations in the image refer to the paper cited, not this thesis. HHG-based sources are shown as hollow red circles. The attosecond XFEL measurement at different photon energies is shown as solid red circles with associated uncertainties. The dashed lines indicate the condition for an x-ray pulse to completely remove all electrons from the K shell of helium (grey), carbon (black), nitrogen (blue) and oxygen (red).

ond to attosecond timescale. Another property of XFELs is that they have tunable photon energies. This is achieved by varying the kinetic energy of the electron beam or the properties of the magnetic fields through which the electrons traverse.[7]

XFELs can also be used with lasers for pump probe experiments. For this reason they have been demonstrated to be useful to a wide range of scientific fields. For some examples, XFELs have been used for scientific studies in solution-phase photochemistry[8–10], protein imaging[11–13], transition-metal complexes[14–17], shockwave interactions[18,19] and condensed-phase thermodynamics[20,21].

Additionally, the Ångström-scale wavelengths of hard x-rays provide the spatial resolution necessary to study atomic structures in condensed matter via x-ray diffraction imaging techniques.[22,23] This has been successfully achieved in chemical systems,[24] biological systems[25] and has been used to image phonon propagation in matter.[26–28]

Since the first operational XFEL, the Linac Coherent Light Source (LCLS) at SLAC National Accelerator Laboratory, a growing number of facilities worldwide have come online, including SACLA (Hyogo, Japan), European XFEL, FLASH-I & FLASH-II (Hamburg, Germany), SwissFEL (Villigen, Switzerland), FERMI (Trieste, Italy), PAL-XFEL (Pohang, Korea), and SHINE (Shanghai, P.R.C.)[29,30]

In more recent years, the available pulse repetition rate at XFELs has increased with the use of superconducting linear accelerators. These can be delivered in bunch trains (e.g. FLASH and European XFEL) or continuously (e.g. LCLS-II). The increase in average brightness offered by superconducting accelerators can allow the detection of signals that originate from low cross-section interactions that would be impractical to detect with a lower repetition rate machine. Additionally, superconducting cavities can offer greater electron beam stability than their normal-conducting counterparts.[31] A comparison of a normal-conducting (LCLS) versus a superconducting (LCLS-II) machine is shown in Table IV.1.

IV.1 Self Amplified Spontaneous Emission

The majority of currently operational XFELs rely on the principle of self-amplified spontaneous emission (SASE) to generate x-ray pulses. This process involves a relativistic electron bunch, produced by an electron gun and then sent through a linear accelerator. More advanced variants of SASE exist and are used to generate x-ray pulse with different properties

Table IV.1: A comparison of the capabilities offered by the normal conducting LCLS and the recently upgraded and made available to users superconducting LCLS-II.[32,33]

Feature	**LCLS**	**LCLS-II**
Electron source	Normal-conducting RF linac (up to ~15 GeV)	Superconducting RF linac (4 GeV, upgrade path to ~8 GeV)
Repetition rate	120 Hz	33 kHz demonstrated, planned up to 1 MHz (continuous wave)
Photon energy range	0.25–25 keV (soft and hard x-ray undulators)	0.25–5 keV (soft x-ray), 1–20 keV (hard x-ray, pending upgrade)
Pulse duration	Tens of fs (down to a sub fs in special modes)	Tens of fs (down to a sub fs in special modes)
Peak brightness	Extremely high (single-shot SASE FEL)	Comparable per pulse; much higher *average* brightness
Average x-ray flux	Limited by 120 Hz operation	Over $10^4\times$ higher *average* flux
Linac Technology	Normal-conducting, continuous wave	Superconducting, continuous wave
Significance	First hard x-ray FEL; enabled ultrafast x-ray science	First CW high-throughput FEL, superior statistics

that are more appropriate for some experiments. One of these will be expanded on later. For now, we focus on the most basic x-ray generation method, SASE.

Initially, electrons are injected into the linear accelerator by an electron gun. This consists of a pulsed laser impinging on a photocathode, typically UV.[34] This electron generating process exploits the photoelectric effect. Each pulse of the laser frees many electrons from the photocathode. Each collection of electrons freed is known as a bunch.[IV.2] These bunches typically contain on the order of hundreds of pC of charge.[35,36]

The linear accelerator consists of multiple radio-frequency (RF) cavity modules. These are resonant electromagnetic structures designed to subject the electrons to an accelerating Lorentz force at all times during its acceleration. To accomplish this, the RF cavities' field polarity must alternate at radio frequencies. For example, the PAL-XFEL operates with a RF cavity switch rate of 2.856 GHz.[37] The relativistic electron beam is sent through a spatially

[IV.2] Sometimes also known as a bunch train.

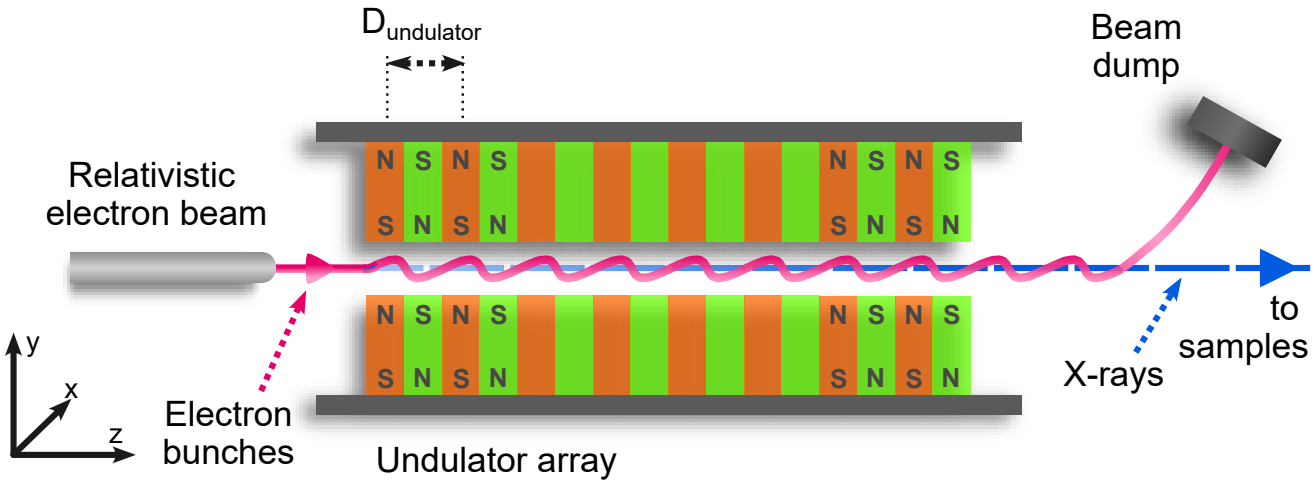

Figure IV.2: A schematic illustration of the principle sections of an x-ray Free Electron laser. The electrons are accelerated by RF cavities in a linear accelerator and enter the diagram from the left. The final energy of the electrons depends on the desired x-ray wavelength and has been demonstrated up to 20 GeV.[29] The electrons enter the undulator field, which has a spatial period $D_{undulator}$ with dipole magnets orientated in the y direction. Here, they interact with a changing magnetic field, resulting in the electrons undergoing a slalom path in the x axis. The acceleration they experience gives rise to radiation emitted primarily in the forward direction. After the undulators, a strong field is used to separate the electron and x-ray beams. In literature, the propagation direction of the electron beam in the undulator and the on axis x-rays is referred to as the longitudinal axis (labeled z in figure).

varying magnetic field. This field is produced by an array of magnets, arranged with alternating polarities as shown in figure IV.2. This array is known as an undulator. The magnets are arranged periodically with a spatial period $D_{undulator}$, typically on the order of centimeters.[38]

In the undulator, electrons are forced onto slalom-like oscillating trajectories and emit incoherent radiation.[39] Consider an electron bunch propagating down the undulator, in the z direction as defined in figure IV.2. The electron bunch will emit a large set of wavelengths, all of different phases.[40] The wavefronts of the forward-traveling radiation will copropagate ahead of the electron bunch. The time taken for the electron bunch to traverse one undulator period is $\Delta t = D_{undulator}/\bar{v}_z$, where $\bar{v}_z$ is the time-averaged speed of the electrons in the longitudinal z direction in an undulator period. The resonant condition within the undulator is

$$\lambda_n = \frac{D_{undulator}}{n} \times \frac{1-\beta}{\beta} \tag{IV.1}$$

for $n = 1, 2, 3\ldots$ and $\beta = \bar{v}_z/c$. Only certain wavelengths will travel an integer number of wavelengths ahead of the electron bunch in the time taken for the electron bunch to traverse one undulator period $D_{undulator}$.[41] Other wavelengths undergo destructive interference.[42] For

technical reasons, only odd harmonics and the fundamental wavelength λ_n for n odd, are emitted along the z axis.[39] Initially, this emission is spontaneous, with random phases.[40] The

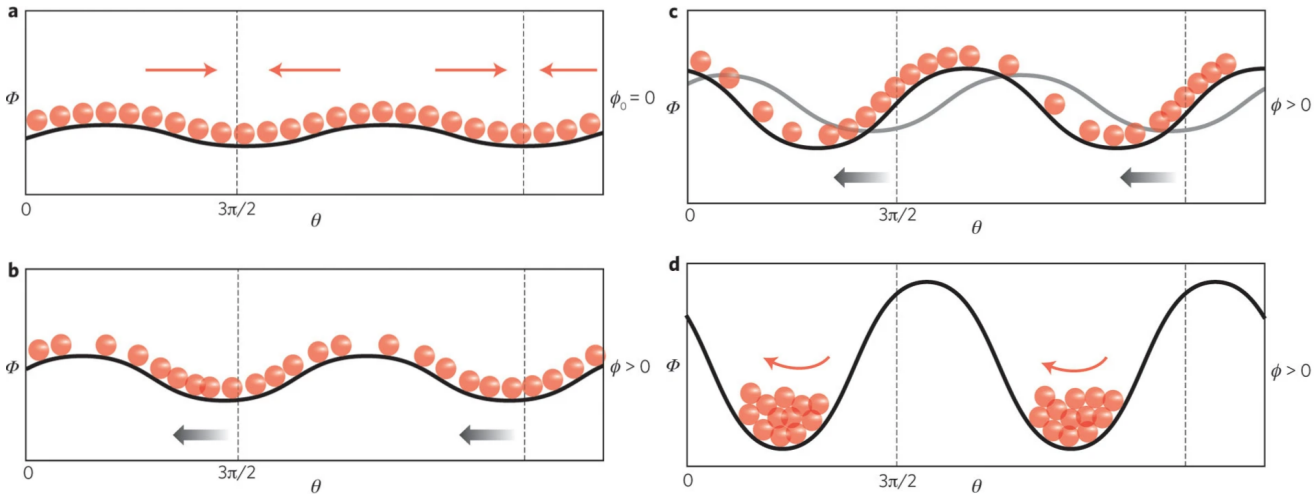

Figure IV.3: Graphical illustrations of how an electron bunch undergoes the self-ordering microbunching process, adapted from McNeil, B., Thompson, N. X-ray free-electron lasers. *Nature Photonics* **4**, 814–821 (2010).[29] The illustration depicts the zero momentum frame of the electron bunch, as it propagates through the undulator. The direction of pondermotive force acting on the electrons due to the field is indicated by the red arrows. The effective potential seen by the electrons is given by the dark line labeled Φ. (**a**) The electron bunch is approximately spatially uniform, with no density modulations. (**b**) The pondermotive forces act to induce a density increase about the phases $\theta = 3\pi/2$ and $7\pi/2$. (**c**) As the radiation moves faster than the electron bunch, the potential outpaces the electrons. Hence, electrons in the wells at $\theta = 3\pi/2$ and $7\pi/2$ gain energy, causing an energy modulation. (**d**) This energy modulation turns into a density modulation as the faster electrons catch up to the slower electrons of the previous well. The result is a microbunching instability within the electron bunch.

interaction between the emitted radiation and the electron bunch leads to an energy modulation of the electrons at the resonant wavelengths shown in equation (IV.1). This energy modulation then leads to a electron density modulation within a bunch as depicted in figure IV.3. The density modulations cause the emitted radiation to gradually start being emitted in phase, causing further energy modulations.

This cyclical self-amplifying effect eventually saturates, and the electron bunch ends up having internal density modulations with a periodic size on the order of the generated x-ray wavelength. This electron self-ordering effect is known as microbunching and results in coherent emission as the initial spatially uniform distribution of the electron bunch is not stable in the undulator's magnetic field.[42,43]

Note that the generated x-rays are essentially resonantly (or spontaneously) amplified noise. Therefore the SASE process produces x-ray pulses in a stochastic manner.[44] This means

that key properties such as pulse energy, spectral width, central photon energy and skewness vary randomly from pulse to pulse. When conducting an experiment that, for example, requires variations in pulse energy, the stochastic nature of the SASE process can be leveraged. In effect, the random variation in the XFEL lasing process is turned into a tool for scanning parameters without moving physical stages or manipulating the electron beam.

IV.2 Special attosecond XFEL modes

In section I.3, the natural timescale of electron dynamics was discussed. For electronic states typically separated in energy by a few eV, coherent superpositions of these states may be formed that show time evolution on the attosecond timescale as seen in equation (I.22). Due to limitations in the coherent bandwidth of SASE x-ray pulses they are limited to full width at half maximum durations of tens of femtoseconds.[45–47] This provides insufficient time resolution to capture the desired electronic motion.

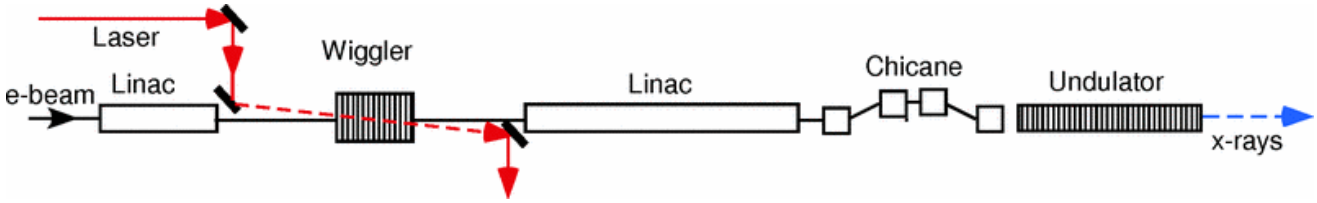

Figure IV.4: A schematic drawing of the enhanced SASE process, from Zholents, A. *Phys. Rev. ST Accel. Beams* **8**, 040701.[48] Prior to entering the undulator the electron beam passes through a wiggler where it interacts with an IR or optical wavelength laser. The laser field causes an energy modulation in the electron bunches. The electron beam after further acceleration passes through a dispersive sequence of magnets known as a chicane.[49] Passing through the chicane, the energy modulation turns into a density modulation in a process called bunch compression.[50] This was proposed as a method to reduce the duration of the produced x-ray pulses from tens of femtoseconds to sub-femtosecond.

In order to shorten the durations of SASE pulses, more advanced x-ray generation schemes based on SASE have been proposed.[48] These schemes rely on manipulating the electron bunch prior to its entry into the undulator and are generally referred to as enhanced SASE.[51]

In one of the schemes using enhanced SASE, the electron bunch interacts with a near infrared or optical laser field after leaving the linear accelerator. This is to introduce an energy modulation into the electron bunch. A magnetic chicane acts dispersively on the electron bunch. Electrons with higher energies take longer paths through the chicane compared to those with lower energy, resulting in density variations within the electron bunch.

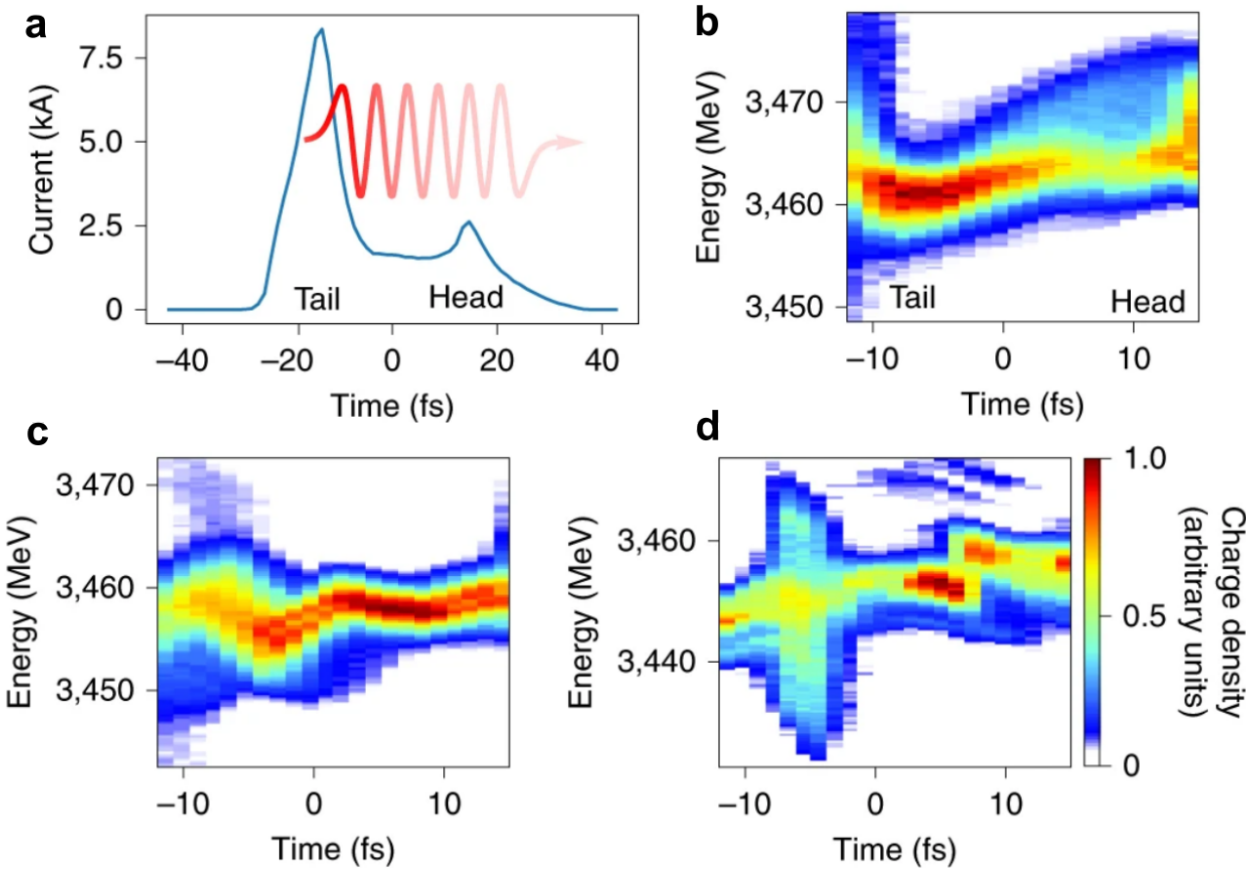

Figure IV.5: Measurements of an electron bunch showing the effect of an energy and density modulation on the electron bunch after exiting the linear accelerator, adapted from Duris, J., Li, S., Driver, T. et al. *Nat. Photonics* **14**, 30–36 (2020).[4] (**a**) Temporal profile of the current of an electron bunch. The current spike in the tail of the distribution generates IR radiation shown by the red wave.[52] (**b**-**e**) The electron bunch is shown resolved in kinetic energy over time (or over the longitudinal axis in space defined in figure IV.2). (**b**) Distribution after the accelerator section. (**c**) After the wiggler, showing an energy modulation. (**d**) After the wiggler and chicane, showing a single current spike at -5 fs.

Both the energy and density modulations can be seen in figure IV.5 in subplots (**c**) and (**d**) respectively. The original eSASE scheme shown in figure IV.4 depicts the overlap of an external infrared laser with the electron bunch in a wiggler. This adds challenges in laser alignment, stability and electron-laser synchronization. A simpler alternative is to use the coherently emitted infrared radiation generated in the wiggler structure by the current spike at the tail of the electron bunch.[52] This radiation is used to produce the single-cycle energy modulation in the electron bunch as shown in figure IV.5 subplot **c**). This mechanism hence enhances the peak current passing through the undulator and results in greater bunch compression when compared to the standard SASE process and is termed x-ray laser-enhanced attosecond pulse

generation (XLEAP).[IV.3]

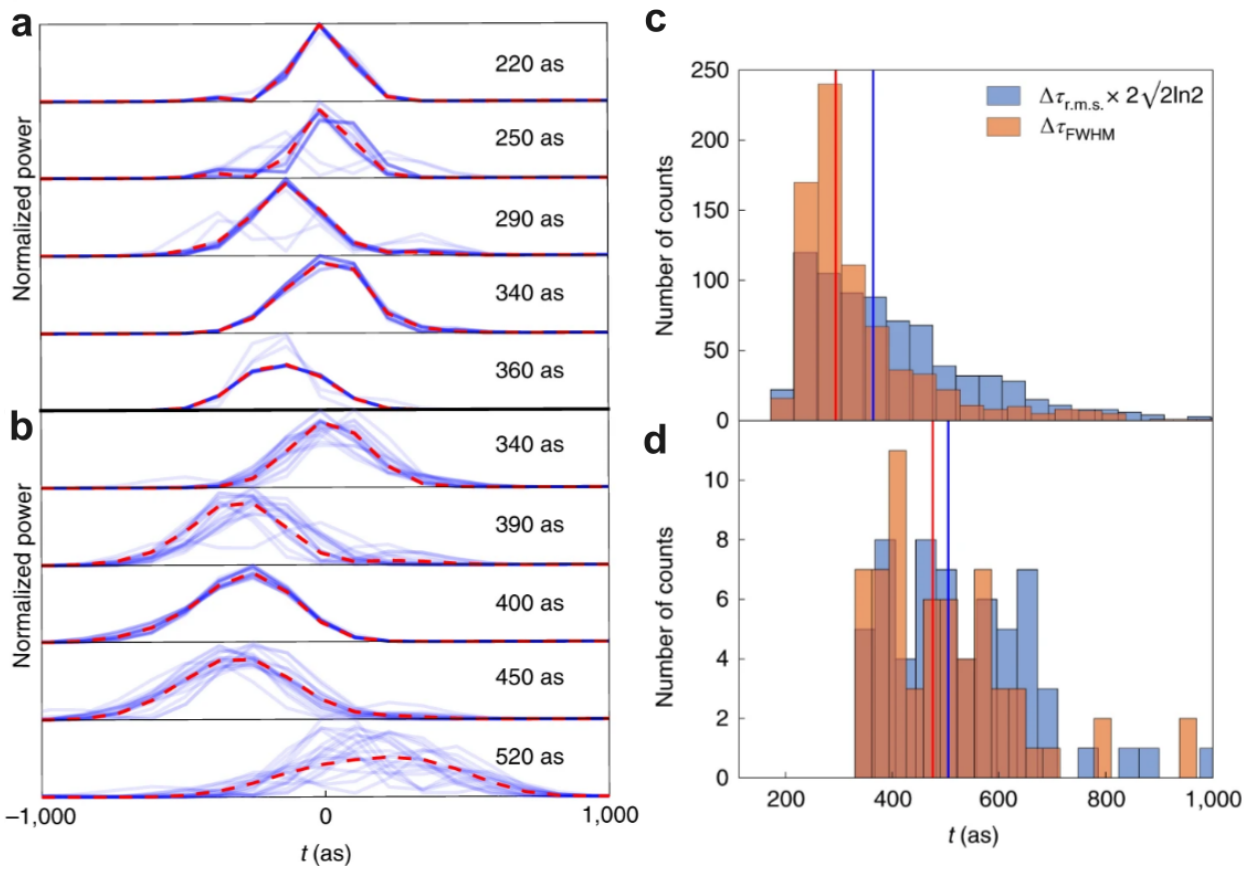

Figure IV.6: Time domain x-ray pulse reconstructions of the attosecond mode at LCLS, XLEAP, adapted from Duris, J., Li, S., Driver, T. et al. *Nat. Photonics* **14**, 30–36 (2020).[4] (**a**) and (**b**) Pulse reconstruction using an algorithm which takes a random input for a series of pulses with increasing duration. The faint blue lines show the reconstruction for different values of the seed input. (**a**) at 905 eV and (**b**) at 570 eV. The red dashed line shows the maximum likelihood solution. (**c**) and (**d**) Histogram of pulse durations for the 905 eV (**c**) and 570 (**d**) cases. Both the scaled root mean square (r.m.s.) and full width at half maximum (FWHM) are shown in blue and red respectively. Their medians are shown by coloured vertical lines.

The enchanced bunch compression reduces the duration of the electron bunch as the imposed energy dispersion (or chirp) combined with the chicane makes faster (higher-energy) electrons catch up to slower ones. As the gain length in an undulator scales inversely with the charge density[53], only high charge density portions of the electron bunch result in x-ray lasing.

In the XLEAP mode, only a very small part of the electron bunch has a high enough charge density to undergo the gain process in the undulator field. For example, the charge density

[IV.3]The term "laser" in XLEAP is kept in the term for historical reasons. The laser was found to be unnecessary to create the electron bunch energy modulation.

spike shown in figure IV.5 subplot **d**). This is different to standard SASE, where multiple parts of the electron bunch meet this condition. The single-spike bunches of XLEAP therefore result in only one part of the electron bunch contributing to lasing. The result is hence single-spike near transform limited pulses of attosecond duration.

Duris et al. used an angular streaking technique to measure the duration of the produced x-ray pulses. X-ray pulses of photon energies 905 eV and 570 eV were impinged onto Neon and CO_2 gas jets respectively. The median FWHM values were found to be 280 and 480 as for the 905 eV and 570 eV measurements respectively, as indicated by the vertical lines on figure IV.6. This indicates that the x-ray pulses are of sufficiently short duration to approach the transform limit, and are stated as approximately 1.5 $\times$ the Fourier transform limit. Some XLEAP pulses with central photon energies in the oxygen pre-edge region are shown in figure IV.7.

To summarize, XLEAP pulses are an example of an enchanced SASE process that allows for the generation of x-ray pulses that can offer sub-femtosecond duration. The peak power density (intensity) of such pulses can enter the 10^{18} Wcm^{-2} regime, enough to drive multiphoton processes.[54] The corresponding coherent bandwidths of the x-rays in the spectral domain is on the order of several electronvolts and is sufficient to excite coherent superpositions of electronic states of molecules.

IV.3 Analysis of XFEL detectors

At XFELs, each x-ray pulse can contribute to a measurement. An event corresponding to an x-ray pulse is referred to as a shot. For each shot, many different parameters are recorded. These can refer to aspects of the linear accelerator such as the electron beam energy and undulator conditions. Additionally, the measured properties of the electron bunch such as bunch charge, and information about the generated x-ray spectrum can be recorded.[55]

Depending on the pulse repetition-rate of the XFEL, these parameters may be recorded on every shot (common on low repetition-rate XFELs) or at a set rate (e.g. at a rate of 10 Hz at high repetition-rate XFELs).[56] These parameters can be used to track the XFEL through scanning measurements, when varying the central photon energy of the XFEL.

Additionally, upstream of the interaction region, there can be photon diagnostic detectors. These measure the number of photons in an x-ray pulse, and therefore the incident pulse energy to the interaction region. This is done by gas detectors.[57] The instrument accepts the x-ray

beam and allows it to ionize a gas. The amount of resulting charged particles detected is then proportional to the number of incident photons.[58] Through a careful selection of the gas and the known interaction cross section for absorption at different x-ray wavelengths, the incident photon flux may be computed, non-invasively on a shot to shot basis.

In more complex schemes, the XFEL is used to create two x-ray pulses of different wavelengths, pump and probe. An example of such a scheme is detailed in section III.3. In this case it is useful to use two different gas detectors GD1 and GD2 in series. A different gas may be placed into each gas monitor detector. From these measurements, it is possible to calculate the amount of pump and probe flux, N_{pump} and N_{probe} respectively, in a single shot. This calculation as compactly expressed as the matrix equation

$$\begin{bmatrix} N_{pump} \\ N_{probe} \end{bmatrix} = \begin{bmatrix} N_{GD1} \\ N_{GD2} \end{bmatrix} \begin{bmatrix} \sigma_{pump,\,GD1} & \sigma_{pump,\,GD2} \\ \sigma_{probe,\,GD1} & \sigma_{probe,GD2} \end{bmatrix}, \tag{IV.2}$$

which can be useful in analysis. Here the σ terms denotes the relation between the number of incident photons and ionized electrons. E.g. $\sigma_{pump,\,GD1}$ denotes the response of gas detector GD1 at the pump photon energy. This will depend on the cross-section of the gas at this photon energy and the density and thickness of the cell. These gases are then chosen to have very different absorption cross-sections to each other at both wavelengths.

Another essential detector for many experiments is an x-ray photon spectrometer. A photon spectrometer is an instrument used to measure the spectral distribution of photons emitted from, transmitted through, or scattered off from a sample. It quantifies the photon intensity as a function of photon energy, or equivalently wavelength. This can provide information on the underlying electronic states populated in the sample, for example in Stimulated Raman Scattering as detailed in section III.4.

Its operation relies on the principle of photon dispersion, i.e. sending light on paths according to their wavelength. This converts the spectral distribution of the x-ray field into a spatial distribution.[59,60] This spatial distribution is then recorded with a position-sensitive detector, typically a CCD camera. The dispersion is commonly achieved by a diffraction grating.[61–63]

An example of the data collected by a photon spectrometer is given in figure IV.7. Shown are wide bandwidth x-ray pulses from the special attosecond mode, XLEAP, available at LCLS. (See section IV.2.) The spectrometer covers a wide range of photon energies (approximately

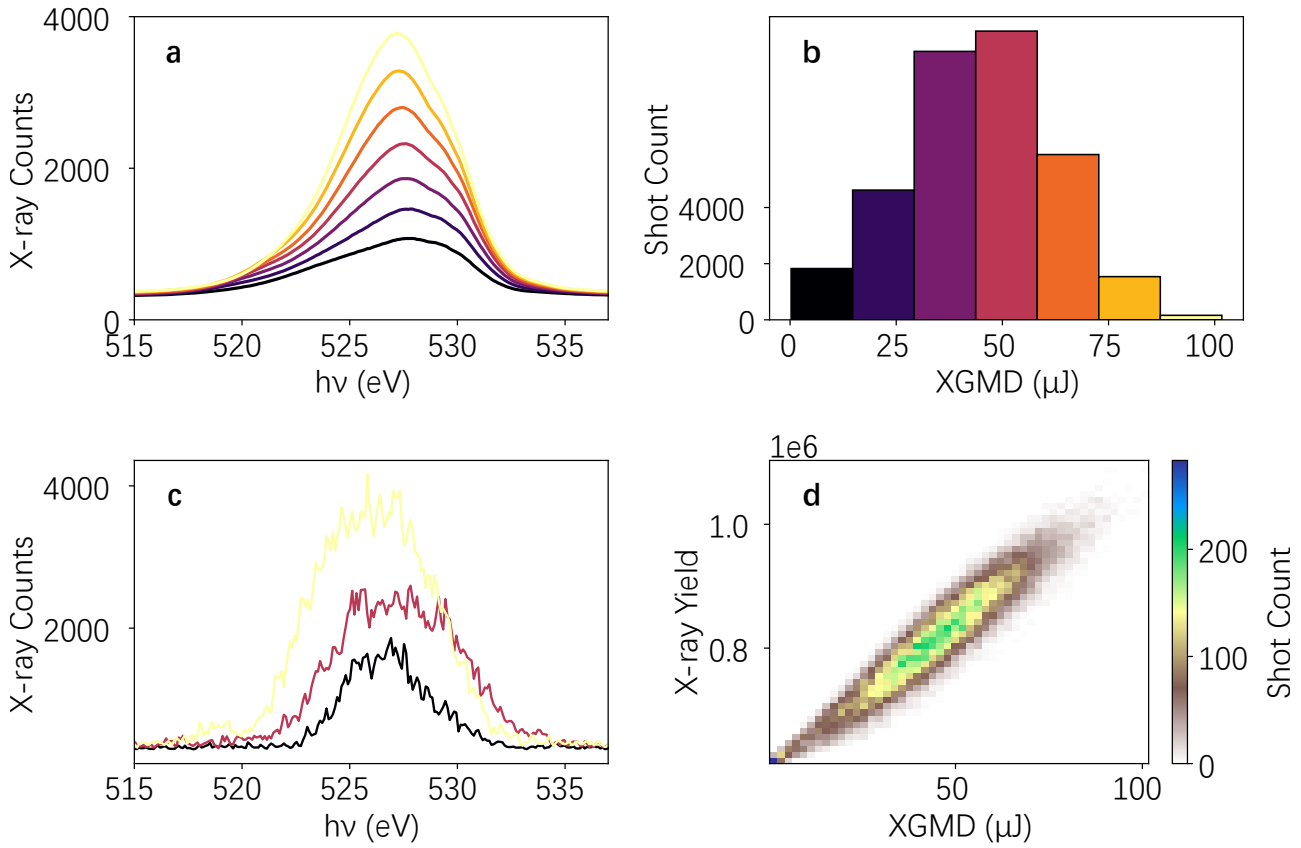

Figure IV.7: Spectrally resolved x-rays imaged on a variable line spacing diffraction grating spectrometer analyzed with a gas detector. (**a**) Average spectra binned on the pulse energy, as measured by the XGMD instrument, a gas detector measuring the pulse energy. (**b**) Statistics for approximately ~34,000 x-ray shots, and how their pulse energy are distributed. The bins are shown are centered at 19, 26, 33, 40, 47, 54 and 61 μJ. (**c**) Three individual shots taken from the lowest, central and highest XGMD bin. (**d**) Correlation map of the total counts on the detector with the XGMD measured pulse energies, showing a correlation coefficient of 0.946.

from 500 to 560 eV.) The x-rays have a central photon energy of 529 eV and a window of 22 eV is shown.

The spectra are grouped (binned) according to an independent measure of their pulse energy. This measure is a gas detector, termed XGMD, upstream of the spectrometer. Approximately 8.1% of shots do not record a valid XGMD value, and have therefore been discarded.[IV.4] Note in subplot (**c**) of IV.7, the shots are close to Gaussian in the spectral domain. This reflects the fact that the attosecond pulses generated are close to the Fourier transform limit.

In order to make use of a photon spectrometer, it is essential to first calibrate it. Namely, it is necessary to find the pixel to energy (or wavelength if preferred) conversion function, before

[IV.4]This occurs when a timing mismatch between the XGMD readout and the rest of the data stream occurs, or dropped network packets.

useful information about the sample can be gained. Normally a calibration involves placing a carefully chosen calibration sample into the interaction region. This calibration sample should have two key properties.

Firstly, the calibration sample should have a well documented x-ray absorption spectrum in literature. Secondly, it should have a sufficiently high cross-section at the photon energy intended to be used in the experiment. This way, XAS features can be observed on the detector at given pixels and mapped to specific energies from literature.

Figure IV.8 illustrates the calibration process used during a beamtime. The XFEL central photon energy is scanned over a resonance known to be strong in the calibration sample. A calibration of the spectrometer for x-rays near to the oxygen K-edge was desired here. Therefore, a calibration sample containing oxygen was used. In this case, the calibration sample chosen was polyethylene terephthalate[IV.5], more commonly known by its commercial name, Mylar.[65]

It shows strong absorption features at the oxygen K-edge and has chemical formula $(C_{10}H_8O_4)_n$ and its structure is shown in figure IV.8. As it has oxygen atoms in slightly different chemical environments, it can show multiple peaks in the oxygen pre-edge region.[66,67]

To measure the XAS, two measurements are taken. The x-rays are impinged onto the calibration sample. The transmitted spectrum is then sent to a diffraction grating. The spectrally resolved light is then imaged onto a camera. This forms a measure of the transmitted spectrum $I_T(\omega)$. The same is done without the sample. This forms a measure of the reference incident spectrum, $I_0(\omega)$. The upper plot in figure IV.8 shows $I_0(\omega)/I_T(\omega)$, a direct measure of XAS similar to the optical density outlined in equation (III.1).

The lower subplot of figure IV.8 shows four spectra. The lower three spectra are measured from polyethylene terephthalate. Three XAS techniques are shown. Scanning transmission x-ray microscope x-ray absorption spectroscopy (STXM-XAS), electron yield x-ray absorption spectroscopy (EY-XAS)[IV.6] and scanning transmission electron microscope electron energy loss spectroscopy (STEM-EELS).

Using the uncalibrated XFEL data, peak finding algorithms can be used to associate a pixel to a specific XAS feature, such as peak finders in python.[68] By identifying the energy associated with each pixel, a pixel to energy conversion may be found. Figure IV.7 subplots (**a**) and

[IV.5] Figure 4.8 abbreviates this molecule as PET.
[IV.6] This XAS technique is detailed in section III.1.

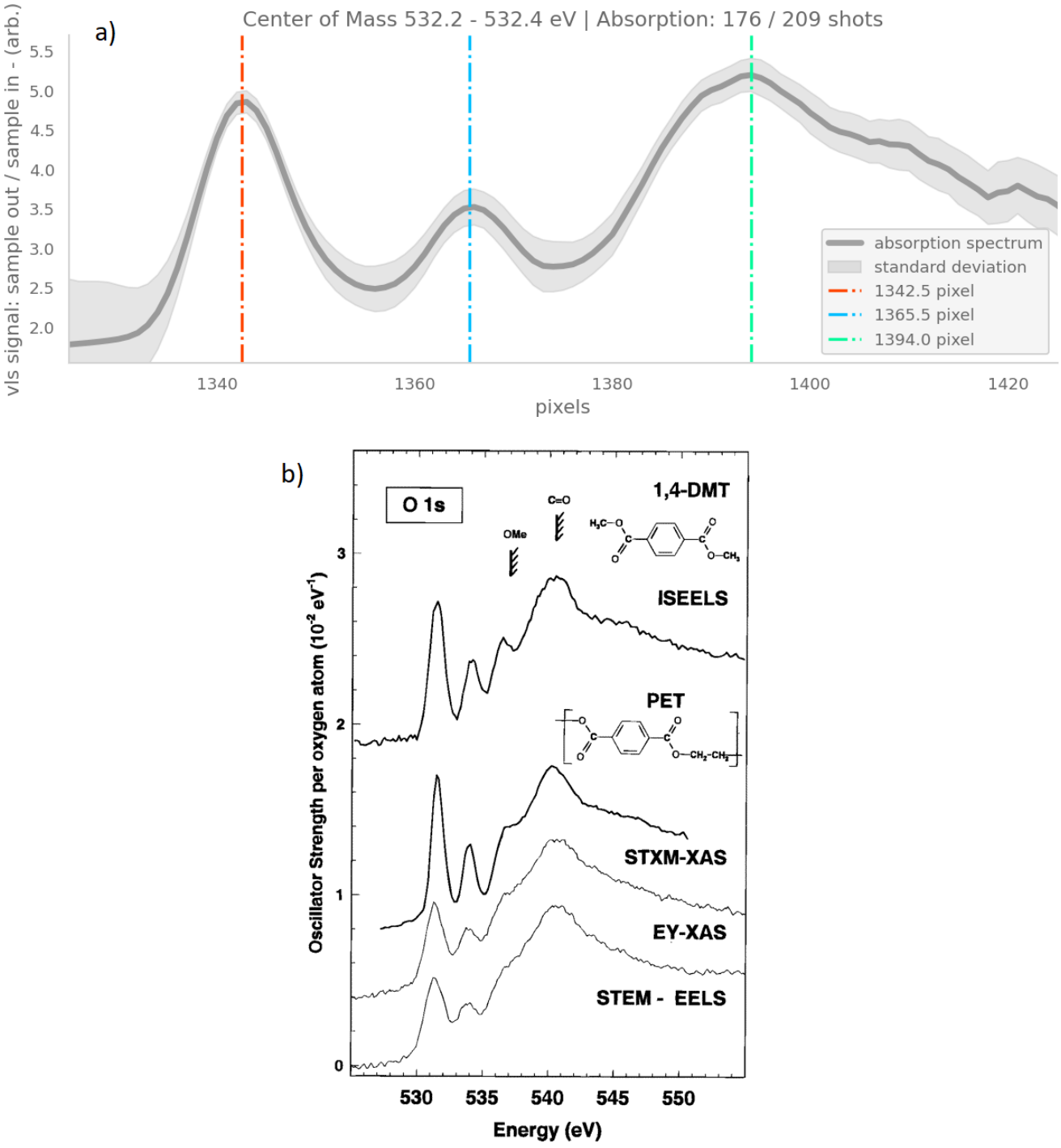

Figure IV.8: Data taken from a beamtime at an attosecond XFEL in order to calibrate a variable line spacing x-ray spectrometer. *Top*: Experimental data taken at the Linac Coherent Light Source, showing $I_0(\omega)/I_T(\omega)$. *Bottom*: Synchrotron data taken from Rightor, E. G. et al. *J. Phys. Chem. B* 1997, **101**, 11, 1950–1960.[64] Shown is data taken in the same part of the electromagnetic spectrum as the top plot, using three different XAS techniques. The calibration sample XAS is shown in the bottom three plots.

(**c**) show the readout of an ANDOR camera of a VLS spectrometer. For this kind of x-ray spectrometer, there is a linear mapping between pixels and photon energy. Therefore, the problem of finding a photon energy calibration reduces to finding the parameters α_1 and α_2 in the expression

$$h\nu = \alpha_1 * px + \alpha_2, \tag{IV.3}$$

where px denotes a pixel number. The parameters α_k can be found via a least squared method.[68] This relation then allows the x-ray spectrometer to be used in the experiment to understand spectral information and the result is the horizontal axis in figure IV.7 subplots (**a**) and (**c**).

In sections III.2 and III.3 measurement schemes that rely on the detection of charged particles are introduced. Time-of-flight (TOF) spectrometers are a widely used diagnostic tool for measuring the energy distribution of charged particles by recording their flight time over a known distance within a chamber.

Their operating principle relies on charged particles emitted from an interaction region with some unknown kinetic energy. They subsequently enter a field-free drift region, and arrive at a detector after a characteristic delay. The time required to traverse the drift region depends on the particle's velocity and thus on its kinetic energy.[69] For a particle of mass m starting from the interaction region with kinetic energy $\mathcal{T}$, it will take a time τ_{drift} to cross a drift region D given by

$$\tau_{drift} = D\left(\frac{2\mathcal{T}}{m}\right)^{-1/2} \quad \Longleftrightarrow \quad \mathcal{T} = \frac{1}{2}mD^2\frac{1}{\tau_{drift}^2}. \tag{IV.4}$$

Here, a non-relativistic expression for the kinetic energy is used.[70,71] This is justified as the electrons detected will have energy not much greater than 500 eV. The relativistic correction to this would change the result by a factor of 1 part in 10^3 and is therefore neglected. The measured arrival time can be directly converted into a kinetic-energy spectrum for a given particle mass (e.g. electrons).

Collection efficiency of detectors is an important factor for experiments as it informs the counting statistics of a measurement and hence the lowest cross-section interactions that can be reasonably detected. It is measure of the fraction of the particles (ions, electrons, photons, etc.) created in the interaction region that are actually detected by the spectrometer. The collection efficiency of a TOF spectrometer depends on its design and the energies of the charged particles being resolved.[72,73]

One design that can increase collection efficiency is the magnetic bottle electron spectrometer (MBES). This type of TOF utilities a strong, inhomogeneous magnetic field to "funnel" electrons into the drift region. It is able to accept electrons with close to a 4π solid angle around the interaction region. Therefore, it has significantly higher collection efficiency than a traditional TOF spectrometer[74,75], but suffers reduced energy resolution.[76] For the x-ray pump x-ray probe described in section III.3, which is a two-photon pump-probe process, collecting large amounts of electrons to maximize the signal-to-noise ratio is critical. Hence, an MBES is used.

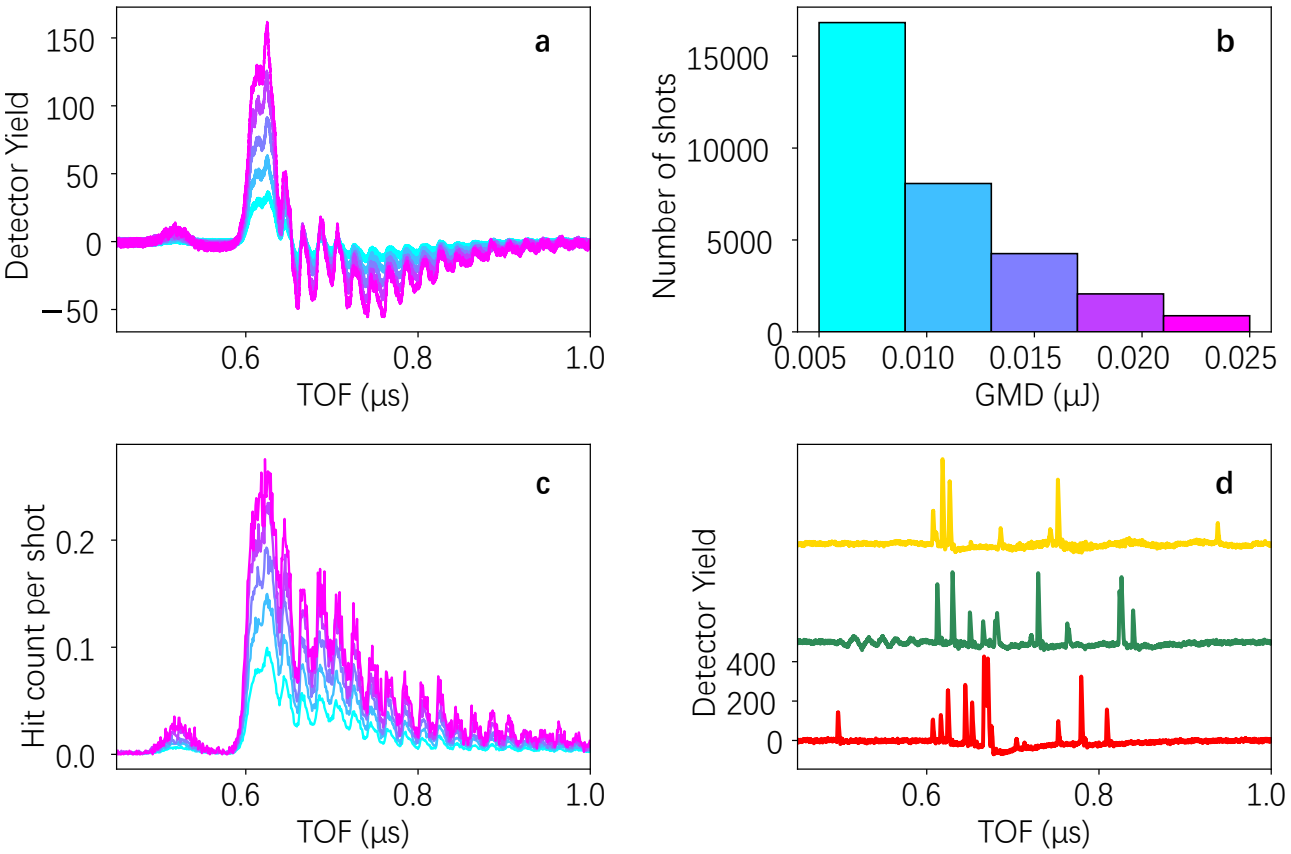

Figure IV.9: Magnetic bottle time-of-flight spectrometry of electrons, following the interaction of an XFEL pulse with a sample. This data was recorded at the TMO instrument at LCLS. The presented data is approximately 32,000 shots binned according to a gas detector known as the GMD. (**a**) Averaged raw waveforms for five GMD bins. (**b**) Statistics and colour legend for each GMD bin. (**c**) A histogram of "hit-found" data, grouped by the same bins used in subplots (**a**). (**d**) Three single shot raw waveforms, plotted with a vertical offset for visibility.

A TOF spectrometer has a minimum time resolution, below which it is unable to distinguish events. At the Time-resolved Atomic, Molecular, and Optical (TMO) instrument at LCLS, the MBES has a time resolution $\delta\tau_{\text{drift}}$ of 0.168353 ns. The relationship between drift

time τ_{drift} and kinetic energy $\mathcal{T}$ is non-linear as stated in equation (IV.4). Therefore, the fractional energy resolution $\delta\mathcal{T}/\mathcal{T}$ of the spectrometer depends on the arrival time of the detected particle.[73,77] It then follows that

$$\delta\mathcal{T} \approx \left|\frac{\partial\mathcal{T}(\tau_{\text{drift}})}{\partial\tau_{\text{drift}}}\right| \delta\tau_{\text{drift}} \quad \Longrightarrow \quad \frac{\delta\mathcal{T}}{\mathcal{T}} \propto \frac{1}{\tau_{\text{drift}}}\delta\tau_{\text{drift}}. \tag{IV.5}$$

This relation shows that the fractional uncertainty in the energy $\delta\mathcal{T}/\mathcal{T}$ gets worse the faster the charged particle reaches the detector. For this reason, XFEL experiments using electron spectrometry to resolve electrons of specific energy commonly used retardation voltages. This involves placing an electric field between the interaction region and the detector. The potential associated with this field is then chosen to significantly reduce the kinetic energy of the particle this increasing its drift time. This has the added benefit of preventing electrons with insufficient energy to reach the detector from showing up in the measured spectrum, acting as a method of background suppression.

In experiments at superconducting XFELs, saving the full waveform is not normally done in tabulated data files. This is due to the fact that data is typically acquired at extremely high repetition rates. At the TMO instrument, each time pixel is assigned an unsigned 16-bit (2 Byte) integer.[IV.7] One waveform is recorded on each x-ray shot. Each waveform contains approximately 12,000 time pixels.[IV.8] Therefore, the waveform on one anode corresponds to 24 kB. With a repetition rate of 33 kHz, one anode on one detector then produces 720 MB in one second of operation. This motivates data reduction techniques, such as hit-finding.

The vast majority of recorded time pixels do not contain a useful interaction signal. Processing all time pixels when conducting analysis would be computationally inefficient. Hit finding is a data reduction process that identifies which time pixels contain a relevant physical event (referred to as a "hit") and which may be discarded. A hit is assigned to a time pixel depending on its amplitude, its value compared to its neighbouring time pixels (prominence), whether is passes a set noise threshold and the width of the potential feature.

An array of the times of time pixels that have been deemed to contain a hit are stored. In the data shown in figure IV.9, an average of 12 hits per shot was recorded. Times are recorded as

[IV.7] They are unsigned integers as the TOF spectrometer output does not give negative values. The data shown in figure IV.9 subplot **a**) is offset.

[IV.8] I use the term time pixel in an analogous sense to a camera pixel. The unit of minium resolvable time resolution of the TOF spectrometer $\delta\tau_{\textbf{drift}}$.

64-bit = 8 Byte floats, corresponding to an average data rate of 96 Bytes per shot. This "lossy" data reduction technique therefore gives a compression rate of approximately 250.

In order to recover meaningful electron spectra from large arrays of times corresponding to hits, the data must be time binned. A start and end time along with a time interval must be picked. The time interval is usually picked to be an integer multiple of the time resolution of the instrument to avoid mismatching sampling and binning intervals. If not done correctly, this can introduce spurious modulations in the reconstructed spectra, such as aliasing.[78,79] In figure IV.9 subplot **c**), hit-found data is shown as a histogram. When attempting to calibrate a TOF spectrometer, it is important to understand at which time the x-ray pulse interacts with the sample. The peak labeled Ar^{X+} could correspond to the first, second, third or possible higher charge state of the Argon ion.[IV.9] To find which, the mass-to-charge ratio is compared to the square of the drift time. The best linear relation is found for the $X = 2$ case, meaning that the annotated peak in (**a**) indicates Ar^{2+}.

The TOF spectrometer is designed to detect charged particles in a short window. Within this window, it is necessary to identify the time when the XFEL pulse interacts with the sample. This time is known as the prompt time t_0. The drift time is then found to be $\tau_{drift} = t_{detected} - t_0$. For this reason, in figure IV.10 subplot (**b**), the x-axes are shifted by t_0. The prompt time is identified by looking for a retardation independent "light peak", corresponding to x-rays scattering from the interaction region, onto the TOF detector. The time of the light peak and the time of the XFEL pulse arrival are very similar, given the typically drift region of TOF spectrometers is on the order of meters.[75]

IV.4 Data collection and real-time reduction at XFELs

The high repetition rates of XFELs present challenges for data management. Large-2D detectors used for diffraction experiments such as the Adaptive Gain Integrating Pixel Detector (AGIPD)[81], the Depleted p-channel field effect transistor Sensor with Signal Compression (DSSC)[82], and the Jungfrau detector at SwissFEL[83] produce images in the range of 10–20 MB per frame. A frame is an exposure period for the detector. It may consist of one or multiple x-ray shots depending on the experiment details and the XFEL repetition rate.

[IV.9] The Ar^{1+} ion is not necessarily the most abundant in the spectrum, as the XFEL pulse contains so many photons that the Ar^{1+} yield can be less than the Ar^{2+} yield.[80]

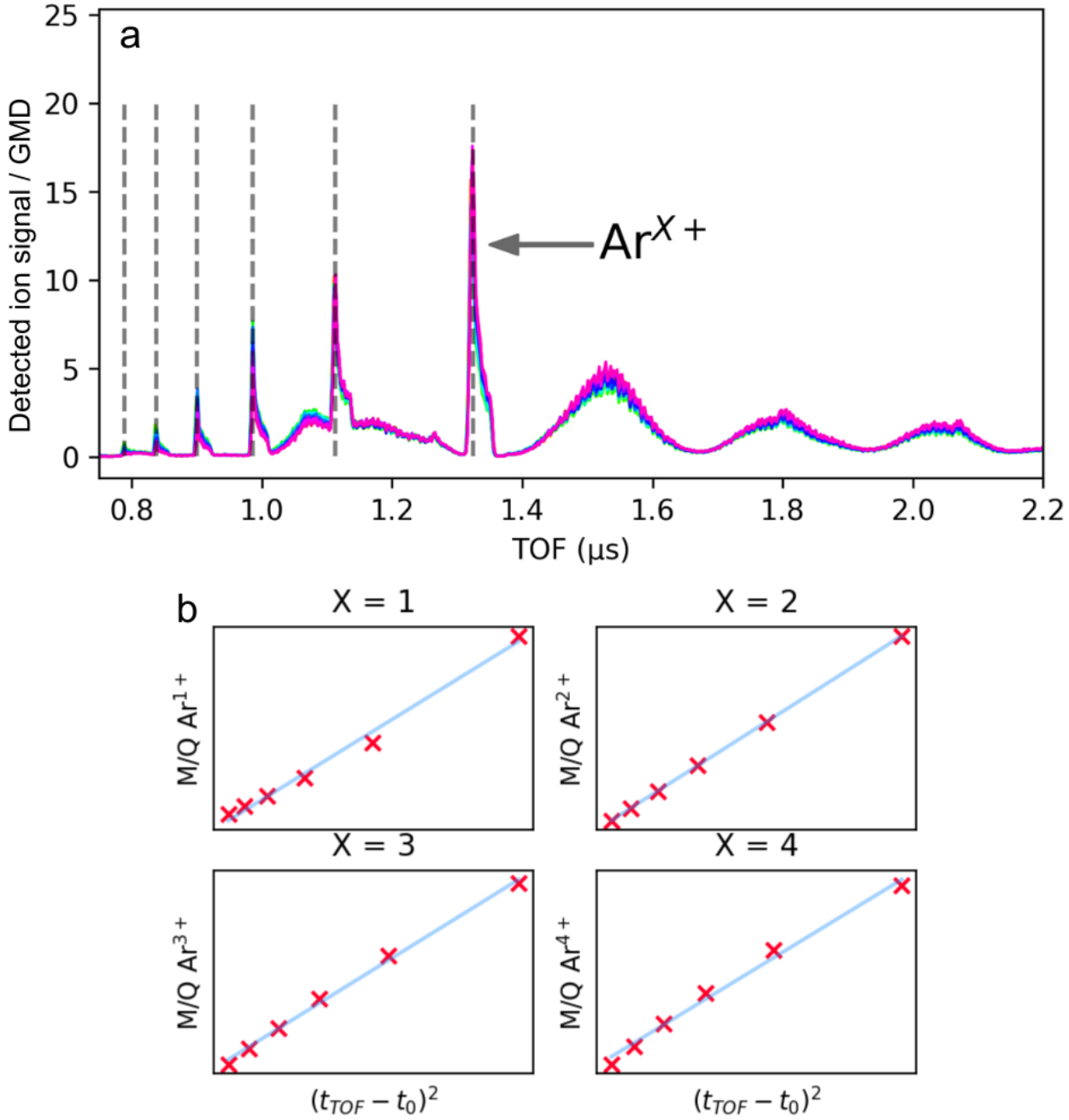

Figure IV.10: Data acquired during an LCLS early science beamtime at the TMO instrument, for the purpose of calibrating the ion TOF spectrometer. The sample is Argon gas. (**a**) A hit-found ion TOF spectrum, showing multiple features. The features marked with dashed lines indicate peaks that have been identified as the argon ion series. The colours indicated spectra taken at different beam focus positions along the z axis. (See figure IV.2 or z axis definition.) The spectra are normalized to the incident photon flux, as measured by the GMD. (**b**) A number of potential fits for the ion spectrum for different values of X. $X = 2$ was deemed to give the fit that best fit the expected relation.

As repetition rates can reach up to several megahertz[84,85], the corresponding data rates can easily reach terabytes per minute of operation. This data volume significantly exceeds that produced at low repetition rate source, necessitating highly specialized data acquisition and storage solutions.

The data flow at XFEL facilities is typically designed around a layered model.[86] In the first stage shown in figure IV.11, data is placed on high-speed local systems to temporarily store the high throughput detector data. This first step involves writing data to special files known as extended tagged containers (XTC). These files have the unique ability to be read into memory before they have finished being written.[87] This is particularly useful when a dataset needs to be inspected during an acquisition period.

In the second stage in figure IV.11, this data is then written to high-performance parallel file systems, which can sustain aggregate large throughputs.[88] Online data reduction techniques are applied to reduce storage demands, including photon and electron hit finding, region-of-interest cropping, event selection / discarding and resolution down-scaling.

Long-term storage is typically implemented on large-scale tape libraries, which provide cost-effective archival capacity, albeit with slower access times. Data are stored in hierarchical formats, the most common of which is HDF5.[89]

Given the scale of the datasets, user access is coupled to on-site computing infrastructure, typically provided by the same system. Most facilities provide dedicated analysis clusters collocated with the storage systems, allowing users to process data without the need for bulk data transfer.[90]

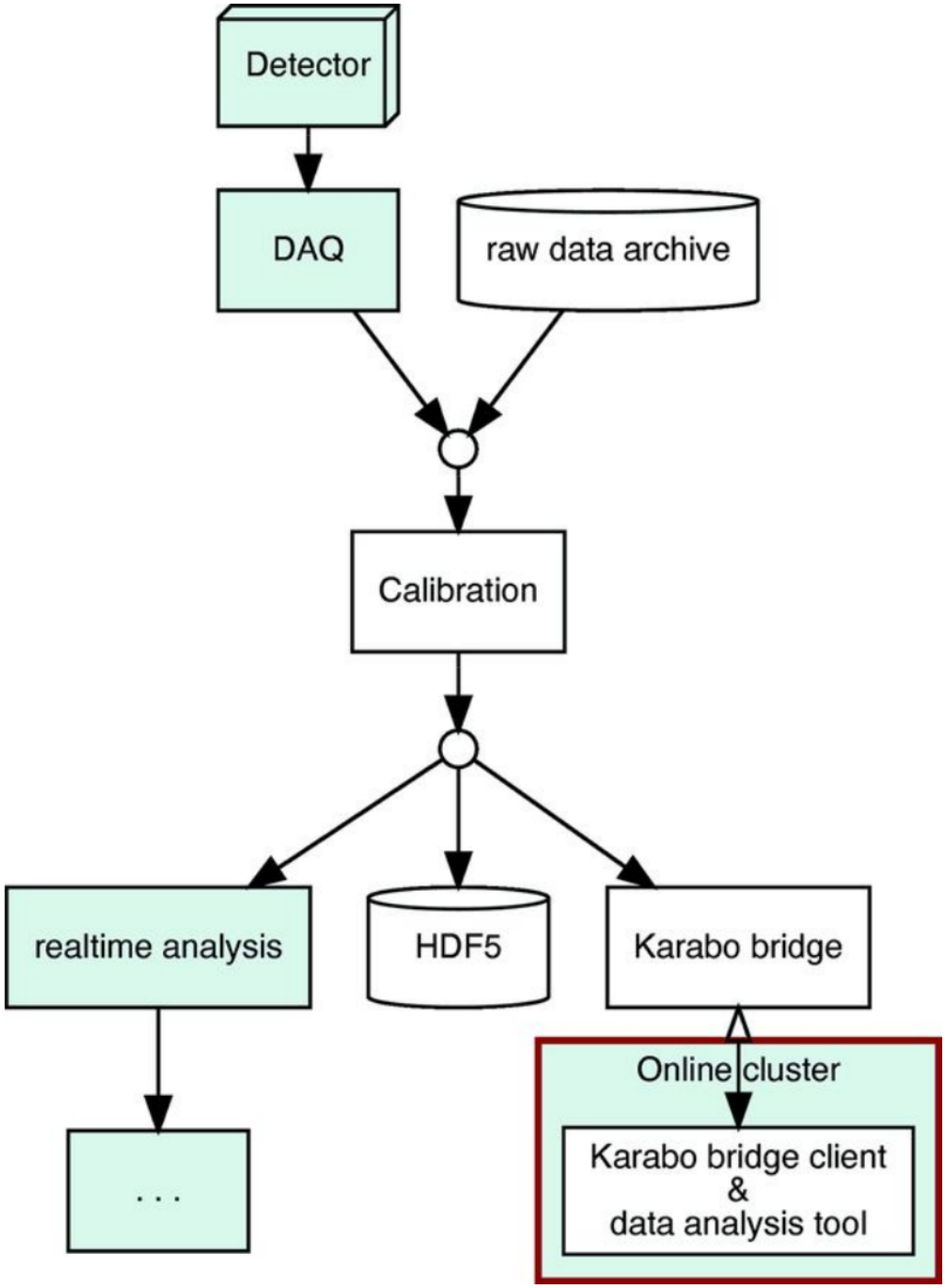

Figure IV.11: A schematic drawing of the data storage and reduction layers, used during an XFEL beamtime adapted from Hauf, S. et al. (2019). *J. Synchrotron Rad.* **26**(5), 1448–1461.[86] A high-data throughput detector, such as a photon 2D detector or a high resolution TOF spectrometer has its output linked to a data acquisition network. It then undergoes calibration. This step may also involve both "lossy" (permanent discard) and "lossless" (file compression without loss) data reduction techniques. This data is then (from *left* to *right*) fed-back to experimentalists for real-time feedback, stored long term in hierarchical formats and available for more detailed analysis.

References

[1] A. D. Shiner, C. Trallero-Herrero, N. Kajumba, H.-C. Bandulet, D. Comtois, F. Légaré, M. Giguère, J.-C. Kieffer, P. B. Corkum, and D. M. Villeneuve, "Wavelength Scaling of High Harmonic Generation Efficiency", Physical Review Letters **103**, ISSN: 1079-7114 (2009) 10.1103/physrevlett.103.073902.

[2] C. Schmidt, Y. Pertot, T. Balciunas, K. Zinchenko, M. Matthews, H. J. Wörner, and J.-P. Wolf, "High-order harmonic source spanning up to the oxygen K-edge based on filamentation pulse compression", Optics Express **26**, 11834, ISSN: 1094-4087 (2018) 10.1364/oe.26.011834.

[3] J. P. Marangos, "Development of high harmonic generation spectroscopy of organic molecules and biomolecules", Journal of Physics B: Atomic, Molecular and Optical Physics **49**, 132001, ISSN: 1361-6455 (2016) 10.1088/0953-4075/49/13/132001.

[4] J. Duris, S. Li, T. Driver, E. G. Champenois, J. P. MacArthur, A. A. Lutman, Z. Zhang, P. Rosenberger, J. W. Aldrich, R. Coffee, G. Coslovich, F.-J. Decker, J. M. Glownia, G. Hartmann, W. Helml, A. Kamalov, J. Knurr, J. Krzywinski, M.-F. Lin, J. P. Marangos, M. Nantel, A. Natan, J. T. O'Neal, N. Shivaram, P. Walter, A. L. Wang, J. J. Welch, T. J. A. Wolf, J. Z. Xu, M. F. Kling, P. H. Bucksbaum, A. Zholents, Z. Huang, J. P. Cryan, and A. Marinelli, "Tunable isolated attosecond X-ray pulses with gigawatt peak power from a free-electron laser", Nature Photonics **14**, 30–36, ISSN: 1749-4893 (2019) 10.1038/s41566-019-0549-5.

[5] N. Huang, H. Deng, B. Liu, D. Wang, and Z. Zhao, "Features and futures of X-ray free-electron lasers", The Innovation **2**, 100097, ISSN: 2666-6758 (2021) 10.1016/j.xinn.2021.100097.

[6] C. Bostedt, S. Boutet, D. M. Fritz, Z. Huang, H. J. Lee, H. T. Lemke, A. Robert, W. F. Schlotter, J. J. Turner, and G. J. Williams, "Linac Coherent Light Source: The first five years", Rev. Mod. Phys. **88**, 015007 (2016) 10.1103/RevModPhys.88.015007.

[7] E. A. Seddon, J. A. Clarke, D. J. Dunning, C. Masciovecchio, C. J. Milne, F. Parmigiani, D. Rugg, J. C. H. Spence, N. R. Thompson, K. Ueda, S. M. Vinko, J. S. Wark, and W. Wurth,

"Short-wavelength free-electron laser sources and science: a review", Reports on Progress in Physics **80**, 115901, ISSN: 1361-6633 (2017) 10.1088/1361-6633/aa7cca.

[8]K. H. Kim, J. G. Kim, S. Nozawa, T. Sato, K. Y. Oang, T. W. Kim, H. Ki, J. Jo, S. Park, C. Song, T. Sato, K. Ogawa, T. Togashi, K. Tono, M. Yabashi, T. Ishikawa, J. Kim, R. Ryoo, J. Kim, H. Ihee, and S.-i. Adachi, "Direct observation of bond formation in solution with femtosecond X-ray scattering", Nature **518**, 385–389, ISSN: 1476-4687 (2015) 10.1038/nature14163.

[9]M. A. Naumova, A. Kalinko, J. W. L. Wong, M. Abdellah, H. Geng, E. Domenichini, J. Meng, S. A. Gutierrez, P.-A. Mante, W. Lin, P. Zalden, A. Galler, F. Lima, K. Kubicek, M. Biednov, A. Britz, S. Checchia, V. Kabanova, M. Wulff, J. Zimara, D. Schwarzer, S. Demeshko, V. Murzin, D. Gosztola, M. Jarenmark, J. Zhang, M. Bauer, M. L. Lawson Daku, W. Gawelda, D. Khakhulin, C. Bressler, F. Meyer, K. Zheng, and S. E. Canton, "Revealing Hot and Long-Lived Metastable Spin States in the Photoinduced Switching of Solvated Metallogrid Complexes with Femtosecond Optical and X-ray Spectroscopies", The Journal of Physical Chemistry Letters **11**, 2133–2141, ISSN: 1948-7185 (2020) 10.1021/acs.jpclett.9b03883.

[10]M. A. Naumova, A. Kalinko, J. W. L. Wong, S. Alvarez Gutierrez, J. Meng, M. Liang, M. Abdellah, H. Geng, W. Lin, K. Kubicek, M. Biednov, F. Lima, A. Galler, P. Zalden, S. Checchia, P.-A. Mante, J. Zimara, D. Schwarzer, S. Demeshko, V. Murzin, D. Gosztola, M. Jarenmark, J. Zhang, M. Bauer, M. L. Lawson Daku, D. Khakhulin, W. Gawelda, C. Bressler, F. Meyer, K. Zheng, and S. E. Canton, "Exploring the light-induced dynamics in solvated metallogrid complexes with femtosecond pulses across the electromagnetic spectrum", The Journal of Chemical Physics **152**, ISSN: 1089-7690 (2020) 10.1063/1.5138641.

[11]J. Tenboer, S. Basu, N. Zatsepin, K. Pande, D. Milathianaki, M. Frank, M. Hunter, S. Boutet, G. J. Williams, J. E. Koglin, D. Oberthuer, M. Heymann, C. Kupitz, C. Conrad, J. Coe, S. Roy-Chowdhury, U. Weierstall, D. James, D. Wang, T. Grant, A. Barty, O. Yefanov, J. Scales, C. Gati, C. Seuring, V. Srajer, R. Henning, P. Schwander, R. Fromme, A. Ourmazd, K. Moffat, J. J. Van Thor, J. C. H. Spence, P. Fromme, H. N. Chapman, and M. Schmidt, "Time-resolved serial crystallography captures high-resolution intermediates of photoactive yellow protein", Science **346**, 1242–1246, ISSN: 1095-9203 (2014) 10.1126/science.1259357.

[12]H. N. Chapman, P. Fromme, A. Barty, T. A. White, R. A. Kirian, A. Aquila, M. S. Hunter, J. Schulz, D. P. DePonte, U. Weierstall, R. B. Doak, F. R. N. C. Maia, A. V. Martin, I. Schlichting, L. Lomb, N. Coppola, R. L. Shoeman, S. W. Epp, R. Hartmann, D. Rolles, A. Rudenko, L. Foucar, N. Kimmel, G. Weidenspointner, P. Holl, M. Liang, M. Barthelmess, C. Caleman, S. Boutet, M. J. Bogan, J. Krzywinski, C. Bostedt, S. Bajt, L. Gumprecht, B. Rudek, B. Erk, C. Schmidt, A. Hömke, C. Reich, D. Pietschner, L. Strüder, G. Hauser, H. Gorke, J. Ullrich, S. Herrmann, G. Schaller, F. Schopper, H. Soltau, K.-U. Kühnel, M. Messerschmidt, J. D. Bozek, S. P. Hau-Riege, M. Frank, C. Y. Hampton, R. G. Sierra, D. Starodub, G. J. Williams, J. Hajdu, N. Timneanu, M. M. Seibert, J. Andreasson, A. Rocker, O. Jönsson, M. Svenda, S. Stern, K. Nass, R. Andritschke, C.-D. Schröter, F. Krasniqi, M. Bott, K. E. Schmidt, X. Wang, I. Grotjohann, J. M. Holton, T. R. M. Barends, R. Neutze, S. Marchesini, R. Fromme, S. Schorb, D. Rupp, M. Adolph, T. Gorkhover, I. Andersson, H. Hirsemann, G. Potdevin, H. Graafsma, B. Nilsson, and J. C. H. Spence, "Femtosecond X-ray protein nanocrystallography", Nature **470**, 73–77, ISSN: 1476-4687 (2011) 10.1038/nature09750.

[13]W. Liu, D. Wacker, C. Gati, G. W. Han, D. James, D. Wang, G. Nelson, U. Weierstall, V. Katritch, A. Barty, N. A. Zatsepin, D. Li, M. Messerschmidt, S. Boutet, G. J. Williams, J. E. Koglin, M. M. Seibert, C. Wang, S. T. A. Shah, S. Basu, R. Fromme, C. Kupitz, K. N. Rendek, I. Grotjohann, P. Fromme, R. A. Kirian, K. R. Beyerlein, T. A. White, H. N. Chapman, M. Caffrey, J. C. H. Spence, R. C. Stevens, and V. Cherezov, "Serial Femtosecond Crystallography of G Protein–Coupled Receptors", Science **342**, 1521–1524, ISSN: 1095-9203 (2013) 10.1126/science.1244142.

[14]G. L. Dakovski, P. Heimann, M. Holmes, O. Krupin, M. P. Minitti, A. Mitra, S. Moeller, M. Rowen, W. F. Schlotter, and J. J. Turner, "The Soft X-ray Research instrument at the Linac Coherent Light Source", Journal of Synchrotron Radiation **22**, 498–502, ISSN: 1600-5775 (2015) 10.1107/s160057751500301x.

[15]A. W. Hahn, B. E. Van Kuiken, V. G. Chilkuri, N. Levin, E. Bill, T. Weyhermüller, A. Nicolaou, J. Miyawaki, Y. Harada, and S. DeBeer, "Probing the Valence Electronic Structure of Low-Spin Ferrous and Ferric Complexes Using 2p3d Resonant Inelastic X-ray Scattering (RIXS)", Inorganic Chemistry **57**, 9515–9530, ISSN: 1520-510X (2018) 10.1021/acs.inorgchem.8b01550.

[16]T. Kroll, C. Weninger, R. Alonso-Mori, D. Sokaras, D. Zhu, L. Mercadier, V. P. Majety, A. Marinelli, A. Lutman, M. W. Guetg, F.-J. Decker, S. Boutet, A. Aquila, J. Koglin, J. Koralek, D. P. DePonte, J. Kern, F. D. Fuller, E. Pastor, T. Fransson, Y. Zhang, J. Yano, V. K. Yachandra, N. Rohringer, and U. Bergmann, "Stimulated X-Ray Emission Spectroscopy in Transition Metal Complexes", Physical Review Letters **120**, ISSN: 1079-7114 (2018) 10.1103/physrevlett.120.133203.

[17]W. Zhang, K. S. Kjær, R. Alonso-Mori, U. Bergmann, M. Chollet, L. A. Fredin, R. G. Hadt, R. W. Hartsock, T. Harlang, T. Kroll, K. Kubiček, H. T. Lemke, H. W. Liang, Y. Liu, M. M. Nielsen, P. Persson, J. S. Robinson, E. I. Solomon, Z. Sun, D. Sokaras, T. B. van Driel, T.-C. Weng, D. Zhu, K. Wärnmark, V. Sundström, and K. J. Gaffney, "Manipulating charge transfer excited state relaxation and spin crossover in iron coordination complexes with ligand substitution", en, Chem. Sci. **8**, 515–523 (2017).

[18]D. S. Hodge, A. F. T. Leong, K. Kurzer-Ogul, S. Pandolfi, D. S. Montgomery, J. Shang, H. Aluie, S. Marchesini, Y. Liu, K. Li, A. Sakdinawat, E. C. Galtier, B. Nagler, H. J. Lee, E. F. Cunningham, T. E. Carver, C. A. Bolme, K. J. Ramos, D. Khaghani, P. M. Kozlowski, A. E. Gleason, and R. L. Sandberg, "Single-shot in-line x-ray phase-contrast imaging of void-shockwave interactions in fusion energy materials", Physics of Plasmas **32**, ISSN: 1089-7674 (2025) 10.1063/5.0272820.

[19]B. Heuser, A. Bergermann, M. G. Stevenson, D. Ranjan, Z. He, J. Lütgert, S. Schumacher, M. Bethkenhagen, A. Descamps, E. Galtier, A. E. Gleason, D. Khaghani, G. D. Glenn, E. F. Cunningham, S. H. Glenzer, N. J. Hartley, J.-A. Hernandez, O. S. Humphries, K. Katagiri, H. J. Lee, E. E. McBride, K. Miyanishi, B. Nagler, B. Ofori-Okai, N. Ozaki, S. Pandolfi, C. Qu, P. T. May, R. Redmer, C. Schoenwaelder, K. Sueda, T. Yabuuchi, M. Yabashi, B. Lukic, A. Rack, L. M. V. Zinta, T. Vinci, A. Benuzzi-Mounaix, A. Ravasio, and D. Kraus, "Release dynamics of nanodiamonds created by laser-driven shock-compression of polyethylene terephthalate", Scientific Reports **14**, ISSN: 2045-2322 (2024) 10.1038/s41598-024-62367-7.

[20]Z. Moldabekov, T. D. Gawne, S. Schwalbe, T. R. Preston, J. Vorberger, and T. Dornheim, "Ultrafast Heating-Induced Suppression of d-Band Dominance in the Electronic Excitation

Spectrum of Cuprum", ACS Omega **9**, 25239–25250, ISSN: 2470-1343 (2024) 10.1021/acsomega.4c02920.

[21]T. G. White, T. D. Griffin, D. Haden, H. J. Lee, E. Galtier, E. Cunningham, D. Khaghani, A. Descamps, L. Wollenweber, B. Armentrout, C. Convery, K. Appel, L. B. Fletcher, S. Goede, J. B. Hastings, J. Iratcabal, E. E. McBride, J. Molina, G. Monaco, L. Morrison, H. Stramel, S. Yunus, U. Zastrau, S. H. Glenzer, G. Gregori, D. O. Gericke, and B. Nagler, "Superheating gold beyond the predicted entropy catastrophe threshold", Nature **643**, 950–954, ISSN: 1476-4687 (2025) 10.1038/s41586-025-09253-y.

[22]I. Gabalski, A. Green, P. Lenzen, F. Allum, M. Bain, S. Bhattacharyya, M. A. Britton, E. G. Champenois, X. Cheng, J. P. Cryan, T. Driver, R. Forbes, D. Garratt, A. M. Ghrist, M. Graßl, M. F. Kling, K. A. Larsen, M. Liang, M.-F. Lin, Y. Liu, M. P. Minitti, S. Nelson, J. S. Robinson, P. H. Bucksbaum, T. J. A. Wolf, N. H. List, and J. M. Glownia, "Imaging Valence Electron Rearrangement in a Chemical Reaction Using Hard X-Ray Scattering", Physical Review Letters **135**, ISSN: 1079-7114 (2025) 10.1103/53h3-vykl.

[23]T. N. Malla, L. Aldama, V. Leon, D. Feliz, H. Hu, I. Thomas, A. Cellini, W. Y. Wahlgren, A. Nimmrich, S. Botha, R. Sierra, M. S. Hunter, F. Poitevin, S. Lisova, A. Batyuk, G. Gate, R. Jernigan, C. J. Kupitz, P. Maj, P. Meszaros, M. Kurttila, L. Monrroy, F. Luo, S. Owada, J. Kang, C. Slavov, M. Maj, C. Gautier, M. Kashipathy, A. Tolstikova, V. Mariani, A. Barty, F. Moss, P. Schwander, H. Liu, S. Boutet, P. Fromme, H. Takala, J. A. Ihalainen, U. Weierstall, S. Westenhoff, E. A. Stojković, and M. Schmidt, "Observation of early events in the photoactivation of Myxobacterial phytochrome using time-resolved serial femtosecond crystallography", Communications Chemistry **8**, ISSN: 2399-3669 (2025) 10.1038/s42004-025-01578-z.

[24]E. Biasin, T. B. van Driel, K. S. Kjær, A. O. Dohn, M. Christensen, T. Harlang, P. Chabera, Y. Liu, J. Uhlig, M. Pápai, Z. Németh, R. Hartsock, W. Liang, J. Zhang, R. Alonso-Mori, M. Chollet, J. M. Glownia, S. Nelson, D. Sokaras, T. A. Assefa, A. Britz, A. Galler, W. Gawelda, C. Bressler, K. J. Gaffney, H. T. Lemke, K. B. Møller, M. M. Nielsen, V. Sundström, G. Vankó, K. Wärnmark, S. E. Canton, and K. Haldrup, "Femtosecond X-Ray Scattering Study of Ultrafast Photoinduced Structural Dynamics in Solvated $[Co(terpy)_2]^{2+}$", Physical Review Letters **117**, ISSN: 1079-7114 (2016) 10.1103/physrevlett.117.013002.

[25]T. Ekeberg, D. Assalauova, J. Bielecki, R. Boll, B. J. Daurer, L. A. Eichacker, L. E. Franken, D. E. Galli, L. Gelisio, L. Gumprecht, L. H. Gunn, J. Hajdu, R. Hartmann, D. Hasse, A. Ignatenko, J. Koliyadu, O. Kulyk, R. Kurta, M. Kuster, W. Lugmayr, J. Lübke, A. P. Mancuso, T. Mazza, C. Nettelblad, Y. Ovcharenko, D. E. Rivas, M. Rose, A. K. Samanta, P. Schmidt, E. Sobolev, N. Timneanu, S. Usenko, D. Westphal, T. Wollweber, L. Worbs, P. L. Xavier, H. Yousef, K. Ayyer, H. N. Chapman, J. A. Sellberg, C. Seuring, I. A. Vartanyants, J. Küpper, M. Meyer, and F. R. N. C. Maia, "Observation of a single protein by ultrafast X-ray diffraction", Light: Science &; Applications **13**, ISSN: 2047-7538 (2024) 10.1038/s41377-023-01352-7.

[26]K. Kurzer-Ogul, B. M. Haines, D. S. Montgomery, S. Pandolfi, J. P. Sauppe, A. F. T. Leong, D. Hodge, P. M. Kozlowski, S. Marchesini, E. Cunningham, E. Galtier, D. Khaghani, H. J. Lee, B. Nagler, R. L. Sandberg, A. E. Gleason, H. Aluie, and J. K. Shang, "Radiation and heat transport in divergent shock–bubble interactions", Physics of Plasmas **31**, ISSN: 1089-7674 (2024) 10.1063/5.0185056.

[27]T. Kluge, M. Bussmann, E. Galtier, S. Glenzer, J. Grenzer, C. Gutt, N. J. Hartley, L. Huang, A. L. Garcia, H. J. Lee, E. E. McBride, J. Metzkes-Ng, M. Nakatsutsumi, I. Nam, A. Pelka, I. Prencipe, L. Randolph, M. Rehwald, C. Rödel, M. Rödel, T. Toncian, L. Yang, K. Zeil, U. Schramm, and T. E. Cowan, "Probing shock dynamics inside micro-wire targets after high-intensity laser irradiation using small angle x-ray scattering of a free-electron laser", New Journal of Physics **25**, 103036, ISSN: 1367-2630 (2023) 10.1088/1367-2630/acfab5.

[28]A. Descamps, B. K. Ofori-Okai, O. Bistoni, Z. Chen, E. Cunningham, L. B. Fletcher, N. J. Hartley, J. B. Hastings, D. Khaghani, M. Mo, B. Nagler, V. Recoules, R. Redmer, M. Schörner, D. G. Senesky, P. Sun, H.-E. Tsai, T. G. White, S. H. Glenzer, and E. E. McBride, "Evidence for phonon hardening in laser-excited gold using x-ray diffraction at a hard x-ray free electron laser", Science Advances **10**, ISSN: 2375-2548 (2024) 10.1126/sciadv.adh5272.

[29]B. W. J. McNeil and N. R. Thompson, "X-ray free-electron lasers", Nature Photonics **4**, 814–821, ISSN: 1749-4893 (2010) 10.1038/nphoton.2010.239.

[30]H. Lv, G. Chen, J. Ding, J. Chen, C. Yu, P. Yu, H. Miao, Q. Mi, H. Zhao, D. Bai, H. Wang, M. Li, Q. Xiao, X. Liu, R. Wang, Y. Gao, and Y. Yan, "The control system for SHINE", Nuclear

Engineering and Technology **57**, 103714, ISSN: 1738-5733 (2025) 10.1016/j.net.2025.103714.

[31]G. Ciovati, G. Cheng, U. Pudasaini, and R. A. Rimmer, "Multi-metallic conduction cooled superconducting radio-frequency cavity with high thermal stability", Superconductor Science and Technology **33**, 07LT01, ISSN: 1361-6668 (2020) 10.1088/1361-6668/ab8d98.

[32]J. Galayda, "The LCLS-II: A High Power Upgrade to the LCLS", en, Proceedings of the 9th Int. Particle Accelerator Conf. **IPAC2018**, Canada (2018) 10.18429/JACOW-IPAC2018-MOYGB2.

[33]H. Zhang, S. Gilevich, A. Miahnahri, S. C. Alverson, A. Brachmann, J. Duris, P. Franz, A. Fry, J. Hirschman, K. Larsen, R. Lemons, S. Li, B. Lu, A. Marinelli, M. Martinez, J. May, E. Milshtein, K. Murari, N. Neveu, J. Robinson, J. Schmerge, L. Sun, T. Vecchione, C. Xu, F. Zhou, and S. Carbajo, "The Linac Coherent Light Source II photoinjector laser infrastructure", High Power Laser Science and Engineering **12**, ISSN: 2052-3289 (2024) 10.1017/hpl.2024.33.

[34]T. Vecchione, A. Brachmann, J. Corbett, M. Ferreira, S. Gilevich, E. Jongewaard, H. Loos, J. Sheppard, S. Weathersby, and F. Zhou, "SLAC RF GUN PHOTOCATHODE TEST FACILITY*", in (Jan. 2013).

[35]H.-S. Kang, H. Yang, G. Kim, H. Heo, I. Nam, C.-K. Min, C. Kim, S. Y. Baek, H.-J. Choi, G. Mun, B. R. Park, Y. J. Suh, D. C. Shin, J. Hu, J. Hong, S. Jung, S.-H. Kim, K. Kim, D. Na, S. S. Park, Y. J. Park, J.-H. Han, Y. G. Jung, S. H. Jeong, M. J. Kim, H. G. Lee, S. Lee, W.-W. Lee, B. Oh, H. S. Suh, K.-H. Park, H.-S. Lee, D. Z. Khan, T. O. Raubenheimer, and J. Wu, "FEL performance achieved at PAL-XFEL using a three-chicane bunch compression scheme", Journal of Synchrotron Radiation **26**, 1127–1138, ISSN: 1600-5775 (2019) 10.1107/s1600577519005861.

[36]C. Davut, O. Apsimon, B. R. Hounsell, B. L. Militsyn, L. S. Cowie, F. Yaman, A. D. Brynes, and P. H. Williams, "Balance of bunch compression and emittance preservation for high-brightness x-ray free electron laser injectors", Physical Review Accelerators and Beams **28**, ISSN: 2469-9888 (2025) 10.1103/rjns-vqzt.

[37]C. Kim, S.-J. Park, C.-K. Min, J. Hu, S.-H. Kim, Y. Joo, H. Heo, D.-E. Kim, S. Lee, H.-S. Kang, I. S. Ko, M. Chae, and J. Lee, "Review of technical achievements in PAL-XFEL", AAPPS Bulletin **32**, ISSN: 2309-4710 (2022) 10.1007/s43673-022-00045-4.

[38]T. Tanikawa, S. Karabekyan, S. Kovalev, S. Casalbuoni, V. Asgekar, S. Serkez, G. Geloni, and M. Gensch, "A superradiant THz undulator source for XFELs", Journal of Instrumentation **14**, P05024–P05024, ISSN: 1748-0221 (2019) 10.1088/1748-0221/14/05/p05024.

[39]J. A. Clarke, *The Science and Technology of Undulators and Wigglers* (Oxford University Press, July 2004), ISBN: 9780198508557, 10.1093/acprof:oso/9780198508557.001.0001.

[40]M. R. Howells and B. M. Kincaid, "The Properties of Undulator Radiation", in *New Directions in Research with Third-Generation Soft X-Ray Synchrotron Radiation Sources* (Springer Netherlands, 1994), pp. 315–358, ISBN: 9789401108683, 10.1007/978-94-011-0868-3_13.

[41]Z. Huang and K.-J. Kim, "Review of x-ray free-electron laser theory", Physical Review Special Topics - Accelerators and Beams **10**, ISSN: 1098-4402 (2007) 10.1103/physrevstab.10.034801.

[42]I. B. Bernstein and J. L. Hirshfield, "Amplification on a relativistic electron beam in a spatially periodic transverse magnetic field", Physical Review A **20**, 1661–1670, ISSN: 0556-2791 (1979) 10.1103/physreva.20.1661.

[43]A. D. Brynes, I. Akkermans, E. Allaria, L. Badano, S. Brussaard, G. D. Ninno, D. Gauthier, G. Gaio, L. Giannessi, N. S. Mirian, G. Penco, G. Perosa, P. Rebernik, I. Setija, S. Spampinati, C. Spezzani, M. Trovò, M. Veronese, P. H. Williams, A. Wolski, and S. D. Mitri, "Characterisation of microbunching instability with 2D Fourier analysis", Scientific Reports **10**, ISSN: 2045-2322 (2020) 10.1038/s41598-020-61764-y.

[44]P. Emma, R. Akre, J. Arthur, R. Bionta, C. Bostedt, J. Bozek, A. Brachmann, P. Bucksbaum, R. Coffee, F.-J. Decker, Y. Ding, D. Dowell, S. Edstrom, A. Fisher, J. Frisch, S. Gilevich, J. Hastings, G. Hays, P. Hering, Z. Huang, R. Iverson, H. Loos, M. Messerschmidt, A. Miahnahri, S. Moeller, H.-D. Nuhn, G. Pile, D. Ratner, J. Rzepiela, D. Schultz, T. Smith, P. Stefan, H. Tompkins, J. Turner, J. Welch, W. White, J. Wu, G. Yocky, and J. Galayda, "First lasing and operation of an ångstrom-wavelength free-electron laser", Nature Photonics **4**, 641–647, ISSN: 1749-4893 (2010) 10.1038/nphoton.2010.176.

[45]I. J. Bermúdez Macias, S. Düsterer, R. Ivanov, J. Liu, G. Brenner, J. Rönsch-Schulenburg, M. K. Czwalinna, and M. V. Yurkov, "Study of temporal, spectral, arrival time and energy fluctuations of SASE FEL pulses", Optics Express **29**, 10491, ISSN: 1094-4087 (2021) 10.1364/oe.419977.

[46]A. A. Lutman, Y. Ding, Y. Feng, Z. Huang, M. Messerschmidt, J. Wu, and J. Krzywinski, "Femtosecond x-ray free electron laser pulse duration measurement from spectral correlation function", Physical Review Special Topics - Accelerators and Beams **15**, ISSN: 1098-4402 (2012) 10.1103/physrevstab.15.030705.

[47]R. R. Robles, A. Halavanau, D. Cesar, A. Lutman, and G. Stupakov, "Reconstruction of x-ray free-electron laser pulse duration and energy chirp from spectral intensity fluctuations", Physical Review Accelerators and Beams **26**, ISSN: 2469-9888 (2023) 10.1103/physrevaccelbeams.26.030701.

[48]A. A. Zholents, "Method of an enhanced self-amplified spontaneous emission for x-ray free electron lasers", Physical Review Special Topics - Accelerators and Beams **8**, ISSN: 1098-4402 (2005) 10.1103/physrevstab.8.040701.

[49]P. J. Emma, *A New Bunch Compressor Chicane for the SLAC Linac to Produce 30-FSEC, 30-KA, 30-GeV Electron Bunches* (May 2002), 10.2172/799087.

[50]B. E. Carlsten and S. J. Russell, "Subpicosecond compression of 0.1-1 nC electron bunches with a magnetic chicane at 8 MeV", Physical Review E **53**, R2072–R2075, ISSN: 1095-3787 (1996) 10.1103/physreve.53.r2072.

[51]S. Huang, Y. Ding, Y. Feng, E. Hemsing, Z. Huang, J. Krzywinski, A. A. Lutman, A. Marinelli, T. J. Maxwell, and D. Zhu, "Generating Single-Spike Hard X-Ray Pulses with Nonlinear Bunch Compression in Free-Electron Lasers", Physical Review Letters **119**, ISSN: 1079-7114 (2017) 10.1103/physrevlett.119.154801.

[52]J. P. MacArthur, J. Duris, Z. Zhang, A. Lutman, A. Zholents, X. Xu, Z. Huang, and A. Marinelli, "Phase-Stable Self-Modulation of an Electron Beam in a Magnetic Wiggler", Physical Review Letters **123**, ISSN: 1079-7114 (2019) 10.1103/physrevlett.123.214801.

[53]G. Marcus, E. Hemsing, and J. Rosenzweig, "Gain length fitting formula for free-electron lasers with strong space-charge effects", Physical Review Special Topics - Accelerators and Beams **14**, ISSN: 1098-4402 (2011) 10.1103/physrevstab.14.080702.

[54]J. T. O'Neal, E. G. Champenois, S. Oberli, R. Obaid, A. Al-Haddad, J. Barnard, N. Berrah, R. Coffee, J. Duris, G. Galinis, D. Garratt, J. M. Glownia, D. Haxton, P. Ho, S. Li, X. Li, J. MacArthur, J. P. Marangos, A. Natan, N. Shivaram, D. S. Slaughter, P. Walter, S. Wandel, L. Young, C. Bostedt, P. H. Bucksbaum, A. Picón, A. Marinelli, and J. P. Cryan, "Electronic Population Transfer via Impulsive Stimulated X-Ray Raman Scattering with Attosecond Soft-X-Ray Pulses", Physical Review Letters **125**, ISSN: 1079-7114 (2020) 10.1103/physrevlett.125.073203.

[55]L. R. Dalesio, J. O. Hill, M. Kraimer, S. Lewis, D. Murray, S. Hunt, W. Watson, M. Clausen, and J. Dalesio, "The experimental physics and industrial control system architecture: past, present, and future", Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment **352**, 179–184, ISSN: 0168-9002 (1994) 10.1016/0168-9002(94)91493-1.

[56]A. Wallace, D. Flath, M. Ghaly, T. Johnson, K. Lauer, Z. Lentz, R. Tang-Kong, and J. Yin, "The LCLS-II Experiment Control System", en, (2024) 10.18429/JACOW-ICALEPCS2023-TH2BCO03.

[57]P. Walter, T. Osipov, M.-F. Lin, J. Cryan, T. Driver, A. Kamalov, A. Marinelli, J. Robinson, M. H. Seaberg, T. J. A. Wolf, J. Aldrich, N. Brown, E. G. Champenois, X. Cheng, D. Cocco, A. Conder, I. Curiel, A. Egger, J. M. Glownia, P. Heimann, M. Holmes, T. Johnson, L. Lee, X. Li, S. Moeller, D. S. Morton, M. L. Ng, K. Ninh, J. T. O'Neal, R. Obaid, A. Pai, W. Schlotter, J. Shepard, N. Shivaram, P. Stefan, X. Van, A. L. Wang, H. Wang, J. Yin, S. Yunus, D. Fritz, J. James, and J.-C. Castagna, "The time-resolved atomic, molecular and optical science instrument at the Linac Coherent Light Source", Journal of Synchrotron Radiation **29**, 957–968, ISSN: 1600-5775 (2022) 10.1107/s1600577522004283.

[58]A. A. Sorokin, Y. Bican, S. Bonfigt, M. Brachmanski, M. Braune, U. F. Jastrow, A. Gottwald, H. Kaser, M. Richter, and K. Tiedtke, "An X-ray gas monitor for free-electron lasers", Journal of Synchrotron Radiation **26**, 1092–1100, ISSN: 1600-5775 (2019) 10.1107/s1600577519005174.

[59]J. Chen, W. Chen, G. Zhang, H. Lin, and S.-C. Chen, "High-resolution compact spectrometer based on a custom-printed varied-line-spacing concave blazed grating", Optics Express **25**, 12446, ISSN: 1094-4087 (2017) 10.1364/oe.25.012446.

[60]M. C. Hettrick and S. Bowyer, "Variable line-space gratings: new designs for use in grazing incidence spectrometers", Applied Optics **22**, 3921, ISSN: 1539-4522 (1983) 10.1364/ao.22.003921.

[61]E. N. Ragozin, E. A. Vishnyakov, A. O. Kolesnikov, A. S. Pirozhkov, and A. N. Shatokhin, "Soft X-ray spectrometers based on aperiodic reflection gratings and their application", Physics-Uspekhi **64**, 495–514, ISSN: 1468-4780 (2021) 10.3367/ufne.2020.06.038799.

[62]E. A. Vishnyakov, A. O. Kolesnikov, A. A. Kuzin, D. V. Negrov, E. N. Ragozin, P. V. Sasorov, and A. N. Shatokhin, "Imaging diffraction VLS spectrometer for a wavelength range $\lambda > 120$ Å", Quantum Electronics **47**, 54–57, ISSN: 1468-4799 (2017) 10.1070/qel16261.

[63]T. Callcott, W. O'Brien, J. Jia, Q. Dong, D. Ederer, R. Watts, and D. Mueller, "A simple variable line space grating monochromator for synchrotron light source beamlines", Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment **319**, 128–134, ISSN: 0168-9002 (1992) 10.1016/0168-9002(92)90543-d.

[64]E. G. Rightor, A. P. Hitchcock, H. Ade, R. D. Leapman, S. G. Urquhart, A. P. Smith, G. Mitchell, D. Fischer, H. J. Shin, and T. Warwick, "Spectromicroscopy of Poly(ethylene terephthalate): Comparison of Spectra and Radiation Damage Rates in X-ray Absorption and Electron Energy Loss", The Journal of Physical Chemistry B **101**, 1950–1960, ISSN: 1520-5207 (1997) 10.1021/jp9622748.

[65]J. R. Shah, M. T. Bengtson, S. Collman, R. C. Hoffmann, D. C. Ferguson, D. P. Engelhart, and E. A. Plis, "Characterization of polyethylene terephthalate (PET) materials under high-energy electron exposure", CEAS Space Journal **16**, 275–283, ISSN: 1868-2510 (2023) 10.1007/s12567-023-00495-y.

[66]J. Wang, G. A. Botton, M. M. West, and A. P. Hitchcock, "Quantitative Evaluation of Radiation Damage to Polyethylene Terephthalate by Soft X-rays and High-energy Electrons", The Journal of Physical Chemistry B **113**, 1869–1876, ISSN: 1520-5207 (2009) 10.1021/jp808289e.

[67]F. Frati, M. O. J. Y. Hunault, and F. M. F. de Groot, "Oxygen K-edge X-ray Absorption Spectra", Chemical Reviews **120**, 4056–4110, ISSN: 1520-6890 (2020) 10.1021/acs.chemrev.9b00439.

[68]P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, A. Vijaykumar, A. P. Bardelli, A. Rothberg, A. Hilboll, A. Kloeckner, A. Scopatz, A. Lee, A. Rokem, C. N. Woods, C. Fulton, C. Masson, C. Häggström, C. Fitzgerald, D. A. Nicholson, D. R. Hagen, D. V. Pasechnik, E. Olivetti, E. Martin, E. Wieser, F. Silva, F. Lenders, F. Wilhelm, G. Young, G. A. Price, G.-L. Ingold, G. E. Allen, G. R. Lee, H. Audren, I. Probst, J. P. Dietrich, J. Silterra, J. T. Webber, J. Slavič, J. Nothman, J. Buchner, J. Kulick, J. L. Schönberger, J. V. de Miranda Cardoso, J. Reimer, J. Harrington, J. L. C. Rodríguez, J. Nunez-Iglesias, J. Kuczynski, K. Tritz, M. Thoma, M. Newville, M. Kümmerer, M. Bolingbroke, M. Tartre, M. Pak, N. J. Smith, N. Nowaczyk, N. Shebanov, O. Pavlyk, P. A. Brodtkorb, P. Lee, R. T. McGibbon, R. Feldbauer, S. Lewis, S. Tygier, S. Sievert, S. Vigna, S. Peterson, S. More, T. Pudlik, T. Oshima, T. J. Pingel, T. P. Robitaille, T. Spura, T. R. Jones, T. Cera, T. Leslie, T. Zito, T. Krauss, U. Upadhyay, Y. O. Halchenko, and Y. Vázquez-Baeza, "SciPy 1.0: fundamental algorithms for scientific computing in Python", Nature Methods **17**, 261–272, ISSN: 1548-7105 (2020) 10.1038/s41592-019-0686-2.

[69]L. N. Lobanov, K. A. Sharypov, V. G. Shpak, S. A. Shunailov, M. I. Yalandin, and N. M. Zubarev, "Time-of-flight technique for estimation of the energy of runaway electron bunches formed in magnetized gas diodes", Physics of Plasmas **31**, ISSN: 1089-7674 (2024) 10.1063/5.0213344.

[70]D. J. Graham and L. J. Gamble, "Back to the basics of time-of-flight secondary ion mass spectrometry of bio-related samples. I. Instrumentation and data collection", Biointerphases **18**, ISSN: 1559-4106 (2023) 10.1116/6.0002477.

[71]S. Fearn, *An Introduction to Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS) and its Application to Materials Science* (Morgan & Claypool Publishers, Oct. 2015), ISBN: 9781681740249, 10.1088/978-1-6817-4088-1.

[72]H. Schmidt-Böcking, J. Ullrich, R. Dörner, and C. L. Cocke, "The COLTRIMS Reaction Microscope—The Spyhole into the Ultrafast Entangled Dynamics of Atomic and Molecular Systems", Annalen der Physik **533**, ISSN: 1521-3889 (2021) 10.1002/andp.202100134.

[73]A. De Fanis, M. Ilchen, A. Achner, T. M. Baumann, R. Boll, J. Buck, C. Danilevsky, S. Esenov, B. Erk, P. Grychtol, G. Hartmann, J. Liu, T. Mazza, J. Montaño, V. Music, Y. Ovcharenko, N. Rennhack, D. Rivas, D. Rolles, P. Schmidt, H. Sotoudi Namin, F. Scholz, J. Viefhaus, P. Walter, P. Ziółkowski, H. Zhang, and M. Meyer, "High-resolution electron time-of-flight spectrometers for angle-resolved measurements at the SQS Instrument at the European XFEL", Journal of Synchrotron Radiation **29**, 755–764, ISSN: 1600-5775 (2022) 10.1107/s1600577522002284.

[74]M. Mucke, M. Förstel, T. Lischke, T. Arion, A. M. Bradshaw, and U. Hergenhahn, "Performance of a short "magnetic bottle" electron spectrometer", Review of Scientific Instruments **83**, ISSN: 1089-7623 (2012) 10.1063/1.4729256.

[75]K. Borne, J. T. O'Neal, J. Wang, E. Isele, R. Obaid, N. Berrah, X. Cheng, P. H. Bucksbaum, J. James, A. Kamalov, K. A. Larsen, X. Li, M.-F. Lin, Y. Liu, A. Marinelli, A. M. Summers, E. Thierstein, T. J. A. Wolf, D. Rolles, P. Walter, J. P. Cryan, and T. Driver, "Design and performance of a magnetic bottle electron spectrometer for high-energy photoelectron spectroscopy", Review of Scientific Instruments **95**, ISSN: 1089-7623 (2024) 10.1063/5.0223334.

[76]N. Kurahashi, S. Thürmer, S. Y. Liu, Y.-i. Yamamoto, S. Karashima, A. Bhattacharya, Y. Ogi, T. Horio, and T. Suzuki, "Design and characterization of a magnetic bottle electron spectrometer for time-resolved extreme UV and X-ray photoemission spectroscopy of liquid microjets", Structural Dynamics **8**, ISSN: 2329-7778 (2021) 10.1063/4.0000107.

[77]A. Kothe, J. Metje, M. Wilke, A. Moguilevski, N. Engel, R. Al-Obaidi, C. Richter, R. Golnak, I. Y. Kiyan, and E. F. Aziz, "Time-of-flight electron spectrometer for a broad range of kinetic energies", Review of Scientific Instruments **84**, ISSN: 1089-7623 (2013) 10.1063/1.4791792.

[78]J. van der Gracht, G. W. Euliss, and V. P. Pauca, "Experimental investigation of aliasing in the assessment of sampled images", in Visual Information Processing XI, Vol. 4736, edited by Z.-u. Rahman, R. A. Schowengerdt, and S. E. Reichenbach (July 2002), pp. 99–106, 10.1117/12.477570.

[79]S. Cho, R. Grazioso, N. Zhang, M. Aykac, and M. Schmand, "Digital timing: sampling frequency, anti-aliasing filter and signal interpolation filter dependence on timing resolution", Physics in Medicine and Biology **56**, 7569–7583, ISSN: 1361-6560 (2011) 10.1088/0031-9155/56/23/015.

[80]L. Young, E. P. Kanter, B. Krässig, Y. Li, A. M. March, S. T. Pratt, R. Santra, S. H. Southworth, N. Rohringer, L. F. DiMauro, G. Doumy, C. A. Roedig, N. Berrah, L. Fang, M. Hoener, P. H. Bucksbaum, J. P. Cryan, S. Ghimire, J. M. Glownia, D. A. Reis, J. D. Bozek, C. Bostedt, and M. Messerschmidt, "Femtosecond electronic response of atoms to ultra-intense X-rays", Nature **466**, 56–61, ISSN: 1476-4687 (2010) 10.1038/nature09177.

[81]A. Allahgholi, J. Becker, A. Delfs, R. Dinapoli, P. Goettlicher, D. Greiffenberg, B. Henrich, H. Hirsemann, M. Kuhn, R. Klanner, A. Klyuev, H. Krueger, S. Lange, T. Laurus, A. Marras, D. Mezza, A. Mozzanica, M. Niemann, J. Poehlsen, J. Schwandt, I. Sheviakov, X. Shi, S. Smoljanin, L. Steffen, J. Sztuk-Dambietz, U. Trunk, Q. Xia, M. Zeribi, J. Zhang, M. Zimmer, B. Schmitt, and H. Graafsma, "The Adaptive Gain Integrating Pixel Detector at the European XFEL", Journal of Synchrotron Radiation **26**, 74–82, ISSN: 1600-5775 (2019) 10.1107/s1600577518016077.

[82]S. Maffessanti, K. Hansen, S. Aschauer, A. Castoldi, F. Erdinger, C. Fiorini, P. Fischer, P. Kalavakuru, H. Klär, M. Manghisoni, C. Reckleben, L. Strüder, and M. Porro, "A 64k pixel CMOS-DEPFET module for the soft X-rays DSSC imager operating at MHz-frame rates", Scientific Reports **13**, ISSN: 2045-2322 (2023) 10.1038/s41598-023-38508-9.

[83]M. Sikorski, M. Ramilli, R. de Wijn, V. Hinger, A. Mozzanica, B. Schmitt, H. Han, R. Bean, J. Bielecki, G. Bortel, T. Dietze, G. Faigel, K. Kharitonov, C. Kim, J. C. P. Koliyadu, F. H. M. Koua, R. Letrun, L. M. Lopez, N. Reimers, A. Round, A. Sarma, T. Sato, M. Tegze, and M. Turcato, "First operation of the JUNGFRAU detector in 16-memory cell mode at European XFEL", Frontiers in Physics **11**, ISSN: 2296-424X (2023) 10.3389/fphy.2023.1303247.

[84]J. Xiao, J. Li, Z. Guo, H. Jiang, and Y. Wang, "High-repetition-rate photoelectron spectroscopic diagnostics for SHINE XFEL using time-of-flight spectrometers", Review of Scientific Instruments **96**, ISSN: 1089-7623 (2025) 10.1063/5.0267436.

[85]M. Diez, H. Kirchberg, A. Galler, S. Schulz, M. Biednov, C. Bömer, T.-K. Choi, A. Rodriguez-Fernandez, W. Gawelda, D. Khakhulin, K. Kubicek, F. Lima, F. Otte, P. Zalden, R. Coffee, M. Thorwart, and C. Bressler, "A sensitive high repetition rate arrival time monitor for X-ray free electron lasers", Nature Communications **14**, ISSN: 2041-1723 (2023) 10.1038/s41467-023-38143-y.

[86]S. Hauf, B. Heisen, S. Aplin, M. Beg, M. Bergemann, V. Bondar, D. Boukhelef, C. Danilevsky, W. Ehsan, S. Essenov, R. Fabbri, G. Flucke, D. Fulla Marsa, D. Göries, G. Giovanetti, D. Hickin, T. Jarosiewicz, E. Kamil, D. Khakhulin, A. Klimovskaia, T. Kluyver, Y. Kirienko, M. Kuhn, L. Maia, D. Mamchyk, V. Mariani, L. Mekinda, T. Michelat, A. Münnich, A. Padee, A. Parenti, H. Santos, A. Silenzi, M. Teichmann, K. Weger, J. Wiggins, K. Wrona, C. Xu, C. Youngman, J. Zhu, H. Fangohr, and S. Brockhauser, "The Karabo distributed control system", Journal of Synchrotron Radiation **26**, 1448–1461, ISSN: 1600-5775 (2019) 10.1107/s1600577519006696.

[87]J. Thayer, D. Damiani, C. Ford, M. Dubrovin, I. Gaponenko, C. P. O'Grady, W. Kroeger, J. Pines, T. J. Lane, A. Salnikov, D. Schneider, T. Tookey, M. Weaver, C. H. Yoon, and A. Perazzo, "Data systems for the Linac coherent light source", Advanced Structural and Chemical Imaging **3**, ISSN: 2198-0926 (2017) 10.1186/s40679-016-0037-7.

[88]Y. Joti, T. Kameshima, M. Yamaga, T. Sugimoto, K. Okada, T. Abe, Y. Furukawa, T. Ohata, R. Tanaka, T. Hatsui, and M. Yabashi, "Data acquisition system for X-ray free-electron laser experiments at SACLA", Journal of Synchrotron Radiation **22**, 571–576, ISSN: 1600-5775 (2015) 10.1107/s1600577515004506.

[89]M. Folk, G. Heber, Q. Koziol, E. Pourmal, and D. Robinson, "An overview of the HDF5 technology suite and its applications", in Proceedings of the EDBT/ICDT 2011 Workshop on Array Databases, EDBT/ICDT '11 (Mar. 2011), pp. 36–47, 10.1145/1966895.1966900.

[90]LSST Data Management, *Batch Resources*, (2025) https://developer.lsst.io/usdf/batch.html (visited on 09/15/2025).

Chapter V

Impulsive Stimulated X-ray Raman Scattering

In this chapter, the central part of the original work in this thesis will be detailed. Also its context in the surrounding literature and the novel aspect of the measurement will be described. The interaction process was discussed in sections III.4 and III.5. Here, the measurement procedures, analysis processes, data interpretation, and outlook will be described.

Impulsive stimulated x-ray Raman scattering (ISXRS) is a nonlinear light–matter interaction.[1] It involves an ultrashort x-ray pulse coupling a ground state to valence-excited states through a stimulated Raman interaction process. In contrast to spontaneous Raman scattering, where the inelastic signal arises from single-photon excitation, the stimulated process relies on the high photon flux of modern x-ray free-electron lasers to drive the transition.

In the case where the x-ray pulse duration is shorter than the characteristic timescales of electronic motion, the interaction is said to be in the impulsive limit. In this regime, the x-ray pulse can create a coherent superposition of valence-excited states (i.e. an electronic wavepacket) while avoiding populating real core excited states, thereby circumventing rapid Auger decay and associated loss channels.[V.1]

ISXRS retains one of the most important advantages of XAS compared to UV/Visible spectroscopy - elemental specificity, due to core-level resonances. It can therefore act as a local probe of the valence electron dynamics. In the time-domain, the impulsive nature of the process enables the study of ultrafast charge migration and correlated electron dynamics on the

[V.1] The stimulated Raman process may also be mediated by virtual core-excited states.

attosecond timescale.

To summarize, ISXRS offers a unique tool for preparing and measuring coherent dynamics of electron in the valence shell with high spatial[V.2] and temporal resolution.

ISXRS creates excitations in the valence shell by excitation. This is different to other schemes to measure charge migration which use ionization as an initial step.[2,3] This means that ISXRS enables the observation of electron dynamics in neutral systems. It opens pathways to explore fundamental processes such as the crossing of conical intersections[4], the evolution of vibronic coherences[5,6] and potential future measurement schemes such as transient redistribution of ultrafast electronic coherences in attosecond Raman signals (TRUECARS) and its derivatives.[7]

ISXRS is similar to the Stimulated X-ray Raman process discussed in section III.4. However, instead of using separate "pump" and "dump" pulses at different photon energies to drive population into valence-excited states[8,9], ISXRS uses a single pulse. The pulse used is an attosecond x-ray pulse generated by the electron bunch modulating XLEAP scheme at LCLS[10], discussed in section IV.2, sometimes known as enhanced SASE (eSASE). This pulse was measured to have spectral widths of approximately 6 eV full width at half-maximum at the photon energies used in the experiment, 520-540 eV.

The energy structure of water in all phases of matter has been well studied at the oxygen K-edge via many different XAS techniques.[12–17] It has also been studied using x-ray Raman scattering. T. Tokushima et al.[11] use non-resonant excitation to drive a 1s electron from the oxygen atom into the continuum. The x-rays were provided by the SPring-8 synchrotron facility and were tuned to 550 eV. The resulting core-hole can relax by spontaneous decay (flouresence). The radiative decay then gives the energy structure of the occupied states, from which an electron fills the core hole, emitting an x-ray via a spontaneous emission process.

Figure V.1 shows the result of spontaneous x-ray Raman scattering. The peaks in the x-ray emission spectra are due to electrons in the HOMO, HOMO-1 and HOMO-2 orbitals relaxing into the oxygen 1s state. This results in inelastic emission at 526 eV. The oxygen K-edge in liquid phase H_2O is at approximately 535 eV and a pre-edge feature is observed at approximately 533 eV. This pre-edge absorption corresponds to excitation into a state similar in energy to the $4a_1$ state in gas-phase H_2O.[19]

Therefore an ISXRS scheme can be driven in water by the following scheme. A "pump"

[V.2] In hetero-atomic systems, elemental specificity becomes atomic-site specificity, leading to the ability to create wavepackets with Angstrom precision.

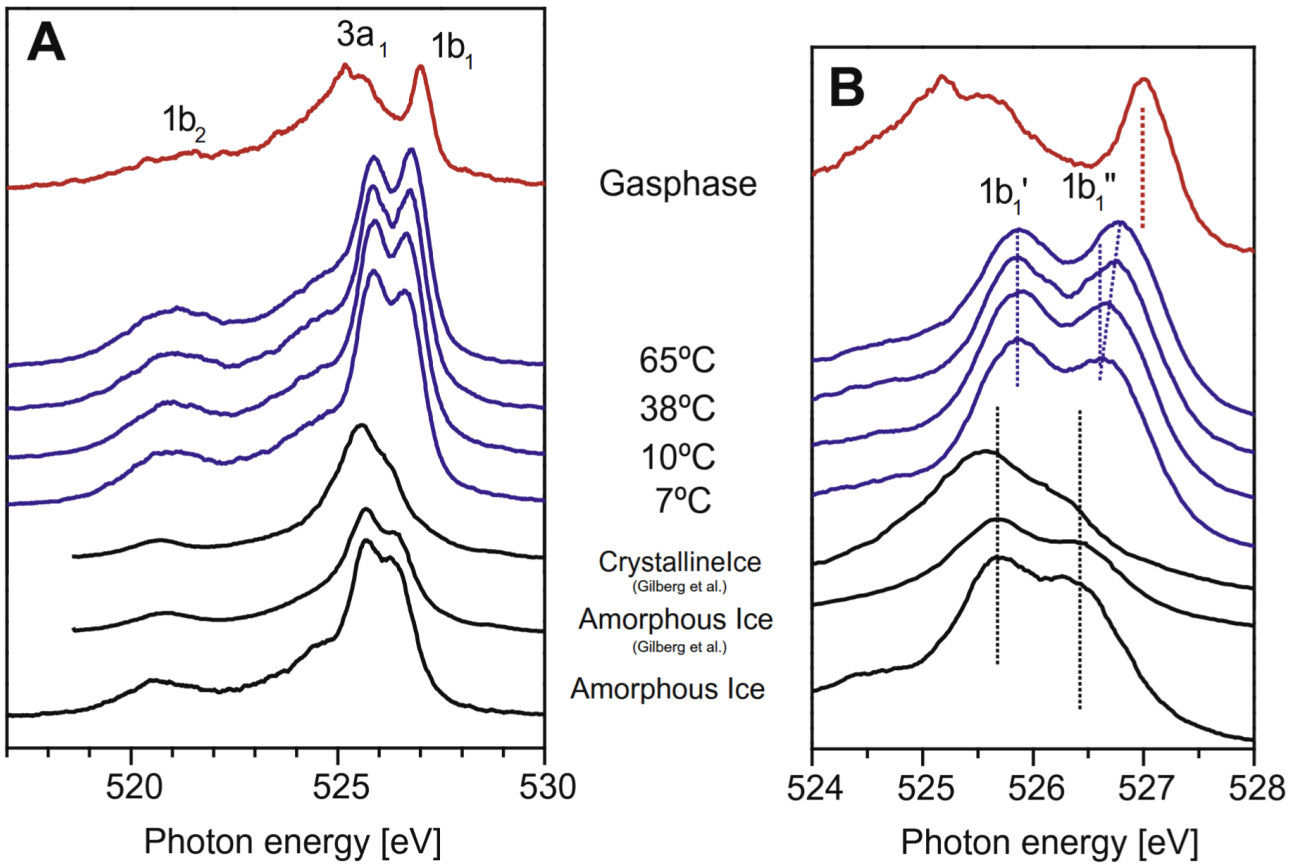

Figure V.1: X-ray emission spectra of the H_2O molecule under different conditions and in solid (black), liquid (blue) and gas (red) phase, from T. Tokushima et al., *Chemical Physics Letters* **460** (2008) 387–400.[11] Incident x-rays at 550 eV about 10-15 eV above the oxygen K-edge and therefore ionize core electron via a non-resonant interaction. The emitted photons come from electrons from the HOMO, HOMO-1 and HOMO-2 relaxing into the core-hole. (**a**) The emission spectra contain information about the occupied states after the x-ray absorption event. The labels $1b_2$, $3a_1$ and $1b_1$ are molecular orbitals, labeled by their symmetry. (**b**) A zoomed in view into region of the spectrum corresponding to $1b_1$ relation. A temperature dependent shift in the liquid phase from 526.5 $\rightarrow$ 527 eV is shown.

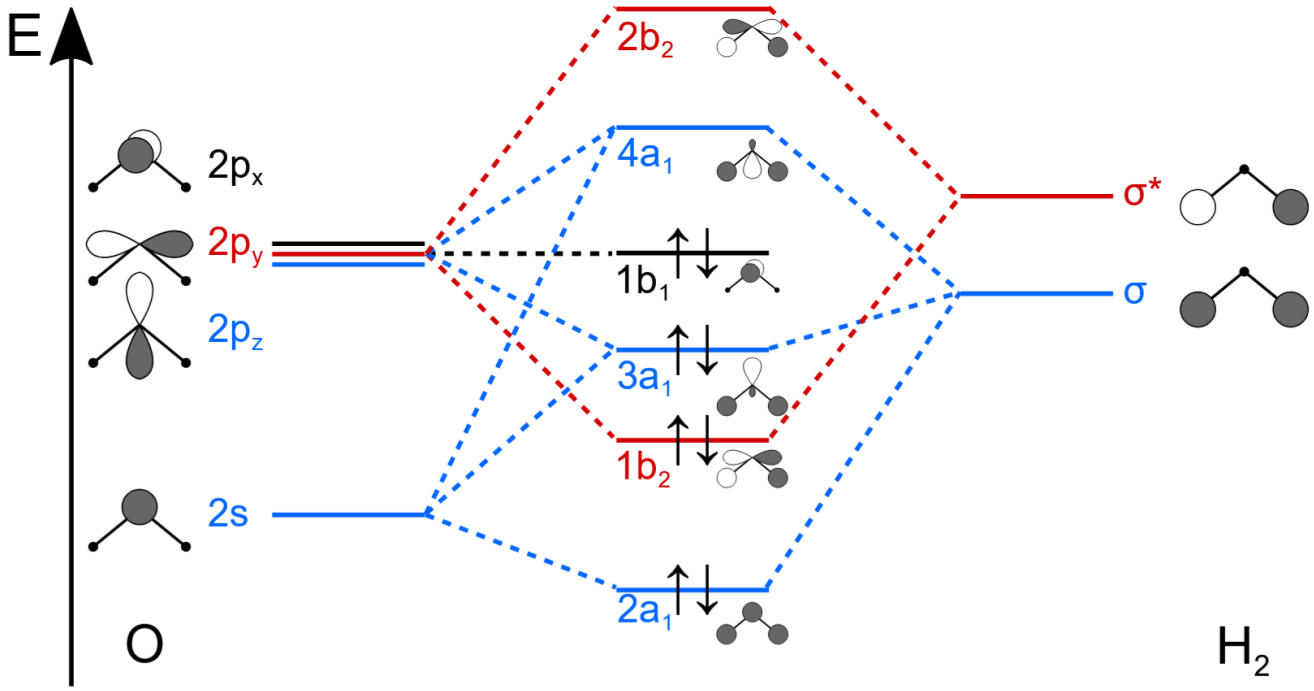

Figure V.2: A qualitative illustration of the group theory labeled energy levels of the water molecule, adapted from Chemistry LibreTexts Chem 110B: Physical Chemistry II.[18] The atomic orbitals are coming from the oxygen atom are shown on the left. The bonding σ and anti-bonding σ* orbitals from H_2 are shown on the right. Both contribute to the molecular orbitals shown in the center. These are illustrated according to their symmetry. The $1b_1$ orbital corresponds to the HOMO. The $4a_1$ orbital corresponds to the LUMO.

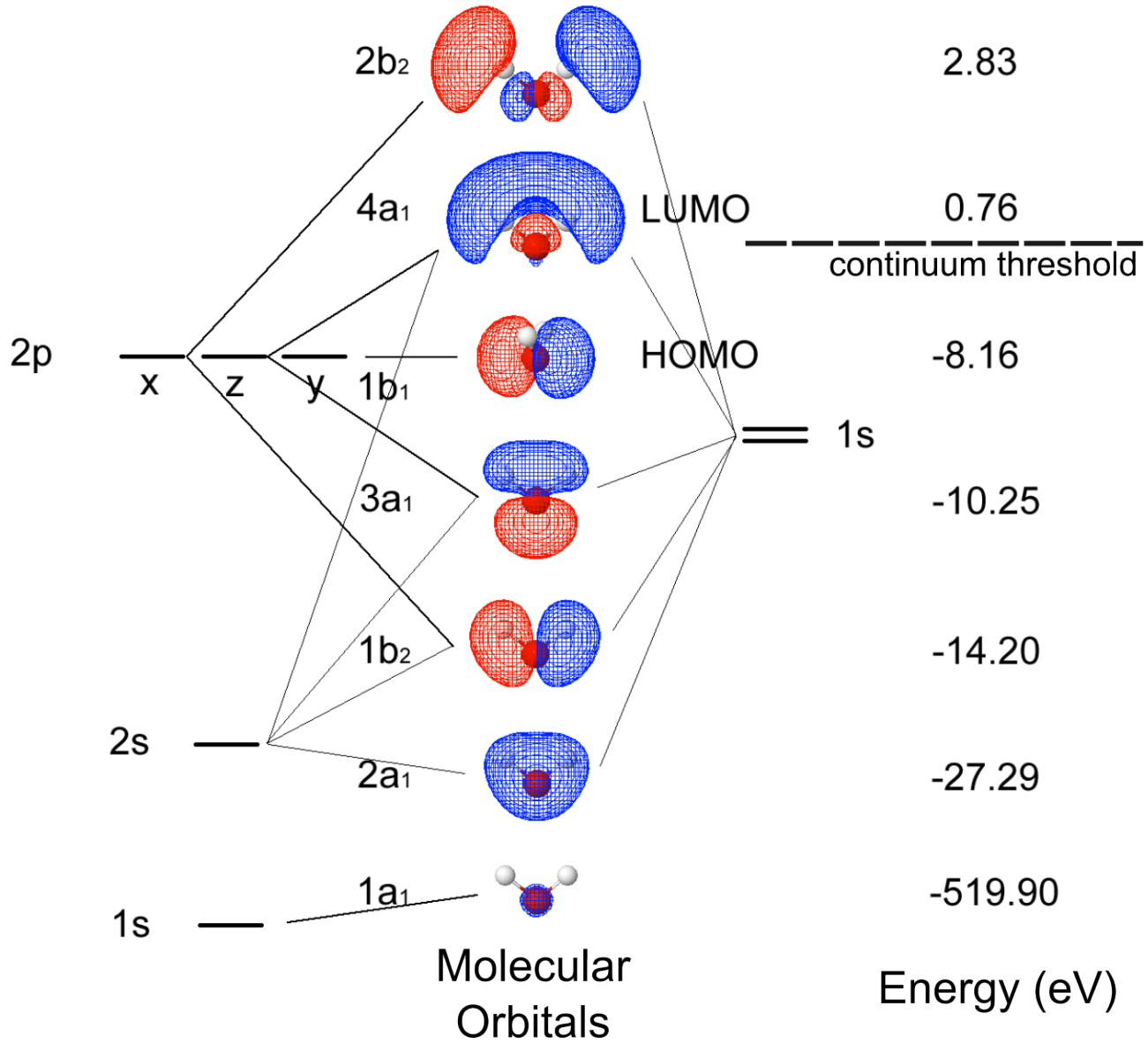

Figure V.3: A render of the seven lowest energy molecular orbitals in H_2O, adapted from the Wikimedia Commons Library.[20] The contribution of the oxygen atomic orbitals to the molecular orbitals is shown on the left side. The colors of the orbitals indicate the phase of the wavefunction. The labels a_1, b_1 and b_1 indicate the symmetry of the wavefunction. The energies relative to the continuum threshold are shown on the right.

transition at 533 eV takes the system into a core-excited state, by promoting an electron from a core state into an unoccupied molecular orbital. This core-excited configuration is illustrated in the central drawing in figure III.6. Then, a subsequent "Stokes" transition at 526 eV drives an electron from an occupied valence orbital into the 1s state, destroying the core-hole via stimulated emission. This valence-hole configuration is shown in the right side of figure III.6.

A single wide bandwidth XLEAP pulse of 6 eV coherent bandwidth can contain enough spectral intensity on both transition resonances to provide photons for both transitions. This spectral width enables the system to be driven to a superposition of valence-excited states. As the interaction takes advantage of a highly localised core-excited state as an intermediate state, the resulting electronic wavepacket is also highly localised immediately after its preparation by the ISXRS process.

The original work in this thesis was to demonstrate the ISXRS interaction in a condensed phase sample using the transmitted x-ray spectrum as an observable. Previous work demonstrated the ISXRS process in the gas-phase using cation yield spectroscopy. This will be now introduced.

V.1 Demonstration in isolated molecules

The Impulsive SXRS process has been demonstrated at LCLS, specifically at the TMO instrument. This work is highly relevant to and motivates the original work discussed in section V.2. **Disclaimer: The work that follows in this section was not conducted by me.** A target of gas-phase nitrogen oxide (NO) was used. In this experiment, attosecond x-ray pulses generated by the XLEAP mode at LCLS[10] were used. They were measured to have coherent bandwidths of 5.5 eV full width at half maximum. X-rays tuned to slightly below the oxygen K-edge at 533-540 eV were used to promote electrons from the $1s_O$ state to the $2\pi^*$ orbital.

Within the same pulse, the electrons are driven via stimulated emission into a superposition of valence-excited states. This corresponds to an electron leaving the 1π orbital, filling the core-vacancy in the core $1s_O$ orbital.

The molecular orbital occupancies for the ground state, intermediate core-excited state and valence-excited states are given in Table V.1. The orbitals that directly participate in the ISXRS process (i.e. lose and gain electrons) are highlighted in lime green.[V.3]

[V.3] In principle, an electron leaving any of the 3σ, $4\sigma^*$, $5\sigma^*$ and 1π orbitals would lead to a valence-excited state.

Table V.1: A table of the electron occupancies for the states in isolated NO molecules.[21] Note that in the ground state configuration, the $2\pi^*$ orbital is only partially filled. This vacancy is leveraged to create a core-excited state. The states within the bandwidth of the x-ray pulse are indicated in green.

Molecular Configuration	Electron Orbital Occupancies
Ground State	$(1s_O)^2(1s_N)^2(3\sigma)^2(4\sigma^*)^2(5\sigma)^2(1\pi)^4(2\pi^*)^1$
Core-Excited State	$(1s_O)^1(1s_N)^2(3\sigma)^2(4\sigma^*)^2(5\sigma)^2(1\pi)^4(2\pi^*)^2$
Valence-Excited State	$(1s_O)^2(1s_N)^2(3\sigma)^2(4\sigma^*)^2(5\sigma)^2(1\pi)^3(2\pi^*)^2$

The demonstration of ISXRS in isolated gas-phase molecules was done using a pump-probe scheme. Initially, an attosecond x-ray pulse is used to drive the ISXRS process, leaving the molecular system in a valence-exited state. A UV pulse delayed by 20 ns with respect to the x-ray pulse then ionizes the molecule.

The ions and electrons were recorded in a double-sided velocity map imaging spectrometer. This allowed the detected ion fragments to be resolved by their charge-to-mass ratio. These fragments are shown in figure V.4 subplot (**c**). The experimental observable was the yield of NO^+ cations.

In order to perform a measurement of the background count rate of NO^+ cations, the pump probe order was inverted. This background measurement scheme is depicted in the lower part of figure V.4, subplot (**b**). This inversion allows for the quantification of count rate due to ions from the x-ray pump and any ions that originate from the UV probe independent of the ISXRS process. Subtracting the background from the "correct" pump-probe ordered data allows for the calculation of a differential NO^+ cation-yield. The cross section of UV absorption in the molecule depends on the population of valence-excited states. Therefore, the differential NO^+ cation-yield encodes how many excess ions are measured due to UV ionization following the ISXRS process.

In figure V.5, the experimental data showing the differential NO^+ cation-yield is shown as a solid dark blue line. We see that when the central photon energy is tuned to approximately the $1s_O \rightarrow 2\pi^*$ resonance, there are more cations detected when the system is pumped with x-rays. This is the expected result of successfully driving population into valence-excited states. Note that this is no longer the case when the central photon energy of the XFEL is tuned significantly above this resonance. In this regime, the attosecond x-ray pulse likely does not have sufficient bandwidth to drive the Stokes transition, resulting in less UV absorption.

The 1π pathway was found to be dominant.

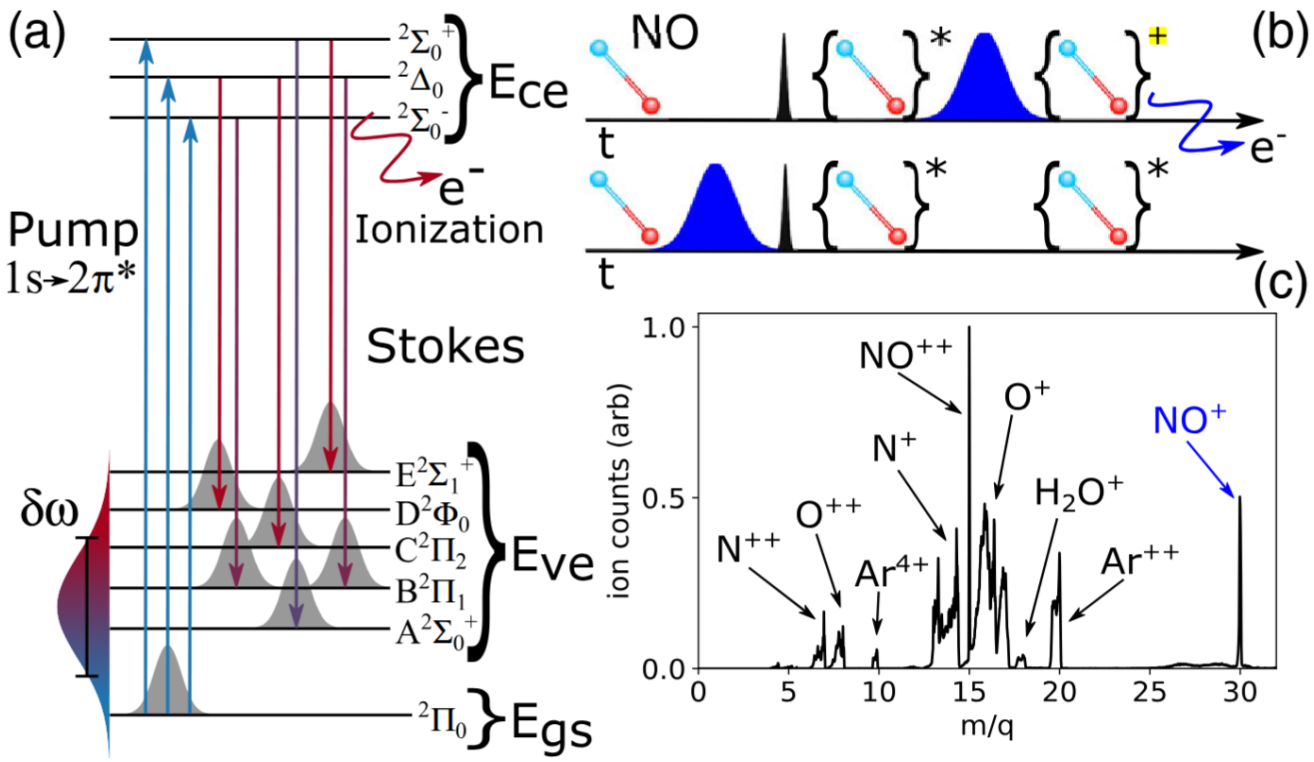

Figure V.4: The experimental x-ray pump, UV probe scheme used to demonstrate the ISXRS process in gas-phase at LCLS, from O'Neal, J. et al. (2020) *Physical Review Letters*, **125**(7), 073203.[22] (**a**) A wide bandwidth XLEAP attosecond x-ray pulse is tuned to be simultaneously resonant with the pump (blue) and Stokes (red) transition resonances in NO. As a result, population is driven into a coherent superposition of valence-excited states. (**b**) An x-ray pump drives the system into an excited state via the ISXRS process. A UV pump then subsequently ionizes the molecule. (**c**) Ion spectrometry is performed on the products of the reaction. The NO^+ cation yield is indicated in blue.

This method of demonstrating ISXRS relies on the detection of a specific ion, identified by and selected due to its charge-to-mass ratio. There are difficulties in attempting an analogous measurement in condensed-phase systems. This can be due to difficulties in sample damage due to the high power density of attosecond x-ray pulses from an XFEL (up to 1×10^{18} W cm^{-2}). Such damage can be caused by the energy from the x-ray being thermally dissipating to the sample, causing chemical modification. This will result in the probe potentially being absorbed by a species that may not be the intended sample, valence-excited NO. Additionally, difficulties in measurement exist, as ions generated in the bulk (interior) of a condensed-phase sample may interact non-elastically with other sample molecules before detection.[23,24] This can cause changes to the detected mass spectrum.[25,26]

For this reason, in order to demonstrate ISXRS in a condensed-phase sample, we are motivated to use spectral modifications to the incident x-ray field as out observable. This is as

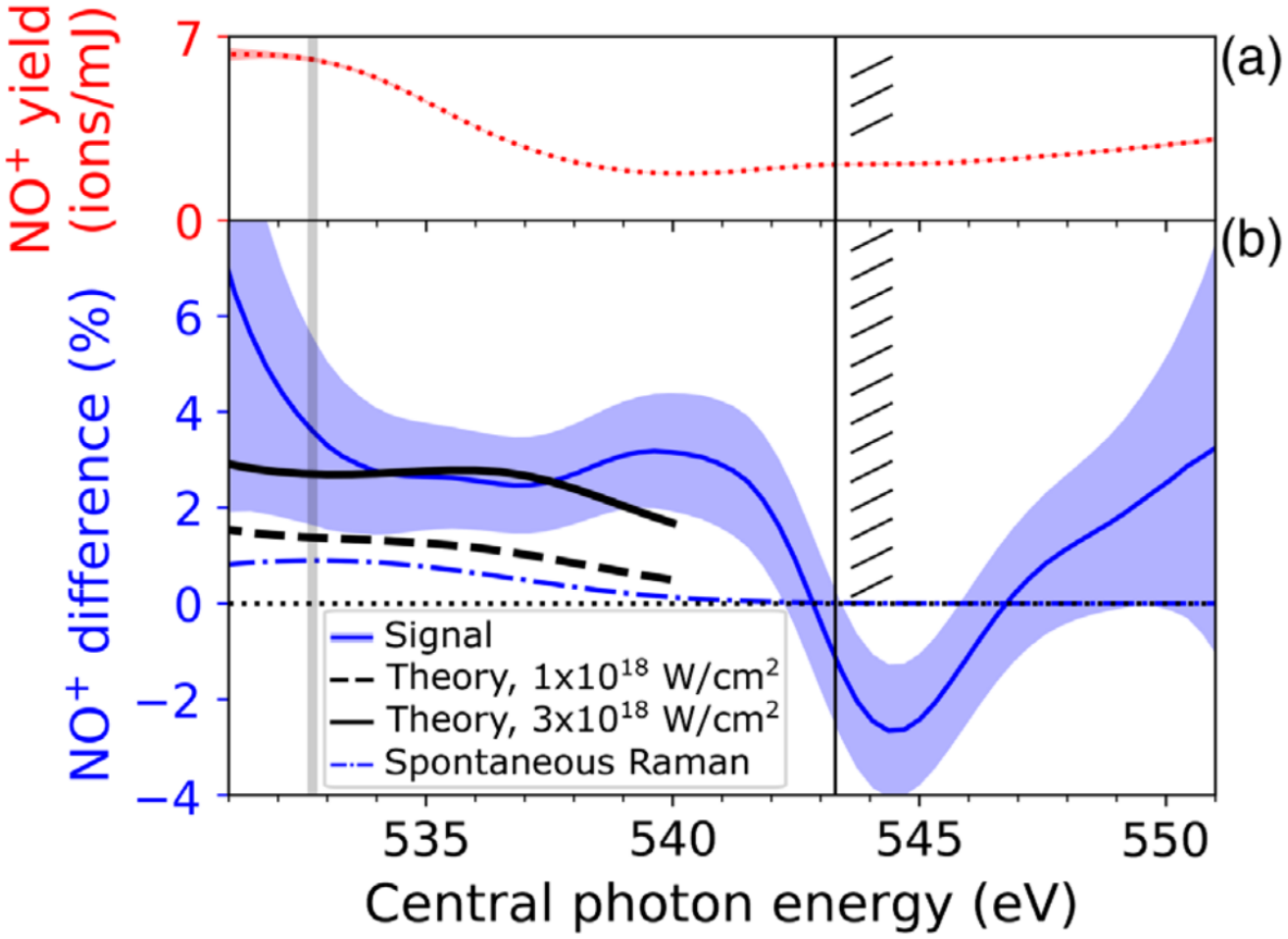

Figure V.5: Experimental data supporting the claim of demonstrated ISXRS in isolated NO molecules looking at the ion yield following UV absorption, from O'Neal, J. et al. (2020) *Physical Review Letters*, **125**(7), 073203.[22] (**a**) Partial ion yield of NO^+ cations, measured as a function of central photon energy of the XFEL. The location of the $1s_O \rightarrow 2\pi^*$ pre-edge resonance feature is indicated by a vertical grey line at 532.6 eV. (**b**) Differential NO^+ cation-yield also as a function of the XFEL central photon energy. The collected data is smoothed by a Gaussian kernel of 1.5 eV width and shown in solid blue. A $\pm 1\sigma$ uncertainty level is shaded in light blue. The oxygen K-edge is shown by a dark grey dashed vertical line at approximately 543 eV. Simulations of the ion yield due to spontaneous decay and ISXRS for the intensities shown are indicted by blue dashed and black lines respectively.

the complications arising from the Coulomb interaction when conducting electron and ion spectroscopy do not apply to the x-ray field.

V.2 Experimental scheme

The experiment to demonstrate ISXRS in the condensed phase was performed at the chemRIXS endstation at LCLS. chemRIXS is an endstation designed for experiments using soft x-rays in condensed-phase targets.[27–29]

Attosecond x-ray pulses generated by the XFEL in the XLEAP mode were focused by a pair of two adjustable Kirkpatrick–Baez mirrors and impinged onto a flat sheet liquid jet.[30] The transmitted spectrum through the jet was then spectrally resolved by a diffraction grating and subsequently imaged on an ANDOR camera via a charge-coupled device (labeled CCD in figure V.6).

ISXRS is two-photon process as illustrated by figure III.8. This means the cross-section for ISXRS has a nonlinear dependence on the intensity of the field.[V.4] Therefore, a high intensity was essential for the measurement.

Additionally, the intensity-dependence of the ISXRS process allows for a control measurement to be made. If the field intensity at the interaction region can be somehow modified, single-photon effects may be isolated from two-photon effects. This may be achieved at the chemRIXS endstation via the deformation of the Kirkpatrick–Baez mirrors. In this way, the focal point of the beam can be translated at will downstream of the interaction point. In this experiment, to quantify the background linear effects (forming a control measurement), the focus was moved 10 cm downstream of the interaction region. This resulted in the estimated lowering of the field intensity by a factor of 13.

The experiment was conducted by taking measurements of the transmitted x-ray spectrum. This was done via a VLS spectrometer. Examples of the form of the recorded spectra are shown in figure IV.7. Data was collected for four separate conditions in the interaction point, shown in figure V.7. Recording $I_{0,\,\mathrm{defoc}}$ and $I_{T,\,\mathrm{defoc}}$ allows for a calculation of the optical density of the sample in the low intensity condition as described in equation (III.1). This will contain both single photon effects (such as core to valence promotion, valence ionization) and higher order

[V.4] This can be reasoned from the result for the interaction Hamiltonian and transition matrix element in second-order perturbation theory shown in equations (III.6) and (III.8) respectively. The field intensity I obeys $I \propto \mathbf{A}\cdot\mathbf{A}$. Therefore, the probability per unit time of a two photon process $\mathcal{P}^{(2)}$ satisfies $\mathcal{P}^{(2)} \propto |\mathcal{M}_{fi}^{(2)}|^2 \propto I^2$.

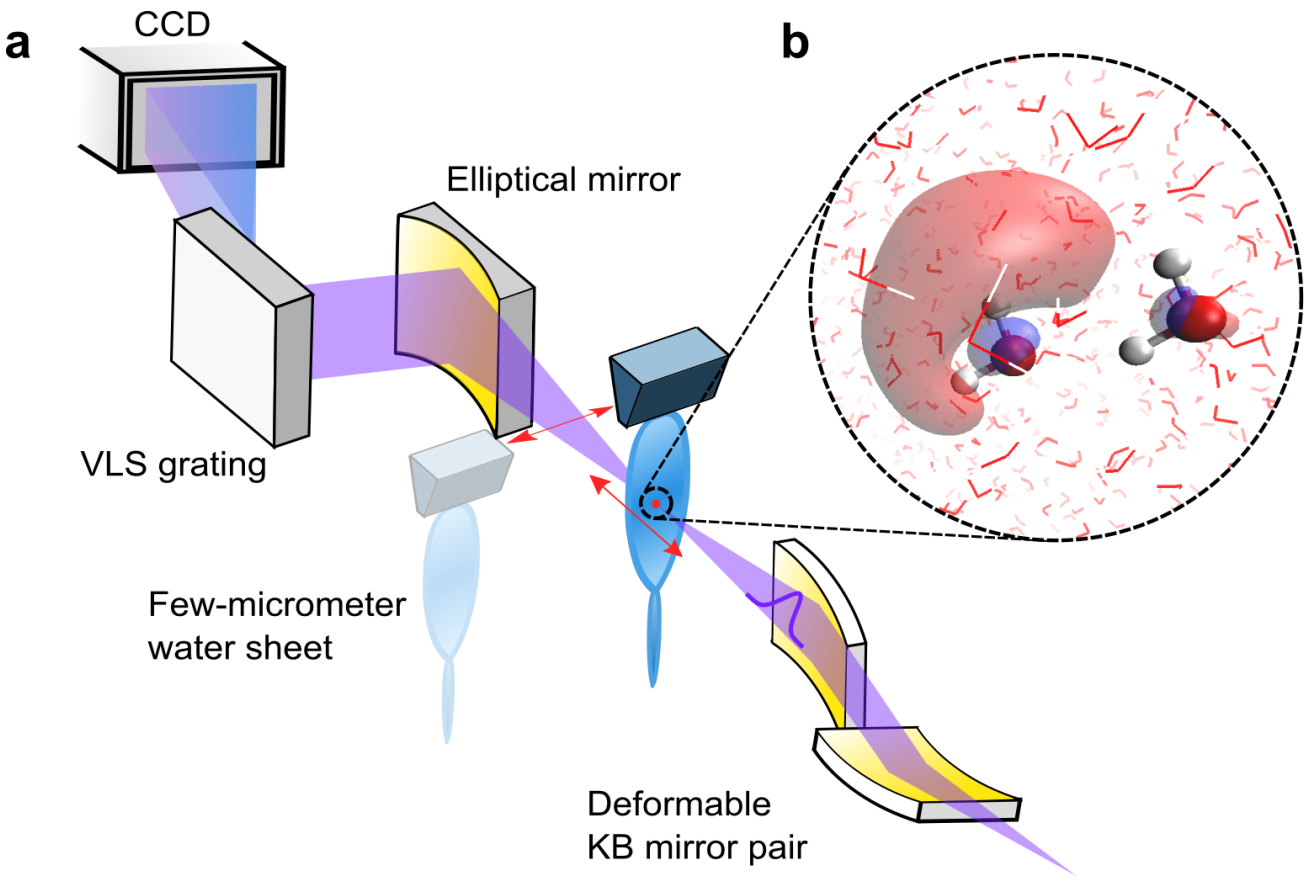

Figure V.6: An illustration of the experimental configuration at the endstation at the chemRIXS instrument at LCLS, adapted from Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] The attosecond x-ray beam coming from the XFEL starts in the bottom-right. The beam is then guided by a pair of adjustable Kirkpatrick–Baez mirrors. These mirrors can be spatially varied to change the location of the focus of the beam, as indicated by the red arrows. The x-rays impinge on a liquid flat-sheet jet.[30] The scattered photons are spectrally resolved by a diffraction grating (variable-line spacing type) and detected onto a CCD camera.

effects (such as elastic scattering, inelastic scattering and two-photon absorption).

In order to isolate the higher order effects (one of which is the ISXRS interaction) two further measurements are taken. These are labeled in figure V.7 as $I_{0,foc}$ and $I_{T,foc}$ and correspond to measurements taken when the beam is focused at the interaction point. These are shown in figure V.8. A second optical density calculation can be performed for this configuration.

Note how a calculation for the optical density takes into account and normalizes to the incident field flux I_0. See equation (III.1). Therefore in principle, single-photon processes should show the same contribution to the measured optical density in both cases. In contrast, the higher order multiphoton effects are expected to contribute more to the optical density in the higher intensity (focused) case. In comparing the optical densities in and out of focus, the

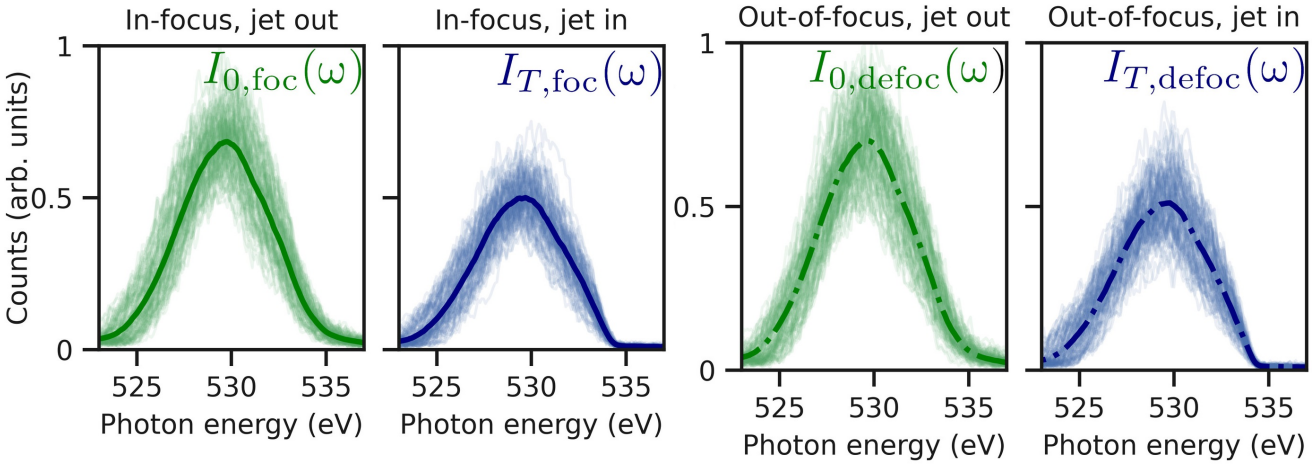

Figure V.7: Data from each of the four experimental conditions used to allow the isolation of and hence demonstrate the ISXRS process, adapted from Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] The data shown is the x-ray spectrum recorded on the CCD detector. In each case, many single-shot spectra are shown with increased transparency. The mean spectrum is drawn with solid / dash-dotted lines. Here, $\mathrm{I}(\omega)$ represents an x-ray spectrum. The subscript label $0/\mathrm{T}$ indicates whether the spectrum is the incident spectrum or the transmitted spectrum respectively. The "foc" / "defoc" label indicates the position of the focus is at the interaction point (focused or "foc") or 10 cm downstream (defocused or "defoc") respectively.

multiphoton effects may be distinguished.

To formalize this, we introduce the nonlinear emission $\mathrm{B}(\omega)$. It is defined as

$$\mathrm{B}(\omega) = \log_{10}\left(\frac{\mathrm{I}_{0,\,\mathrm{defoc}}(\omega)}{\mathrm{I}_{\mathrm{T},\,\mathrm{defoc}}(\omega)}\right) - \log_{10}\left(\frac{\mathrm{I}_{0,\,\mathrm{foc}}(\omega)}{\mathrm{I}_{\mathrm{T},\,\mathrm{foc}}(\omega)}\right) = \mathrm{OD}_{\text{out of focus}} - \mathrm{OD}_{\text{in focus}}, \tag{V.1}$$

where the traditional definition of the optical density has been used.[32] $\mathrm{B}(\omega)$ is a useful spectroscopic metric to identify intensity-dependent effects. The ISXRS is a two photon scattering process, which absorbs photons at the pump resonance and re-emits them via stimulated emission at the Stokes resonance. This means that the field mode corresponding to forward scattered photons at the Stokes resonance should be enchanced in the transmitted x-ray spectrum. This enhancement grows with intensity. Therefore, due to the ISXRS process, at the Stokes transition resonance ω_{Stokes} we might expect

$$\begin{aligned} \mathrm{OD}_{\text{out of focus}}(\omega_{\mathrm{Stokes}}) &> \mathrm{OD}_{\text{in focus}}(\omega_{\mathrm{Stokes}}) \\ \mathrm{B}(\omega_{\mathrm{Stokes}}) &> 0. \end{aligned} \tag{V.2}$$

This is because an enhancement of the x-ray spectrum at the Stokes transition resonance results

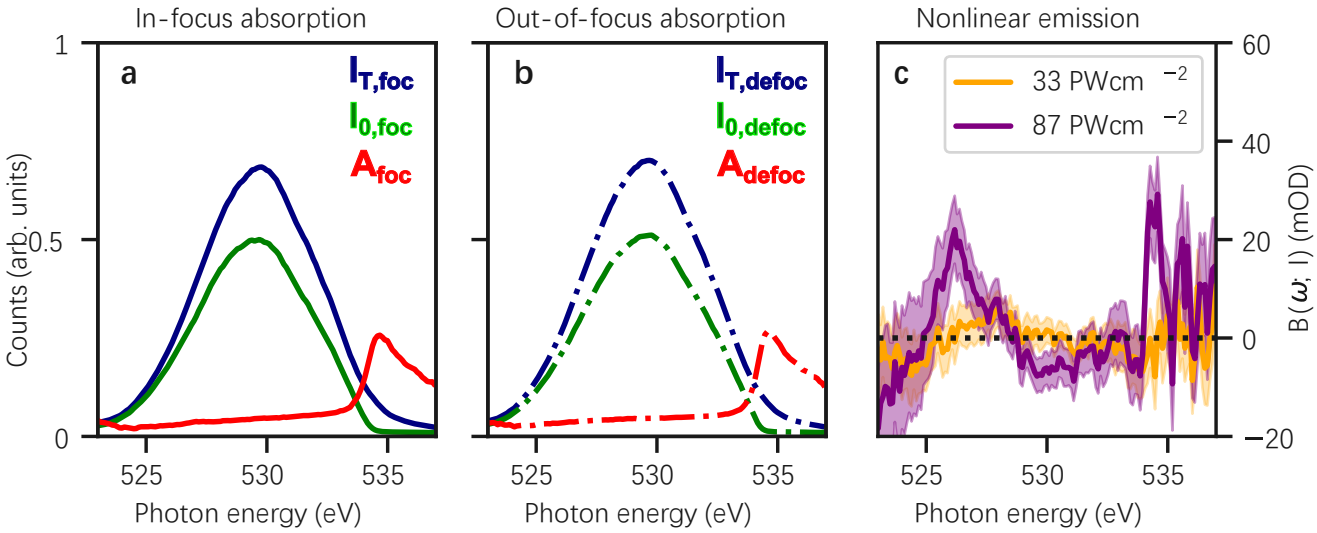

Figure V.8: The optical densities and how they related to the recorded x-ray spectra taken near the oxygen pre-edge, adapted from Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] Here, the optical densities are written with the notation A_{foc} and A_{defoc} displayed for both the focus at interaction point and defocused 10 cm downstream cases, and correspond directly to the terms in equation (V.1). The recorded spectra are colour coded in the same way as figure V.7. (**a**) Spectra collected in the focused case. (**b**) Spectra collected in the out of focus case. (**c**) The nonlinear emission as a function of photon energy for two different intensities. The shaded regions indicate the $\pm 1\sigma$ uncertainty level, found via bootstrap resampling.

in an increase in the transmitted light in that part of the spectrum. This therefore translates into a reduction in the measured optical density at that photon energy.

Positive values of nonlinear emission $B(\omega)$ can be interpreted as a decrease in optical density when the focal point is located in the interaction region. Equivalently, these positive values indicate more photons being emitted from the sample in the higher intensity case compared to a lower intensity case. Therefore, a measurement of $B(\omega_{Stokes}) > 0$ located at the Stokes transition resonance is the expected signature of the ISXRS process.

V.3 Analysis techniques

In addition to varying the focal position of the beam in to and out of the interaction region, a secondary method of analyzing the effect of the field intensity was performed. This method takes advantage of the stochastic nature of the XFEL lasing process discussed in sections IV.1 and IV.2 which results in shot-to-shot fluctuations in properties of the x-ray pulses, such as the pulse energy.

Two krypton-filled gas detectors (GD) were employed to measure the total energy of each pulse, or the incident number of photons.[33,34] These are referred to as gas detector 1 (GD1) and gas detector 2 (GD2). Each detector provides two measurement channels: the average intensity and the root-mean-square electron sum measured by the electrodes. This was discussed in section IV.3.

The average intensity channel constitutes a slow measurement with a time constant of approximately 25 s, whereas the root-mean-square channel is capable of providing single-shot measurement after calibration against the average intensity output. Following calibration, the mean values obtained from the RMS electron sum and the average intensity are highly consistent. The deviation between the two averages is less than 1.5% for both detectors. The measured correlation of the gas detector GD1 versus the x-ray counts yielded from the transmitted spectrum is shown in figure V.9 subplot (**b**).

Another shot-to-shot pulse energy measurement is provided by the fluorescence intensity monitors (FIMS).[35] These are non-invasive intensity monitors that are also capable of performing an intensity measurement for each shot, recording at 120 Hz. They comprise of a photodiode imaging the secondary reflection of the Kirkpatrick-Baez mirrors. See figure V.6. The output of the FIMS on each shot is a waveform corresponding to a photodiode response.

Data was recorded in the four experimental conditions that were discussed in section V.2. In the left-most and center right plots of figure V.7, data was recorded with the jet translated out of the beam. Hence, these are measurements of the incident x-ray spectrum. A polynomial model was used to translate the output of the FIMS into the recorded incident x-ray spectrum. The results of the fit are shown in figure V.9, subplots (**a**) and (**c**). As the FIMS are located upstream of the interaction point, the FIMS can provide a measure of the incident photon flux to the jet, even when the transmitted x-ray beam appearing on the CCD detector has undergone spectral modification due to interaction with the sample.

We can see that the FIMS prediction of the flux of the incident x-ray spectrum is superior to the prediction provided by the GMD. This is evident in the correlation maps in figure V.9 subplots (**a**) and (**b**). The Pearson correlation coefficient for these 2D-distributions are 0.901 and 0.99 respectively. We also see that in V.9 subplot (**c**) the FIMS prediction is quantitatively more accurate (closer to unity) over a wide range of x-ray counts.

The calculation of nonlinear emission as specified in equation V.1 requires two comparisons of two recorded x-ray spectra. When calculating the relevant optical densities, any minor

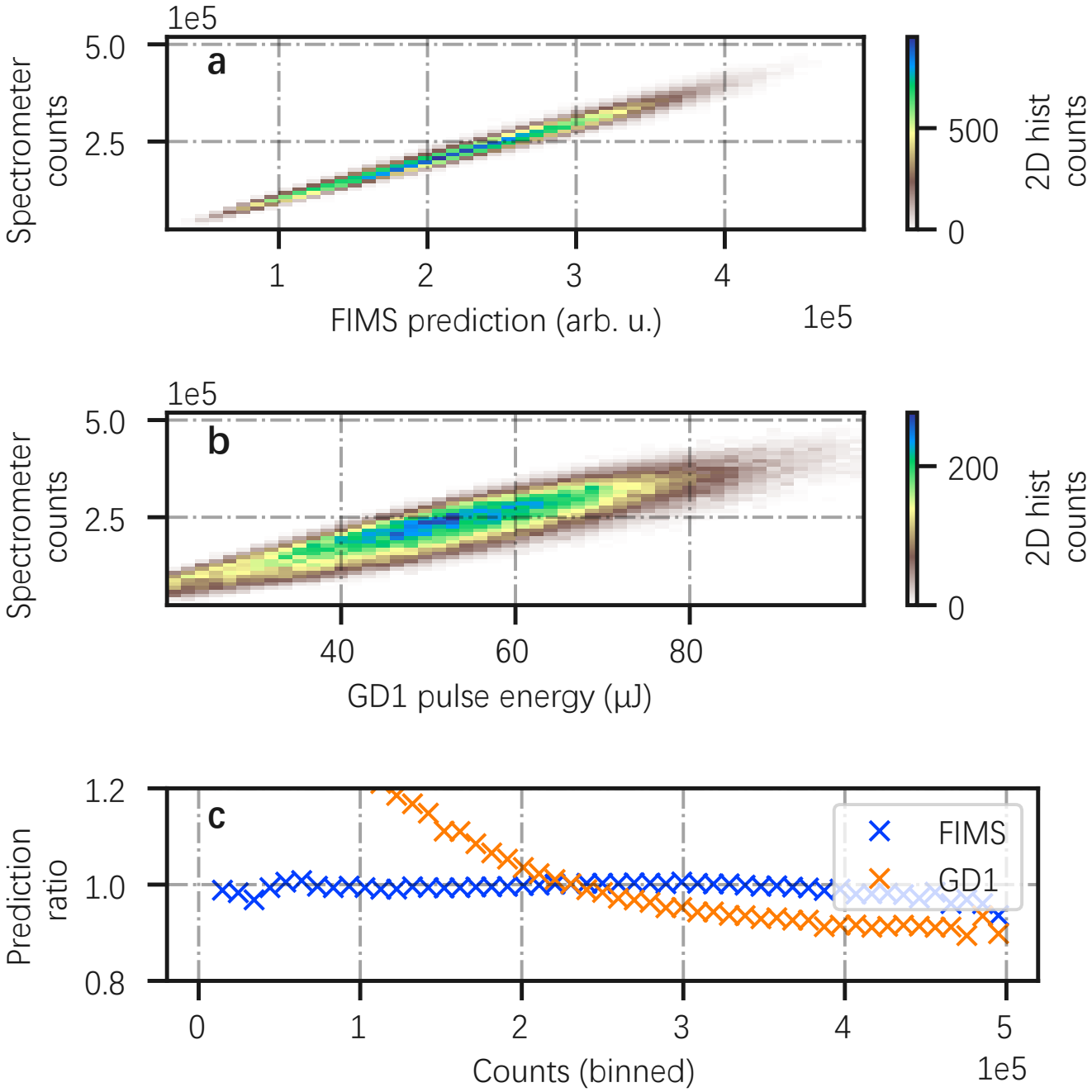

Figure V.9: A comparison of the measurements for the incident photon flux performed during the experiment, adapted from the supplementary material of Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] (**a**) Statistical 2D-correlation plot of the polynomial model predicted incident flux versus the measured x-ray yield on the CCD detector, via the VLS spectrometer. (**b**) An analogous correlation plot but for the calibrated output of the gas detector GD1. (**c**) The ratio of the anticipated counts from the FIMS (blue) and GD1 (orange) divided by the actual counts, as a function of x-ray counts. A fully ideal measure of the incident counts will show a predicted ratio of 1, for all x-ray yields.

change in a spectrum can cause an apparent feature in the resulting calculation for $B(\omega)$. Such a minor change can result from an effect resulting from a field interaction with the sample.

However, unfortunately, the XFEL is known to exhibit drifts. Drifts are changes XFEL photon beam, often caused by instabilities in the x-ray generating mechanism, magnet alignment variation, temperature fluctuations and other environment changes.[36,37] These changes do not always correlate to measured parameters of the machine that may be used to correct for the effect of drifts. Consider comparing a spectrum I_A taken at time t_A to a spectrum I_B taken at time t_B under what is thought to be identical conditions.

Suppose between times t_A and t_B a significant drift in the XFEL occurs. If an optical density $\log_{10}(I_A/I_B)$ is calculated, it may not show zero. Rather, it may show apparent "features". Now suppose the spectra I_A and I_B are taken without and with a sample respectively. An optical density may show real features due to sample interactions in addition to apparent features due to drifts. If the drift is not carefully accounted for, it is easy to misassign physical for peaks that are in fact acting as false-positives.

One way to counter this is to only compare spectra with similar properties. One property of an x-ray pulse spectrum is its center of mass. The concept of a moment for a non-normalized distribution $f(x)$ in statistics is defined as

$$\langle (x - x_0)^N \rangle = \frac{\int dx' \, (x' - x_0)^N f(x')}{\int dx' \, f(x')} \longrightarrow \frac{\sum_k (x_k - x_0)^N f(x_k)}{\sum_k f(x_k)}, \tag{V.3}$$

where x_0 is zero for standard moments and $\langle x \rangle$ for central moments.[38] The discretized version is given on the right. See figure IV.7 subplot (**c**) for single-shot x-ray spectra. The first standard moment is the center of mass of the distribution given by

$$x_{COM} = \langle x \rangle = \frac{\int dx' \, x' f(x')}{\int dx' \, f(x')} \longrightarrow \frac{\sum_k x_k f(x_k)}{\sum_k f(x_k)}, \tag{V.4}$$

and is used to quantity variations in the central photon energy of the x-ray pulse.

The second central moment is the variance of the distribution. It is found by the expression

$$\begin{aligned} \sigma^2 = \langle (x - \langle x \rangle)^2 \rangle = & \frac{\int dx' \, (x' - \langle x \rangle)^2 f(x')}{\int dx' \, f(x')} \\ \longrightarrow \quad & \frac{\sum_k (x_k - \langle x \rangle)^2 f(x_k)}{\sum_k f(x_k)}, \end{aligned} \tag{V.5}$$

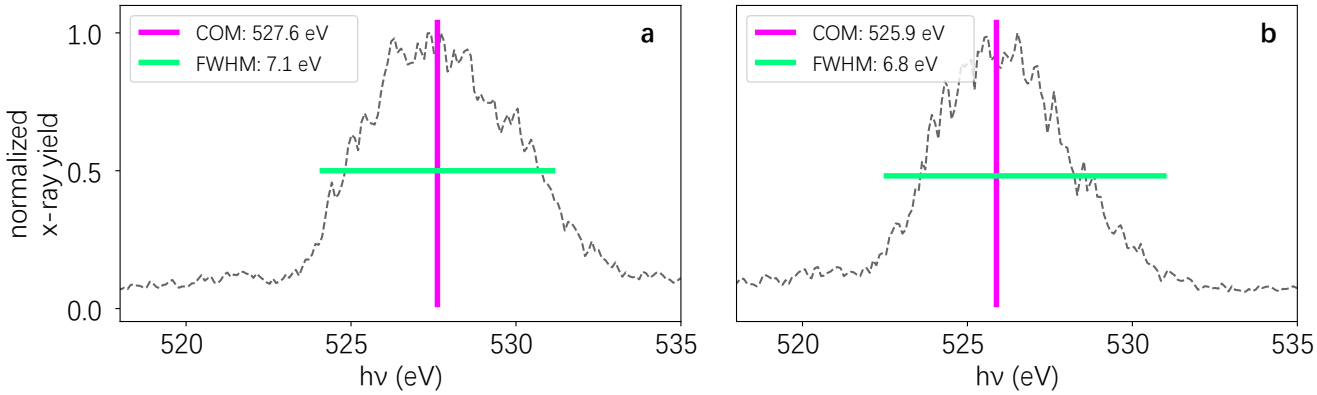

Figure V.10: Two examples of the incident x-ray spectrum. Displayed are two randomly selected normalized single shot spectra as grey dashed lines in subplots (**a**) and (**b**). The first standard moment measures the center of mass of the spectrum, This is a useful measure of the central photon energy, shown in bright pink. The full width at half maximum calculated via the second central moment characterizes the spectral width and is shown in bright green. The values for both spectral parameters are show in the upper right of both subplots (**a**) and (**b**).

where x_0 has been replaced by $\langle x \rangle$ coming from equation (V.3). Again, the discretized version of the calculation is given in the lower line. This parameter is used to calculate and quantify variations in the x-ray pulse bandwidth. We now assume the almost transform-limited XLEAP pulses are approximately Gaussian distributions with noise. See the spectral shape in figure V.10 for justification.[V.5] Given this, the variance σ^2 can be related to the full-width at half maximum bandwidth x_{FWHM} by

$$x_{FWHM} = 2\sigma\sqrt{2\ln 2} = \sigma \times 2.3548... \tag{V.6}$$

A value for x_{COM} and x_{FWHM} can be assigned to each recorded x-ray spectrum. The results of assigning values for x_{COM} and x_{FWHM} to two randomly chosen single-shot x-ray spectra are shown in figure V.10. The result of grouping together and averaging x-ray spectra with common center of masses and FWHM spectral widths is shown in figure V.11.[V.6]

The spectral width characterized by the FWHM shown in subplot (**c**) in figure V.11 can be used to infer further information about the XLEAP pulses. Consider a Gaussian pulse in the time domain with a root-mean-square time width σ_t. Its intensity profile in the time domain

[V.5] This is as fully transform-limited pulses must be Gaussian in both the spectral and temporal domains. Technically, one could consider a more complex function such as $H_n(x)\exp(-x^2)$ where H_n is a Hermite polynomial. However, the additional degree of freedom added by H_n is not justified given the spectral form shown in figure V.10.

[V.6] The procedure of grouping together and averaging x-ray spectra with common properties such as central photon energy, spectral width and pulse energy is referred to as binning.

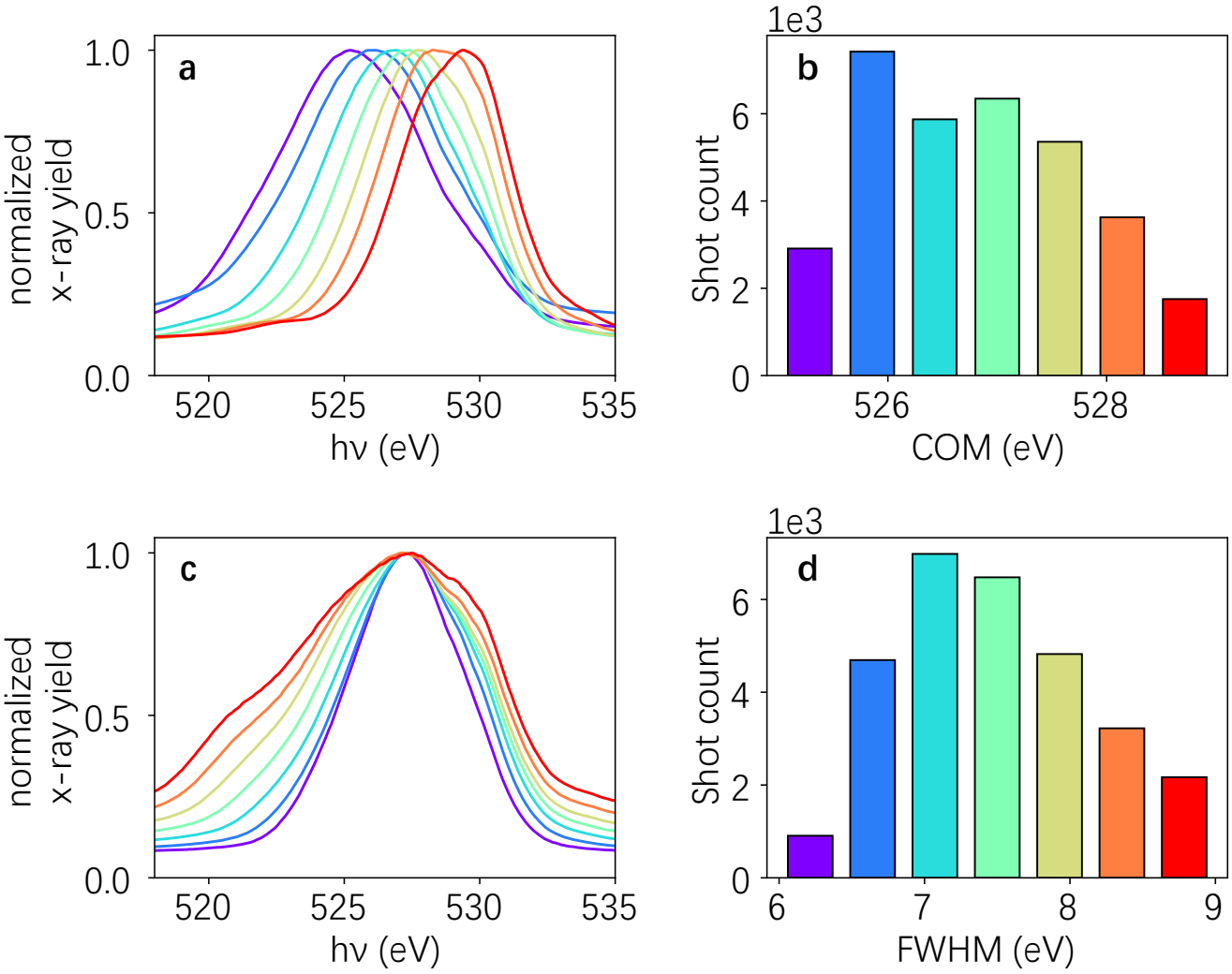

Figure V.11: Averages of the transmitted x-ray spectrum resolved by the VLS spectrometer binned by different parameters. This data was acquired without the liquid jet running. Therefore, the data shown is representative of the incident x-ray beam $I_0(\omega)$. (**a**) Mean spectra, binned by the calculated center of mass. (**b**) Number of shots per bin for the spectra shown in (**a**). (**c**) Mean spectra binned by spectra width, characterized by their FWHM. (**d**) Similarly to (**b**), statistics for the plot shown in (**c**). Subplots (**a**) and (**c**) are normalized to their respective maxima for presentation. In reality, the total yield for different central photon energies will be different.

will have the form

$$I(t) \propto \exp\left(-\frac{t^2}{2\sigma_t^2}\right). \tag{V.7}$$

Applying the Fourier transform (here denoted by $\mathcal{F}$) will result in a Gaussian form and ignoring normalization factors we find

$$\mathcal{F}\{I(t)\} \equiv \widetilde{I}(\omega) \propto \sigma_t \exp\left(-2\pi^2\sigma_t^2\omega^2\right). \tag{V.8}$$

Therefore the spectral domain intensity profile $\widetilde{I}(\omega)$ is also a Gaussian.[39] It has spectral-domain root-mean-square width

$$\sigma_\omega = \frac{1}{2\pi\sigma_t} \implies \sigma_t\sigma_\omega = \frac{1}{2\pi}. \tag{V.9}$$

From this, the time-frequency product is found between the root mean square widths. We convert to the more easily visualized FWHM in both time and spectral domains via the Gaussian factor

$$\Delta t_{FWHM} = 2\sqrt{2\ln 2}\sigma_t \quad \text{and} \quad \Delta\omega_{FWHM} = 2\sqrt{2\ln 2}\sigma_\omega. \tag{V.10}$$

The time-frequency product now becomes

$$\Delta t_{FWHM} \times \Delta\omega_{FWHM} = \left(2\sqrt{2\ln 2}\sigma_t\right) \times \left(2\sqrt{2\ln 2}\sigma_\omega\right) = \frac{4}{\pi}\ln 2 = 0.44127... \quad [s \cdot Hz] \tag{V.11}$$

in units of second-Hertz. As we will record frequencies as energies in eV and times in femtoseconds a unit conversion gives

$$\Delta t_{FWHM} \times \Delta E_{FWHM} = \Delta t_{FWHM} \times \hbar\Delta\omega_{FWHM} = \frac{4\hbar\ln 2}{\pi} = 1.8238... \quad [fs \cdot eV]. \tag{V.12}$$

From previous measurements of XLEAP, the x-ray pulses are known to be 1.5 times the transform limit. Therefore, having characterized the spectral width on each shot, an estimation of the pulse duration may be obtained. Using this reasoning, we find that the pulse durations during this experiment varied mostly between 300-400 as.

A measurement of the beam spatial profile was performed by a knife-edge measurement. The beam waist at the focal point was found to be $x_{waist} = (10.17 \pm 0.44)\mu m \times y_{waist} = (9.16 \pm 0.27)\ \mu m$ in the x and y directions. See figure IV.2 for the definition of the x-y orientation. The

time-dependent intensity profile of the beam I_0 may be written as

$$I(x, y, t) = I_0 \exp\left[-4 \ln 2 \left(\frac{t}{\Delta t_{FWHM}}\right)^2\right] \times \exp\left[-4 \ln 2 \left\{\left(\frac{x}{x_{waist}}\right)^2 + \left(\frac{y}{y_{waist}}\right)^2\right\}\right]. \quad (V.13)$$

This can be related to the total pulse energy E by the expression

$$E = \int dx\, dy\, dt\, I(x, y, t). \quad (V.14)$$

This leads to the result for the peak intensity I_0

$$I_0 = E \left(\frac{4 \ln 2}{\pi}\right)^{\frac{3}{2}} \times \frac{1}{x_{waist} y_{waist} \Delta t_{FWHM}}. \quad (V.15)$$

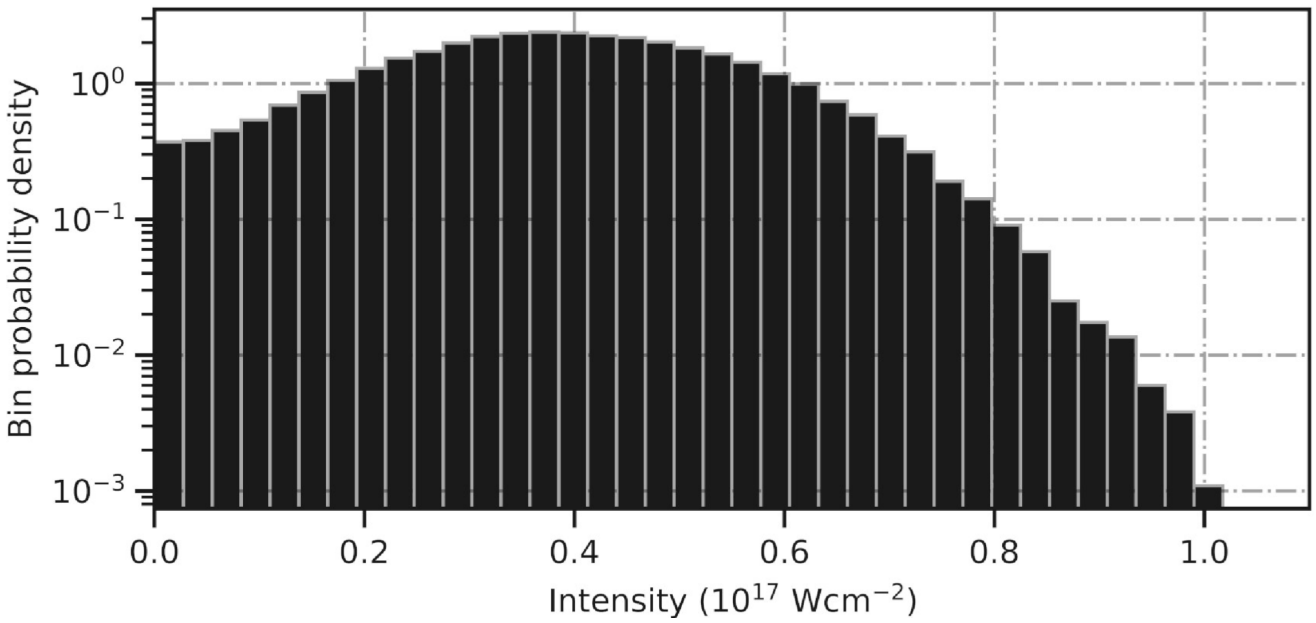

Figure V.12: A logarithmically scaled histogram of calculated pulse intensities of a dataset from the supplementary material of Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] The data shown was recorded with the focal point in the interaction region, with a measured focal spot size of $x_{waist} = (10.17 \pm 0.44)\ \mu m \times y_{waist} = (9.16 \pm 0.27)\ \mu m$.[31] The intensities in the interaction point when in focus were found to be up to 100 PW.

x_{waist} and y_{waist} are assumed to remain constant throughout the experiment. Δt_{FWHM} fluctuates on a shot to shot basis due to the nature of the XFEL lasing process. The incident pulse energy E is found via the FIMS prediction shown in figure V.9, Therefore, an intensity may be

estimated and associated with every shot. Given this, it is possible to bin the data with respect to intensity.

When calculating an experimental measure that combines the results of multiple measurements in a complex manner, it can become difficult to quantify uncertainties using standard error propagation techniques. In addition, quantifying the measurement error on a detector can involved complexities such as measuring detector thermal noise when no x-rays are impinging on the sample.

An alternative technique which can bypass these complications is bootstrap resampling.[40] This is an uncertainty estimation technique that relies on sampling multiple randomly selected partitions of the data. This can be done with or without replacement. The experimental measure is then calculated multiple times, once per partition. The variance of the calculated values is then found, and gives uncertainties in the desired quantity.

The errors in figure V.13 subplot (**b**) and figure V.16 subplot (**b**) were found via a linear approximation to a bootstrap process, known as jack-knifing.[41] Jack-knifing allows an estimator of the variance of a statistic by resampling data points with replacement. In traditional jack-knifing, if N observations are made, the desired statistic must be recalculated N times. As our datasets from super-conducting XFELs can contain up to 10^7 shots, this is not computationally viable. Therefore, a variant of jack-knifing intended for large datasets known as the delete-m jack-knife was used.[42]

The delete m jack-knife works by calculating multiple times the statistic Q for which an uncertainty is needed. Data points $\{x_j\}$ with $j = 1, 2...N$ in this case are multiple detected x-ray spectra. The data is split up into G not necessarily equal-size groups. Each group therefore has approximately $m = \text{floor}(N/G)$ data-points. A value for Q using only a partition of the dataset Q_i^{partial} is calculated. Formally, this is computed from values x_j where $j = 0, 1, 2...i-1, i+m, i+(m+1), ...N$. Notice how m data points have been excluded, namely $j = i, i+1, i+2..., i+(m-1)$. An estimate for the variance in the target statistic Q is then given by

$$\sigma_Q^2 = \frac{G-1}{G} \sum_i \left(Q_i^{\text{partial}} - \langle Q \rangle \right)^2. \tag{V.16}$$

Here, $\langle Q \rangle \equiv \frac{1}{G} \sum_i Q_i^{\text{partial}}$, and is essentially the value for Q calculated with all data points included. Using this process, the uncertainties were calculated. The delete m jack-knife process measures how much the target statistic Q depends on a random subset of the data. Similarly

to more standard error propagation techniques, this method can only account for random errors, and cannot provide an accurate estimate of more complex systematic errors that may be present in the experiment.

The choice of G can have an effect on the calculated uncertainty. For the data collected in this experiment, the target statistic Q was the nonlinear emission measure $B(\omega)$ defined in equation (V.1). σ_Q^2 was found to converge for $G \gtrsim 50$ groups.

V.4 Spectral modifications to the X-ray Field

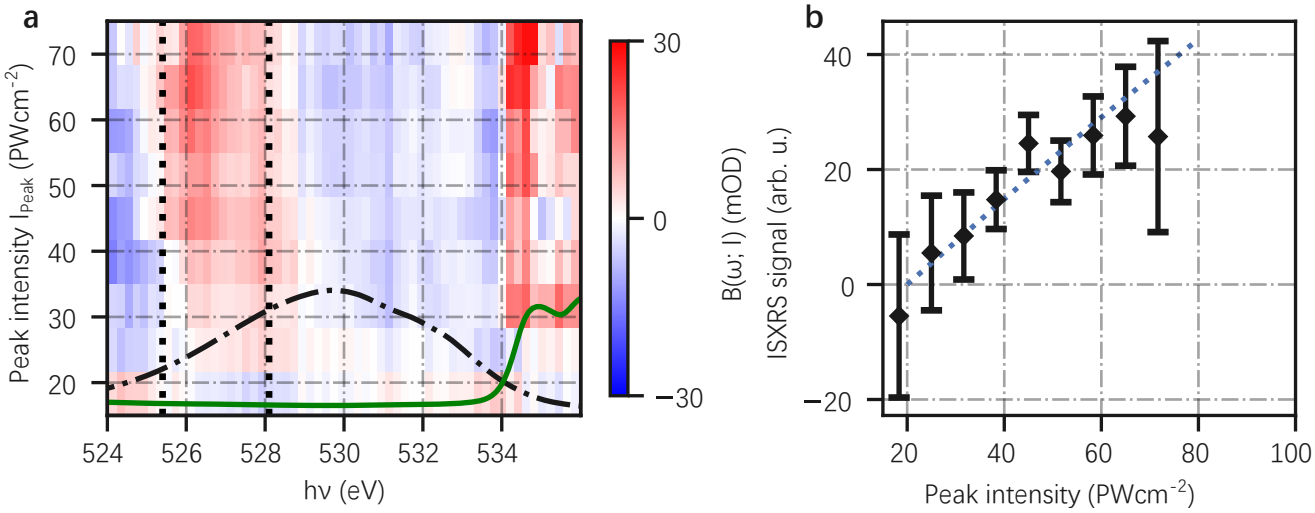

Figure V.13: Nonlinear emission $B(\omega)$ measurements resulting from data recorded from the four experimental geometries shown in figure V.7, from Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] The x-ray spectrum is tuned to have a central photon energy approximately 6 eV below the oxygen K-edge. (**a**) Spectrally resolved data. The average incident XLEAP x-ray spectrum $I_{0,defoc}(\omega)$ is approximately a Gaussian form and is shown by a black dash-dotted line centered at approximately 529.5 eV. The optical density of the sample in the defocused case is shown in green. This forms a standard transmission x-ray absorption spectrum of liquid water as discussed in section III.1. A pre-edge absorption feature is visible at 535 eV in agreement with literature.[16] A false colourmap shows the nonlinear emission in units of mOD resolved vertically as a function of peak intensity $B(\omega; I)$. A region with $B(\omega) > 0$ is indicated by vertical dotted black lines. This is consistent with the Stokes field enhancement expected of the ISXRS proceeds. (**b**) Total spectrum indicated by the dash lines in subplot (**a**). A line of best fit is shown by a blue dotted line. Uncertainties are shown at the $\pm 1\sigma$ level. These uncertainties are computed via bootstrap resampling methods.

Figure V.13 shows the field modifications to the x-ray spectrum after passing through the interaction region. The field modifications are quantified by the nonlinear emission measure defined in equation (V.1). Positive (negative) values of nonlinear emission $B(\omega)$ may be in-

terpreted as a reduction (increase) in optical density when in focus.

However, it is more relevant for this work to interpret nonlinear emission in how it relates to spectral modifications of the x-ray field. From this viewpoint, positive (negative) values of nonlinear emission at an energy $\hbar\omega^*$ correspond to more (less) photons in the $\hbar\omega^*$ mode of the x-ray field exiting the interaction region at higher intensities compared to lower intensities.

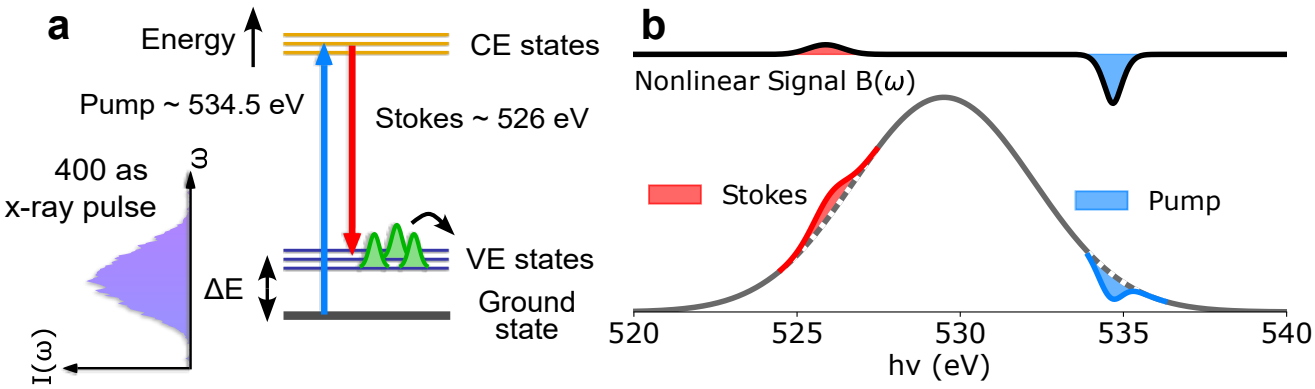

Figure V.14: An illustration of the field modifications that result in photonic emission at the Stokes resonance due to an ISXRS interaction, adapted from Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] (**a**) Energy level scheme of the ISXRS process. In relation to figure V.13, the central photon energy of the XLEAP pulse is centered at 529.5 eV with a spectral width of approximately 6 eV. The pump and Stokes transition resonances shown were found from the experimental data shown in figure V.13. (**b**) A schematic representation of how the ISXRS induced field modifications show up in the nonlinear emission measure. An idealized incident XLEAP spectrum is shown in grey. The depletion of the pump channel and the enhancement of the Stokes channel is shown on the XLEAP spectrum in blue and red respectively. How this translates to nonlinear emission $B(\omega)$ is shown in the nonlinear signal in the upper right.

In principle, the spectral modifications to the x-ray field for an idealized Stokes type ISXRS interaction are drawn in figure V.14. The scattering process annihilates photons at the pump channel and creates them in the Stokes channel. The energy difference between the absorbed and emitted photons results in an excitation of the molecular system, written as ΔE in figure V.14. This is the characteristic modification of the field that corresponds to the ISXRS interaction.

The part of the spectrum corresponding to the Stokes emission [red transition shown in figure V.14 subplots (**a**) and (**b**)] for the water molecule shows as a positive nonlinear emission feature in figure V.13 subplot (**a**) at approximately 526 eV. The boundaries of this feature are indicated by black dotted lines overlaid on the false colourmap. The total signal in this region is indicated as an "ISXRS signal" shown in V.13 subplot (**b**). This ISXRS signal is calculated by $\int d(h\nu)\, B(h\nu)$, with integration limits indicated by the vertical lines in subplot (**a**).

Note that the calculation of nonlinear emission automatically normalizes the transmitted spectrum to the incident spectrum. This means that a single-photon effect that would be expected to have a signal proportional to the incident intensity is then normalized to the incident flux, resulting in an intensity-independent effect. (See the linear absorption effect in figure V.15 at 534 for an example.) For analogous reasons, a two-photon effect such as ISXRS shows a linear increase with intensity, instead of a quadratic increase. For this reason, the two-photon ISXRS effect is fitted to a linear function in figure V.13 subplot (**b**).

During the experiment, a vapour was observed originating from the interaction region when the focal spot is located at the interaction point.[V.7] We therefore measure some gas-phase effects in the transmitted x-ray spectrum. In particular, we observe an absorption feature at 534 eV that is consistent with a gas-phase isolated H_2O absorption resonance corresponding to the $1a_1 \rightarrow 4a_1$ excitation consistent with values documented in literature.[43–46] This transition is shown in figures V.2 and V.3.

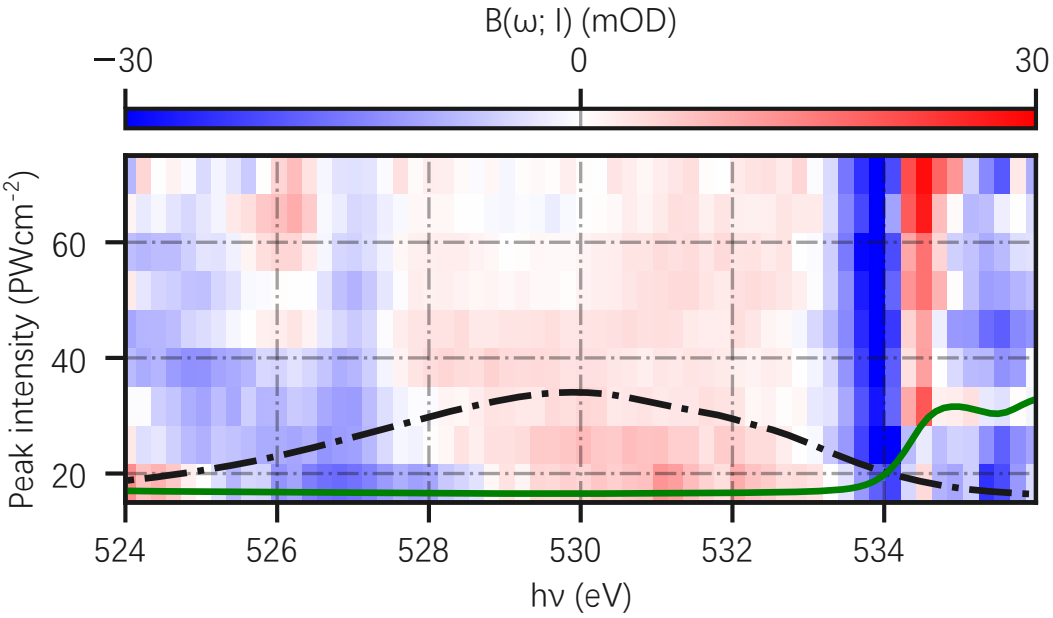

Figure V.15: Nonlinear Emission data taken using x-rays with a central photon energy of 529.5 eV, adapted from the supplementary material of Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] Shown is a false colour plot of intensity resolved nonlinear emission calculated from raw uncorrected data. Similarly to figure V.13 subplot (**a**), the green solid line indicates a linear absorption spectrum taken with the focal spot out of the interaction region. The black dash-dotted line shows the incident x-ray spectrum.

[V.7] Note that in figure V.15, the green solid line shows transmission XAS of the sample when the focal point is moved out of the interaction region. It does not display the absorption feature as the vapour is only produced in significant quantities when the beam is focused into the interaction point.

This absorption effect can be clearly seen in figure V.15 as a vertical blue feature at all intensities at 534 eV. As we wish to demonstrate ISXRS in the condensed phase, this feature is considered as an artifact. This feature arises from a single photon interaction and is therefore expected to not vary with intensity after normalization to the incident photon flux. Therefore, in the nonlinear emission false colourmap shown in figure V.13, a removal of gas-phase effects has been performed by subtracting the lowest intensity bin from the remaining data.

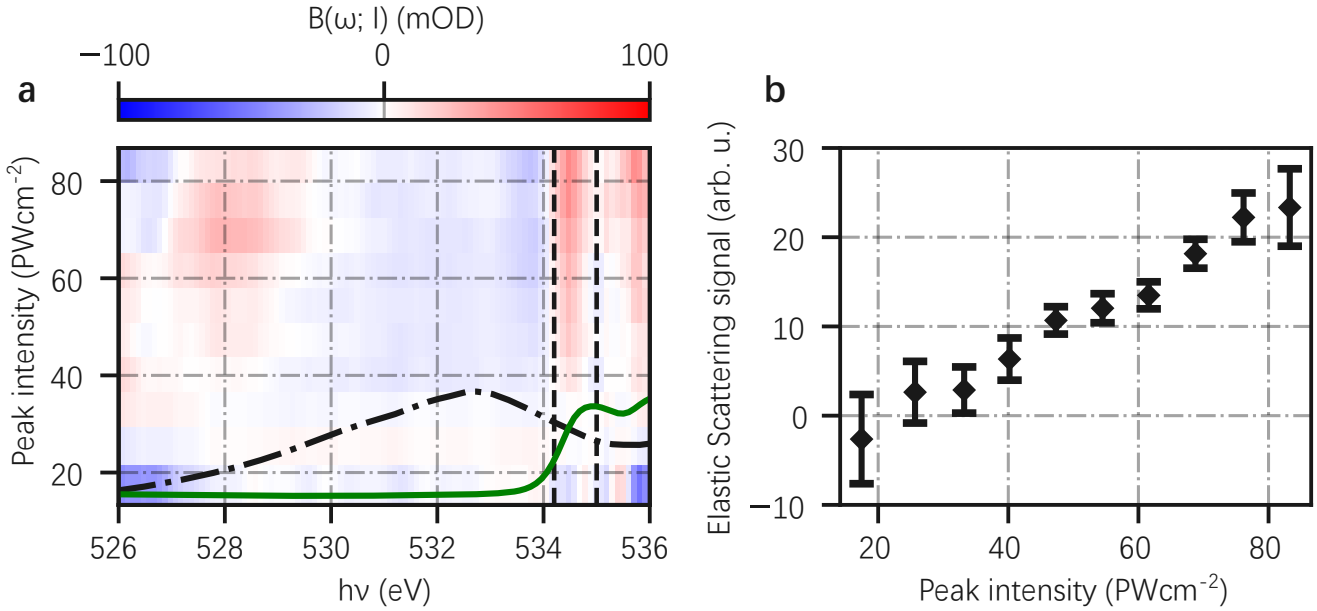

Figure V.16: Nonlinear Emission data taken using x-rays with a central photon energy of 533 eV, adapted from the supplementary material of Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] This data is the same type of data presented in figures V.13 and V.15 but with the XFEL tuned 4 eV higher. (**a**) A false colourmap of nonlinear emission resolved with respect to peak intensity. The incident spectrum is shown as a black dashed line. The defocused optical density is shown in solid green. The vertical dashed lines show the location of the pump transition resonance. (**b**) The total stimulated x-ray elastic scattering signal between the vertical black dashed lines shown in subplot (**a**) resolved with respect to peak intensity. Uncertainties are shown at the $\pm 1\sigma$ level. These uncertainties are computed via bootstrap resampling methods.

During the experiments, evidence of some additional effects were observed. Again, modifications to the transmitted x-ray field in different intensity regimes were the observables used. Therefore, these additional effects are quantified again by the nonlinear emission measure, $B(\omega)$ defined in equation (V.1).

One additional effect was stimulated x-ray elastic scattering.[V.8] This effect is similar to the ISXRS process. It is a two-photon scattering interaction in which the molecular system is taken

[V.8] Elastic photon scattering is sometimes also known as Rayleigh scattering in some contexts.

from its ground state back to its ground state via an intermediate core-excited state. (See figure III.6.) As a result, the elastic scattering process results in no net energy being transferred to the molecule after the interaction. The elastic x-ray scattering cross-section does not require photons on resonance with two separate molecular transitions. Therefore, it is typically larger than its inelastic counterpart in most experiments.[47,48]

Field modifications consistent with stimulated x-ray elastic scattering were observed and are shown in figure V.16. Here, the false colourmap in subplot (**a**) shows a feature with positive nonlinear emission at a photon energy of approximately 535 eV, indicated by the vertical black dashed lines. The growth of this feature with respect to intensity is shown in subplot (**b**). For similar reasons to the ISXRS process, the growth of the signal shown in subplot (**b**) grows linearly with peak intensity despite being a two-photon interaction.

ISXRS was not observed in the dataset shown in figure V.16. This is due to the fact that the x-ray pulses had a spectral width of 6-8 eV. This means that at a central photon energy of 533 eV, there is very little spectral intensity at 526 eV. Therefore the Stokes-type ISXRS interaction cannot be stimulated as no photons at the Stokes resonance are available from the incident x-ray field to drive the process.

An additional effect was identified via a different field modification. This effect is consistent with cationic resonant absorption. In this process, an x-ray photon is non-resonantly absorbed by a valence electron which is promoted into the continuum. The resulting valence photoionization process leaves a valence-hole and a remaining H_2O^+ cation. A second photon promotes an electron from the oxygen core state to the newly created valence hole, leaving the cation in a core-excited state. The spectral imprint of the cationic resonant absorption is shown in figure V.17. Subplots (**a**) and (**b**) both show features with negative $B(\omega)$ at approximately 527 eV. A negative nonlinear emission value implies the optical density of the sample increases when the intensity of the field increases. This is consistent with the proposed explanation. The second resonant x-ray absorption event at 527 eV corresponding to a core to valence promotion can only occur if the initial valence ionization has created the required valence-hole. Hence the field is more likely to be depleted in the 527 channel at high intensities. This can be seen in figure V.17 subplot (**c**). An absorption feature is observed only in the high intensity case.

The data shown in figure V.17 shows some potential evidence of ISXRS, which for the liquid sample was measured to show Stokes enhancement at approximately 526 eV. There is a feature showing positive nonlinear emission in this region. However, for this dataset, the XFEL was

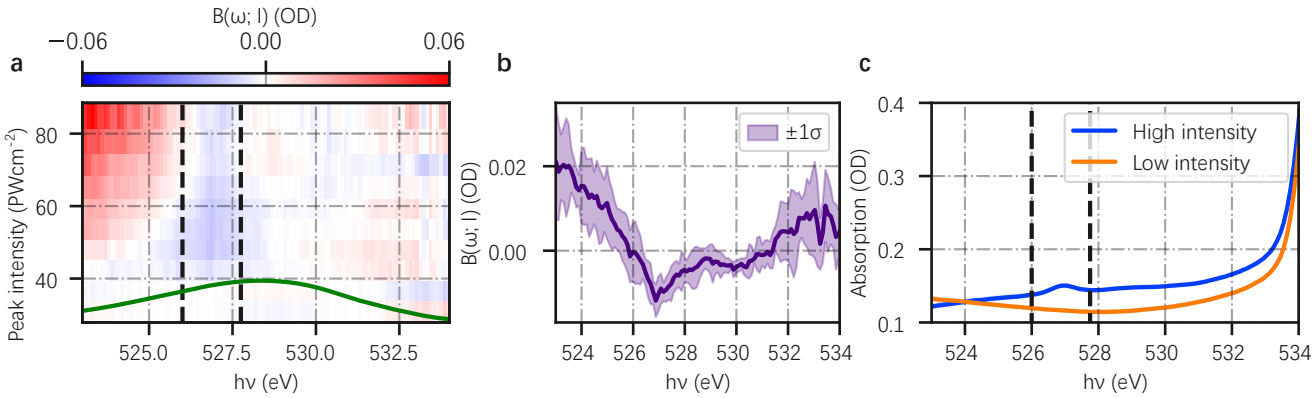

Figure V.17: Nonlinear Emission data taken using x-rays with a central photon energy of 527 eV, adapted from the supplementary material of Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] This dataset shows the expected signature of the creation of a core-excited cation. (**a**) False colourmap of nonlinear emission resolved by peak intensity. The vertical dashed lines show the position of the resonant absorption. (**b**) A lineout of the mean of the two highest intensities shown in subplot (**a**). Uncertainties are shown at the $\pm 1\sigma$ level. These uncertainties are computed via bootstrap resampling methods. (**c**) The optical densities measured via transmission XAS for the focused (labeled as high intensity) and defocused (labeled as low intensity) configurations described in figure V.7. These are shown in blue and orange respectively. The vertical black dashed lines indicate the same energies as in subplot (**a**).

tuned with a central photon energy 8 eV away from the pump resonance. The bandwidth of the XFEL at this photon energy showed typical values in the range of 6-8 eV. Therefore, there was little spectral intensity provided at the pump resonance. Also, the expected ISXRS Stokes emission at 526 eV energy would have partial spectral overlap with the cation absorption signature. These reasons make resonant ISXRS a less likely explanation for this feature compared to data taken with a central photon energy of 530 eV, shown in figure V.13.

V.5 Modeling of field-polarization interactions

To accompany the experimental work, theoretical work was also performed. This theory was used to mimic the experimental conditions to give a better understanding of the ISXRS process. **Disclaimer: The work that follows in this section was not conducted by me**. It is the work of Laura Rego of Consejo Superior de Investigaciones Científicas and Ana Martinez Gutierrez and Antonio Picón of Departamento de Química, Universidad Autónoma de Madrid. I will briefly discuss the key results of the theory and its relevance to the experimental work on ISXRS.

The theory work focused on calculating how the polarization of the water molecules is

affected by the x-ray field. It then modeled the coupling of the induced polarization in the sample with the x-ray field, giving rise to spectral modifications.

These field modifications are then propagated through the longitudinal axis. This means that the polarization response of the molecule and the field modifications were calculated for different propagation lengths. In our experiment, the sample thickness was estimated to be approximately 3 µm.

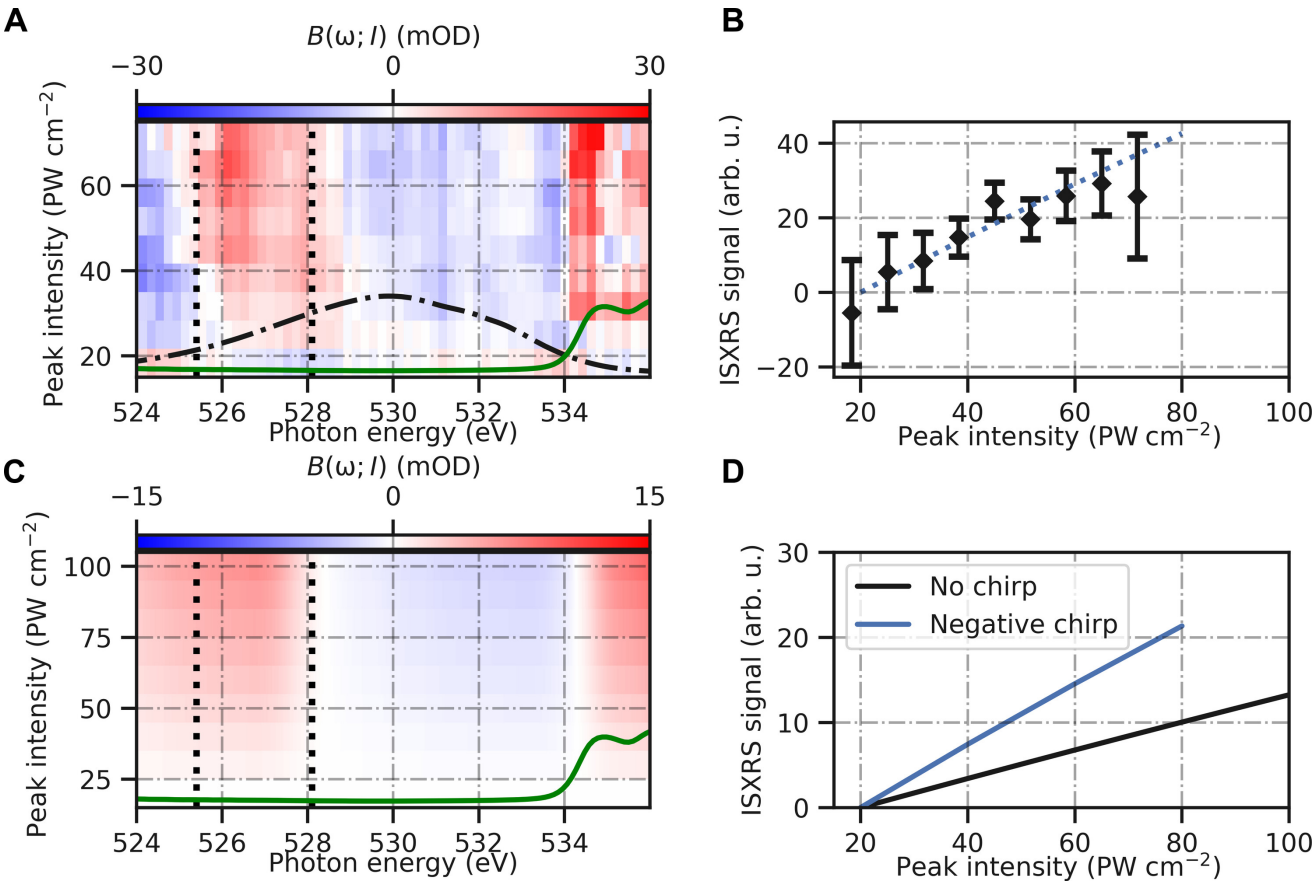

Figure V.18: Nonlinear Emission data at a central photon energy of 529.5 eV and total signal in the Stokes emission region for both experiment and theory, from Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] (**a**) and (**b**) contain the same data as subplots (**a**) and (**b**) in figure V.13. (**c**) The result of the polarization response calculation resolve with respect to peak intensity. The experimental transmission XAS in defocused conditions is shown by a solid green line. The black vertical lines indicate the Stokes emission and show the same energies as in subplot (**a**). (**d**) The total signal in the Stokes emission region without and negative chirp shown in black and blue respectively.

The results of the simulations are shown in figure V.18 reveal a prominent absorption feature near 535 eV and an emission feature centered around 526-527 eV, with a maximum at 527 eV. Both of these features are consistent with the experimental data.

XLEAP pulses are not expected to contain any chirp. The total signal shown in V.18 subplot

(**d**) for zero chirp does not agree well with the experimental measure. This is likely because the simulation of the x-ray molecule interaction did not take into account coupling to the continuum. However, at a similar photon energy o 527 eV, we identify an effect that relies on valence ionization, shown in figure V.17. An increased ISXRS signal is observed when a negative chirp is simulated compared to without chirp, shown in figure V.18 subplot (**d**). Note that both cases only show a nonzero signal beyond approximately 20 PW cm^{-2}. This is due to the vapour-phase subtraction discussed in section V.4. The result in figure V.18 subplots (**c**) and (**d**) is after a propagation distance of 3 μm.

In order to understand the electronic structure of the liquid phase H_2O, the system was modeled as pairs of H_2O molecules interacting via hydrogen bonding. These configurations are known as dimers.[32,49] Multiple different dimer geometries were considered. The ground state electronic configuration of one such dimer configuration is given by

$$\begin{aligned}&(O(1)1s)^2(O(2)1s)^2(O(1)2s)^2(O(2)2s)^2\\&(b_2)^2(b_2/a_1)^2\left((b_2/a_1)^*\right)^2(a_1/b_1)^2\left((a_1/b_1)^*\right)^2(b_1)^2(4a_1(1))^0(4a_1(2))^0.\end{aligned} \tag{V.17}$$

The states are written by the same labeling system used for gas-phase H_2O. a_1, b_1, and b_2, represent the symmetries of the molecule. [V.9] The $4a_1$ states form the LUMO. The bracketed labels (1) and (2) indicate which molecule the orbital is attached to.

The dimer wavefunctions, and associated eigen-energies were calculated by a combination of Hartree-Fock and complete active space self-consistent field methods in order to model the effective electron-electron interactions.[50–52] The oxygen 1s orbitals were treated as fully atomic orbitals. All other orbitals were allowed to interact with the orbitals based on the other water molecule within the dimer. In figure V.19, the combined label (a_1/b_1) of the water dimer is formed by the contribution of an a_1 orbital from molecule (1) and a b_1 orbital from molecule (2).

93 dimer geometries were geometry (internal molecule alignment) and orientation (x-ray polarization - dimer alignment) averaged to produce the final result shown in figure V.20. Each one of these geometries had their own set of calculated orbitals using OpenMolcas.[53] The result for one dimer geometry is shown in figure V.19.

The Time-Dependent Schrödinger Equation for fixed nuclei was used to understand how

[V.9] By convention a_j (b_j) indicates the molecular orbital is symmetric (anti-symmetric) under reflection about the principal C_2 rotation axis.

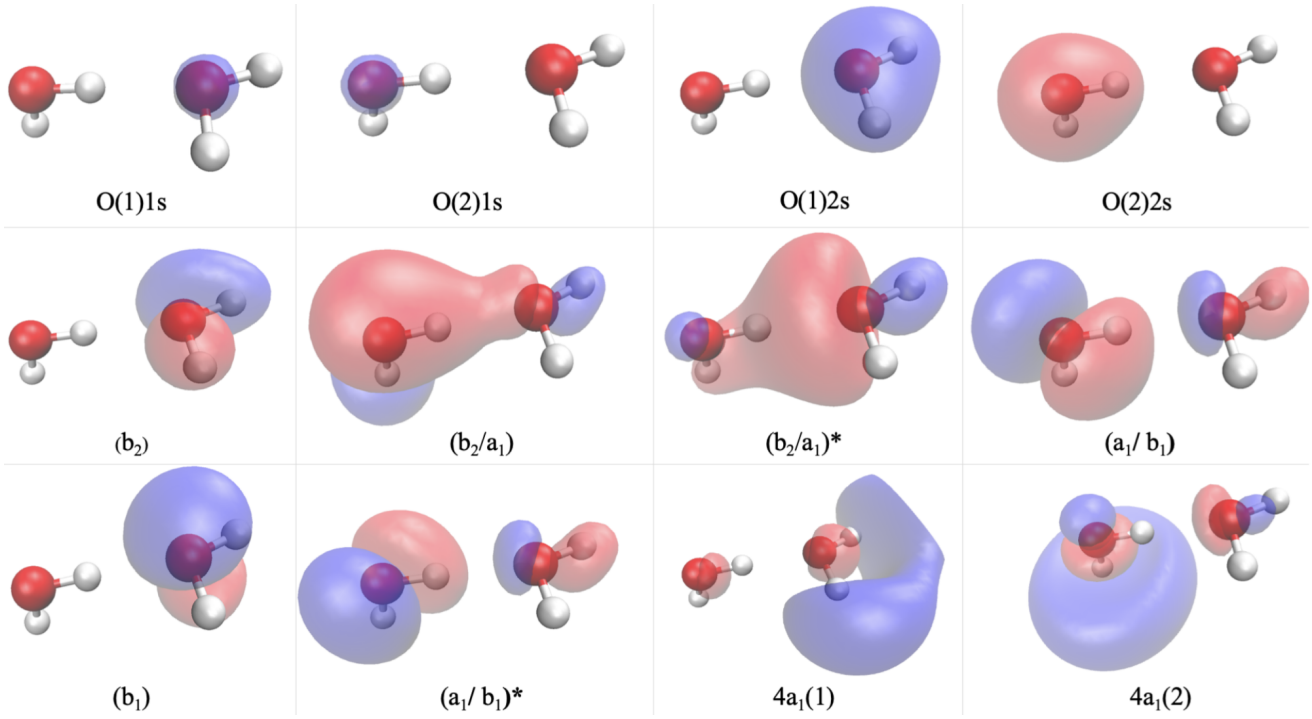

Figure V.19: The wavefunctions of the dimer configuration shown in expression (V.17) from the supplementary material of Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] The wavefunctions are shown from left to right and then in descending rows in increasing energy. Note how well localised the core oxygen 1s orbitals are in comparison to the valence orbitals. The red and blue lobes indicate the sign of the wavefunction.

the population dynamics of the various involved core and valence-excited states. The assumption of fixed nuclei (the static Born-Oppenheimer approximation discussed in section II.1) was justified by the impulsive nature of the short x-ray pulse, which acts on a timescale significantly faster than any nuclear dynamics. From this, the polarization of the dimer $\mathcal{P}_{\text{average}}(z,\omega)$ can be found. The effect of the induced polarization of the sample on the x-ray field $\mathcal{E}(z,\omega)$ was calculated via the field propagation equation derived from the frequency domain maxwell equations, limited to the one dimension, the z axis.[54] This field propagation equation reads

$$\partial_z\mathcal{E}(z,\omega) = \underbrace{ik_z\mathcal{E}(z,\omega)}_{\text{sinusoid}} + \underbrace{\frac{i\omega^2}{2\epsilon_0 c^2 k_z}\mathcal{P}_{\text{average}}(z,\omega)}_{\text{molecular polarization}} - \underbrace{\frac{\omega}{2\epsilon_0 c^2 k_z}j(z,\omega)}_{\text{free charges}}, \tag{V.18}$$

where the abbreviation $\partial_z \equiv \partial/\partial z$ has been used for the partial derivative and k_z is the wavenumber of the field. In this spatial differential equation, $ik_z\mathcal{E}(z,\omega)$ is the standard sinusoidal term that describes the propagation of the field in free space. $j(z,\omega)$ is the free charge current density and the corresponding $-\omega j/2\epsilon_0 c^2 k_z$ term represents the contribution of free charges to

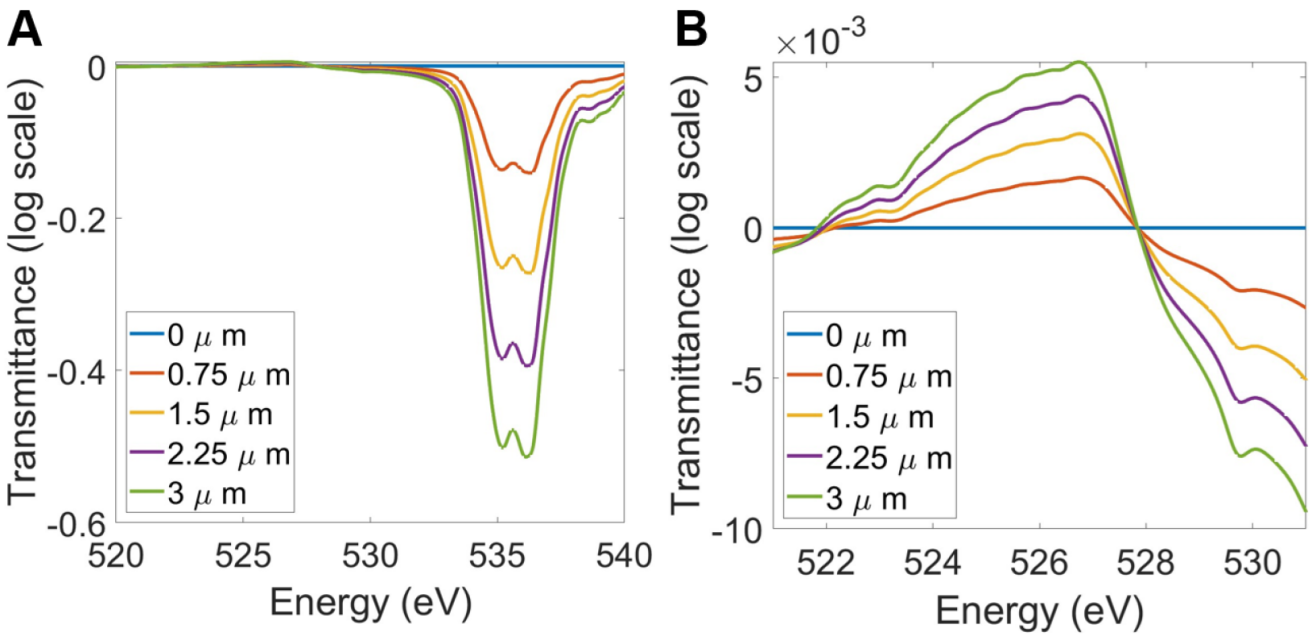

Figure V.20: The suppression and enhancement of the incident x-ray spectrum as a function of propagation distance, from the supplementary material of Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] Here, negative transmittance indicates absorption. Positive transmittance indicates emission. (**a**) The transmittance for the whole spectrum showing strong resonant absorption at 535 eV. (**b**) Zooming in on the Stokes-Raman emission region of interest.

the field. This term was neglected as free charges were not considered in the calculations. $\mathcal{P}_{\text{average}}(z, \omega)$ is the overall polarization of the dimer after orientation and geometry averaging, calculated from the result of the TDSE and $i\omega^2\mathcal{P}_{\text{average}}/2\epsilon_0 c^2 k_z$ is its corresponding contribution. This leads to temporal reshaping of the x-ray field, as shown in figure V.21.

In summary, the theory modeled how the x-ray field moves population out of the ground state and into a variety of excited states. The Auger-Meitner decay of core-excited states was modeled. This is the reason for the long term (after than 500 as) exponential decay of the core-excited states after the x-ray pulse has largely ended in figure V.21 subplot (**b**). The resulting population dynamics correspond to a time dependent distribution of charge that results in a time dependent polarization of the water dimer. This polarization then makes an imprint on the field.

In the temporal domain, this can be seen in the field reshaping shown in figure V.21 subplot (**a**). In the spectral domain it is evident by the transfer of energy from 535 eV to 526 eV in the inelastic scattering process encoded by ISXRS shown in figure V.18 subplot (**c**).

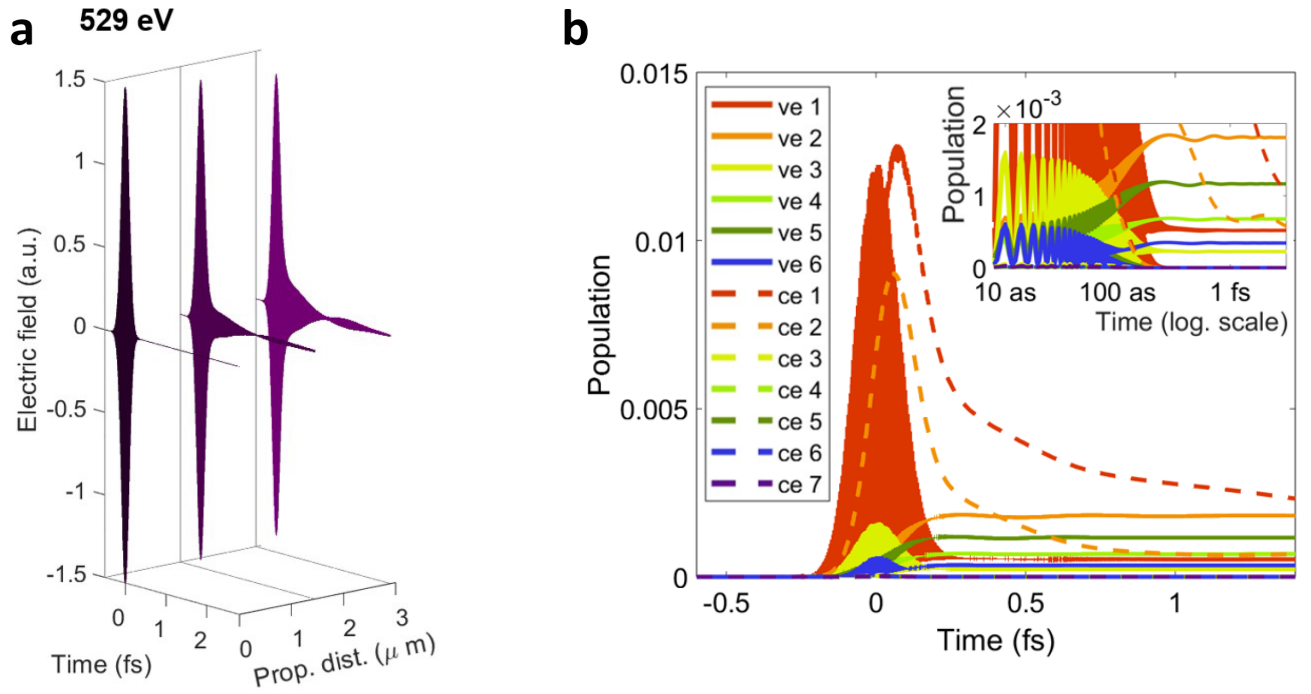

Figure V.21: Further results from the modeling of the dimer system with the x-ray pulse, adapted from the supplementary material of Oliver Alexander et al. *Sci. Adv.* **10**, eadp0841 (2024).[31] (**a**) The effect of the induced polarization on the incident x-ray pulse as a function of propagation distance through the sample. The electric field of the incident spectrum is shown, and is a Gaussian form. Further propagation distances result in revivals of the field. (**b**) Population dynamics of the water dimer after geometry and orientation averaging. Core (valence) excited states are rendered in dashed (solid) lines. Note the order of magnitude difference between the core and valence excited states. The incident pulse has its maximum power density at time 0 fs.

References

[1]S. Mukamel, D. Healion, Y. Zhang, and J. D. Biggs, "Multidimensional Attosecond Resonant X-Ray Spectroscopy of Molecules: Lessons from the Optical Regime", Annual Review of Physical Chemistry **64**, 101–127, ISSN: 1545-1593 (2013) 10.1146/annurev-physchem-040412-110021.

[2]B. Cooper, P. Kolorenč, L. J. Frasinski, V. Averbukh, and J. P. Marangos, "Analysis of a measurement scheme for ultrafast hole dynamics by few femtosecond resolution X-ray pump–probe Auger spectroscopy", Faraday Discuss. **171**, 93–111, ISSN: 1364-5498 (2014) 10.1039/c4fd00051j.

[3]T. Barillot, O. Alexander, B. Cooper, T. Driver, D. Garratt, S. Li, A. Al Haddad, A. Sanchez-Gonzalez, M. Agåker, C. Arrell, M. J. Bearpark, N. Berrah, C. Bostedt, J. Bozek, C. Brahms, P. H. Bucksbaum, A. Clark, G. Doumy, R. Feifel, L. J. Frasinski, S. Jarosch, A. S. Johnson, L. Kjellsson, P. Kolorenč, Y. Kumagai, E. W. Larsen, P. Matia-Hernando, M. Robb, J.-E. Rubensson, M. Ruberti, C. Sathe, R. J. Squibb, A. Tan, J. W. G. Tisch, M. Vacher, D. J. Walke, T. J. A. Wolf, D. Wood, V. Zhaunerchyk, P. Walter, T. Osipov, A. Marinelli, T. J. Maxwell, R. Coffee, A. A. Lutman, V. Averbukh, K. Ueda, J. P. Cryan, and J. P. Marangos, "Correlation-Driven Transient Hole Dynamics Resolved in Space and Time in the Isopropanol Molecule", Physical Review X **11**, ISSN: 2160-3308 (2021) 10.1103/physrevx.11.031048.

[4]M. Kowalewski, K. Bennett, K. E. Dorfman, and S. Mukamel, "Catching Conical Intersections in the Act: Monitoring Transient Electronic Coherences by Attosecond Stimulated X-Ray Raman Signals", Physical Review Letters **115**, ISSN: 1079-7114 (2015) 10.1103/physrevlett.115.193003.

[5]D. Keefer, V. M. Freixas, H. Song, S. Tretiak, S. Fernandez-Alberti, and S. Mukamel, "Monitoring molecular vibronic coherences in a bichromophoric molecule by ultrafast X-ray spectroscopy", Chemical Science **12**, 5286–5294, ISSN: 2041-6539 (2021) 10.1039/d0sc06328b.

[6]D. Keefer, S. M. Cavaletto, J. R. Rouxel, M. Garavelli, H. Yong, and S. Mukamel, "Ultrafast X-Ray Probes of Elementary Molecular Events", Annual Review of Physical Chemistry **74**, 73–97, ISSN: 1545-1593 (2023) 10.1146/annurev-physchem-062322-051532.

[7]S. M. Cavaletto, D. Keefer, and S. Mukamel, "High Temporal and Spectral Resolution of Stimulated X-Ray Raman Signals with Stochastic Free-Electron-Laser Pulses", Physical Review X **11**, ISSN: 2160-3308 (2021) 10.1103/physrevx.11.011029.

[8]C. Weninger, M. Purvis, D. Ryan, R. A. London, J. D. Bozek, C. Bostedt, A. Graf, G. Brown, J. J. Rocca, and N. Rohringer, "Stimulated Electronic X-Ray Raman Scattering", Physical Review Letters **111**, ISSN: 1079-7114 (2013) 10.1103/physrevlett.111.233902.

[9]N. Rohringer, "X-ray Raman scattering: a building block for nonlinear spectroscopy", Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences **377**, 20170471, ISSN: 1471-2962 (2019) 10.1098/rsta.2017.0471.

[10]J. Duris, S. Li, T. Driver, E. G. Champenois, J. P. MacArthur, A. A. Lutman, Z. Zhang, P. Rosenberger, J. W. Aldrich, R. Coffee, G. Coslovich, F.-J. Decker, J. M. Glownia, G. Hartmann, W. Helml, A. Kamalov, J. Knurr, J. Krzywinski, M.-F. Lin, J. P. Marangos, M. Nantel, A. Natan, J. T. O'Neal, N. Shivaram, P. Walter, A. L. Wang, J. J. Welch, T. J. A. Wolf, J. Z. Xu, M. F. Kling, P. H. Bucksbaum, A. Zholents, Z. Huang, J. P. Cryan, and A. Marinelli, "Tunable isolated attosecond X-ray pulses with gigawatt peak power from a free-electron laser", Nature Photonics **14**, 30–36, ISSN: 1749-4893 (2019) 10.1038/s41566-019-0549-5.

[11]T. Tokushima, Y. Harada, O. Takahashi, Y. Senba, H. Ohashi, L. Pettersson, A. Nilsson, and S. Shin, "High resolution X-ray emission spectroscopy of liquid water: The observation of two structural motifs", Chemical Physics Letters **460**, 387–400, ISSN: 0009-2614 (2008) 10.1016/j.cplett.2008.04.077.

[12]L.-Å. Näslund, J. Lüning, Y. Ufuktepe, H. Ogasawara, P. Wernet, U. Bergmann, L. G. M. Pettersson, and A. Nilsson, "X-ray Absorption Spectroscopy Measurements of Liquid Water", The Journal of Physical Chemistry B **109**, 13835–13839, ISSN: 1520-5207 (2005) 10.1021/jp052046q.

[13]P. Wernet, D. Nordlund, U. Bergmann, M. Cavalleri, M. Odelius, H. Ogasawara, L. Å. Näslund, T. K. Hirsch, L. Ojamäe, P. Glatzel, L. G. M. Pettersson, and A. Nilsson, "The Structure of the First Coordination Shell in Liquid Water", Science **304**, 995–999, ISSN: 1095-9203 (2004) 10.1126/science.1096205.

[14]K. R. Wilson, M. Cavalleri, B. S. Rude, R. D. Schaller, A. Nilsson, L. G. M. Pettersson, N. Goldman, T. Catalano, J. D. Bozek, and R. J. Saykally, "Characterization of hydrogen bond acceptor molecules at the water surface using near-edge x-ray absorption fine-structure spectroscopy and density functional theory", Journal of Physics: Condensed Matter **14**, L221–L226, ISSN: 1361-648X (2002) 10.1088/0953-8984/14/8/107.

[15]J. W. Smith and R. J. Saykally, "Soft X-ray Absorption Spectroscopy of Liquids and Solutions", Chemical Reviews **117**, 13909–13934, ISSN: 1520-6890 (2017) 10.1021/acs.chemrev.7b00213.

[16]A. Nilsson and L. Pettersson, "Perspective on the structure of liquid water", Chemical Physics **389**, 1–34, ISSN: 0301-0104 (2011) 10.1016/j.chemphys.2011.07.021.

[17]S. Li, L. Lu, S. Bhattacharyya, C. Pearce, K. Li, E. T. Nienhuis, G. Doumy, R. D. Schaller, S. Moeller, M.-F. Lin, G. Dakovski, D. J. Hoffman, D. Garratt, K. A. Larsen, J. D. Koralek, C. Y. Hampton, D. Cesar, J. Duris, Z. Zhang, N. Sudar, J. P. Cryan, A. Marinelli, X. Li, L. Inhester, R. Santra, and L. Young, "Attosecond-pump attosecond-probe x-ray spectroscopy of liquid water", Science **383**, 1118–1122, ISSN: 1095-9203 (2024) 10.1126/science.adn6059.

[18]C. LibreTexts, *10.3: Why is BeH_2 Linear and H_2O Bent?*, Shared under CC BY-NC-SA 4.0 license, LibreTexts, (2019) https://chem.libretexts.org/Courses/University_of_California_Davis/Chem_110B%3A_Physical_Chemistry_II/Text/10%3A_Bonding_in_Polyatomic_Molecules/10.3%3A_Why_is_BeH2_Linear_and_H2O_Bent (visited on 09/25/2025).

[19]J. A. Sellberg, S. Kaya, V. H. Segtnan, C. Chen, T. Tyliszczak, H. Ogasawara, D. Nordlund, L. G. M. Pettersson, and A. Nilsson, "Comparison of x-ray absorption spectra between water and ice: New ice data with low pre-edge absorption cross-section", The Journal of Chemical Physics **141**, ISSN: 1089-7690 (2014) 10.1063/1.4890035.

[20]ChiralJon, *Molecular Orbitals for Water*, Licensed under CC BY 2.0, (May 25, 2019) https://commons.wikimedia.org/wiki/File:Molecular_Orbitals_for_Water.png.

[21]J. Berkowitz, "PARTIAL CROSS SECTIONS", in *Photoabsorption, Photoionization, and Photoelectron Spectroscopy* (Elsevier, 1979), pp. 155–357, ISBN: 9780120916504, 10.1016/b978-0-12-091650-4.50012-8.

[22]J. T. O'Neal, E. G. Champenois, S. Oberli, R. Obaid, A. Al-Haddad, J. Barnard, N. Berrah, R. Coffee, J. Duris, G. Galinis, D. Garratt, J. M. Glownia, D. Haxton, P. Ho, S. Li, X. Li, J. MacArthur, J. P. Marangos, A. Natan, N. Shivaram, D. S. Slaughter, P. Walter, S. Wandel, L. Young, C. Bostedt, P. H. Bucksbaum, A. Picón, A. Marinelli, and J. P. Cryan, "Electronic Population Transfer via Impulsive Stimulated X-Ray Raman Scattering with Attosecond Soft-X-Ray Pulses", Physical Review Letters **125**, ISSN: 1079-7114 (2020) 10.1103/physrevlett.125.073203.

[23]G. Mao, J. Guo, W. Liu, R. Wu, J. Zhao, C. Shen, H. Mou, J. Li, and G. Du, "Study on the Lateral Scattering of High-Energy Heavy Ions Penetrating through Solids", physica status solidi (a) **220**, ISSN: 1862-6319 (2023) 10.1002/pssa.202200515.

[24]S. A. Cruz, "Ion stopping in nano-layered materials, physical-phase effects and surface scattering potentials: role of spatial confinement", Radiation Effects and Defects in Solids **164**, 389–401, ISSN: 1029-4953 (2009) 10.1080/10420150902945603.

[25]T. Yeung and D. Watt, "Multiple scattering corrections to energy loss measurements for slow ions", Nuclear Instruments and Methods **173**, 379–390, ISSN: 0029-554X (1980) 10.1016/0029-554x(80)90796-x.

[26]J. G. Pearce, Z. Shaar, and R. E. Crosbie, "Scattering of energetic ions by solids — a simulation", SIMULATION **29**, 97–104, ISSN: 1741-3133 (1977) 10.1177/003754977702900403.

[27]A. Sopena Moros, S. Li, K. Li, G. Doumy, S. H. Southworth, C. Otolski, R. D. Schaller, Y. Kumagai, J.-E. Rubensson, M. Simon, G. Dakovski, K. Kunnus, J. S. Robinson, C. Y. Hampton, D. J. Hoffman, J. Koralek, Z.-H. Loh, R. Santra, L. Inhester, and L. Young, "Tracking Cavity Formation in Electron Solvation: Insights from X-ray Spectroscopy and Theory", Journal of the American Chemical Society **146**, 3262–3269, ISSN: 1520-5126 (2024) 10.1021/jacs.3c11857.

[28]R. Plumley, S. R. Chitturi, C. Peng, T. A. Assefa, N. Burdet, L. Shen, Z. Chen, A. H. Reid, G. L. Dakovski, M. H. Seaberg, F. O'Dowd, S. A. Montoya, H. Chen, A. Okullo, S. Mardanya, S. D. Kevan, P. Fischer, E. E. Fullerton, S. K. Sinha, W. Colocho, A. Lutman, F.-J. Decker, S. Roy, J. Fujioka, Y. Tokura, M. P. Minitti, J. A. Johnson, M. Hoffmann, M. E. Amoo, A. Feiguin, C. Yoon, J. Thayer, Y. Nashed, C. Jia, A. Bansil, S. Chowdhury, A. M. Lindenberg, M. Dunne,

E. Blackburn, and J. J. Turner, "On ultrafast x-ray scattering methods for magnetism", Advances in Physics: X **9**, ISSN: 2374-6149 (2024) 10.1080/23746149.2024.2423935.

[29]M. Bluschke, N. K. Gupta, H. Jang, A. A. Husain, B. Lee, M. Kim, M. Na, B. Dos Remedios, S. Smit, P. Moen, S.-Y. Park, M. Kim, D. Jang, H. Choi, R. Sutarto, A. H. Reid, G. L. Dakovski, G. Coslovich, Q. L. Nguyen, N. G. Burdet, M.-F. Lin, A. Revcolevschi, J.-H. Park, J. Geck, J. J. Turner, A. Damascelli, and D. G. Hawthorn, "Orbital-selective time-domain signature of nematicity dynamics in the charge-density-wave phase of $La_{1.65}Eu_{0.2}Sr_{0.15}CuO_4$", Proceedings of the National Academy of Sciences **121**, ISSN: 1091-6490 (2024) 10.1073/pnas.2400727121.

[30]C. J. Crissman, M. Mo, Z. Chen, J. Yang, D. A. Huyke, S. H. Glenzer, K. Ledbetter, J. P. F. Nunes, M. L. Ng, H. Wang, X. Shen, X. Wang, and D. P. DePonte, "Sub-micron thick liquid sheets produced by isotropically etched glass nozzles", Lab on a Chip **22**, 1365–1373, ISSN: 1473-0189 (2022) 10.1039/d1lc00757b.

[31]O. Alexander, F. Egun, L. Rego, A. M. Gutierrez, D. Garratt, G. A. Cárdenas, J. J. Nogueira, J. P. Lee, K. Zhao, R.-P. Wang, D. Ayuso, J. C. T. Barnard, S. Beauvarlet, P. H. Bucksbaum, D. Cesar, R. Coffee, J. Duris, L. J. Frasinski, N. Huse, K. M. Kowalczyk, K. A. Larsen, M. Matthews, S. Mukamel, J. T. O'Neal, T. Penfold, E. Thierstein, J. W. G. Tisch, J. R. Turner, J. Vogwell, T. Driver, N. Berrah, M.-F. Lin, G. L. Dakovski, S. P. Moeller, J. P. Cryan, A. Marinelli, A. Picón, and J. P. Marangos, "Attosecond impulsive stimulated X-ray Raman scattering in liquid water", Science Advances **10**, ISSN: 2375-2548 (2024) 10.1126/sciadv.adp0841.

[32]P. Atkins, J. De Paula, and J. Keeler, *Atkins' physical chemistry*, en, 11th ed. (Oxford University Press, London, England, Dec. 2017), ISBN: 9780198847816.

[33]K. Tiedtke, J. Feldhaus, U. Hahn, U. Jastrow, T. Nunez, T. Tschentscher, S. V. Bobashev, A. A. Sorokin, J. B. Hastings, S. Möller, L. Cibik, A. Gottwald, A. Hoehl, U. Kroth, M. Krumrey, H. Schöppe, G. Ulm, and M. Richter, "Gas detectors for x-ray lasers", Journal of Applied Physics **103**, ISSN: 1089-7550 (2008) 10.1063/1.2913328.

[34]K. Tiedtke, A. A. Sorokin, U. Jastrow, P. Juranić, S. Kreis, N. Gerken, M. Richter, U. Arp, Y. Feng, D. Nordlund, R. Soufli, M. Fernández-Perea, L. Juha, P. Heimann, B. Nagler, H. J. Lee, S. Mack, M. Cammarata, O. Krupin, M. Messerschmidt, M. Holmes, M. Rowen, W. Schlotter, S. Moeller, and J. J. Turner, "Absolute pulse energy measurements of soft x-rays at the Linac

Coherent Light Source", Optics Express **22**, 21214, ISSN: 1094-4087 (2014) 10.1364/oe.22.021214.

[35]P. Heimann, A. Reid, Y. Feng, and D. Fritz, "Fluorescence intensity monitors as intensity and beam-position diagnostics for X-ray free-electron lasers", Journal of Synchrotron Radiation **26**, 358–362, ISSN: 1600-5775 (2019) 10.1107/s1600577519001802.

[36]M. Czwalinna, R. Boll, H. Kirkwood, J. Koliyadu, J. Kral, B. Lautenschlager, R. Letrun, J. Liu, J. Müller, F. Pallas, D. Rivas, T. Sato, H. Schlarb, S. Schulz, and B. Steffen, "Beam Arrival Stability at the European XFEL", en, Proceedings of the 12th International Particle Accelerator Conference **IPAC2021**, Brazil (2021) 10.18429/JACOW-IPAC2021-THXB02.

[37]T. Togashi, S. Owada, T. Yabuuchi, and M. Yabashi, "Long-term timing stabilization for pump–probe experiments at SACLA", Journal of Synchrotron Radiation **32**, 269–274, ISSN: 1600-5775 (2025) 10.1107/s1600577524011974.

[38]A. DasGupta, *Fundamentals of Probability: A First Course* (Springer New York, 2010), ISBN: 9781441957801, 10.1007/978-1-4419-5780-1.

[39]K. Gröchenig, *Foundations of Time-Frequency Analysis* (Birkhäuser Boston, 2001), ISBN: 9781461200031, 10.1007/978-1-4612-0003-1.

[40]G. Dikta and M. Scheer, *Bootstrap Methods: With Applications in R* (Springer International Publishing, 2021), ISBN: 9783030734800, 10.1007/978-3-030-73480-0.

[41]C. F. J. Wu, "Jackknife, Bootstrap and Other Resampling Methods in Regression Analysis", The Annals of Statistics **14**, ISSN: 0090-5364 (1986) 10.1214/aos/1176350142.

[42]F. M. T. A. Busing, E. Meijer, and R. V. d. Leeden, "Delete-m Jackknife for Unequal m", Statistics and Computing **9**, 3–8, ISSN: 0960-3174 (1999) 10.1023/a:1008800423698.

[43]F. Frati, M. O. J. Y. Hunault, and F. M. F. de Groot, "Oxygen K-edge X-ray Absorption Spectra", Chemical Reviews **120**, 4056–4110, ISSN: 1520-6890 (2020) 10.1021/acs.chemrev.9b00439.

[44] A. Sankari, C. Stråhlman, R. Sankari, L. Partanen, J. Laksman, J. A. Kettunen, I. F. Galván, R. Lindh, P.-Å. Malmqvist, and S. L. Sorensen, "Non-radiative decay and fragmentation in water molecules after $1a_1^{-1}4a_1$ excitation and core ionization studied by electron-energy-resolved electron–ion coincidence spectroscopy", The Journal of Chemical Physics **152**, ISSN: 1089-7690 (2020) 10.1063/1.5141414.

[45] Z.-H. Loh, G. Doumy, C. Arnold, L. Kjellsson, S. H. Southworth, A. Al Haddad, Y. Kumagai, M.-F. Tu, P. J. Ho, A. M. March, R. D. Schaller, M. S. Bin Mohd Yusof, T. Debnath, M. Simon, R. Welsch, L. Inhester, K. Khalili, K. Nanda, A. I. Krylov, S. Moeller, G. Coslovich, J. Koralek, M. P. Minitti, W. F. Schlotter, J.-E. Rubensson, R. Santra, and L. Young, "Observation of the fastest chemical processes in the radiolysis of water", Science **367**, 179–182, ISSN: 1095-9203 (2020) 10.1126/science.aaz4740.

[46] I. Hjelte, M. Piancastelli, R. Fink, O. Björneholm, M. Bässler, R. Feifel, A. Giertz, H. Wang, K. Wiesner, A. Ausmees, C. Miron, S. Sorensen, and S. Svensson, "Evidence for ultra-fast dissociation of molecular water from resonant Auger spectroscopy", Chemical Physics Letters **334**, 151–158, ISSN: 0009-2614 (2001) 10.1016/s0009-2614(00)01434-2.

[47] R. H. Temperton, S. T. Skowron, K. Handrup, A. J. Gibson, A. Nicolaou, N. Jaouen, E. Besley, and J. N. O'Shea, "Resonant inelastic X-ray scattering of a Ru photosensitizer: Insights from individual ligands to the electronic structure of the complete molecule", The Journal of Chemical Physics **151**, ISSN: 1089-7690 (2019) 10.1063/1.5114692.

[48] D. V. Rao, S. M. Seltzer, J. H. Hubbell, R. Cesareo, A. Brunetti, and G. E. Gigante, "Measurements of elastic and inelastic scattering cross-sections using monoenergetic Kα radiation", in Penetrating Radiation Systems and Applications II, Vol. 4142, edited by F. P. Doty, H. B. Barber, H. Roehrig, and E. J. Morton (Dec. 2000), p. 124, 10.1117/12.410577.

[49] G. A. Jeffrey, *An introduction to hydrogen bonding*, en, Topics in Physical Chemistry (Oxford University Press, New York, NY, Apr. 1997).

[50] P.-Å. Malmqvist and B. O. Roos, "The CASSCF state interaction method", Chemical Physics Letters **155**, 189–194, ISSN: 0009-2614 (1989) 10.1016/0009-2614(89)85347-3.

[51] J. Olsen, "The CASSCF method: A perspective and commentary", International Journal of Quantum Chemistry **111**, 3267–3272, ISSN: 1097-461X (2011) 10.1002/qua.23107.

[52]S. Larsson, "Applications of CASSCF", International Journal of Quantum Chemistry **111**, 3424–3430, ISSN: 1097-461X (2011) 10.1002/qua.23016.

[53]F. Aquilante, J. Autschbach, A. Baiardi, S. Battaglia, V. A. Borin, L. F. Chibotaru, I. Conti, L. De Vico, M. Delcey, I. Fdez. Galván, N. Ferré, L. Freitag, M. Garavelli, X. Gong, S. Knecht, E. D. Larsson, R. Lindh, M. Lundberg, P. Å. Malmqvist, A. Nenov, J. Norell, M. Odelius, M. Olivucci, T. B. Pedersen, L. Pedraza-González, Q. M. Phung, K. Pierloot, M. Reiher, I. Schapiro, J. Segarra-Martí, F. Segatta, L. Seijo, S. Sen, D.-C. Sergentu, C. J. Stein, L. Ungur, M. Vacher, A. Valentini, and V. Veryazov, "Modern quantum chemistry with [Open]Molcas", The Journal of Chemical Physics **152**, ISSN: 1089-7690 (2020) 10.1063/5.0004835.

[54]M. Kolesik and J. V. Moloney, "Nonlinear optical pulse propagation simulation: From Maxwell's to unidirectional equations", Physical Review E **70**, ISSN: 1550-2376 (2004) 10.1103/physreve.70.036604.

Chapter VI

Valence Dynamics in cations

Time-resolved X-ray pump–X-ray probe spectroscopy is a technique that allows for the measurement of electronic and structural dynamics in matter with potentially sub-femtosecond temporal and atomic spatial resolution. In this approach, an intense x-ray pulse first excites the system, creating an electronic or structural perturbation. This initial pulse is known as the pump. A subsequent time-delayed x-ray pulse then interrogates the resulting transient state by mapping some quantity of the system on to an observable.[1] The second pulse is known as the probe.

In this work - the observable will be absorption as a function of the pump–probe delay. By varying this delay, it is possible to reconstruct the temporal evolution of the excited-state dynamics, revealing how electrons and nuclei respond on their natural timescales. X-ray pump X-ray probe spectroscopy is often performed with X-ray Free-Electron Lasers (XFELs) due to their high intensities and femtosecond pulse durations[VI.1], critical to time-resolved experiments.

An additional advantage of using x-ray free electron lasers is the high intensities of the x-rays produced[2,3] is spread over a sufficiently short time to allow the probe to interrogate the state of the sample prior to x-ray induced damage occurring. For example, this has been shown to occur on the femtosecond timescale in crystal samples by measuring the yield of Bragg diffracted x-rays as a function of pulse duration.[4] An example of extracting information (in this case atomic spacing within diamond) before x-ray induced disruption of the atomic order is given by I. Inoue et. al.[5]

[VI.1] Potentially sub-femtosecond in the case of lasing schemes that exploit density modulations in the electron bunch.

In order to apply an x-ray pump x-ray probe scheme to measure electron migration, it is necessary to find a way to allow the probe pulse to map some critical property of the transient state prepared by the pump pulse to an observable. One example of this is shown in figure VI.1. The process shown in figure VI.1 is proposed for very short XFEL pulses with a

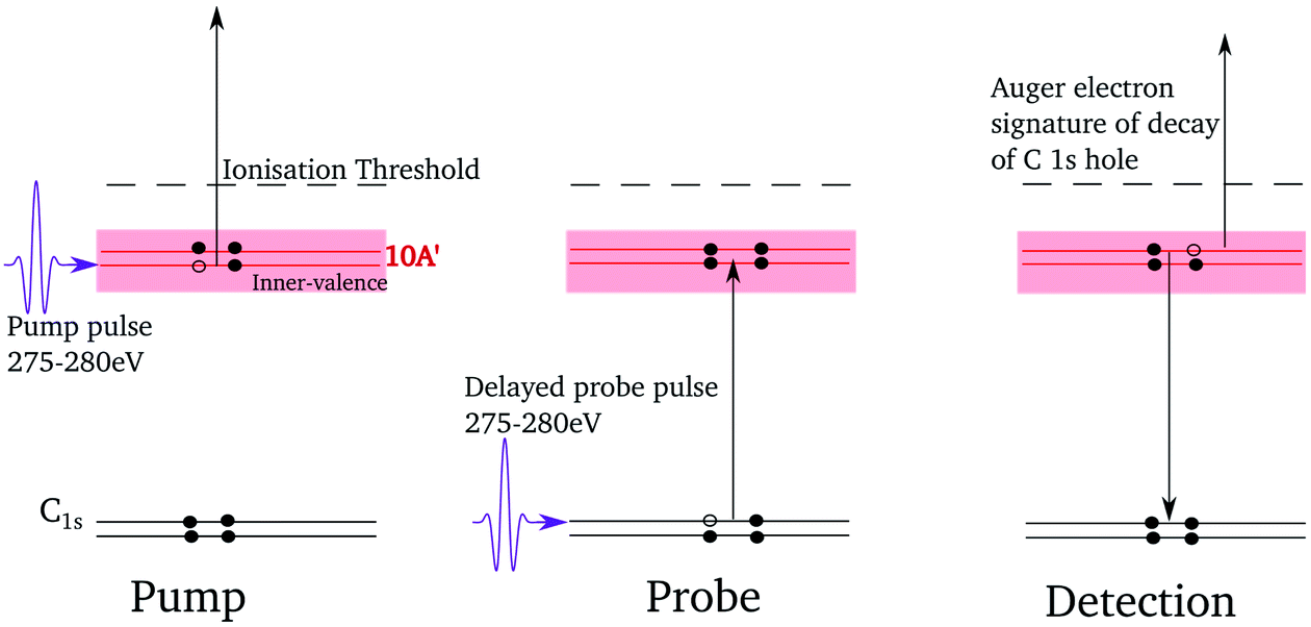

Figure VI.1: A schematic of a proposed x-ray pump x-ray probe scheme for measuring charge migration at x-ray free electron lasers, from B. Cooper et. al., *Faraday Discuss.*, 2014, **171**, 93.[6] The example system here is an isolated glycine molecule. *Left*: An initial pump pulse ionizes an electron from the inner valence of the molecule. This prepares a cation with a valence-hole. *Center:* After a controlled time-delay, a probe pulse excites an electron from the carbon 1s core-state to the valence, destroying the valence-hole in the process and creating a core-hole state. *Right:* Finally the core-hole state relaxes via the resonant Auger-Meitner process which results in the promotion of an Auger-Meitner electron into the continuum.

correspondingly large bandwidth. Given this, the pump step may ionize electrons from two energetically distinct eigenstates within the 10a′ inner valence manifold shown in pink. This results in the preparation of a coherent superposition of energy eigenstates, with a expected oscillatory behaviour with a period of approximately 15 fs.

Figure VI.1 shows the case where the pump and probe pulses have the same photon energy. It is useful to note that in principle this is not a requirement of the process. Critically, two conditions must hold for this scheme to work. Firstly, the pump pulse must have insufficient energy to ionize the carbon 1s core electrons. Otherwise, the Auger-Meitner electron emission on which the measurement relies would also be created by the pump pulse, in addition to the probe pulse. Secondly, the probe pulse must be on resonance with the core-valence transition

of the target atom.

The cross section for the absorption of the probe pulse is sensitive to the valence-hole density in the vicinity of the resonant atom(s). This is because the absorption step involves a core electron in a well localised 1s state. In this case, the core-state corresponds to the carbon sites of glycine.

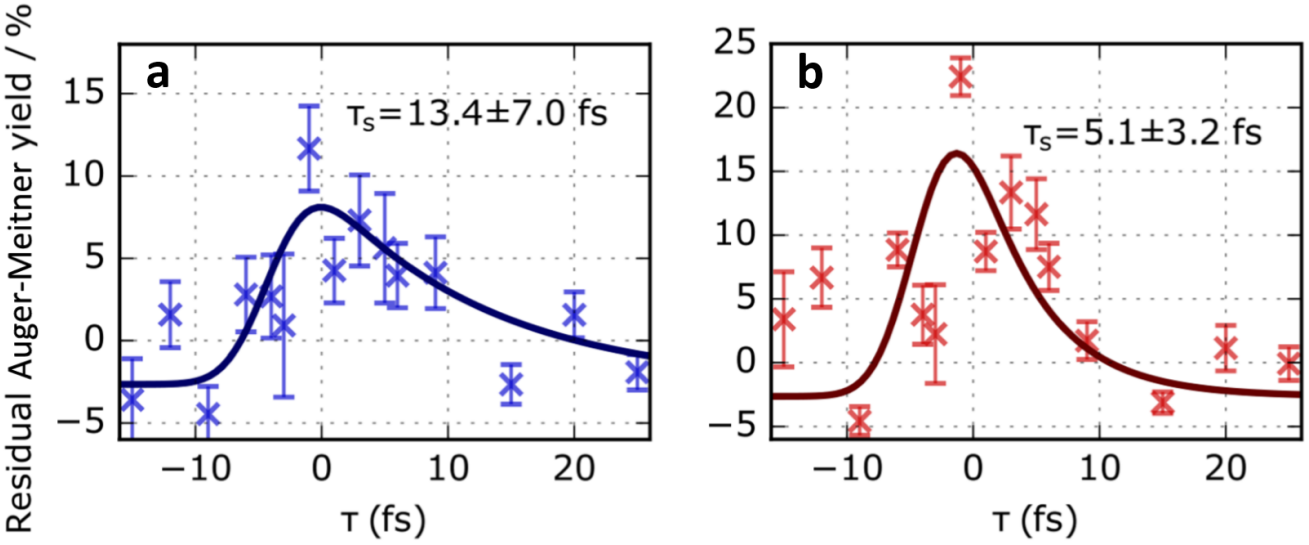

Figure VI.2: Measurements of valence hole decay in isolated isopropanol molecules using a similar x-ray pump x-ray probe scheme to the one shown in figure VI.1, adapted from Barillot et. al. *Phys. Rev. X* **11**, 031048.[7] Subplots (**a**) and (**b**) show the valence hole lifetime for two different inner valence hole states of the cation, created by the pump: the $7a^{-1}$ and $6a^{-1}$ states respectively. These are measured by tuning the probe pulse to slightly different resonances. The probe was matched to the 1s → 7a resonance at 516.3 eV and 1s → 6a resonance at 514.1 eV subplots for (**a**) and (**b**) respectively.

The data in figure VI.2 was collected using a similar scheme to that described in figure VI.1. However, in this case, the site at which the local hole density was probed was at the oxygen atom. In this experiment, the pump pulse was centered at approximately 502 eV. In this case, while clearly sufficient to ionize the carbon core-electrons, it is not sufficient to ionize from the oxygen core-electrons. Therefore, only the probe pulse can create the core-hole leading to the observable. In this case, the observable is Auger-Meitner electrons corresponding to the oxygen site.

In figure VI.3, a special XFEL fresh slice mode is used which is capable of producing 2.5 fs pulses.[8] Due to the lower available Auger-Meitner electron counting statistics available in this dataset, the plotted yield is a differential yield relative to Auger-Meitner counts recorded off the 6a resonance. The fitting function is an exponential decay convolved with a Gaussian to

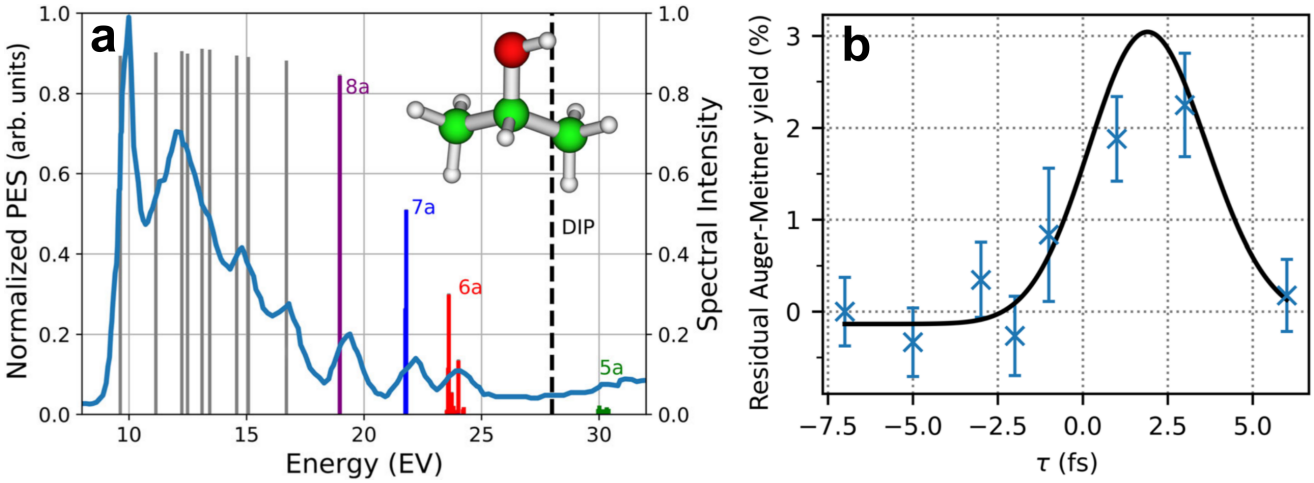

Figure VI.3: Photoelectron spectrum and further measurements of the valence hole decay in the 6a orbital in isolated isopropanol molecules, adapted from Barillot et. al. *Phys. Rev. X* **11**, 031048.[7] (**a**) A synchrotron measured photoelectron spectrum in solid blue. The calculated energies of the 6a and 7a orbitals measured in figure VI.2 are shown in red and blue respectively. The double ionization potential is indicated by a vertical dashed line. (**b**) Hole decay of the 6a orbital measured with higher time resolution. Data points (blue crosses) are fitted with fitting function shown in solid black.

model the instrument response.

Figures VI.2 and VI.3 show a measurement of hole migration. Shown is the transient hole decay for the 7a and 6a valence-hole states and their associated characteristic femtosecond timescale lifetimes. These inner valence hole states are energetically forbidden from undergoing the normal Auger-Meitner process. Instead a frustrated Auger-Meitner process was proposed to explain the decay mechanism.[9–11]

VI.1 Gas-phase glycine cation dynamics

A relevant experiment to this work was conducted at the FLASH XFEL at DESY. This work is summarized to provide context. **Disclaimer: It was not conducted or contributed to by me.** This work looked at electron migration by using the detection scheme described in figure VI.1. In this experiment the XFEL spectral bandwidth was estimated to approximately 1.2 eV. This was sufficient to allow a pump pulse to excite electrons from two energy eigenstates within the 10a′ to the continuum, shown as E_α and E_β in figure VI.5 subplot (**3a**). This forms a cation in a coherent superposition with an expected oscillation of 15 fs.[6]

The experimental geometry is shown in figure VI.4. A single-spike self amplified spontaneous emission (SASE) is generated by the FLASH XFEL.[12] The x-ray pulse is separated into

a pump and probe pulse by a split and delay unit (SDU).[13] The pulse energy is measured upstream of the SDU, hence the energy of both the pump and probe pulses is known.

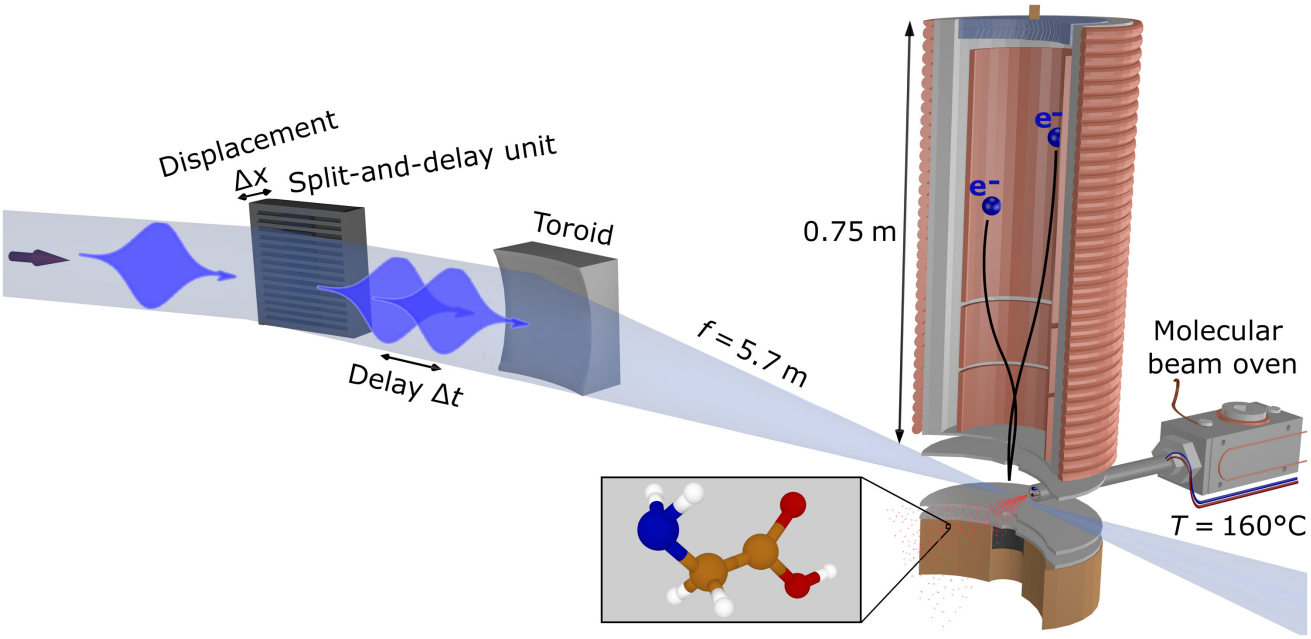

Figure VI.4: A schematic view of the experiment used to detect the electron migration in isolated glycine molecules, from David Schwickert et al., *Sci. Adv.* **8**, eabn6848 (2022).[14] The FLASH XFEL is used to create few femtosecond x-ray pulses using its single spike SASE mode. The x-ray beam impinges onto the SDU which creates the pump and probe pulses.[13] These pulses are focused into the interaction region. A magnetic bottle electron and an ion time of flight spectrometer are used to record the reaction products.

One of the measurements taken in this experiment was the collection of electrons and their resolution in energy. This is needed as the Auger-Meitner electrons which are emitted if resonant absorption to the targeted orbital occurs have a specific energy. (230 eV - 260 eV depending on which orbital is studied.) Two types of electron emission can be distinguished in figure VI.6. The vertical features show an electron kinetic energy independent of the incident photon energy. This is consistent with Auger-Meitner emission.[VI.2]

In contrast, there are diagonal features with gradient of magnitude one. This is consistent with the single photon and sequential double ionization processes, illustrated in figure VI.5. Both of these photoelectron type features display dispersive behaviour. Note that the yield of these photoelectrons is approximately 2-3 times greater than the Auger-Meitner electrons.

[VI.2] Technically, in order to create the required core-hole for this Auger-Meitner process, the probe pulse photons must have enough energy to excite electrons from the core to the inner-valence hole, making it a threshold process by conservation of energy. However in figure VI.6, the probe photon energy does not go sufficiently low to see the onset of this threshold.

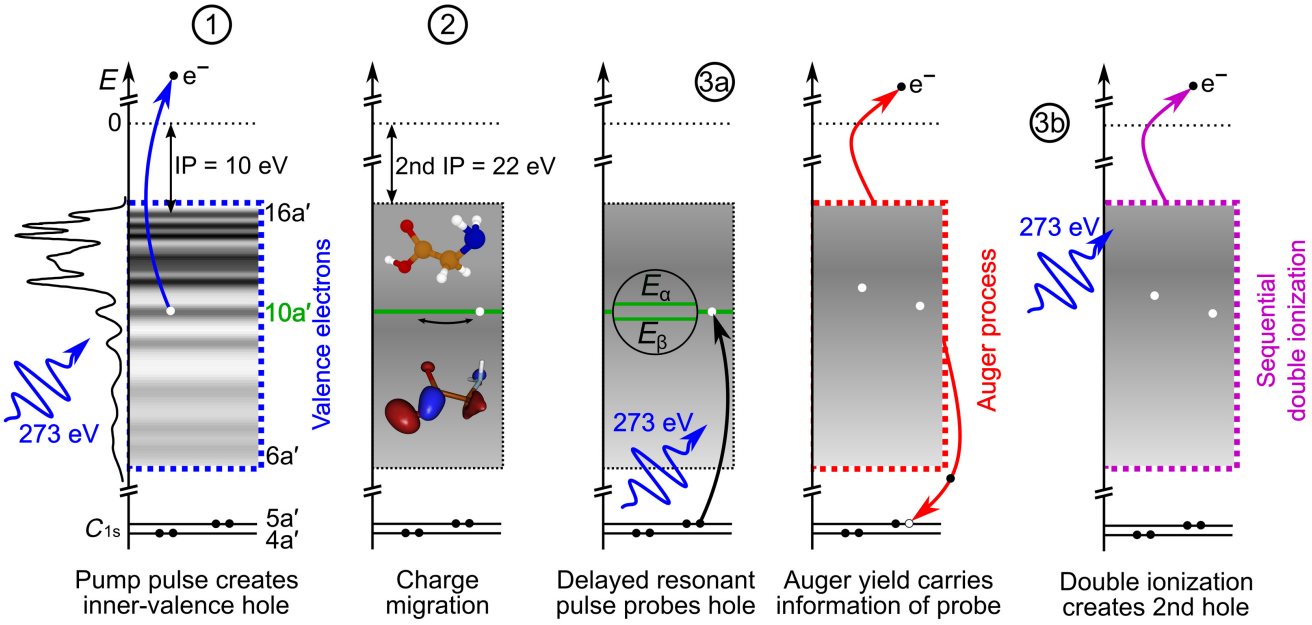

Figure VI.5: The measurement scheme used to measure hole migration in glycine, adapted from David Schwickert et al., *Sci. Adv.* **8**, eabn6848 (2022).[14] From left to right: (**1**) The measured density of states for glycine is shown in greyscale and the sideways plot on the extreme left.[15] A pump pulse prepares a glycine cation in an excited state by removing an electron from the inner valence, resulting in a coherent superposition. (**2**) Hole migration is then anticipated. (**3a**) After a delay, a probe pulse attempts to fill the hole via resonant absorption at a photon energy of 273 eV. This results in resonant Auger-Meitner decay. (**3b**) When the probe photon energy is tuned off resonance with the carbon 1s $\rightarrow$ 10a′ transition, sequential double ionization dominates the resonant Auger-Meitner process.

Each of the pump and probe processes result in the removal of an electron in the channels described in figure VI.5. Therefore, the result of this desired measurement channel is a doubly charged glycine molecule. In order to resolve this fragment, the experiment detected ions via an ion time of flight spectrometer in addition to electrons.

The isolation of the doubly-charged glycine cation allowed for the probe induced Auger-Meitner electron emission to be related to the resultant doubly-charged cation. The experimental data shown in figure VI.7 is the result of coincidence detection of the initial photoelectron originating from valence ionization caused by the pump and the Auger-Meitner electron caused by the probe correlated to the doubly charged glycine species.

Subplot (**a**) of figure VI.7 shows the variation of the correlated electron-ion yield. Oscillations on the scale of 50% - 100% are visible approximately in the initial 40 fs following the pump interaction. This is reflected for the same pump-probe delays at a red region in the false colourmap shown in subplot (**b**). It shows that a temporal oscillation of the observable which occurs with a period of 19.6 fs. Another oscillation is seen to occur after the 40 fs, with a period

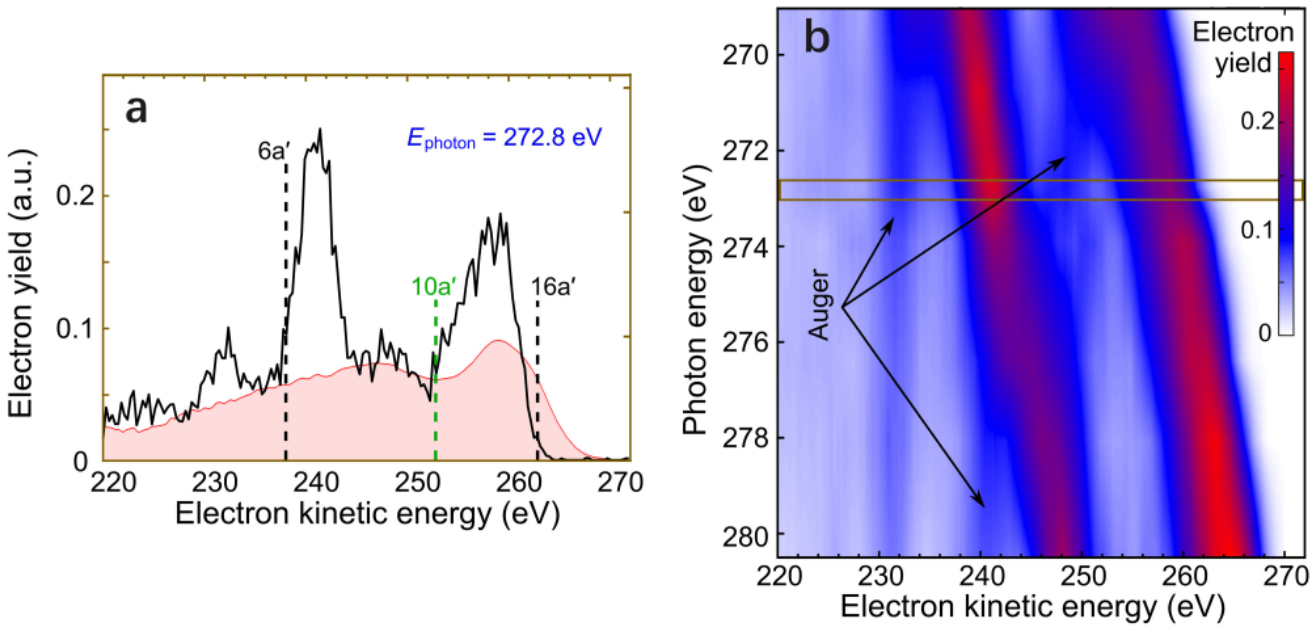

Figure VI.6: X-ray photoelectron spectroscopy following the interaction of isolated glycine molecules and a varied XFEL central photon energy, adapted from David Schwickert et al., *Sci. Adv.* **8**, eabn6848 (2022).[14] (**a**) The measured kinetic energy of electrons at a central photon energy of 272.8 eV are plotted in solid black. In red is an independently measured Auger-Meitner spectrum.[16] The vertical dashed lines indicate the expected Auger-Meitner energies of electrons emitted after the probe fills the labeled orbitals and the subsequent core-hole relaxation. (**b**) A scan of recorded electron kinetic energies over different central photon energies. Both photoelectrons and Auger-Meitner electrons are observed. The central photon energy singled out in subplot (**a**) is shown by the golden box.

of 29.3 fs.

A cone of influence is shown by a shaded grey region. The cone of influence indicates the region in the wavelet spectrum where edge effects become significant because the time duration of the wavelet extends beyond the boundaries of the data.[17] Inside the cone of influence, the results are more reliable than outside it, where the strength of any oscillations may be artificially reduced or distorted by boundary-effects.[18]

In summary, this experiment was able to use an x-ray pump x-ray probe scheme to measure electron hole migration in glycine. This measurement revealed dynamics on the femtosecond timescale. This represents a step forwards from previous work undertaken by Calegari et al. who measured electron migration via a nuclear observables, i.e. cation yield.[19] This study directly images the dynamics via an electronic observable, i.e. coincidence detection of photoelectrons and Auger-Meitner electrons.

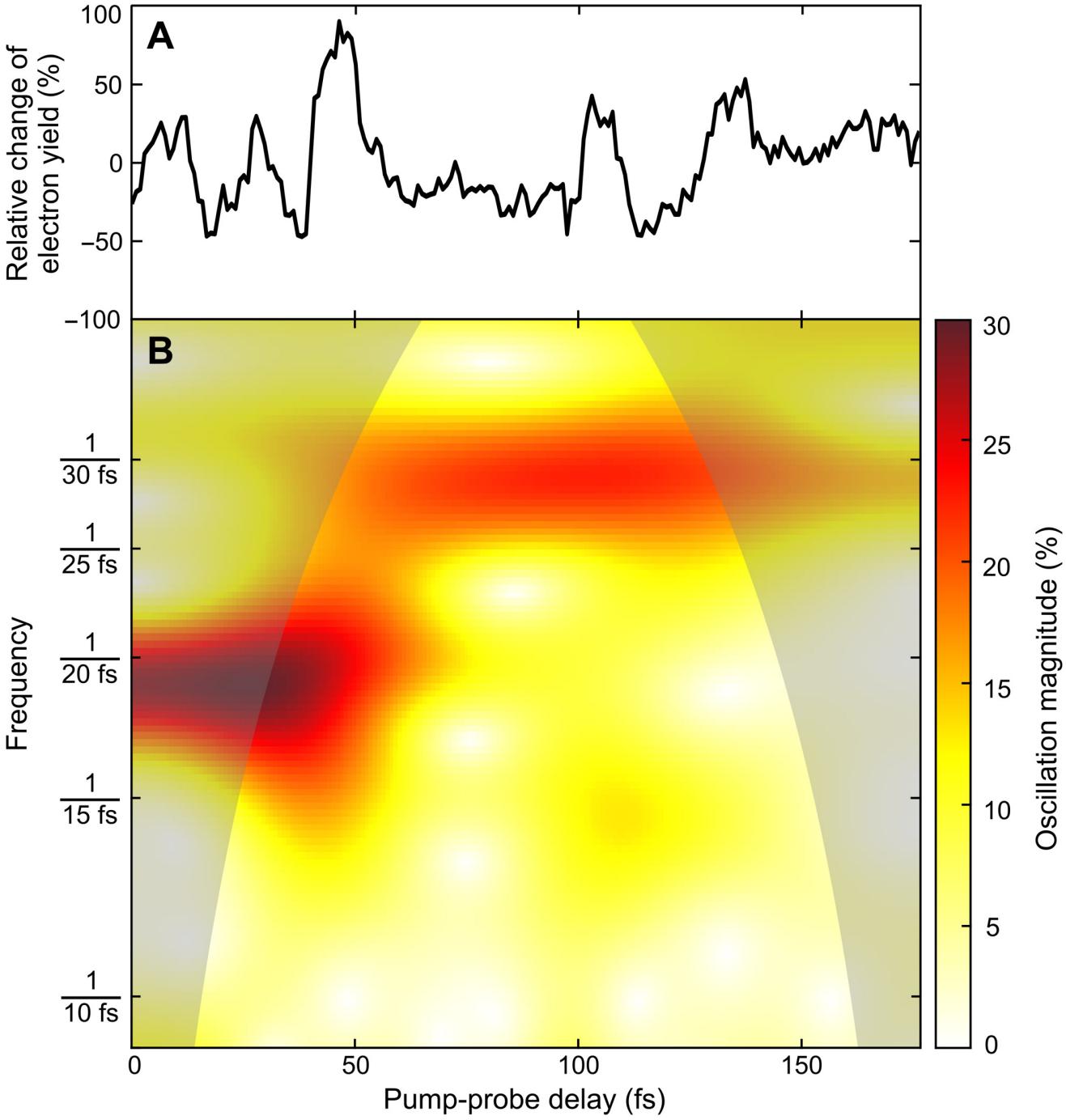

Figure VI.7: Time resolved Auger-Meitner electron yield following excitation with photons tuned to the 10a′ resonance at 272.7 eV, adapted from David Schwickert et al., *Sci. Adv.* **8**, eabn6848 (2022).[14] (**a**) Relative change in the 10a′ orbital originating valence photoelectron - Auger-Meitner electron coincidence counts correlated with the doubly charged glycine cation resolved by pump-probe delay. (**b**) False colourmap showing frequency domain analysis of the coincidence relative change shown in subplot (**a**) found via a continuous wavelet transform. The cone of influence is shown in a dark grey shade.

VI.2 Attosecond electron migration in para-aminophenol

X-ray pump x-ray probe measurements were also conducted at the Linac Coherent Light source (LCLS). The electron bunch shaping XLEAP mode was used, capable of providing x-ray pulses of ~500 as duration at the nitrogen K edge.[VI.3] This experiment formed one of the first scientific uses of the superconducting soft x-ray undulator line at the facility, and took advantage of the increased repetition rate compared to the normal-conducting accelerator. At LCLS, the x-ray generation process works slightly differently to the corresponding process at FLASH. At FLASH, a single x-ray pulse is generated in the single-spike SASE mode. This single pulse is then sent through a split and delay unit, resulting in two pulses with a controllable time delay.

At LCLS however, both the pump and probe pulses are lased in separate undulators. This process is illustrated in figure VI.8. This means that the pump and probe do not show simple correlations. For this reason, it is necessary to have an independent measure of both the total flux of the pump and the probe pulses. This is accomplished by three detectors.

Two are gas detectors each containing a different gas, chosen to have different absorption cross sections for the pump (ω) and probe (2ω). Lastly, the probe is spectrally resolved on a Fresnel zone plate.[20] From this, not only the total flux but the components of the probe pulses can be recorded. Both the gas-detectors and Fresnel zone plate provide a readout on a pulse-resolved basis.

The two-colour lasing scheme used to generate the x-ray pump and x-ray probe pulses creates attosecond pulses and is shown in figure VI.8. This is accomplished by the density modulation to energy modulation process discussed in section IV.2 and shown in figure IV.5. A magnetic chicane is used to change the path length of the electrons with respect to the generated x-ray radiation. This allows for control of the time delay between the pulses.

In any pump-probe experiment, it is necessary to compare unpumped data with pumped data, where the unpumped data acts as a control dataset. In this x-ray pump x-ray probe scheme, it is not possible to generate the probe pulse alone with the same properties as the probe pulse in the ω and 2ω scheme. This is as the lasing of the pump may induce changes in the electron bunch which have an impact on the lasing of the probe pulse.

Instead, one compares shots recorded with high pump flux and compares them to shots with low pump flux. The number of photons contained within the pump pulse can be cal-

[VI.3] This number is an estimate based on the fact that XLEAP pulses have been demonstrated to be approximately 1.5 times the Fourier Transform Limit by streaking measurements, discussed in section IV.2.

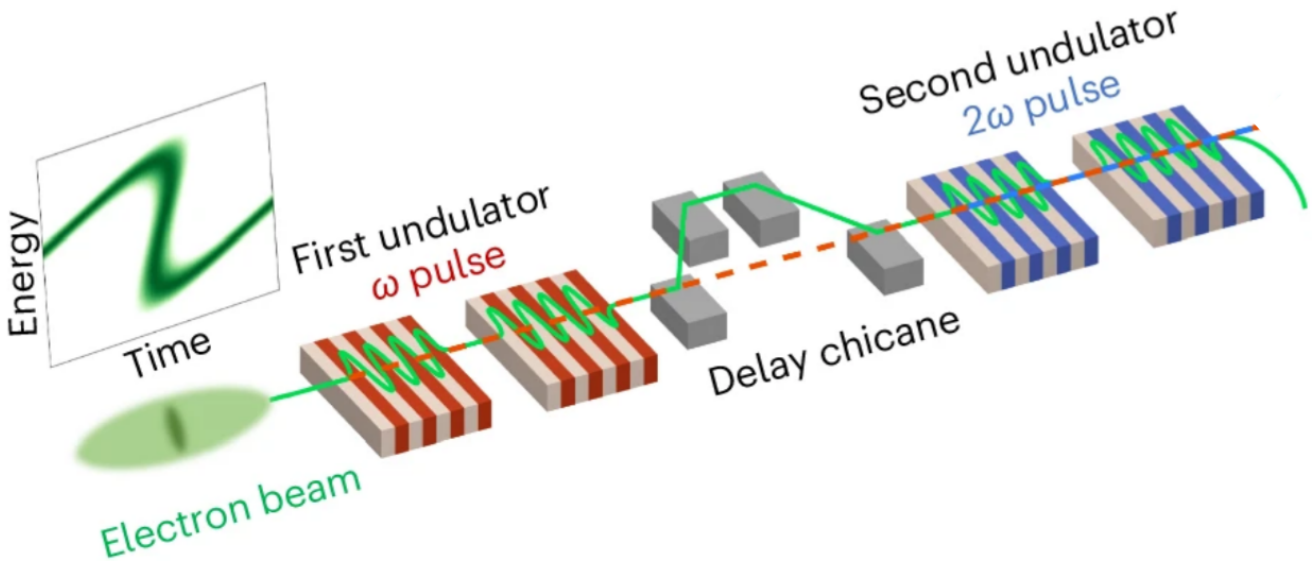

Figure VI.8: An illustration of the two colour x-ray generating process used at LCLS, adapted from Guo, Z., Driver, T., Beauvarlet, S. et al. *Nat. Photon.* **18**, 691–697 (2024).[21] An energy modulated electron bunch (shown in green) is sent through two undulators. These generate radiation in the forward direction shown in red and blue. A magnetic delay chicane allows for a variable pump-probe delay. The generated radiation in this scheme forms harmonics and are labeled as ω and 2ω.

culated from the readout of the two gas detectors. This procedure is described in equation (IV.2).

During the collection of data, the pump and probe photon energies were simultaneously scanned in the vicinity of the nitrogen K edge. A magnetic bottle electron spectrometer was used to record electrons exiting the interaction region with a high collection efficiency. Electron-yield x-ray absorption spectroscopy was then performed, focusing on the Auger-Meitner electrons emitted at approximately 390 eV.

Evidence of the opening up of resonant absorption channels due to the creation of a valence hole by the pump is shown in figure VI.11. In the colour scale the number of pump photons increases. Up until saturation, this corresponds to an increasing probability of a valence-hole being created in the para-aminophenol cation. The resonant absorption is therefore expected to grow with increasing pump flux and to occur below the nitrogen K-edge. A feature consistent with both of these properties is visible in figure VI.11 subplot (**b**) at approximately 401 eV.

It is reasonable to assume that prior to any interaction with any x-rays, the sample should be in its electronic ground state. By conservation of energy, this would imply that the highest energy electrons that could be measured would have energy $2\hbar\omega$, neglecting any effect of

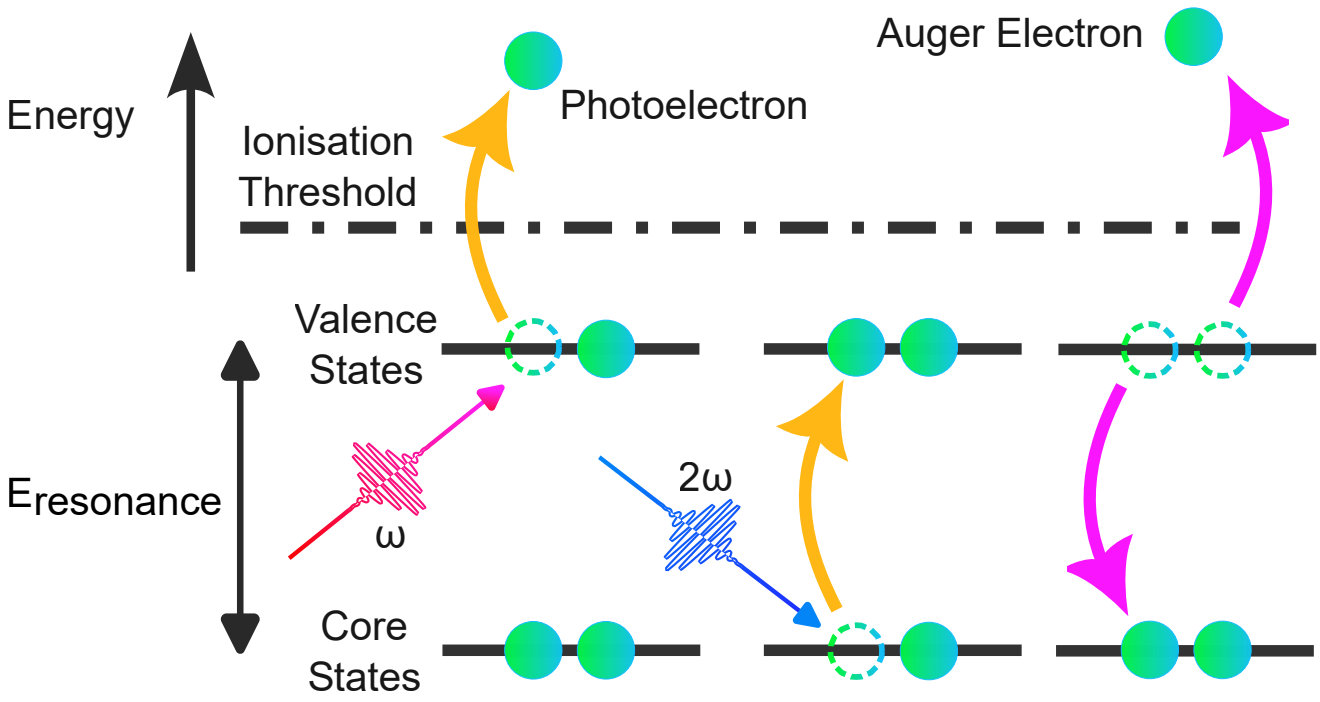

Figure VI.9: An illustration of the measurement process shown in figure VI.1 but adapted for the $\omega/2\omega$ x-ray pump x-ray probe scheme. The pump and probe have energy $\hbar\omega$ and $2\hbar\omega$ respectively. An essential difference in this scheme compared to the one drawn in figures VI.1 and VI.5 is that here, the pump has insufficient energy to allow for resonant absorption into the valence-hole state. Similar however to the previous scheme, the observable is still the emission of an electron following a resonant Auger-Meitner process.

binding energies. However, this is not the case for the data shown in figure VI.11 subplot (**a**). Electrons with significantly lower times of flights are detected.

This can be explained by the presence of the third harmonic of the pump, with energy $3\hbar\omega$. This additional light is able to cause valence ionization and hence eject photoelectrons, which is the likely explanation for the fastest peak at approximately 1.6 µs, and would be expected to appear at slightly less than 600 eV for this dataset.[VI.4]

The second fastest feature at 1.7 µs is consistent with oxygen Auger-Meitner electrons. These are emitted due to the third harmonic causing core ionization of the oxygen 1s orbital. Note that the scale in figure VI.11 subplot (**a**) is base 10 logarithmic. Therefore approximately an order of magnitude exists between the number of third harmonic valence photoelectrons and the oxygen Auger-Meitner electron. Another order of magnitude separates the non-resonant oxygen 1s core ionization from the resonant absorption causing nitrogen

[VI.4] Although a light peak due to scattered x-rays would arrive in the detector earlier, it produces so few counts that it is barely detectable above the detector noise threshold, and typically does not survive hit-finding.

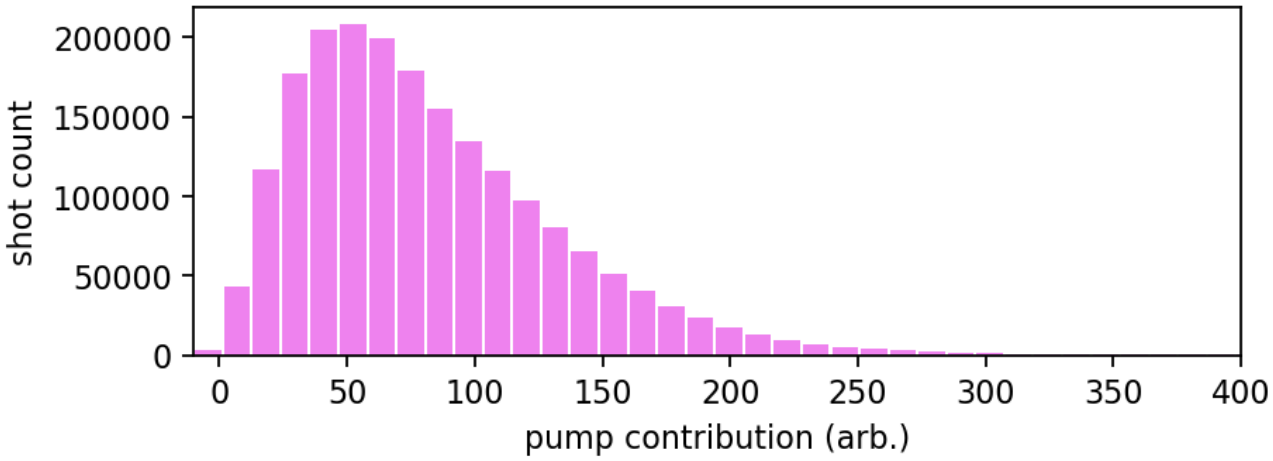

Figure VI.10: Distribution of a measure of the pump flux. Its units are arbitrary, but is directly proportional to the number of photons in the pump pulse N_{pump}, defined in equation (IV.2). Binning on this variable allows for the effect of the pump on the sample to be identified and studied.

Auger-Meitner emission.[VI.5]

An additional way of measuring XAS is by recording the number of ions (or charged fragments of a parent molecule) produced when a sample absorbs x-rays and fragments. It is an indirect detection technique that measures a nuclear observable. As the valence photoionization caused by the pump and subsequent Auger-Meitner decay initiated by the probe each remove an electron, we focus on doubly charged fragments. Doubly-charged nitrogen N^{2+} and NH_2^{2+} cations were clearly resolvable and are shown in figure VI.12 subplot (**a**). Using this ion yield XAS technique, a feature consistent with the electron yield XAS data is observed, also at 401 eV. Another feature which appears to show the signature of a valence hole can be seen at 404 eV. The latter is however not seen in the electron-yield XAS.[VI.6]

Another way to analyze the resonant Auger-Meitner emission is to study how it is correlated to the detection of various ion channel fragments. This can be done by using of covariance matrices. Covariances are typically calculated for single numbers (scalars).[23] However this concept may be generalized for application to arrays of numbers, which can correspond to a spectrum. This results in a covariance matrix. Following Leszek J Frasinski 2016 *J. Phys.*

[VI.5] Also, the third harmonic of the pump in this scheme with photon energy $3\hbar\omega$ can result in non-resonant (normal) Auger-Meitner emission. This is a false-positive to the probe induced resonant Auger-Meitner channel. For this measurement is is essential to quantify the non-resonant Auger-Meitner count rate and subtract from the probe induced count rate.

[VI.6] The signature of a valence hole in these measures of XAS is an increase in XAS in a region below the pre-edge that becomes more pronounced with increasing pump flux.

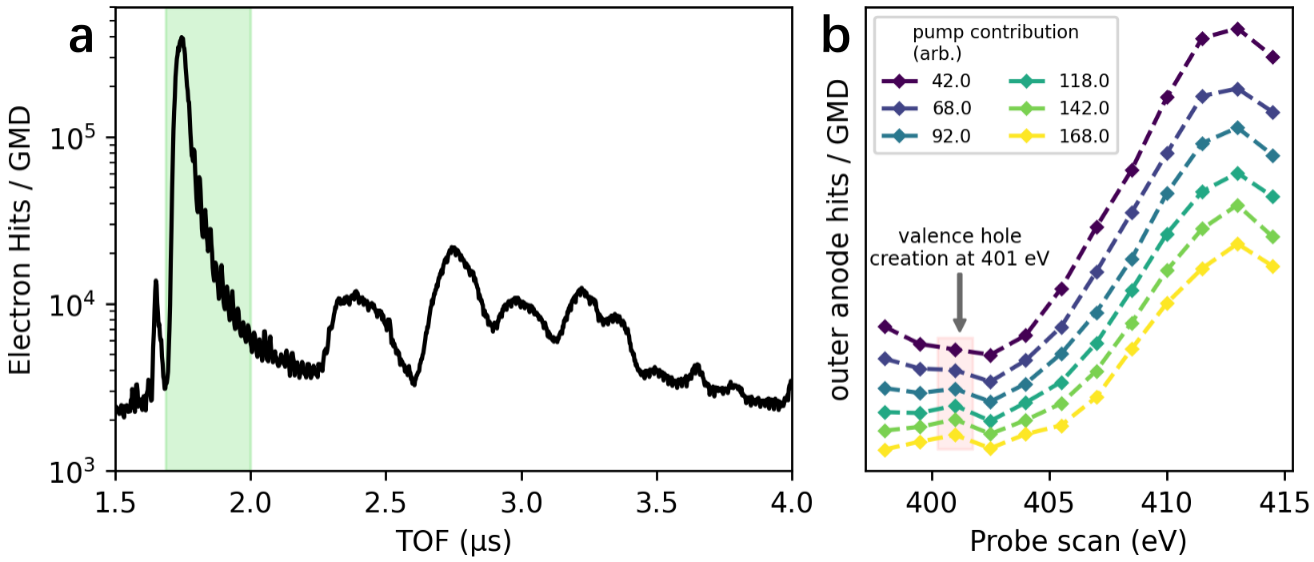

Figure VI.11: Auger-Meitner electron yield XAS recorded in para-aminophenol at the nitrogen K-edge using the $\omega/2\omega$ scheme at LCLS. For this dataset, the pump-probe delay was 900 as. (**a**) Logarithmic electron time of flight spectra collected by the magnetic bottle electron spectrometer. This data is hitfound. The region indicated by the green shading indicates the nitrogen Auger-Meitner peak of interest. (**b**) Electron yield XAS plot over a 17 eV photon energy range. The data is binned according to the pump flux, shown in figure VI.10.

B: At. Mol. Opt. Phys. **49** 152004[24], for two different spectra $\mathbf{X}$ and $\mathbf{Y}$ the covariance matrix is defined as

$$\mathrm{cov}(\mathbf{X},\mathbf{Y}) = \langle(\mathbf{Y}-\langle\mathbf{Y}\rangle)(\mathbf{X}-\langle\mathbf{X}\rangle)\rangle = \langle\mathbf{YX}\rangle - \langle\mathbf{Y}\rangle\langle\mathbf{X}\rangle. \tag{VI.1}$$

Here, $\mathbf{X}$ and $\mathbf{Y}$ are 2D arrays e.g. $\mathbf{X} = [X_1, X_2 \ldots, X_N]$, where X_i = is the ith spectrum and may contain n_x data points.[VI.7] The resulting covariance map has dimensions $n_y \otimes n_x$. Common mode correlations with a third variable I can cause the covariance to over-estimated correlations between variables. If this third variable is measured, the over-estimation may be calculated and subtracted away. This more advanced technique is known as partial covariance and is computed as

$$\mathrm{pcov}(\mathbf{X},\mathbf{Y}) = \mathrm{cov}(\mathbf{X},\mathbf{Y}) - \frac{\mathrm{cov}(\mathbf{Y},I)\times\mathrm{cov}(I,\mathbf{X})}{\mathrm{cov}(I,I)}. \tag{VI.2}$$

This also has dimensions $n_y \otimes n_x$. Note that the self covariance term $\mathrm{cov}(I,I)$ is is given by $\langle I^2\rangle - \langle I\rangle^2$ and can therefore be recognized simply as the variance of I.

Figure VI.13 shows a partial covariance map for the same data presented in figures VI.11

[VI.7] These n_x data points correspond to the time of flight, energy or mass-to-charge ratio for example.

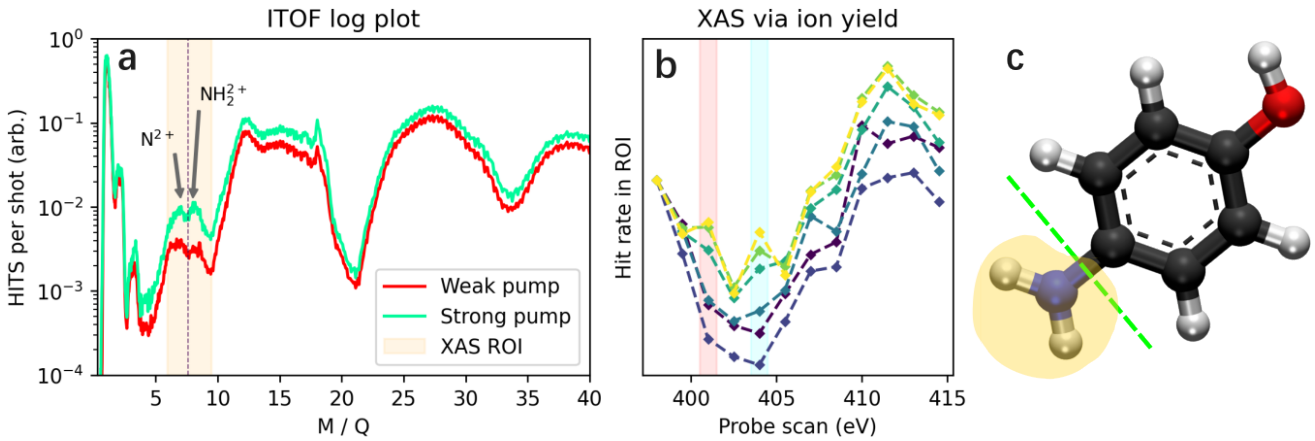

Figure VI.12: Doubly-charged cation yield XAS recorded in para-aminophenol at the nitrogen K-edge using the $\omega/2\omega$ scheme at LCLS. This data was recorded simultaneously with the data shown in figure VI.11 and a largely analogous analysis procedure has been performed. The pump-probe delay here is also 900 as. (**a**) Logarithmic ion time of flight spectra collected by the ion time of flight spectrometer. This data is hit-found. The region indicated by the red shading indicates the doubly charged fragments used to study the XAS. (**b**) Ion yield XAS plot over a 17 eV photon energy range. The data is again binned according to the pump flux, shown in figure VI.10. (**c**) A ball and stick model illustrating the basic layout of the para-aminophenol molecule, adapted from the rendered structure of 4-aminophenol, by Jynto, CC0, *Wikimedia Commons*.[22] The fragmentation leading to the detection of NH_2^{2+} cations is indicated.

and VI.12. The region of the spectrum highlighted in figure VI.11 subplot (**a**) corresponding to the resonant Auger-Meitner emission is indicated. Here, the common mode parameter I used to correct the partial covariance map was the probe flux as measured by the Fresnel zone plate. Due to instabilities in the superconducting XFEL, only one delay point was able to be measured during this experiment.

In figure VI.13, the time-of-flight region corresponding to the nitrogen Auger-Meitner electron emission is indicated by two black dashed lines. The circled features in this region at M/Q = 7-8 and M/Q = 16-17 show positive covariance, indicating a correlated process. This feature can be explained by resonant absorption of the probe due to an electron hole in the valence leading to fragmentation of the molecule. Such a fragmentation process would result in NH_2^{2+} fragments and possibly N^{2+} after further fragmentation. This bond-breaking process is shown in figure VI.12 subplot (**c**).

The times of flight of the ion spectrometer were converted into mass-to-charge ratios in order to allow for identification of specific molecular fragments and their associated branching ratios. This conversion was done by sending argon gas into the interaction region. A progressive ionization series Ar^{1+}, Ar^{2+}, Ar^{3+} etc... was recorded. A highest ionization state of Ar^{7+}

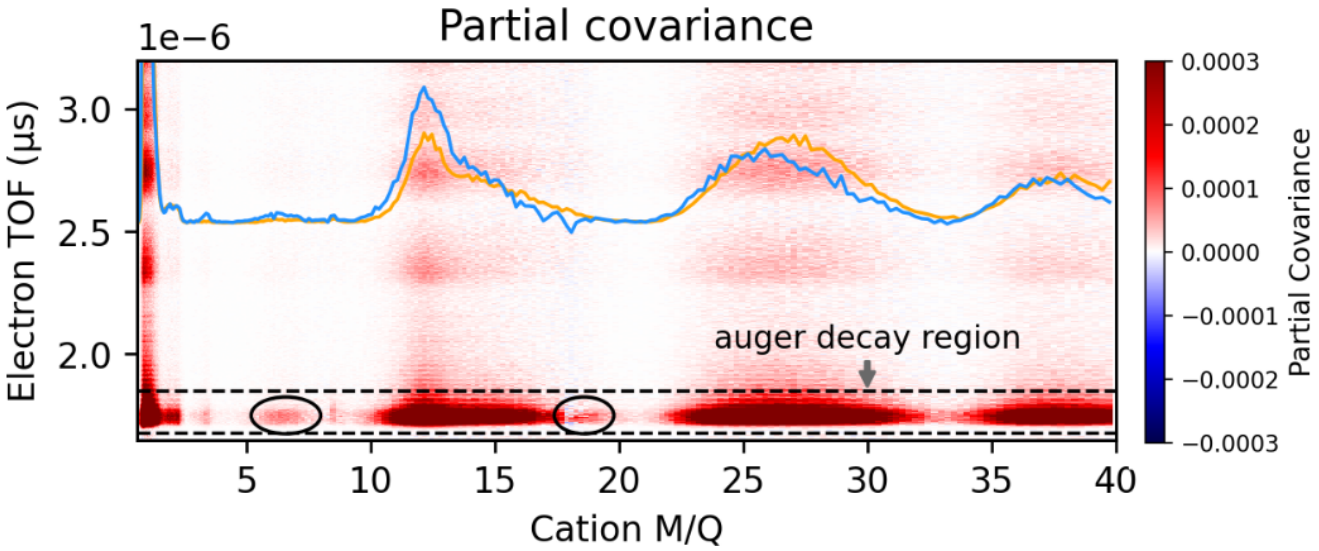

Figure VI.13: False colour map of the partial covariance matrix of ion and electron time of flight spectrometers, calculated according to equation (VI.2). The blue → red colour scale is unbiased, i.e. zero shows as exactly white. The region of the electron spectrum corresponding to nitrogen Auger-Meitner emission is indicated by the horizontal black dashed lines. Lineouts of the map for both high (blue) and low (orange) pump flux are shown. Features indicating a correlation between the nitrogen K-edge Auger-Meitner emission and some resolvable fragments are circled. These are the N^{2+} and NH_2^{2+} features at M/Q = 7-8 and the NH_2^+ and OH^+ at M/Q = 16-17.

was observed with with the highest yield state being Ar^{2+}. As the charge to mass ratio for this series is easily found, a calibration of the ion time of flight spectrometer was found. The result of this procedure is shown in figure IV.10.

Further work using the LCLS $\omega/2\omega$ x-ray pump x-ray probe scheme has since been undertaken in the organic molecules of tryptophan and phenylalanine. This work is not included as analysis for this dataset is ongoing.

VI.3 Time resolved XAS in condensed-phase glycine

In a continuation of the gas-phase work in glycine, another experiment using a similar setup was performed. In this experiment, the target was liquid phase glycine molecules. The sample was maintained at approximately 175°C. The sample was delivered by a flat sheet liquid jet at a rate of 0.3 mL/min. The size of the sheet was measured to be 22 μm×33 μm, with an estimated thickness of less than 5 μm. The x-rays were focused to a spot size 35 μm in diameter. Note that this is significantly larger than the focal size in the experiment described in chapter V. This is because unlike that experiment, the measurement scheme shown in figure VI.9 is always single

photon in each pulse. Therefore lower field intensities in the interaction region may be used.

Single-spike SASE pulses were sent into a split and delay unit (SDU). This aperture splits the incident x-ray beam into two pulses with a controllable time-delay. See figure VI.14. As both the pump and probe pulses originate from the same lightsource, they always have the same photon energy.

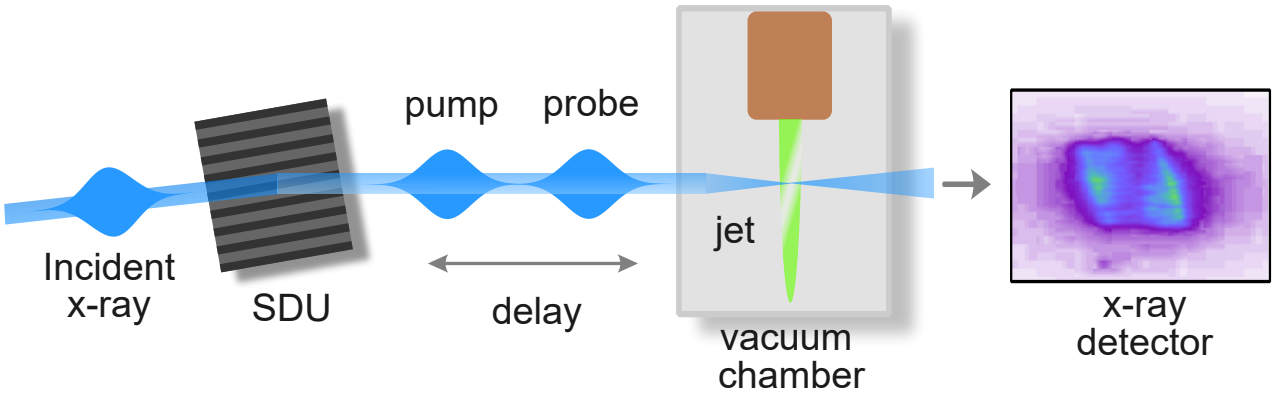

Figure VI.14: A drawing outlining the experimental setup used at FLASH to measure time resolved XAS in liquid phase glycine. The pump and probe pulses are created by the SDU. The delay between these pulses is controllable and was varied to obtain time-resolved data. The sample was provided with a liquid jet nozzle. After passing through the interaction point, the x-rays are recorded via a camera.

The FLASH XFEL generates x-ray pulses in groups of 400 pulses. Each group of 400 pulses is referred to as a bunch train. The facility produces these bunch trains with a repetition rate of up to 10 Hz. Therefore, the effective pulse repetition rate is 4 kHz.

The x-ray detector shown on the right side of figure VI.4 was a 2D camera. This camera was in effect an integrating detector. This is because it is not capable of outputting an x-ray yield on each shot due to data throughput limitations. Instead it was setup to output a 2D image of the x-ray transmission for every bunch train. Hence, each image of the camera corresponded to the total transmitted x-ray yield through the sample of 400 pulses.

However, the pulse energy monitor, which measures the incident photon flux to the interaction region is capable of providing a pulse resolved measurement. Therefore, in order to combine the bunch train resolved transmission measure with the incident flux pulse resolved measure, the pulse energy measure was averaged over each bunch train.

Figure VI.15 shows a correlation map for approximately 10,000 bunch trains. Plotted is a blue heatmap of the average (mean) pulse energy over 10,000 bunch trains versus the total x-ray yield recorded for each exposure period. The alpha (transparency) at each point encodes the number of counts. A clear positive correlation is shown across all incident fluxes. This is

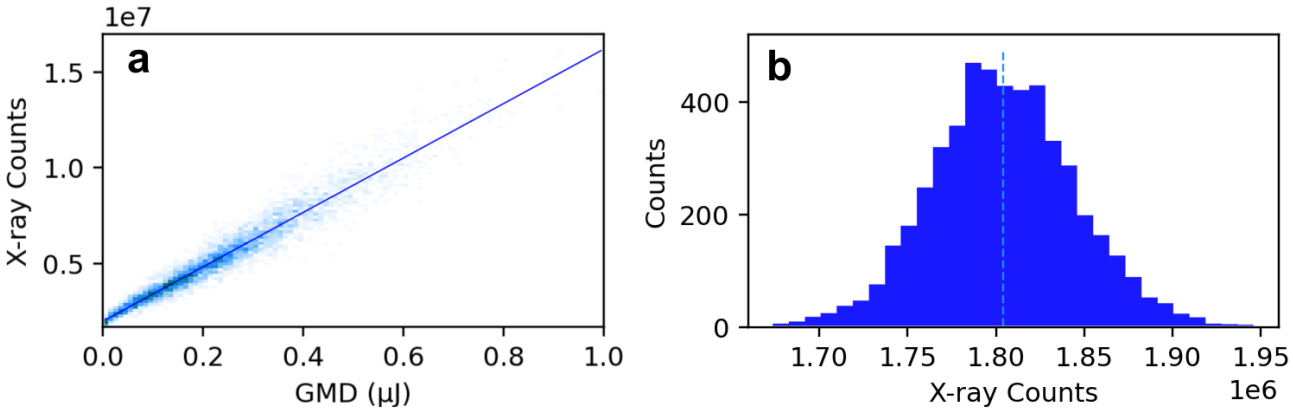

Figure VI.15: Correlation map and statistics for the measured x-ray pulse energy. (a) Correlation map of the pulse energy as measured by the gas monitor detector (GMD) upstream of the interaction point and the number of x-ray counts (scaled by 10^7) recorded on the x-ray detector downstream of the interaction point. A line of best fit is shown in solid blue. (**b**) Distribution of the number of x-ray counts, showing a Gaussian profile. The mean number of counts is indicated by a vertical blue dashed line.

a sign of a good transmission measurement without saturation effects.[VI.8]

In order to ensure the detectors were properly calibrated, the correlation map shown in figure VI.15 subplot (**a**) should pass through the point (0,0). Namely, when no incident photons are detected, there should be no transmitted x-ray counts recorded on the camera. This was found to not be the case in this experiment. In figure VI.15 subplot (**a**), the correlation map shows zero incident photons corresponded to an x-ray yield of approximately 1.67×10^7. This was likely due to dark counts appearing on the camera. These events are treated as false positives and may be caused by thermal noise, leakage currents, electronic fluctuations or stray light being collected by the detector. In order to remove this effect, the pulse energy values were offset from the raw measured values. Explicitly,

$$I_0^{\text{corrected}} = I_0^{\text{raw}} - \text{offset} \propto \text{Detected x-ray counts.} \tag{VI.3}$$

In order to measure transmission XAS in this experiment, a calculation of the optical density must be made. The total x-ray yield for every bunch train corresponds to one 1024×1024 pixel camera image. An example of a camera image is shown on the rightmost side of figure VI.14.

[VI.8] If the correlation were to become worse in the high incident flux regime, this can indicate saturation: i.e. the x-ray detector is overwhelmed by the number of transmitted photons and cannot be sensitive to changes caused by sample absorption in this regime. This is to be avoided, especially during an experiment whose goal is to measure an observable relying on the high flux regime to induce two-photon absorption.

The transmission XAS for a single bunch train is found by a sum over all pixels p_{ij} as

$$I_T(\omega) = \sum_{i,j} p_{ij}(\omega). \tag{VI.4}$$

Cropping the camera image to a restricted region of interest was found to have no impact on the calculated XAS but comprised the signal to noise ratio. Therefore, it was decided to use the entire camera image in the sum. Therefore, using the optical density definition given in equation (III.1) we define the optical density in this experiment as

$$OD(\omega) = \log_{10}\left(\frac{I_0(\omega)}{I_T(\omega)}\right) = \log_{10}\left(\frac{\overline{GMD}(\omega)}{\sum_{i,j} p_{ij}(\omega)}\right). \tag{VI.5}$$

Here, $\overline{GMD}$ denotes the mean pulse-resolved incident flux over a bunch train. During the experiment, the photon energy was varied allowing the measurement of an XAS spectrum.

The observable for this experiment is the absorption of the probe pulse. In the gas-phase experiment, this absorption event was correlated to Auger-Meitner electron emission. In contrast, this condensed phase experiment would use transmission XAS to indicate probe absorption, as electron spectroscopy in the condensed phase is known to suffer from space charge effects[25,26] and bulk interaction effects such as inelastic electron scattering[27] which would add unnecessary difficulties in interpretation.[VI.9]

Additionally, transmission XAS can be very difficult to perform in gas-phase due to the comparatively low number density of sample molecules in the interaction region. This can lead to a very small fraction of photons being absorbed compared to condensed-phase samples, which do not suffer this problem.

Returning to the x-ray pump x-ray probe scheme illustrated in figure VI.5, the absorption of the probe pulse is related to the hole density in the 10a′ state. The $1s \rightarrow 10a'$ resonance is at a lower energy than the carbon K-edge by approximately 10 eV. As the 10a′ state is occupied when the molecule is in its ground state, the probe absorption of interest is also expected to show up at an energy below any pre-edge features.

The entire pump-probe process relies on a two photon absorption event. In this experiment, both the pump and probe pulses originate from a single source. Therefore the cross section

[VI.9] The mean free paths of inelastic scattering in condensed phase are on the order of nanometers.[28] As liquid jets typically are micrometers in thickness, one expected many such inelastic scattering event to occur even in the smallest of interaction regions.

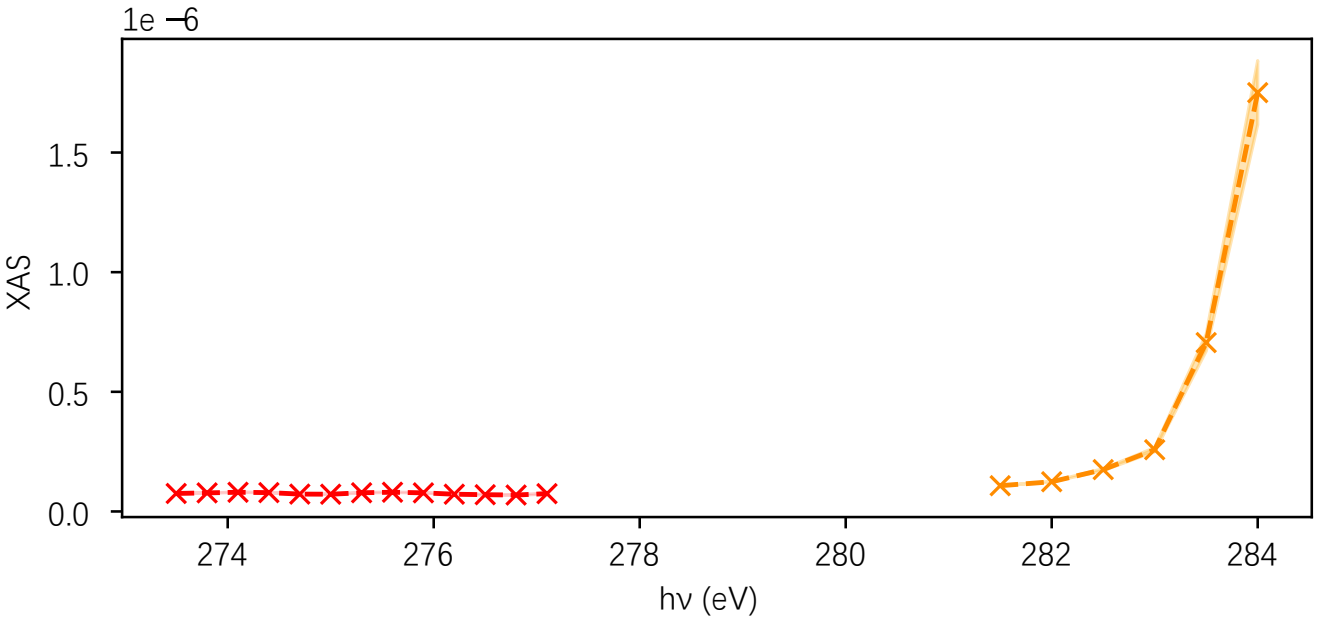

Figure VI.16: Experimental data showing XAS measurements taken in two different parts of the x-ray absorption near edge structure (XANES) region. The optical density calculated by the procedure outlined in equation (VI.5). From 273.5 eV - 277.1 eV in (red crosses) and 281.5 eV - 284 eV (orange crosses). This spectrum is not pulse energy resolved. The initial onset of the carbon K-edge is visible at 283 eV. This is consistent with XAS measurements for similar systems.[29]

of the process will depend nonlinearly on the incident pulse energy of the pump and probe pulses. Therefore the expected absorption signal is expected to grow with increasing incident photon flux.

Figure VI.16 may miss out some pre-edge features corresponding to open excitation pathways from the 1s to an array of unoccupied states, i.e. LUMO, LUMO+1, LUMO+2 etc... These states have been measured by UV-Vis-NIR spectroscopy to sit approximately 1 - 5 eV above the HOMO.[30]

The spectral region corresponding to the opening of valence hole by the pump is 273.5 eV - 277.1 eV. The XAS measurement for this region is shown in detail in figure VI.17. In order to distinguish this feature from other linear effects, the data is grouped by the incident flux.

It is expected that the feature will grow with increasing pulse energy. This is the behaviour shown by two parts of the spectrum. One feature at 274 eV and another at 275.5 eV. These valence-holes are about 1 eV below and above the valence hole seen in the previous gas-phase glycine work respectively.[14] Such changes are to be expected by the difference in intermolecular interactions from gas-phase to condensed-phase[31], especially due to hydrogen-bonding.[32]

Unfortunately, the time-resolved part of this experiment did not yield any reliable oscilla-

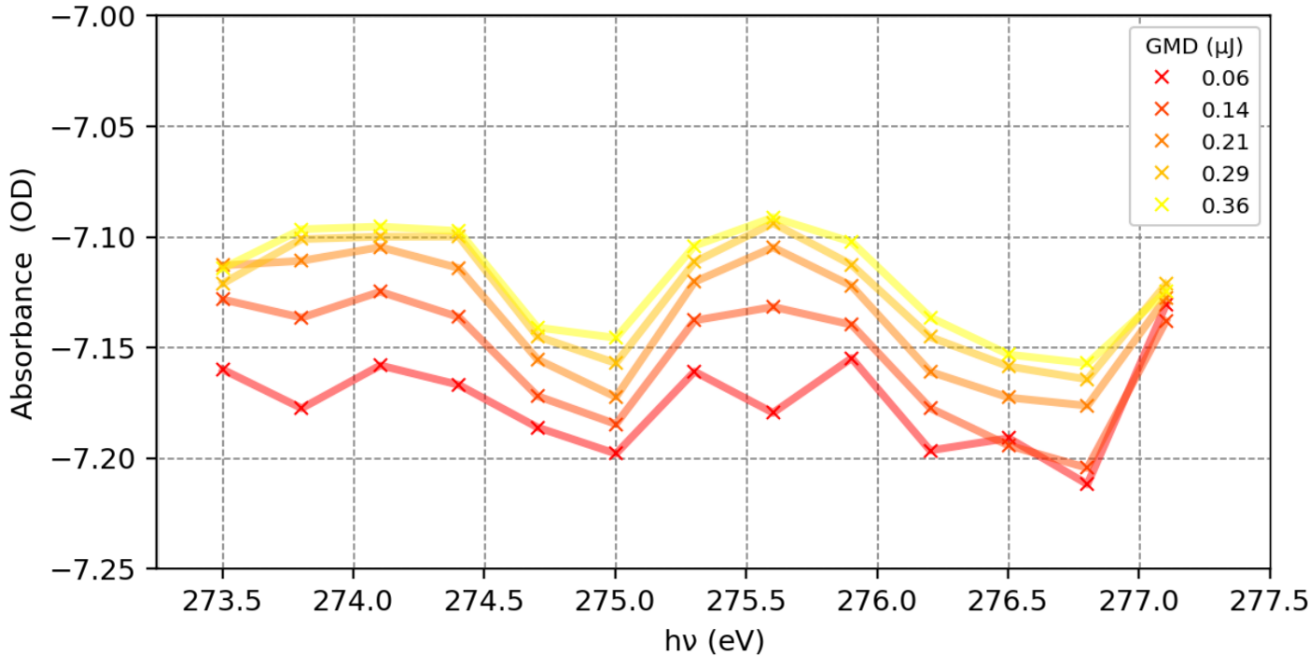

Figure VI.17: Experimental data showing the same XAS measurements taken in figure VI.16 zooming in on the lower photon energy region from 273.5 eV - 277.1 eV. This time, the data is binned according to the measured incident pulse energy, increasing from red to yellow as shown in the legend. The vertical axis shows a log plot of base 10 and shows the same units as figure VI.16 allowing for a direct comparison. Two features consistent with the opening of channels for probe absorption to a freed up valence state are visible at 274 and 275.6 eV.

tions that would be expected of a coherent beating between valence-hole states opened up by the pump pulse. Two delay scan runs taken at the higher energy resonance at 275.6 eV are shown in figure VI.18. The gradual increase in absorption at delays beyond approximately 80 fs is due to partial blocking of the incident beam by the SDU. This reduction in flux then causes an apparent increase in the calculated optical density. (See figure VI.14). This effect occurs when no sample was in the interaction region. It is therefore ruled out from being a relevant dynamical effect.

One effect that could cause this would be XFEL drifts. An improvement of this experiment could attempt to better resolve the incident spectrum. The only current quantification of the incident spectrum is the GMD, which does not provide spectral information about the x-ray light. Instead, an photon spectrometer upstream of the interaction region would allow any drifts in the spectrum to be characterized and removed.

Additionally, the best single-spike SASE pulses could be identified in the frequency domain. This would allow for multi-spike SASE pulse which do not have optimum temporal

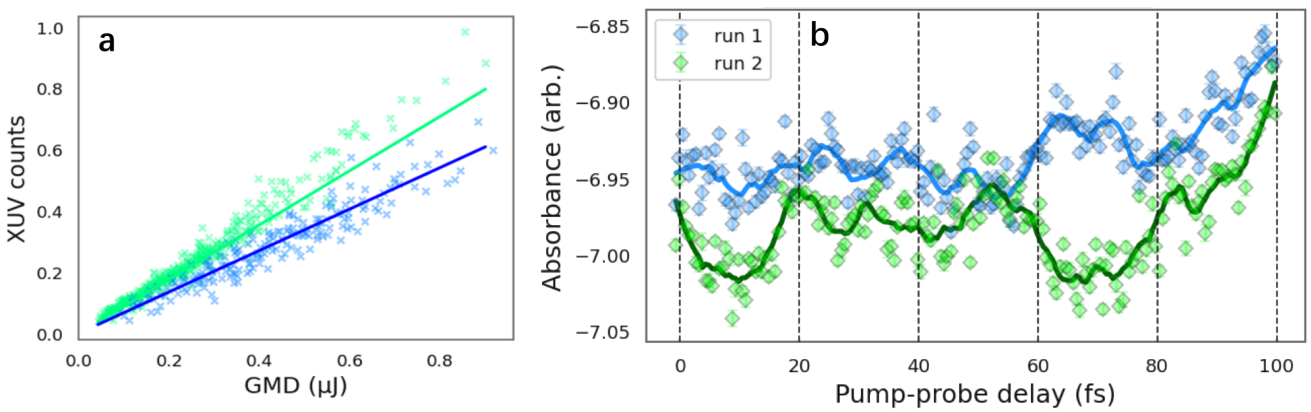

Figure VI.18: Experimental data showing pump-probe delay resolved XAS in liquid glycine. (**a**) Correlation map showing the incident x-ray counts versus the transmitted x-ray counts. Two different pump-probe delays are shown in blue and green. Best fit parameters for the data are also indicated by solid lines. (**b**) The time-resolved data for the first 100 fs. Data from two runs are shown in blue and green which were direct repeats under the same experimental conditions. Uncertainties are shown at the 1σ level and are calculated via bootstrap resampling. Trend lines for this data were found by use of the Savitzky–Golay technique.[33]

durations to be filtered out of the data. This may allow for a potential improvement in the time resolution of the pump-probe delay scan data at the cost of statistics.

Finally, the recording of spectrally resolved transmission data can allow for correlation effects to be measured. These can be studied using techniques such as simple, partial and contingent covariance[24,34] and coincidence detection techniques.[35] These have been used to identify processes in the gas phase and an analogous technique may be used in the condensed phase.[36,37]

References

[1]N. Berrah, J. Cryan, R. Robles, T. Driver, A. Marinelli, and P. Bucksbaum, "Attosecond X-ray sources, methods, and applications at present and future free-electron lasers: tutorial", Advances in Optics and Photonics **17**, 623, ISSN: 1943-8206 (2025) 10.1364/aop.540527.

[2]A. Picón, C. S. Lehmann, C. Bostedt, A. Rudenko, A. Marinelli, T. Osipov, D. Rolles, N. Berrah, C. Bomme, M. Bucher, G. Doumy, B. Erk, K. R. Ferguson, T. Gorkhover, P. J. Ho, E. P. Kanter, B. Krässig, J. Krzywinski, A. A. Lutman, A. M. March, D. Moonshiram, D. Ray, L. Young, S. T. Pratt, and S. H. Southworth, "Hetero-site-specific X-ray pump-probe spectroscopy for femtosecond intramolecular dynamics", Nature Communications **7**, ISSN: 2041-1723 (2016) 10.1038/ncomms11652.

[3]J. Duris, S. Li, T. Driver, E. G. Champenois, J. P. MacArthur, A. A. Lutman, Z. Zhang, P. Rosenberger, J. W. Aldrich, R. Coffee, G. Coslovich, F.-J. Decker, J. M. Glownia, G. Hartmann, W. Helml, A. Kamalov, J. Knurr, J. Krzywinski, M.-F. Lin, J. P. Marangos, M. Nantel, A. Natan, J. T. O'Neal, N. Shivaram, P. Walter, A. L. Wang, J. J. Welch, T. J. A. Wolf, J. Z. Xu, M. F. Kling, P. H. Bucksbaum, A. Zholents, Z. Huang, J. P. Cryan, and A. Marinelli, "Tunable isolated attosecond X-ray pulses with gigawatt peak power from a free-electron laser", Nature Photonics **14**, 30–36, ISSN: 1749-4893 (2019) 10.1038/s41566-019-0549-5.

[4]A. Barty, C. Caleman, A. Aquila, N. Timneanu, L. Lomb, T. A. White, J. Andreasson, D. Arnlund, S. Bajt, T. R. M. Barends, M. Barthelmess, M. J. Bogan, C. Bostedt, J. D. Bozek, R. Coffee, N. Coppola, J. Davidsson, D. P. DePonte, R. B. Doak, T. Ekeberg, V. Elser, S. W. Epp, B. Erk, H. Fleckenstein, L. Foucar, P. Fromme, H. Graafsma, L. Gumprecht, J. Hajdu, C. Y. Hampton, R. Hartmann, A. Hartmann, G. Hauser, H. Hirsemann, P. Holl, M. S. Hunter, L. Johansson, S. Kassemeyer, N. Kimmel, R. A. Kirian, M. Liang, F. R. N. C. Maia, E. Malmerberg, S. Marchesini, A. V. Martin, K. Nass, R. Neutze, C. Reich, D. Rolles, B. Rudek, A. Rudenko, H. Scott, I. Schlichting, J. Schulz, M. M. Seibert, R. L. Shoeman, R. G. Sierra, H. Soltau, J. C. H. Spence, F. Stellato, S. Stern, L. Strüder, J. Ullrich, X. Wang, G. Weidenspointner, U. Weierstall, C. B. Wunderer, and H. N. Chapman, "Self-terminating diffraction gates femtosecond X-ray nanocrystallography measurements", Nature Photonics **6**, 35–40, ISSN: 1749-4893 (2011) 10.1038/nphoton.2011.297.

[5]I. Inoue, Y. Inubushi, T. Sato, K. Tono, T. Katayama, T. Kameshima, K. Ogawa, T. Togashi, S. Owada, Y. Amemiya, T. Tanaka, T. Hara, and M. Yabashi, "Observation of femtosecond X-ray interactions with matter using an X-ray–X-ray pump–probe scheme", Proceedings of the National Academy of Sciences **113**, 1492–1497, ISSN: 1091-6490 (2016) 10.1073/pnas.1516426113.

[6]B. Cooper, P. Kolorenč, L. J. Frasinski, V. Averbukh, and J. P. Marangos, "Analysis of a measurement scheme for ultrafast hole dynamics by few femtosecond resolution X-ray pump–probe Auger spectroscopy", Faraday Discuss. **171**, 93–111, ISSN: 1364-5498 (2014) 10.1039/c4fd00051j.

[7]T. Barillot, O. Alexander, B. Cooper, T. Driver, D. Garratt, S. Li, A. Al Haddad, A. Sanchez-Gonzalez, M. Agåker, C. Arrell, M. J. Bearpark, N. Berrah, C. Bostedt, J. Bozek, C. Brahms, P. H. Bucksbaum, A. Clark, G. Doumy, R. Feifel, L. J. Frasinski, S. Jarosch, A. S. Johnson, L. Kjellsson, P. Kolorenč, Y. Kumagai, E. W. Larsen, P. Matia-Hernando, M. Robb, J.-E. Rubensson, M. Ruberti, C. Sathe, R. J. Squibb, A. Tan, J. W. G. Tisch, M. Vacher, D. J. Walke, T. J. A. Wolf, D. Wood, V. Zhaunerchyk, P. Walter, T. Osipov, A. Marinelli, T. J. Maxwell, R. Coffee, A. A. Lutman, V. Averbukh, K. Ueda, J. P. Cryan, and J. P. Marangos, "Correlation-Driven Transient Hole Dynamics Resolved in Space and Time in the Isopropanol Molecule", Physical Review X **11**, ISSN: 2160-3308 (2021) 10.1103/physrevx.11.031048.

[8]A. A. Lutman, M. W. Guetg, T. J. Maxwell, J. P. MacArthur, Y. Ding, C. Emma, J. Krzywinski, A. Marinelli, and Z. Huang, "High-Power Femtosecond Soft X Rays from Fresh-Slice Multistage Free-Electron Lasers", Physical Review Letters **120**, ISSN: 1079-7114 (2018) 10.1103/physrevlett.120.264801.

[9]V. Averbukh, U. Saalmann, and J. M. Rost, "Suppression of Exponential Electronic Decay in a Charged Environment", Physical Review Letters **104**, ISSN: 1079-7114 (2010) 10.1103/physrevlett.104.233002.

[10]P. S. Bagus, R. Broer, and E. S. Ilton, "A new near degeneracy effect for photoemission in transition metals", Chemical Physics Letters **394**, 150–154, ISSN: 0009-2614 (2004) 10.1016/j.cplett.2004.06.120.

[11]J. Tarrant, M. Khokhlova, and V. Averbukh, "Interferometry of quantum revivals", The Journal of Chemical Physics **157**, ISSN: 1089-7690 (2022) 10.1063/5.0098056.

[12]J. Rosenzweig, D. Alesini, G. Andonian, M. Boscolo, M. Dunning, L. Faillace, M. Ferrario, A. Fukusawa, L. Giannessi, E. Hemsing, G. Marcus, A. Marinelli, P. Musumeci, B. O'Shea, L. Palumbo, C. Pellegrini, V. Petrillo, S. Reiche, C. Ronsivalle, B. Spataro, and C. Vaccarezza, "Generation of ultra-short, high brightness electron beams for single-spike SASE FEL operation", Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment **593**, 39–44, ISSN: 0168-9002 (2008) 10.1016/j.nima.2008.04.083.

[13]S. Roling, M. Rollnik, M. Kuhlmann, E. Plönjes, F. Wahlert, and H. Zacharias, "A soft x-ray split-and-delay unit for FLASH II", in Advances in X-ray Free-Electron Lasers Instrumentation IV, Vol. 10237, edited by T. Tschentscher and L. Patthey (June 2017), p. 1023712, 10.1117/12.2265708.

[14]D. Schwickert, M. Ruberti, P. Kolorenč, S. Usenko, A. Przystawik, K. Baev, I. Baev, M. Braune, L. Bocklage, M. K. Czwalinna, S. Deinert, S. Düsterer, A. Hans, G. Hartmann, C. Haunhorst, M. Kuhlmann, S. Palutke, R. Röhlsberger, J. Rönsch-Schulenburg, P. Schmidt, S. Toleikis, J. Viefhaus, M. Martins, A. Knie, D. Kip, V. Averbukh, J. P. Marangos, and T. Laarmann, "Electronic quantum coherence in glycine molecules probed with ultrashort x-ray pulses in real time", Science Advances **8**, ISSN: 2375-2548 (2022) 10.1126/sciadv.abn6848.

[15]O. Plekan, V. Feyer, R. Richter, M. Coreno, M. de Simone, K. C. Prince, and V. Carravetta, "Investigation of the Amino Acids Glycine, Proline, and Methionine by Photoemission Spectroscopy", The Journal of Physical Chemistry A **111**, 10998–11005, ISSN: 1520-5215 (2007) 10.1021/jp075384v.

[16]A. Sanchez-Gonzalez, T. R. Barillot, R. J. Squibb, P. Kolorenč, M. Agaker, V. Averbukh, M. J. Bearpark, C. Bostedt, J. D. Bozek, S. Bruce, S. Montero, S. Carron, R. N. Coffee, B. Cooper, J. P. Cryan, M. Dong, J. H. D. Eland, L. Fang, H. Fukuzawa, M. Guehr, M. Ilchen, A. S. Johnsson, C. Liekhus-S, A. Marinelli, T. Maxwell, K. Motomura, M. Mucke, A. Natan, T. Osipov, C. Östlin, M. Pernpointner, V. S. Petrovic, M. A. Robb, C. Sathe, E. R. Simpson, J. G. Underwood, M. Vacher, D. J. Walke, T. J. A. Wolf, V. Zhaunerchyk, J.-E. Rubensson, N. Berrah, P. H. Bucksbaum, K. Ueda, R. Feifel, L. J. Frasinski, and J. P. Marangos, "Auger electron and photoabsorption spectra of glycine in the vicinity of the oxygen K-edge measured with an

X-FEL", Journal of Physics B: Atomic, Molecular and Optical Physics **48**, 234004, ISSN: 1361-6455 (2015) 10.1088/0953-4075/48/23/234004.

[17]J. M. Lilly and S. C. Olhede, "Generalized Morse Wavelets as a Superfamily of Analytic Wavelets", IEEE Transactions on Signal Processing **60**, 6036–6041, ISSN: 1941-0476 (2012) 10.1109/tsp.2012.2210890.

[18]L. Kralj, M. Hultman, and H. Lenasi, "Wavelet Analysis and the Cone of Influence: Does the Cone of Influence Impact Wavelet Analysis Results?", Applied Sciences **14**, 11736, ISSN: 2076-3417 (2024) 10.3390/app142411736.

[19]F. Calegari, D. Ayuso, A. Trabattoni, L. Belshaw, S. De Camillis, S. Anumula, F. Frassetto, L. Poletto, A. Palacios, P. Decleva, J. B. Greenwood, F. Martín, and M. Nisoli, "Ultrafast electron dynamics in phenylalanine initiated by attosecond pulses", Science **346**, 336–339, ISSN: 1095-9203 (2014) 10.1126/science.1254061.

[20]P. Walter, T. Osipov, M.-F. Lin, J. Cryan, T. Driver, A. Kamalov, A. Marinelli, J. Robinson, M. H. Seaberg, T. J. A. Wolf, J. Aldrich, N. Brown, E. G. Champenois, X. Cheng, D. Cocco, A. Conder, I. Curiel, A. Egger, J. M. Glownia, P. Heimann, M. Holmes, T. Johnson, L. Lee, X. Li, S. Moeller, D. S. Morton, M. L. Ng, K. Ninh, J. T. O'Neal, R. Obaid, A. Pai, W. Schlotter, J. Shepard, N. Shivaram, P. Stefan, X. Van, A. L. Wang, H. Wang, J. Yin, S. Yunus, D. Fritz, J. James, and J.-C. Castagna, "The time-resolved atomic, molecular and optical science instrument at the Linac Coherent Light Source", Journal of Synchrotron Radiation **29**, 957–968, ISSN: 1600-5775 (2022) 10.1107/s1600577522004283.

[21]Z. Guo, T. Driver, S. Beauvarlet, D. Cesar, J. Duris, P. L. Franz, O. Alexander, D. Bohler, C. Bostedt, V. Averbukh, X. Cheng, L. F. DiMauro, G. Doumy, R. Forbes, O. Gessner, J. M. Glownia, E. Isele, A. Kamalov, K. A. Larsen, S. Li, X. Li, M.-F. Lin, G. A. McCracken, R. Obaid, J. T. O'Neal, R. R. Robles, D. Rolles, M. Ruberti, A. Rudenko, D. S. Slaughter, N. S. Sudar, E. Thierstein, D. Tuthill, K. Ueda, E. Wang, A. L. Wang, J. Wang, T. Weber, T. J. A. Wolf, L. Young, Z. Zhang, P. H. Bucksbaum, J. P. Marangos, M. F. Kling, Z. Huang, P. Walter, L. Inhester, N. Berrah, J. P. Cryan, and A. Marinelli, "Experimental demonstration of attosecond pump–probe spectroscopy with an X-ray free-electron laser", Nature Photonics **18**, 691–697, ISSN: 1749-4893 (2024) 10.1038/s41566-024-01419-w.

[22]Jynto, *4-Aminophenol 3D balls model*, Image released under CC0 1.0 Universal (Public Domain Dedication), Wikimedia Commons, (2013) https://commons.wikimedia.org/wiki/File:4-Aminophenol-3D-balls.png.

[23]B. E. Huitema, *The Analysis of Covariance and Alternatives: Statistical Methods for Experiments, Quasi-Experiments, and Single-Case Studies* (Wiley, Oct. 2011), ISBN: 9781118067475, 10.1002/9781118067475.

[24]L. J. Frasinski, "Covariance mapping techniques", Journal of Physics B: Atomic, Molecular and Optical Physics **49**, 152004, ISSN: 1361-6455 (2016) 10.1088/0953-4075/49/15/152004.

[25]G. Greco, A. Verna, F. Offi, and G. Stefani, "Space-charge effect in electron time-of-flight analyzer for high-energy photoemission spectroscopy", Journal of Electron Spectroscopy and Related Phenomena **212**, 86–93, ISSN: 0368-2048 (2016) 10.1016/j.elspec.2016.09.004.

[26]A. Verna, G. Greco, V. Lollobrigida, F. Offi, and G. Stefani, "Space-charge effects in high-energy photoemission", Journal of Electron Spectroscopy and Related Phenomena **209**, 14–25, ISSN: 0368-2048 (2016) 10.1016/j.elspec.2016.03.001.

[27]S. Tanuma and C. J. Powell, "Inelastic scattering of electrons in solids", in *International Tables for Crystallography* (International Union of Crystallography, Nov. 2022), pp. 100–106, ISBN: 9781119433941, 10.1107/s1574870722001586.

[28]K. Iakoubovskii, K. Mitsuishi, Y. Nakayama, and K. Furuya, "Mean free path of inelastic electron scattering in elemental solids and oxides using transmission electron microscopy: Atomic number dependent oscillatory behavior", Physical Review B **77**, ISSN: 1550-235X (2008) 10.1103/physrevb.77.104102.

[29]B. M. Messer, C. D. Cappa, J. D. Smith, K. R. Wilson, M. K. Gilles, R. C. Cohen, and R. J. Saykally, "pH Dependence of the Electronic Structure of Glycine", The Journal of Physical Chemistry B **109**, 5375–5382, ISSN: 1520-5207 (2005) 10.1021/jp0457592.

[30]T. Balakrishnan, R. R. Babu, and K. Ramamurthi, "Growth, structural, optical and thermal properties of γ-glycine crystal", Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy **69**, 1114–1118, ISSN: 1386-1425 (2008) 10.1016/j.saa.2007.06.025.

[31]R. A. Cormanich, L. C. Ducati, and R. Rittner, "Are hydrogen bonds responsible for glycine conformational preferences?", Chemical Physics **387**, 85–91, ISSN: 0301-0104 (2011) 10.1016/j.chemphys.2011.07.006.

[32]D. Valverde, Z. M. da Costa Ludwig, C. R. da Costa, V. Ludwig, and H. C. Georg, "Zwitterionization of glycine in water environment: Stabilization mechanism and NMR spectral signatures", The Journal of Chemical Physics **148**, ISSN: 1089-7690 (2018) 10.1063/1.5006645.

[33]A. Savitzky and M. J. E. Golay, "Smoothing and Differentiation of Data by Simplified Least Squares Procedures.", Analytical Chemistry **36**, 1627–1639, ISSN: 1520-6882 (1964) 10.1021/ac60214a047.

[34]O. Kornilov, M. Eckstein, M. Rosenblatt, C. P. Schulz, K. Motomura, A. Rouzée, J. Klei, L. Foucar, M. Siano, A. Lübcke, F. Schapper, P. Johnsson, D. M. P. Holland, T. Schlathölter, T. Marchenko, S. Düsterer, K. Ueda, M. J. J. Vrakking, and L. J. Frasinski, "Coulomb explosion of diatomic molecules in intense XUV fields mapped by partial covariance", Journal of Physics B: Atomic, Molecular and Optical Physics **46**, 164028, ISSN: 1361-6455 (2013) 10.1088/0953-4075/46/16/164028.

[35]C. Cornaggia, "Statistical analysis of fragmentation channels of small multicharged molecular ions", Journal of Physics B: Atomic, Molecular and Optical Physics **45**, 085602, ISSN: 1361-6455 (2012) 10.1088/0953-4075/45/8/085602.

[36]L. J. Frasinski, V. Zhaunerchyk, M. Mucke, R. J. Squibb, M. Siano, J. H. D. Eland, P. Linusson, P. v.d. Meulen, P. Salén, R. D. Thomas, M. Larsson, L. Foucar, J. Ullrich, K. Motomura, S. Mondal, K. Ueda, T. Osipov, L. Fang, B. F. Murphy, N. Berrah, C. Bostedt, J. D. Bozek, S. Schorb, M. Messerschmidt, J. M. Glownia, J. P. Cryan, R. N. Coffee, O. Takahashi, S. Wada, M. N. Piancastelli, R. Richter, K. C. Prince, and R. Feifel, "Dynamics of Hollow Atom Formation in Intense X-Ray Pulses Probed by Partial Covariance Mapping", Physical Review Letters **111**, ISSN: 1079-7114 (2013) 10.1103/physrevlett.111.073002.

[37]K. Li, C. Ott, M. Agåker, P. J. Ho, G. Doumy, A. Magunia, M. Rebholz, M. Simon, T. Mazza, A. De Fanis, T. M. Baumann, J. Montano, N. Rennhack, S. Usenko, Y. Ovcharenko, K. Chordiya, L. Cheng, J.-E. Rubensson, M. Meyer, T. Pfeifer, M. B. Gaarde, and L. Young, "Super-resolution

stimulated X-ray Raman spectroscopy", Nature **643**, 662–668, ISSN: 1476-4687 (2025) 10.1038/s41586-025-09214-5.

Conclusion and Future Work

This thesis details two ways to observe ultrafast valence electron dynamics in small molecules using attosecond x-ray measurements. In chapter V, Impulsive Stimulated x-ray Raman scattering (ISXRS) is introduced. This two-photon, intensity dependent process is capable of preparing a molecule in an initially well localized valence-excited state. This is a first step towards photon-only observation of valence electron dynamics in neutral systems. Evidence supporting the successful demonstration of this scattering process is given with the accompanying theoretical work by collaborators.

The experiment differed from previous experiments in the nature of its observable. Previous experiments to demonstrate this effect used nuclear observables (cation yield) to identify the process. The work presented here uses the modifications to the transmitted photon field as the observable. This has the advantage of being generalizable to a wide variety of systems in the condensed phase.

It is hoped that this concept can permit the creation of valence-excited states in more complex systems, such as solution phase and transition metal complexes. Additionally, it may be possible to observe ISXRS when detuned from the core-excited intermediate state. This off resonance variant could be mediated by a virtual state, and would have the advantage of not driving population into a core-excited state.

Two other experiments presented in this work used an x-ray pump x-ray probe scheme to probe electron hole migration. These are described in chapter VI One used transmission x-ray absorption spectroscopy to demonstrate the opening up of valence holes below the carbon pre-K edge in liquid phase glycine. Another experiment demonstrated the preparation of valence-hole states by the resonant Auger-Meitner process in gas-phase para-aminophenol at the nitrogen atomic site. Both of these methods rely on photoionization to introduce a time-dependent perturbation, meaning that the valence-hole states produced were in cations.

These particular experiments were somewhat limited. A robust time-dependent absorption signal in the liquid-phase glycine work could not be established, likely due to large drifts in the XFEL that were not characterized. This problem could be solved by spectrally resolving the incident light with an x-ray spectrometer, allowing such drifts to be measured and their effects mitigated.

The LCLS work contained only one delay point due to instabilities in the relatively new high-repetition rate superconducting soft x-ray line of the XFEL at the time. However, since this experiment, much more successful experiments have been made there. As the analysis is still at an early stage, it has not been included in this work.

This thesis demonstrates that perturbations may be successfully introduced in the valence states of molecular systems, and that these perturbations can be detected via the two spectroscopic techniques detailed in chapters V and VI. The natural next step is to use these techniques to observe time-dependent electron migration. It is hoped that from this, insights into vibronic couplings, bond breaking / formation and other complex chemical processes that can be influenced by electron migration can be understood in terms of electron dynamics.